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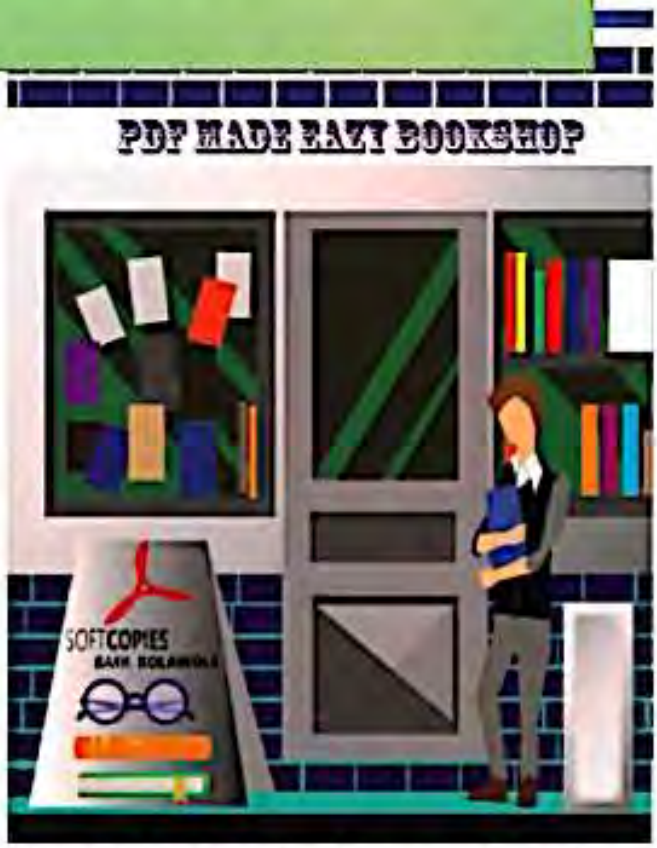

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Biology

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PREFACE

This book is written to assist candidates who are preparing for the SSCE, GCE O/L and JME examinations. The book has a lot of illustrations where necessary and it is written in simple and straightforward language. Efforts have been made to cover the syllabuses in great detail.

Candidates are provided with useful guide and practice exercises so as to attain success. A lot of benefit can be derived from this book by reading the topics carefully be fore attempting the questions. Attempts should be made to answer the questions first before checking the answers.

Every chapter has been provided with questions, answers and explanations to the questions. The answers provided to the questions have been compiled based on experience. it is hoped that through conscientious preparation, combined with through revision of the questions and answers, candidates will achieve the desired success.

The publishers are grateful to Mrs. . B. O. Ijasan, Principal, Federal Government College, Sokoto who painstakingly reviewed the original manuscript and offered many useful suggestions.

G. A. O. Arawomo
A. I. Akinpelu

1 Concept of Living

1.1 Science, Biology and Life

- Science is a process by which we can find out about our world while **biology** is the science that deals with the study of living things.
- **Living things and non-living things** are the two main constituents of everything in the world.

1.2 Characteristics of Living Things

- **Structure:** All living organisms are composed of cells. These can either be unicellular as in *Amoeba* or multicellalar as in man.
- **Form and Sizes:** Living organisms have definite forms and sizes. Hence a chicken in Nigeria has the same and similar form and size as another chicken in Britain.
- **Respiration:** Living things take in oxygen by breathing to break down food and release energy for other life processes.
- **Nutrition:** Living things take in food from which energy can be obtained through respiration.
- **Movement:** All living things move either by themselves as in animals or by wind, water or man as in plants.
- **Growth:** As living things take in food, they increase in size and complexity by incorporating products of the digested food into their bodies.
- **Excretion:** All living organisms must have a means by which they can remove or eliminate unwanted waste products of metabolism from the body.
- **Secretion:** Different parts of the body of living organisms produce useful liquid substances. e.g. In human beings the salivary glands produce saliva which contains an enzyme, ptyalin, while in plants, the stem and root apices of maize plant produce plant hormones, auxins.
- **Reproduction:** Organisms on attaining sexual maturity reproduce their own kind. This can be asexually as in binary fission of *Amoeba* or sexually as in human beings.
- **Irritability:** This is the ability of living things to respond to stimulus. Part of the body of an organism can respond to stimulus as in phototropism in green plants, or the whole body of an organism responding to stimulus as in taxism in animals or some plants.

1.3 Classification of Living Things

- All living things are generally classified into two major kingdoms: plant and animal kingdoms.
- Recent classification divides living organisms into two super kingdoms prokaryote and eukaryote.
- Superkingdom Prokaryote contain only one kingdom: Monera e.g. blue-green algae — *Nostoc;* bacteria — *Bacilli.*
- Superkingdom Eukaryote is subdivided into four kingdoms:
  - (i) Protista e.g. Unicellular green algae — *Euglena*
  - (ii) Protozoa — *Amoeba*
  - (iii) Plantae e.g. higher plants — *Elais guineesis* (oil palm tree)
  - (iv) Fungi e.g. toadstool, *Amania* sp
  - (v) Animalia e.g. worm, *Ascaris* sp, toad, *Bufo* sp
- Kingdom Plantae:
  - (i) Algae e.g. sea weeds, *Spirogyra*
  - (ii) Moses e.g. *Octoblepharum*
  - (iii) Ferns e.g. *Lygodium*
  - (iv) Gymnosperms e.g. Cycads, *Encephalartos*
  - (v) Anglosperms e.g. *maize, Zea mays*
- Kingdom Animalia
  - (a) Invertebrates:
    - (i) Porifera e.g. sponges
    - (ii) Cnidarians e.g. *Hydra*
    - (iii) Flatworms e.g. tapeworm, *Taenia*
    - (iv) Roundworms e.g. *Ascaris*
    - (v) Annelida e.g. earthworm, *Libyodrilus*
    - (vi) Arthropoda e.g. insects: *Anopheles;* crayfish
    - (vii) Mollusca e.g. snail, *Archachatina*
    - (viii) Echinodermata e.g. starfish, *Asterias*
  - (b) Chordates:
    - (i) Protochordates e.g. *Amphioxus*
    - (ii) Vertebrates:
      Pisces e.g. fishes: *Tilapia*
      Amphibia e.g. toads: *Bufo*
      Reptilia e.g. lizards: *Agama*
      Aves e.g. birds, *Gallus*
      Mammalia e.g. rat, man, *Rattus, Homo*

1.4 Differences between Plants and Animals

	Plants	Animals
(i)	They have chlorophyll.	They do not have chlorophyll.
(ii)	They are autotrophic i.e. they feed themselves through photosynthesis.	They are heterotrophic i.e. they depend on plants and other animals for food.
(iii)	Growth is usually terminal and indefinite.	Growth usually occurs at various parts of the body and can stop growing after some time.
(iv)	They have branched bodies and, usually, organs are external.	They have compact bodies and, usually, organs are internal.
(v)	They cannot move from one place to another by themselves.	They can move from one place to another by themselves.
(vi)	They respond to stimuli at the root and shoot apices only.	They respond to stimuli at all parts of the body.

1.5 Organization of life

(a) Levels of organization

(i) **Cell** is the smallest fundamental building unit of the body of living organisms, either plants or animals. The body of man or of a flowering plant is made up of numerous cells. However single-celled organisms such as *Amoeba* are made up of only a single cell.

(ii) **Tissue** is a group of cells that have similar structures and perform a particular function e.g. bone, phloem, blood of man.

(iii) **Organ** consists of several tissues grouped together to make a functional unit. e.g. stem, leaf, eye, heart.

(iv) **System** is a series of organs whose functions are co-ordinated to produce an effective action in an organism e.g. The circulatory system of man comprises the heart, arteries, veins and capillaries.

(v) **Organisms** result from the efficient co-ordination of the organs and systems to produce an individual capable of separate existence and able to perpetuate its own kind e.g. man, goat, mango plant.

In unicellular organisms such as Amoeba, their organization involves not organs but organelles which are all within the single cell.

(b) Complexity of Organization in higher organisms

- In the higher organisms, the variety of tissues and organs found within them are responsible for their efficiency.
- The specialization of constituent organs in the process of expelling nitrogeneous wastes in man, such as the kidney tubules make the excretory system highly efficient during filteration and re-absorption.
- Higher organisms have been successful because they have been able to adapt to changes within their environment.

Such changes can be advantageous but where such changes are too drastic, they may be disadvantageous, e.g. when the environment of a rabbit becomes flooded over a long period of time or when the habitat of a forest tree suddenly becomes dry and less humid, these may be lethal to the organisms.

1.6 Cell Structure and Functions of Cell Components

(a) The Cell Structure

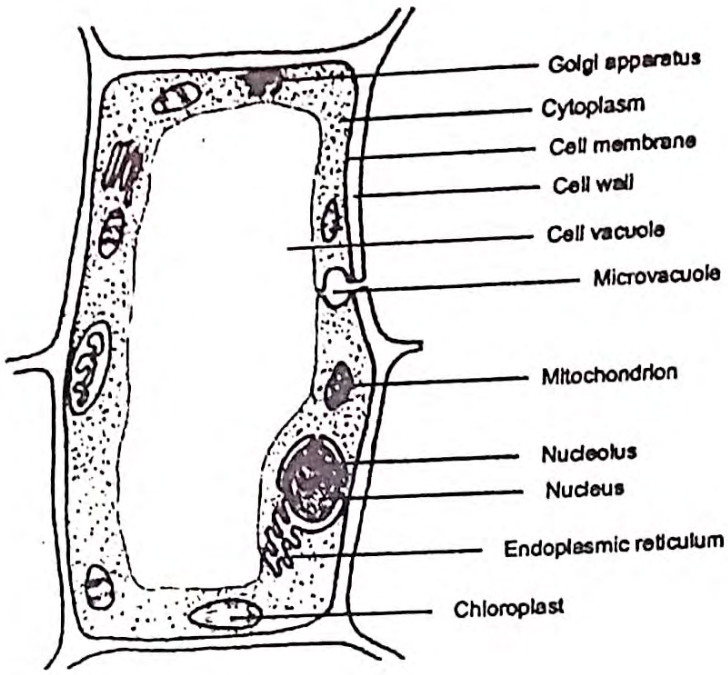
- A cell is a 'single unit' of a living organism. Since both plants and animals differ from one another, the cells that compose the body of a plant or an animal must be different from each other in certain respects.
- Plant cells are of different types and sizes. Most plant cells have regular outlines and fixed shapes. They may be cylindrical, polygonal or rectangular in shape.

The ultra structures of a typical plant cell and an animal cell are shown in Fig: 1.1.

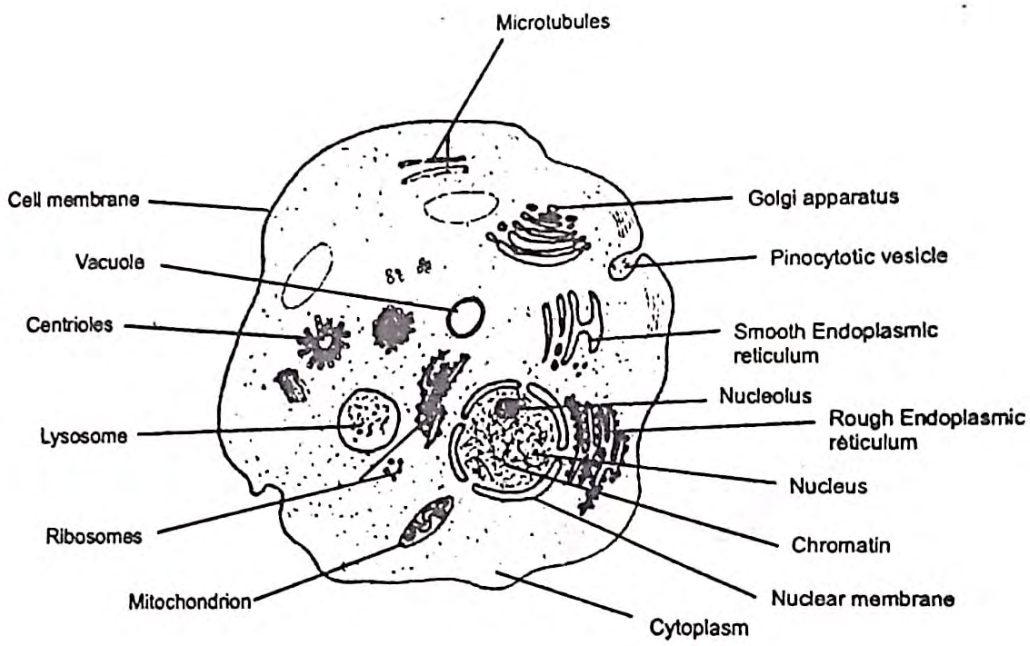
- Animal cells, like plant cells, are of different shapes and sizes. They may be spherical, rectangular, spindle-shaped, cuboidal or irregular in shape as shown in Fig. 1.1b.

(b) Similarities and differences between plant and animal cells

- **Similarities:**
  - They both have cell membranes

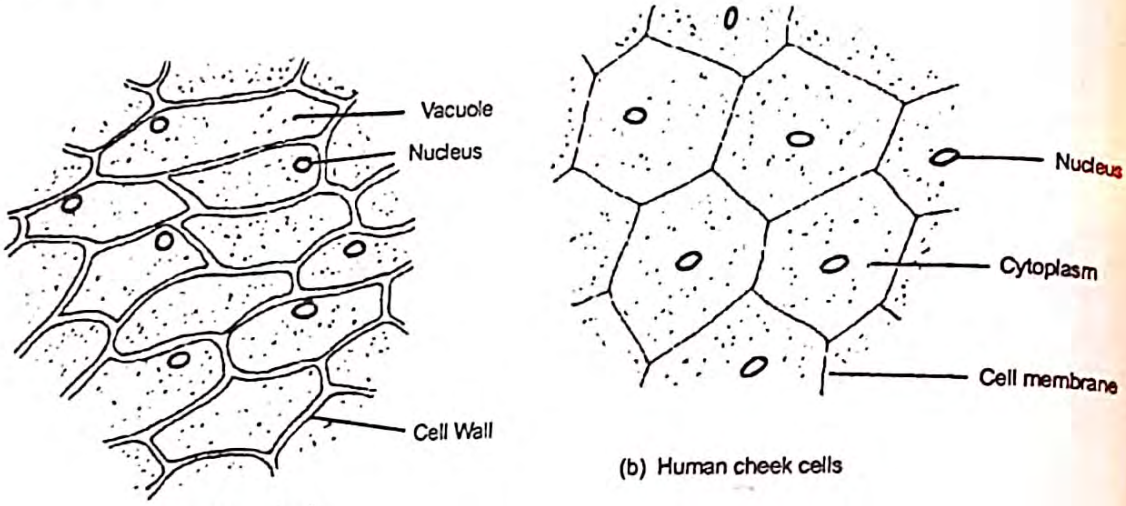


(a)



(b)

Fig. 1.1: Plant cell (a) and animal cell (b) showing the various organelles



(a) Onion epidermal cells

(b) Human cheek cells

Fig. 1.2: Epidermal cells

- They have nuclei.
- They have cytoplasm.
- They both have cytoplasm with high proportions of water mixed with chemical compounds.
- They both undertake cell division.

- **Differences:**
  - Plant cells are usually larger than animal cells.
  - Plant cells have cellulose cell walls which are absent in animal cells (Fig. 1.2).
  - Plant cells have definite shapes but animal cells have none.
  - Plant cells contain conspicuous vacuoles of varying sizes with cell sap but these are usually absent in animal cells and, where present, are usually small and many.
  - Plant cells contain plastids including chlorophyll but these are absent in animal cells.
  - Plant cells have a more watery protoplasm than the protoplasm of animal cells.
  - Plants cells have carbohydrates stored as starch granules but as glycogen in animal cells.

1.7 Forms in Which Living Cells Exist

- Living cells are made up of one or more cells. In many cases, a single cell is the living organism. The Amoeba found in freshwater is an example (Fig. 1.3a). It is a cell of about 300 μm across and it houses the machinery with which it carries out all the functions of life. It feeds itself and exchanges matter and energy with its environment. It responds to stimuli in its environment. It grows and reproduces. It moves by means of finger-like projections of the body wall called pseudopodia.
- The **Staphylococcus** is a spherical cell about 1 um in diameter. It is far too small to be seen without the high magnification of a microscope. Nevertheless, it contains within its single cell all the machinery of life.
- **Mycoplasma** are even smaller and can only be seen with the aid of an electron microscope. They are still living things.
- *Chlamydomonas* (Fig. 1.3b) which is a microscopic green alga found in fresh water is a single-celled organism. Within the single cell are found all organelles required to carry out the various functions of life. It reproduces asexually by producing eight or sixteen daughter cells and sexually by the fusion of two individuals called gametes to form zygote. From the zygote (zygospore) four individuals are produced.
- *Euglena viridis* (Fig. 1.3c) is another microscopic organism found in freshwaters. The spindle-shaped body is broad in the middle. The presence of a gullet, flagellum, eye-spot (photo tactic), contractile vacuole, pellicle and myonemes often qualifies *Euglena* as an animal. Equally the presence of stellate chloroplast and paramylum granules also qualifies it as a plant. It is therefore capable of independent existence.
- Among the flagellated green algae are some colonial forms like *Volvox* (Fig. 1.3e). It is a colonial form because it is made up of clusters of independent cells. The cells of *Volvox* have lost their ability to live

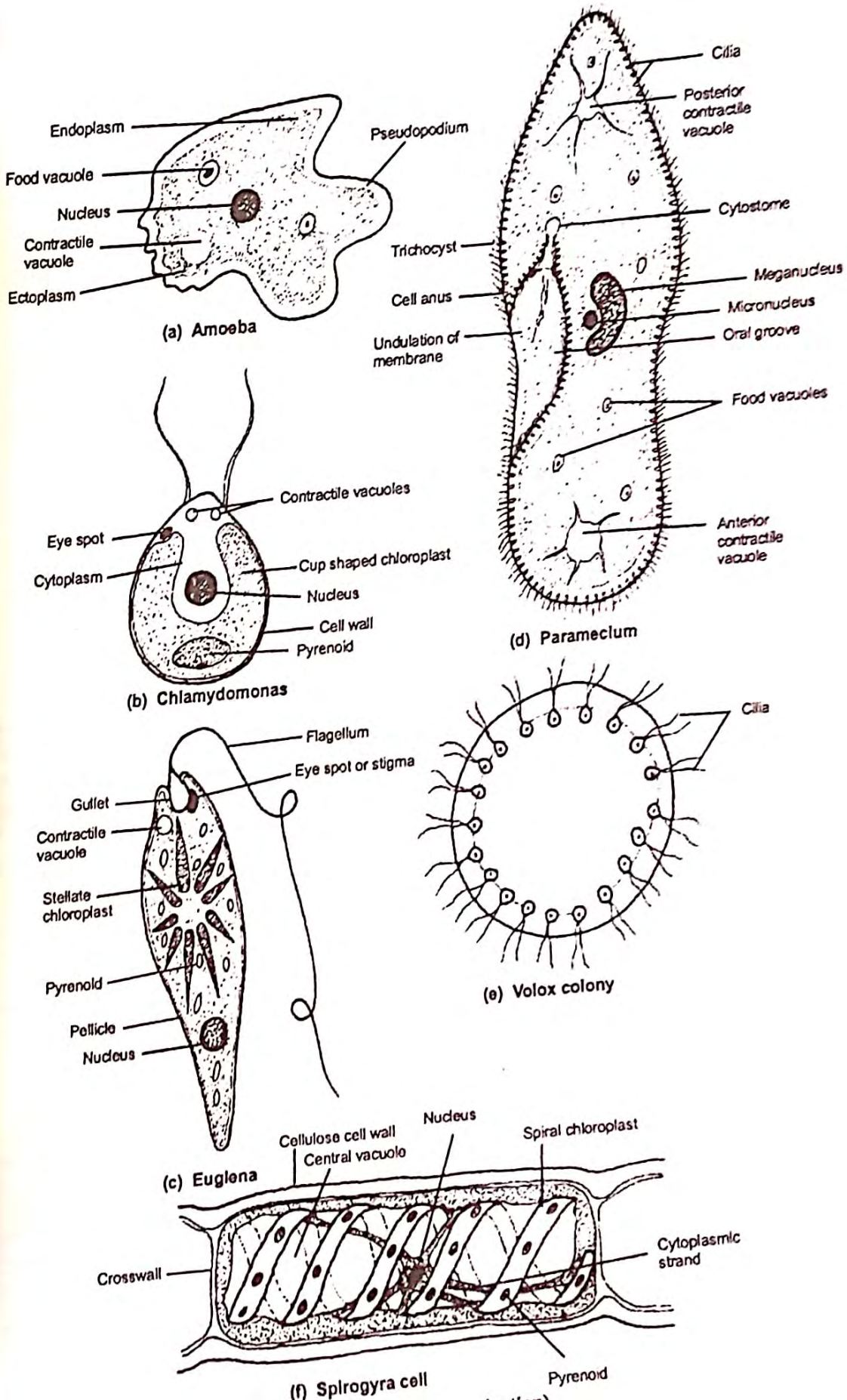


Fig. 1.3: Diversity of free living cells (Cellular level of organization)

independently, i.e. they have become specialized. No longer does every cell carry out all life's functions. Some are responsible for reproduction while others unable to reproduce themselves take care of photosynthesis and locomotion. This division of labour among cells is called differentiation.

- *Spirogyra* (Fig. 1.3c) is a simple multicellular filamentous green alga. It consists of a single row of cylindrical cells joined end to end. All the cells in the filament are identical. *Spirogyra* is able to manufacture its known food with the help of the spiral chloroplasts. Starch grains normally accumulate around the pyrenoids when exposed to light. Each filament shows no differentiation of any cell for a particular function. Each cell is capable of carrying out its own life processes. Thus each cell is physiologically independent but they are not mobile.

  Like *Chlamydomonas* and *Euglena, Amoeba* and *Paramecium caudatum* are free-living unicellular micro-organisms.

- *Paramecium* (Fig. 1.3d) as a unicellular organism has to carry out within one cell all the functions that in a multicellular animal are performed by differentiated cells, tissues and organs. The macronucleus exercises control over metabolic functions, including growth, whilst the micronucleus is necessary for sexual reproduction. The cilia are responsible for locomotion, the contractile vacuoles for osmoregulation. The trichocysts are used for defence. Food particles swept into the oral groove by ciliary action, are taken up into a food vacuole formed at the base of the cytopharynx or gullet. Food vacuoles circulate through the cytoplasm before discharging undigested remains to the exterior through the cell anus called cytoproct.

- Most of other living things are built from many cells. Two easily studied examples are the epithelial cells that line the inner surfaces of cheek and red blood cells. These cells are different from the cells of single-celled organisms like *Amoeba* in that they are not capable of living an independent existence, i.e. they cannot carry out all the functions of the living organism. Each cell is dependent on others to carry out the functions that it cannot perform by itself.

1.8 Functions of Cell Components

Despite the diversity of cell types there are certain structural and functional features that are common to most cells.

(a) Cell membrane (plasma membrane)

This is the outer limiting membrane surrounding the cytoplasm. It is thin, flexible and selectively permeable. The plasma membrane [illegible] cell wall of plant cells. It is only recognisable when such cells are plasmolysed.

(b) Cell Wall

Only very rarely is the cell membrane the outer surface of the cell. Usually some types of exterior coating is present.

- In algae and in all plants, the exterior coating is made of the polysaccharide cellulose. This forms a rigid cellulose cellwall characteristic of plant cells.
- Rigid walls are also found around the cells of bacteria where they are protein in nature, and in fungi where the wall is of chitin.

(c) The Nucleus

The cell nucleus is bounded by a pair of membranes which contain pores. The pores permit materials to pass into and out of the nucleus to the cytoplasm. Within the nuclear membrane is a semi-fluid medium in which are suspended the chromosomes. A nucleolus is usually found in the nucleus.

(d) The Cytoplasm

This term describes everything within the cell except the nucleus. An extraordinary complexity of structures are found in the cytoplasmic region. There are elaborate patterns of membranes and membrane-bounded compartments within the cytoplasm. These clearly defined structures are called organelles. Each organelle carries out a particular function as follows:

(i) *The Mitochondria*: They are spherical or sausage-shaped bodies found in various members in the cytoplasm of cells. Active cells like the liver cells may contain over a thousand of them. They contain enzymes that carry out the oxidation of food substances and they synthesize ATP (adenosine triphosphate), the energy of the cell.

(ii) *The Chloroplasts:* They are formed only in the cells of plants and certain algae. In plant cells, the chloroplasts occur as disc-shaped structures with as many as up to 50 of them per plant cell. Chlorophyll which traps the energy of sunlight enables it to be used for the synthesis of food. Thus chloroplast is the seat of photosynthesis.

(iii) *The Ribosomes:* They are among the smallest structures suspended in the cytoplasm. They are roughly spherical and they are the site of protein synthesis.

(iv) *The Endoplasmic Reticulum:* This is an elaborate system of membranes within the cell. Evenly spaced

ribosomes are found adhering to the surface of some and hence are called rough endoplasmic reticulum (RER). Where they are absent they are called smooth endoplasmic reticulum (SER).

(v) *The Golgi Apparatus*: This is a stack of flat, membrane-bounded sacs. Proteins synthesized by the RER are transferred to the golgi apparatus. Additional carbohydrates may be added to them. These protein-filled sacs then migrate to the surface of the cell and discharge their contents to the outside. Golgi apparatus is also the site where synthesis of polysaccharides take place.

(vi) *Lysosomes*: They are roughly spherical structures bounded by a single membrane. They are produced by golgi apparatus and are filled with proteins. These proteins are made up of many enzymes. Lysosomes play an important role in the death of cells because the enzymes in them bring about autolysis.

(vii) *Vacuoles:* They are cytoplasmic organelles filled with liquid. Food materials or wastes may be found inside vacuoles.A young plant cell contains many small vacuoles, but as the cell matures, these unite to form a large central vacuole (Fig. 1.1a). Dissolved food molecules, waste materials and pigments may be found in it.

(viii) *Centrioles:* Animal cells and cells of other micro-organism and lower plants contain two centrioles. Just before a cell divides, its centrioles duplicates and one pair migrates to the opposite side of the nucleus. Spindle fibres then form between them, i.e. they play a role in the formation of spindle fibres during cell division. (Fig. 1.1b).

SSCE/UME–Type Objective Questions

1. Which of the following structures differentiates an animal cell from a plant cell?
   A. Ribosomes
   B. Cell membrane
   C. Chloroplast
   D. Mitochondrion
   E. Lysosome.

2. Which of the following is a similarity between a typical animal cell and a typical plant cell?
   A. Cellusose cell wall
   B. Chlorophyll
   C. Centrally-placed nucleus
   D. Cell membrane
   E. Large vacuole

3. All living organisms
   A. Photosynthesize
   B. Conjugate
   C. Have plastids
   D. Feed
   E. Transpire

4. The cell is the functional unit of living organisms because
   A. Multicellular Organisms are made up of cells
   B. All the cells in a multicellular organism function as a single cell.
   C. The function of a multicellular organism is the sum total of the functions of its constituent cells.
   D. The function of a single cell is superior to that of multicellular organism.
   E. The cells of a multicellular organism are similar in structure.

5. Which of the following cell constituents is NOT common in both plants and animals?
   A. Mitochodria
   B. Chloroplast
   C. Ribosomes
   D. Golgi apparatus
   E. Vacuoles.

6. Which of the following is characteristic of the animal cell?
   A. Presence of choloroplasts
   B. Possession of cellulose cell wall
   C. Absence of large vacuoles
   D. Presence of large vacuoles
   E. Presence of chromosomes.

7. The vacuole of a Plant cell is
   A. A large empty space
   B. Smaller than that of an animal cell
   C. Filled with air,
   D. An ordinary vacuum
   E. Surrounded by a membrane.

8. Which of the following is not true of the nucleus of a living cell? It contains
   A. Chromosomes
   B. Nucleolus
   C. Nucleoplasm
   D. Chromatids
   E. Ribosomes

9. Amoeba moves by means of
   A. Cilia
   B. Flagella
   C. Pseudopodia
   D. Swimmerets
   E. Setae

10. Where is energy produced in a cell?
    A. Nucleus
    B. Nucleolus
    C. Lysosomes
    D. Chloroplast
    E. Mitochondria

11. The function of ribosomes in cells is
    A. Protein synthesis
    B. Starch synthesis
    C. Transport of materials
    D. Liquid storage
    E. Respiration

12. In which of the following organism does a single cell perform all the functions of active movement, nutrition, growth, excretion and photosynthesis?
    A. Paramecium
    B. Euglena
    C. Amoeba
    D. Spirogyra
    E. Hydra

13. Members of the kingdom Protozoa use the contractile vacuole
    A. To remove excess food
    B. For movement
    C. For digestion
    D. To remove excess water
    E. To remove nitrogenous waste

14. Which of the following organisms does not exist as a single free living cell?
    A. Amoeba
    B. Volvox
    C. Euglena
    D. Paramecium
    E. Chlamydomonas

15. In which of the following is a cellulose cell wall present?
    A. Amoeba
    B. Mammalian white blood cell
    C. Spirogyra
    D. Paramecium
    E. Parasitic worms

16. Which of the following organisms exists as a filament?

A. Euglena B. Amoeba
C. Volvox D. Paramecium
E. Spirogyra

17. Which of the following is not present in the nucleus of a cell?
A. Chromosome B. Nucleolus
C. Mitochondrion D. Genes
E. Chromation

18. Which of the following structures controls the activities of the living cell?
A. Nucleus B. Centrosome
C. Chlorophyll D. Golgi body
E. Mitochondrion

19. The organelle which eliminates water from the body of the protozoa is
A. Plasma membrane B. Contractile vacuole
C. Cell wall D. Protoplasm
E. Nucleus

20. The part of the cell solely responsible for respiration is the
A. Nucleus B. Nucleolus
C. Mitochondria D. Golgi apparatus
E. Endoplasmic reticulum

21. In animals, cilia, flagella, pseudopodia and chaetae can be used for
A. Reproduction B. Fighting enemies
C. Attachment to plants D. Locomotion
E. Locating sensation

22. Which of the following structures is common to Euglena, white blood cell and Amoeba?
A. Vacuole B. Cell wall
C. Cilia D. Cell membrane
E. Chloroplast

23. Which of the following characteristics is common to **Amoeba** and **Paramecium**
A. Oral groove B. Trichocyst
C. Contractile vacuole D. Cilia
E. Pseudopodia

24. Which of the following organelles is used for locomotion in **Paramecium**?
A. Pseudopodium B. Trichocyst
C. Cilium D. Pellicle
E. Contractile vacuole

The diagram (Fig. 1.4) illustrates the structure of **Euglena**. Use it to answer **Questions 25-27.**

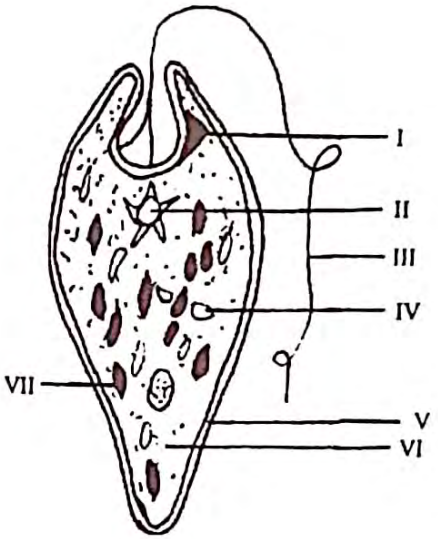


Fig. 1.4

25. Which of the following structures is responsible for osmoregulation?
A. I B. II C. III
D. IV E. V

26. Excretion is carried out through the structure labelled
A. I B. III C. IV
D. V E. VI

27. The part labelled VII is made up of
A. Carbohyrates B. Lipid materials
C. Waste materials D. Protein materials
E. Food particles

28. **Euglena** moves by the
A. Whipping action of its flagelum
B. Beating of its cilia
C. Rotating
D. Pushing out a jet of water from an organulle
E. Beating the pseudopodia and cilea simultaneously

The diagram (Fig. 1.5) illustrates the structure of a cell. Use it to answer Question 29 and 30.

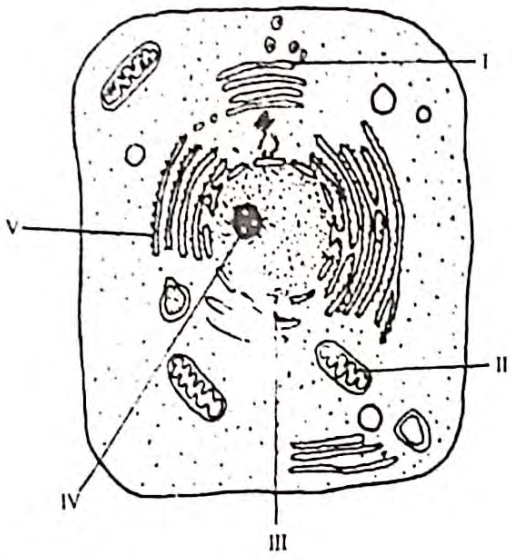


Fig. 1.5

29. The structure that produce the energy required by the cell is labelled
A. I B. II C. III
D. IV E. V

30. The structure labelled V is the
A. Mitochondron B. Ribosome
C. Nucleus D. Centriole
E. Nucleolus

31. The organ which is sensitive to light in **Euglena** is the
A. Gullet B. Chloroplast
C. Eye spot D. Contractile vacuole
E. Flagellum

32. **Spirogyra** is regarded as a multicellular plant because
A. Its cells are linked together by cytoplasmic stand
B. Its cells are joined to form organs
C. The cylindrical cells are linked end to end
D. The filament consists of one cell only
E. It is an alga containing large vacuoles

33. The cilia in **Paramecium** are used for
A. Respiration
B. Locomotion
C. Protection
D. Regulating food intake
E. Excretion

34. **Euglena** may be classified as a plant because it
A. Has chloroplasts
B. Has a jacket
C. Lives in a pond
D. Possess a flagellum
E. Has a pellicle

35. Which of the following statements is not true of **Paramecium**?
A. It is slipper-shaped
B. The anterior end is blunt
C. The gullet is lined with cilia
D. It has an anal pore
E. Its food vacuole is not stationary

36. One of the following statements is NOT true of **Spirogyra**
A. It is a simple multicellular plant
B. During conjugation, its gametes, which are structurally and physiologically similar, fuse to form a zygote
C. It possess spiral chloroplast which enable the plant to photosynthesize
D. Its cells are protected externally by a layer of mucilage
E. There are pyreniods along the chloroplast

37. What is the function of the contractile vacuole in **Paramecium**?
A. Produces enzymes
B. Gets rid of excreta
C. Stores and digests food
D. Gets rid of excess water
E. Controls movement

38. **Spirogyra** reproduces vegetatively by
A. Spore production
B. Fragmentation
C. Multiple fission
D. Budding
E. Binary fission

39. In **Spirogyra**, the pyrenoid
A. Excrete waste products
B. Is suspended by cytoplasmic strands
C. Is mainly used for respiration
D. Usually contains starch
E. Makes the plant slimy to touch

40. In Fig. 1.6 the structure labelled X is the
A. Cellulose cell wall
B. Plasma membrane
C. Nuclear membrane
D. Mucilage wall
E. Ectoplasm

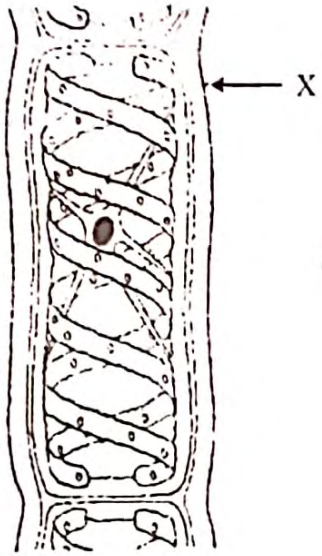


Fig 1.6

41. **Euglena** is an autotrophic organism because it
   A. Has flagella
   B. Has plant and animal feature
   C. Is found in water
   D. Can manufacture its food
   E. Moves fast

42. **Spirogyra, Euglena** and **Chlamydomonas** show many characteristics EXCEPT
   A. Nutrition
   B. Reproduction
   C. Mobility
   D. Irritability
   E. Excretion

43. What is the function of trichocyst in **Paramecium**?
   A. Movement
   B. Defence
   C. Excretion
   D. Reproduction
   E. Feeding

44. One of the features which adapt **Paramecium** to its environment is the possesssion of
   A. A regular shape
   B. Two nuclei
   C. Cilia
   D. A pellicle
   E. A vacuole

Answer Key

1.	C	10.	E	19.	B	28.	A	37.	D
2.	D	11.	A	20.	C	29.	B	38.	B
3.	D	12.	B	21.	D	30.	B	39.	D
4.	A	13.	D	22.	D	31.	C	40.	D
5.	B	14.	B	23.	C	32.	C	41.	D
6.	C	15.	C	24.	C	33.	B	42.	C
7.	E	16.	E	25.	B	34.	A	43.	B
8.	E	17.	C	26.	D	35.	C	44.	C
9.	C	18.	A	27.	A	36.	A		

Explanations to Objective Questions

1. (C) Chloroplasts are disc-shaped objects which contain the green pigment chlorophyll. It absorbs light energy for photosynthesis. Animal cells do not contain chlorophyll.

2. (D) Both plant and animal cells have cell membrane. Cellulose cell wall, chlorophyll, large vacuole are typical of plant cells while animal cell has a centrally placed nucleus.

3. (D) Feeding is one of characteristics of living things. Few organisms conjugate and only plants photosynthesize, transpire and have plastids.

4. (A) The multicellular organisms are made up of cells which are capable of carrying out the functional characteristics of living organisms.

5. (B) Chloroplasts are not common in both plants and animals while the other cell constituents are common in both.

6. (C) Large vacuoles are absent in animal cell but present in plant cell along with chloroplasts and cellulose cell wall while chromosomes occur in both.

7. (E) The vacuole of a plant cell is surrounded by a membrane

8. (E) The ribosome is a structure suspended in the cytoplasm and it is for protein synthesis. The chromosomes, nucleolus, nucleoplasm and chromatids are artefacts of the nucleus of the living cell.

9. (C) Amoeba moves by means of pseudopodia which is a finger-like projection of the body wall.

10. (E) The mitochondria is the source of energy in a cell and this is produced through the synthesis of ATP.

11. (A) The ribosomes are among the structures found in the cytoplasm of cells and the sites of protein synthesis

12. (B) Euglena is a single celled organism capable of performing all the functions of movement, nutrition, growth, excretion and photosynthesis.

13. (D) In the phylum Protozoa, the contractile vacuole is used to remove excess water

14. (B) Volvox is the only organism that does not exist as a single free-living cell because it is colonial and is made up of clusters of independent cells.

15. (C) A cellulose cell-wall is present only in the plant-like Spirogyra while the others are animal-like.

16. (E) Spirogyra is the only organism that exist as a filament. It consists of a single row of cylindrical cells joined end to end

17. (C) Mitochondrion is the only one of the organelles that is not an essential part of the nucleus.

18. (A) The nucleus controls all the activities of the living cell. It contains chromosomes which are hereditary materials.

19. (B) The contractile vacuole is the organelle which eliminates excess water from the body of protozoa

20. (C) The mitochondria is the part of the cell solely responsible for respiration. It contains enzymes that carry out oxidation of food materials.

21. (D) In animals, cilia, flagella, pseudopodia and chaetae are used for locomotion.

22. (D) The cell membrane is the only structure that is common to **Euglena**, white blood cell and Amoeba.

23 (C) Contractile vacuole is the only structure that is common to **Amoeba** and **Paramecium**

24. (C) The cilium is the organelle used for locomotion in Paramecium.

25. (B) The contractile vacuole (II) is the structure responsible for osmoregulation.

26. (D) Excretion is carried out through the structure labelled (V), which is the pellicle by diffusion of CO_2.

27. (A) The part labelled (VII) is the paramylum granule which is a food reserve material consisting of carbohydrates.

28. (A) **Euglena** moves by whipping action of its flagellum

29. (B) The structure that produces the energy required by the cell is labelled (II). It is the mitochondrion which is the source of energy in a cell.

30. (B) The structure labelled (V) is the ribosome which is the site of protein synthesis.

31. (C) The organ which is sensitive to light in **Euglena** is the eye spot.

32. (C) **Spirogyra** is regarded as a multicellular plant because its cells are linked end to end.

33. (B) The cilia in **Paramecium** are used for locomotion.

34. (A) **Euglena** may be classified as a plant because it has chloroplast.

35. (C) The gullet of **Paramecium** is not linked with cilia.

36. (A) **Spirogyra** is not a simple multicellular plant

37. (D) The function of the contractile vacuole in **Paramecium** like any other protozoa is to get rid of excess water.

38. (B) **Spirogyra** reproduces vegetatively by fragmentation.

39. (D) In **Spirogyra** the pyrenoid usually contain starch.

40. (D) In the figure, the structure labelled (X) is the mucilage wall.

41. (D) **Euglena** is an autotrophic organism because it can manufacture its own food by photosynthesis.

42. (C) **Spirogyra, Euglena** and **Chlomydomonas** share many characteristics except mobility.

43. (B) The trichocyst in **Paramecium** is for defence.

44. (C) The possession of cilia is a feature that adapt **Paramecium** to its environment.

2 The Cell and its Environment

Physical and Bio-physical Processes

- Many physical processes occur in living organisms that are vital to their existence. This involves the exchange of materials between the cell and its environment. These processes lead to various metabolic activities which take place in the organism.

2.1 Diffusion

- This is a process which involves the random movement of substances such as gas or liquid from a region of higher concentration to a region of lower concentration
- Diffusion can be demonstrated by dividing a chamber into two compartments by means of a sheet of cellophane (Fig. 2.1). Fill one compartment (A) with a solution of glucose whose concentration can be measured. Fill the other compartment (B) with pure water. After a few minutes, remove a sample from each compartment and measure the concentration of glucose in each. You will find that as time goes by, the glucose concentration drops in compartment A and at the same rate, rises in compartment B.
- Finally, after a time of about three hours, the concentrations in both compartments become equal and remain so.
- The process described above is diffusion and it occurs because of the continual random motion characteristic of molecules that are not held together in a solid state. Each molecule travels in a straight line until it collides with another molecule and in this case another molecule of glucose, of water or of cellophane. After each collision, the molecule rebounds in a different direction. The result is random motion of the molecule. The random motion of glucose molecules in compartment A at the initial state results in many collisions with the cellophane which has pores large enough to allow glucose molecules to pass through to compartment B.
- As long as the concentration of glucose molecules is higher in A than in B (i.e. concentration gradient), there will always be a net movement of glucose molecules from A into B until equilibrium state is reached. The rate of diffusion of substances through cell membrane is also dependent on the size, change, and lipid solubility of the substances in question. Generally, lipid soluble substances diffuse across the membrane more easily than non lipid soluble substances.

2.2 Osmosis

- This is a process which involves the movement of solvent molecules through a semi-permeable membrane or the diffusion of any solvent through a semi-permeable membrane from a region of high concentration to a region of low concentration.
- The concentration here refers to water which is a universal solvent in living organisms.
- Osmosis can be demonstrated by washing an Irish potato and cutting across to form a base. The opposite is similarly treated, but in addition has a cavity cut into it almost to the base. The peel is removed from about one centimeter along the base. Into the cavity strong sugar solution is placed, and the whole peel is put into a trough of water as shown in Fig. 2.2a
- After a time the liquid in the cavity is seen to have risen because osmosis has taken place (Fig. 2.2b) When boiled potato is similarly treated, or when unboiled potato does not have strong sugar solution in its cavity as in Fig 2.2c, osmosis does not take place.
- This can be explained thus: The water concentration in the trough is 100%.
- The water concentration in the sugar solution is less than this because the given volume of glucose contains fewer water molecules than the same volume of distilled water.
- There is, therefore, a net movement of water molecules through the potato into the strong sugar solution.
- The pressure exerted by diffusion of water molecules into the strong sugar solutions is called **osmotic pressure.**

2.3 Effect of Different Media Concentrations on Red Blood Cells

- When human red blood cells are placed in pure water, water molecules begin to diffuse into them.
- Thus osmosis occurs because there is a greater concentration of water molecules outside the cells than within.

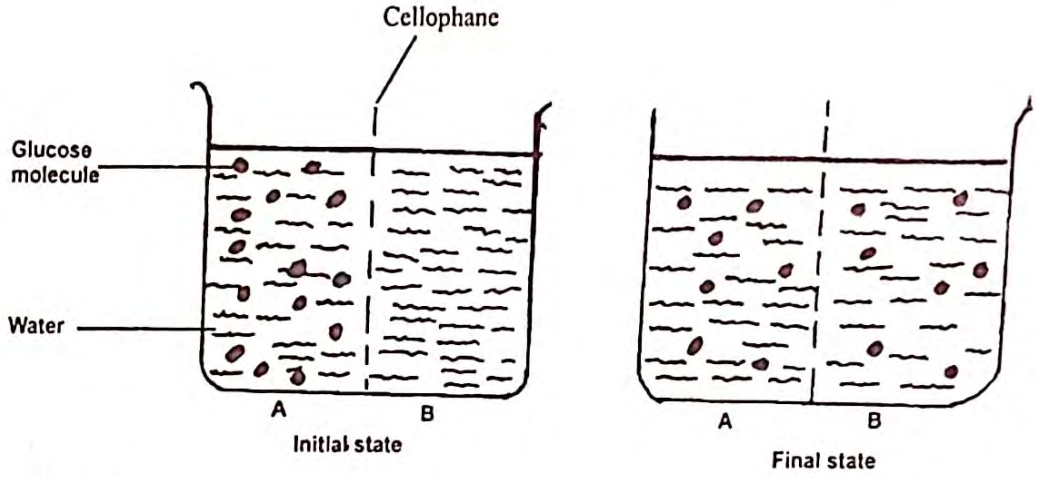


Fig. 2.1: Diffusion

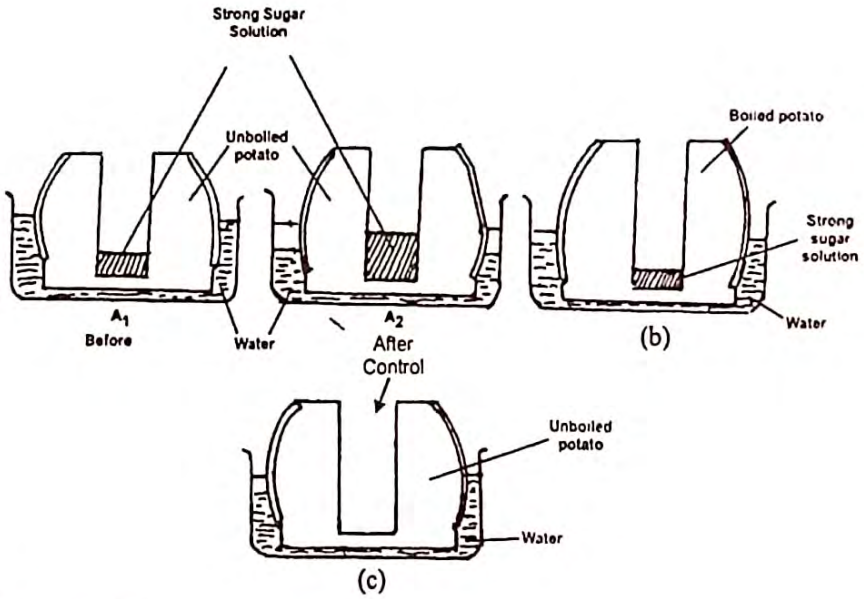


Fig. 2.2: Potato Osmometer

- The surrounding water is said to be hypotonic to the cytoplasm of the cells.
- The membrane of the red cells is very fragile and cannot withstand the build up of pressure within the cell. Consequently the cells become turgid and then burst.
- If the cells are placed in concentrated salt solution or in sea water, water passes out of the cells by osmosis and the cells shrink.
- This is because a given volume of concentrated salt solution contains a smaller number of water molecules than the same volume of cytoplasmic fluid of the red cells.
- The salt solution is said to be hypertonic to the cytoplasm of the cells.
- However, if the red cells are placed in a medium that contains exactly the same concentration of water as their cytoplasm they neither gain nor lose water by osmosis.
- Such a solution is said to be isotonic.

2.4 How Organisms Cope with Hypotonic and Hypertonic Environments

- Water passes into the cells of plants that spend their life in freshwater (e.g. *Spirogyra*) by osmosis. As more water molecules enter, the cell swells up so that it now exerts a pressure against the cellulose cellwall.
- This pressure is called **tugor pressure.**
- The strong cellulose cellwall in turn resists this

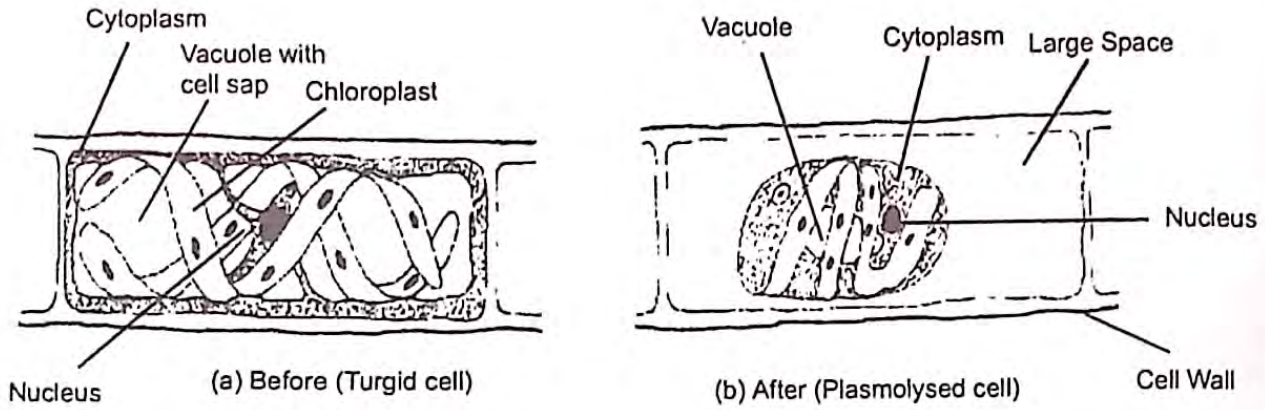


Fig. 2.3: Plasmolysis in *Spirogyra* cell

pressure by an opposing force called **wall pressure.**

- Since the central vacuoles of the cells of terrestrial plants contain a high concentration of solutes, water absorbed from the soil enter into the cells by osmosis and establishes a tugor pressure which makes the cellwall to be hard. When terrestrial plants are unable to get adequate amount of water from the soil their cells lose tugor pressure and the plant wilts.
- Freshwater animals and protozoans are able to cope with the osmotic movement of water into their cells by employing energy and some contractile structures to pump the excess water back out into the environment. For example, *Amoeba, Paramecium* use contractile vacuoles. The water concentration in the cytoplasm of the cells of marine organism (plants and animals) is approximately the same as that of marine water.
- So a state of equilibrium is achieved with respect to water in their surroundings (i.e. isotonic relationship).

2.5 Plasmolysis

- This occurs when live organisms such as freshwater plants, are placed in media that are hypertonic to the cytoplasmic fluid of their cells.
- The cells lose tugor as a result of the continual movement of water out of their vacuoles.
- The cytoplasm then shrinks and the contents pull away from the cellwall.
- The shrinking of the contents of the cells is called *plasmolysis* (Fig. 2.3).

2.6 Active Transport

- This is the movement of ions and molecules against a concentration gradient, that is, the movement of ions and molecules from a lower concentrated solution to a higher concentrated solution.
- This process requires the expenditure of energy (ATP) and the presence of enzymes located in the cell membrane, which serve to move molecules and gas from the side of low concentration to the side of high concentration. For example, the salt glands of some marine birds (herring gull) and fish continuously remove salt from the blood of these animals. The fluid thus excreted is saltier than the blood.

2.7 Properties and Functions of the Living Cell

(a) Nutrition

- **Anabolism:** Anabolic reactions are concerned with building up structures, storage compounds and complex metabolites in the cell from simpler ones.
- The product of anabolic reactions contain more potential energy than the reactants.
- It is regarded as the synthesis of macromolecules such as protein, polysaccharides and nucleic acids from small precursors. This reaction requires energy input and the source is Adenosine Triphosphate (ATP) (i.e. the energy source for all anabolic activities of the cells). The products of anabolism serve a number of necessary functions.
- Glycogen and fats and proteins serve as reserve fuels for later catabolism.
- Synthesis of new nucleic acids enables the organism to store copies of its genetic material.
- All anabolic changes are catalyzed by enzymes.
- **Autotrophic** (Photosynthesis): This is the ability to manufacture food. Simple compounds, namely carbon dioxide, water and salts, are absorbed and built up into complex substances that can be oxidized for energy or synthesized into living protoplasm and cell walls. This process is known as photosynthesis or chemosynthesis in some bacteria.

Heterotrophic

(i) *Holozoic*

- This is the method of feeding that is characteristic of animals.
- Complex food material is broken down by digestion to simple substances which are then absorbed and incorporated into cellular structures or oxidized to obtain energy.

(ii) *Saprophytic*

- Fairly complex chemical compounds are absorbed into plant cells. Some of these compounds are absorbed directly and others are first digested outside the organism by enzymes secreted on or in the material. Decaying organic matter like rotting wood and humus are good materials for saprophytic feeders.

(iii) *Parasitic*

- The organism derives its food from another individual without necessarily killing it.

2.8 Cellular respiration

(i) Anaerobic

- The main source of energy in organisms is carbohydrates
- The energy produced by the breakdown of carbohydrates is stored in the form of adenosine triphosphate (ATP) in living cells.
- All cells retain the enzymatic ability to breakdown glucose (a carbohydrate) without the aid of oxygen (i.e. anaerobic).
- The anaerobic breakdown of glucose is called glycolysis which leads to the production of two molecules of pyruvic acid and two molecules of ATP.
- Yeast cells use glycolysis, otherwise called alcoholic fermentation, to produce ethanol, carbon dioxide and energy.

$$C_6H_{12}O_6 \longrightarrow \underset{(2ATP)}{2C_2H_5OH+2CO_2+105Kj}$$

- An overworked muscle cell uses glycolysis (i.e. anaerobic condition) to produce lactic acid and energy, and the process

$$C_6H_{12}O_6 \longrightarrow \underset{(2ATP)}{2C_3H_6O_3+Energy}$$

is called lactic acid fermentation.

(ii) Aerobic

- Most cells are able to use oxygen (aerobic) to catabolize glucose so as to produce carbon dioxide, water and energy.
- Each molecule of pyruvic acid produced during glycolysis is oxidised in a stepwise manner providing energy for the synthesis ATP molecules in a process called **Krebs cycle.**
- Further oxidative reactions in the presence of oxygen yields more ATP in a process called oxidative phosphorylation because the oxygen carried to the cells reacts with cytochrome (a respiratory enzyme) which in its reduced form produces water. This process takes place in mitochondria.
- This chain series in which reduced cytochrome is the end link involves the transfer of hydrogen (electrons) from substance to substance to produce a great deal of energy in aerobic respiration:

$$C_6H_{12}O_6+6O_2 \longrightarrow \underset{\text{molecules)}}{6CO_2+6H_2O+\text{Energy}\ (38ATP \text{ or } (2827Kj)}$$

2.9 Excretion

- This is the way in which living things return the waste products of their metabolism to the environment.
- For most single-celled organisms which live in a watery environment, the elimination of their metabolic wastes is by diffusion and for most, the chief product of protein metabolism is ammonia.
- Ammonia though toxic is very soluble in water and therefore, readily diffuses out of the cell before reaching dangerous concentration.
- The problem of excess water is solved by the contractile vacuole which probably plays a signifi- cant role in the excretion of other substances.

2.10 Growth

- It involves an increase in size of an organism brought about when the rate at which complex molecules are synthesized is greater than the rate at which they are broken down.
- More molecules are gained from the environment than is given back to it in the form of metabolic wastes.
- Growth may occur as a result of an increase in the size and dry weight of the cells that make up the organism or just an increase in the number of cells that make up the organism.
- Every new cell that is formed increases in size until it reaches a maximum size, stops growing, and takes on special functions.
- Mitosis: Takes place when cells divide during growth and development, and also during asexual

reproduction.

- It takes place in four convenient phases, i.e. **prophase, metaphase anaphase and telophase.** Interphase is the resting phase because the cell is not involved in division.

(i) Prophase

- Centrioles in the cell move to opposite sides of the nucleus.
- A number of delicate threads called asters radiate form each centriole.
- Chromosomes relax their coils and thicken so that their double nature becomes apparent.
- Nuclear membrane and nucleolus disappear.

(ii) Metaphase

- Cytoplasmic fibres radiating from the centrioles meet near the centre of the cell to form a structure known as spindle.
- Chromosomes move to the equatorial plane of the spindle and stand clearly apart from one another
- Each chromosome is made up of two strands or chromatids and each chromosome makes an exact copy of itself by the process called replication.
- The chromosomes become attached to the fibres at their centromeres.

(iii) Anaphase

- Chromatids separate at the centromere.
- Each member of a pair of chromatids move towards opposite end or pole of the spindle.

(iv) Telophase

- The chromatids form a close group at each pole.
- The spindle disappears and a nuclear membrane is formed round each group of chromosomes.
- The chromosomes reorganise as two nuclei.
- In animals the cytoplasm constricts to form two daughter cells but in plants a new cell wall is formed between the two sets of chromatids by the process of cell plate formation.

Regions of fastest growth in plants

- Tissues which are capable of cell division are called **meristems.** The division takes place by the process **mitosis.**
- Meristems are small, thin-walled actively dividing cells densely filled with cytoplasm.
- Apical meristems are responsible for the increase in length of roots and stems and are found at the tips of stems and roots (Fig. 2.4).

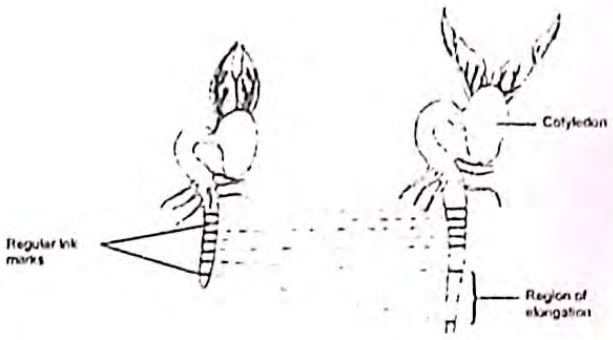


Fig. 2.4: Region of fatest growth in plants (Growth measurement of root ina bean seedling)

- The cambium is the meristem responsible for increase in thickness.
- Only dicotyledonous roots and stems have a cambium. Monocotyledons do not normally grow into shrubs and trees with the exception of the palms.
- Temperature, humidity, light and pH are all factors which affect metabolic process and hence, have an effect on growth.
- The growth of roots and shoots is controlled by auxin (a plant hormone) manufactured in the tips of the roots and shoots and then passed to the growing regions.
- Auxins influence cell division, cell elongation, growth of ovaries into fruits and the development of abscission layer.
- Other plant hormones include **gibberellins** which promote growth of stems and sprouting of buds, and **cytokinins** which when acting together with auxin, strongly stimulate mitosis in meristematic tissues and also slows down aging of plant parts.

2.11 Cell's Reaction to its Environment

- Irritability is the power of living things to respond to changes in their environment which may be internal or external.
- Responsiveness in multicellular organisms requires a proper coordination of parts which in plants is achieved through a system of chemical coordinators called **hormones** and in animals through the nervous system.

- The external stimuli that bring out response in organisms are light, temperature, gravity, touch, water and chemical substances.
- The movement of a whole organism, like *Euglena*, and *Chlamydomonas*, from one place to another in response to an external stimuli is known as **taxis** or **tactic movement. Phototactic** movement is in response to light and **chemotactic** in response to chemicals.
- **Nastic** movements and **tropisms** are growth movements in response to outside stimuli.
- A movement brought about by a response that is independent of the direction (non-directional) from which an external stimulus strikes the organism is called nastic movement, e.g. opening of certain flowers after sunrise.
- A growth movement whose direction is determined by the direction from which the stimulus strikes the plant is called tropism.

(i) Phototropism

- Growth curvature brought about by the unequal distribution of light.
- The growing shoot and leaves show a positive response by growing towards light while growing roots generally show a negative response by growing away from light.

(ii) Hydrotropism

- Growth curvature is caused by the unequal distribution of moisture (water).
- Roots are positively hydrotropic while shoots are negatively hydrotropic.

(iii) Geotropism

- Growth curvature caused by gravitational pull. Roots are positively geotropic while shoots are negatively geotropic. But when seedlings are attached to Klinostat the plumule (shoot) and radicle (root) will not show any curvature.

2.12 Movement

- Cilia and flagella are whip-like appendages found on the exterior surface of many cells.
- They move the extracellular fluid surrounding the cell thereby propelling the cell through the medium.
- Cilia and flagella are locomotor organelles for many single cells but in Amoeba pseudopodia is the locomotory organelle.

2.13 Reproduction

2.13.1 Asexual Reproduction

- New off springs are produced by a single parent by the process of mitosis when the environment is favourable.
- The offspring will be identical to the parent.
- The division of the parent body into smaller daughter cells is called *fission*. In *Amoeba* it is binary fission, in *Paramecium* transverse binary fission, in *Euglena* longitudinal binary fission, and where the parent nucleus divides several times within the cytoplasm to form several nuclei before the cytoplasm divides, it is called multiple fission as found in encysted cells during unfavourable conditions.

(i) Budding

- Asexual reproduction where offsprings develop as growths on the body of the parent.
- In the Yeast the parent nucleus divides into two and one of the nuclei moves toward a side so that a bulge is formed which gradually increases in size before, it either breaks away or remains attached to the parent stock (Fig. 2.5).

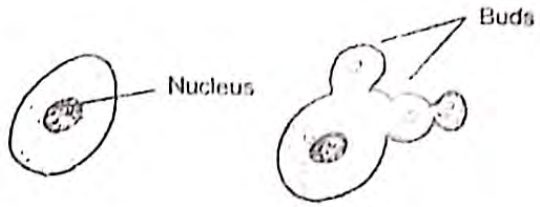
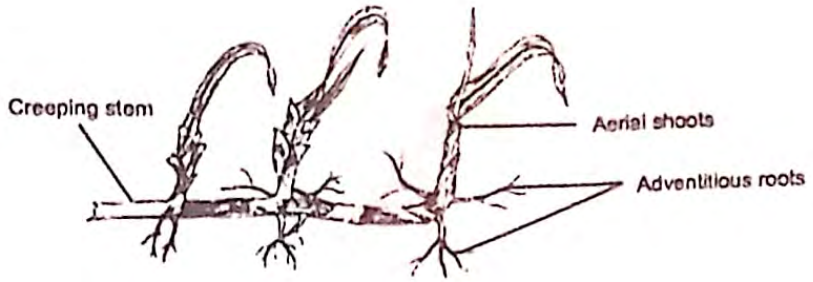


Fig. 2.5: Budding in Yeast

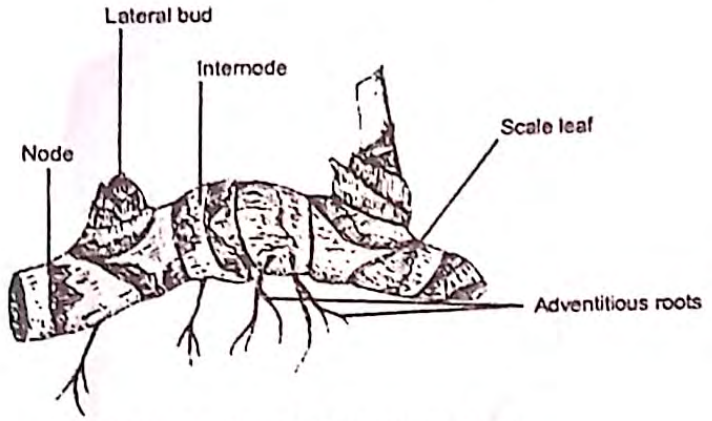
- In tapeworms **proglottids** are budded off from the neck region.

(ii) Vegetative propagation

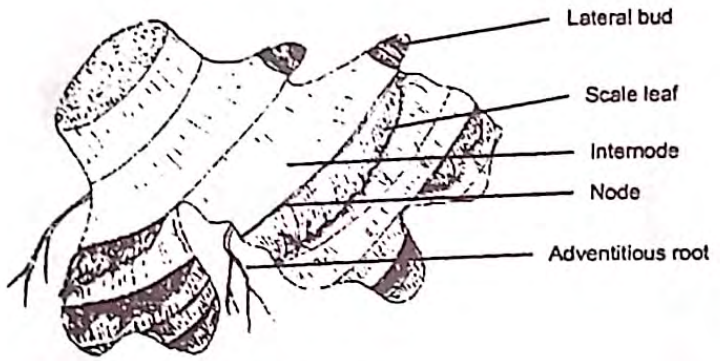
- When any part of a plant apart from the seeds are used in producing new plants, the method is said to be **vegetative.**
- Natural vegetative propagation occurs by means of creeping stems (Elephant grass), underground stems (Rhizomes of cannality and ginger, corms (*Colocasia*), stem tubers (Yam and Irish potato, Bulb (Onion), adventitious buds (*Bryophyllum* leaf), Bulbils (pineapple) and Root tubers (sweet potato) (Fig. 2.6 (a–i)).
- Artificial vegetative propagation is by stem cuttings (cassava, *Hibiscus*, rose), from twigs which have become woody, marcotting, layering, grafting buds and stems.



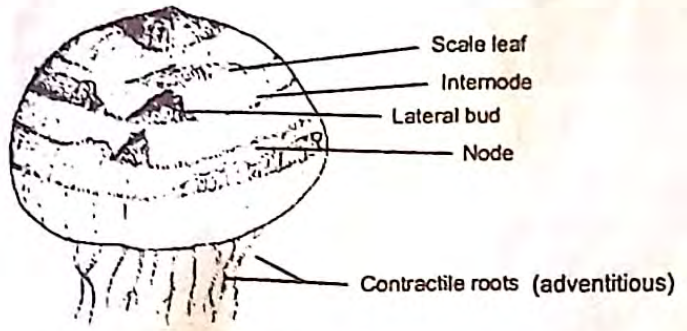
(a) Creeping stem of *Imperata cylindenca* (Elephant grass)



(b) Rhizome of Canna lily (Underground stem)

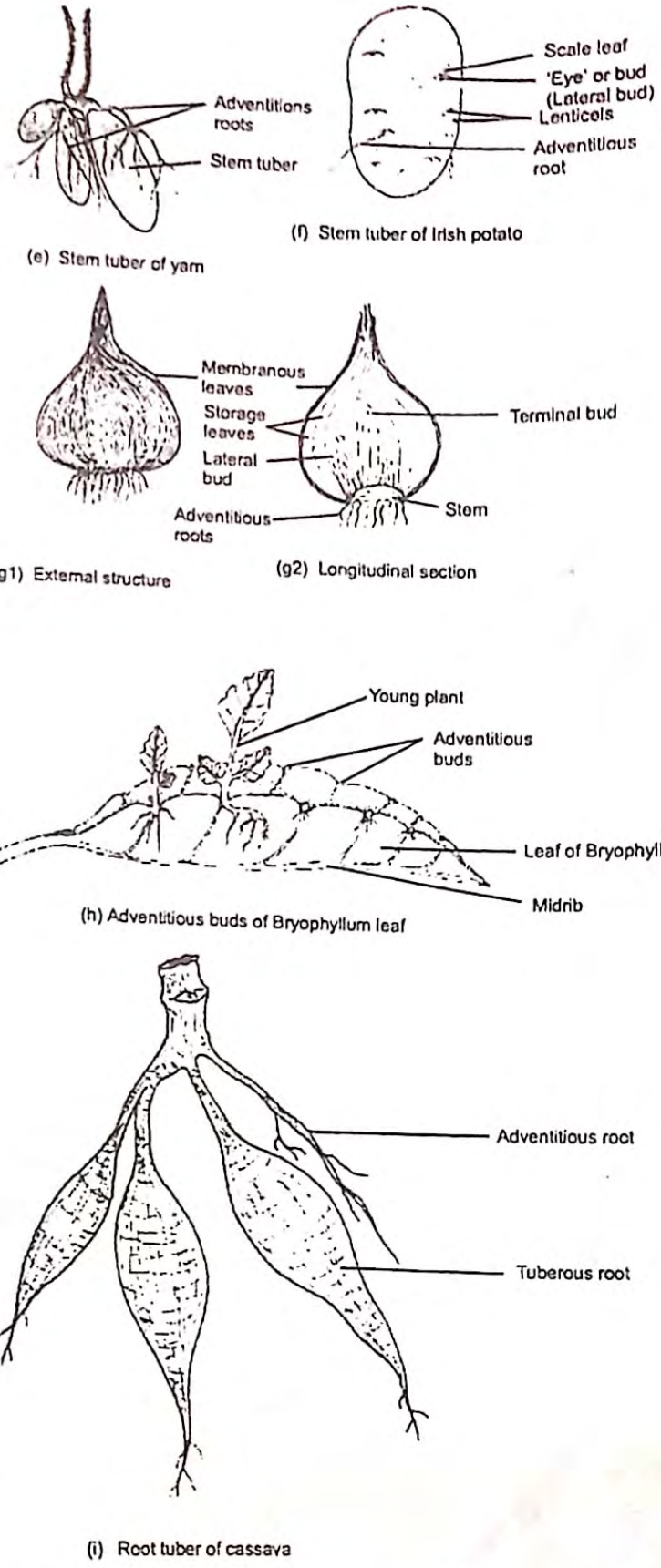


(c) Rhizome of Ginger (Underground stem)



(d) Corm of Gladiolus (underground stem)

Fig. 2.6: Diversity of vegetative propagation (a-d)



(e) Stem tuber of yam

(f) Stem tuber of Irish potato

(g1) External structure

(g2) Longitudinal section

(h) Adventitious buds of Bryophyllum leaf

(i) Root tuber of cassava

Fig. 2.6: Diversity of vegetative propagation (e-l)

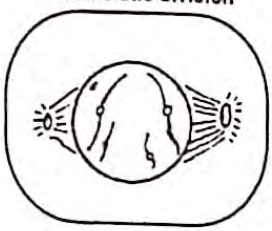
2.13.2 Sexual Reproduction

- When two cells join together with their cell contents fusing, they are said to conjugate as in *Paramecium*. Each cell taking part in the process is called **conjugant**.

(ii) Formation of gamates

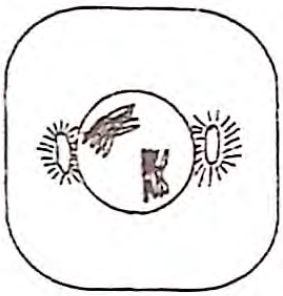
- Gamates are formed through the process of meiosis (Reduction division).
- A gamete is a cell that cannot develop further until it fuses with another gamate.
- In animals heterogamates are produced, i.e. the male gamate (sperm) is small and motile while the female gamate (egg or ovum) is larger and filled with food reserves.
- During fertilization the nuclei of spermatozoon and an egg fuse to form the nucleus of a zygote.
- During meiosis the chromosome number in gamates becomes half (haploid) of the original mother cell (diploid).
- Fusion of male (haploid) and female (haploid) gamates restores the diploid chromosome number in the nucleus of the zygote (Fig 2.7).

First Meiotic division



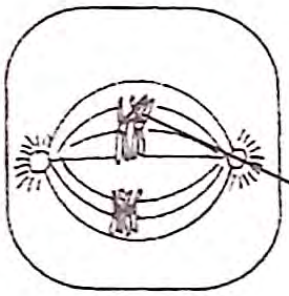
— Chromosomes contract
— Become more visible
— Nucleolus shrinks

(a) Early Prophase I



— Homologous chromosomes come together (synapsis) to form a bivalent.
— Each chromosome is a pair of chromatids.
— Chromatids of 2 homologous chromosome remain in contact at points called chiasmata ⇨ exchange of genetic material.

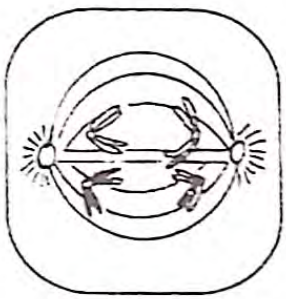
(b) Late Prophase I



— Chromosomes arrange themselves on equator of spindle.

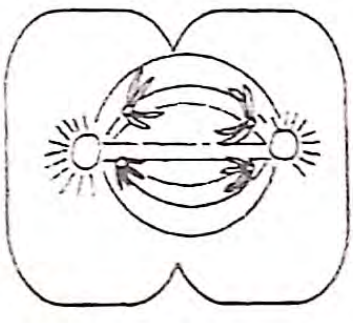
Chiasma

(c) Metaphase I



— Homologous chromosomes separate
— They move towards opposite poles of the cell

(d) Anaphase I

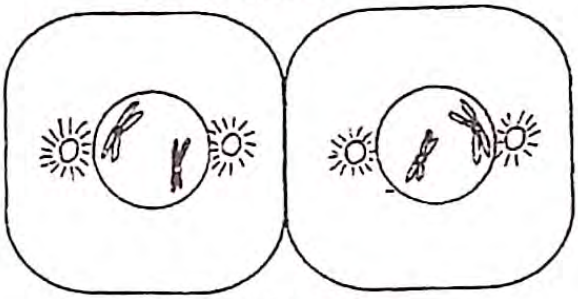


— Chromosomes at opposite poles
— Cell constricts across the middle as in mitosis

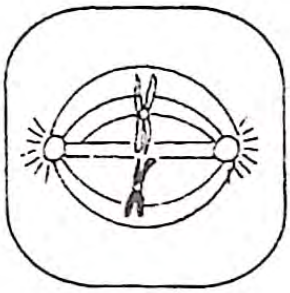
(e) Telophase I

Second Meiotic division

— The two daughter cells prepare for the second meiotic division
— Centrole, have replicated
— New spindle is formed.

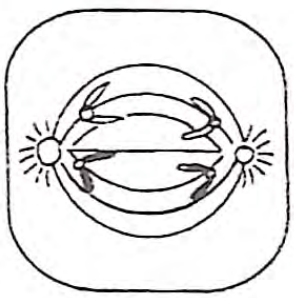


(f) Prophase II



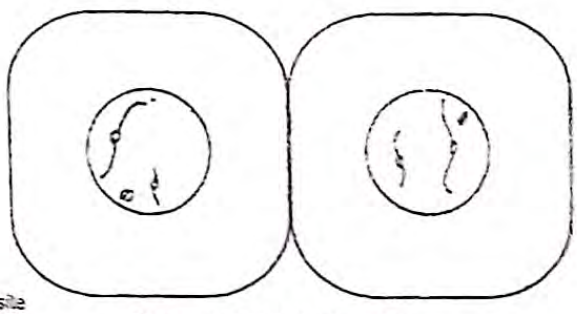
— Chromosomes arrange themselves on the equator of the spindle.
— Centromeres divide.

(g) Metaphase II



— Chromatids separate and move towards opposite poles of the cell.

(h) Anaphase II



— Cell constrict across the middle.
— Nuclear membranes and nucleoli are reformed
— Chromosomes regain their thread-like appearance.

(i) Telephase II

Fig. 2.7: Meoisis in Animals

(d) Water molecules from a dilute solution to a concentrated solution through a semi-permeable membrane
(e) Water molecules from a concentrated solution to a dilute solution through a semi-permeable membrane

12. In cellular respiration, energy is stored in the form of
(a) Adenosine diphosphate (ADP)
(b) Adenosine monophosphate (AMP)
(c) Adenosine triphosphate (ATP)
(d) Heat energy
(e) Electric energy

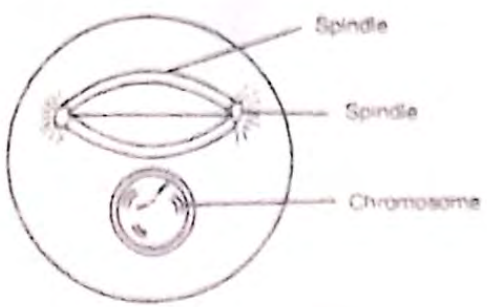
13. Which of the following is the medium of transportation of nutrients within unicellular organism?
(a) Blood (b) Serum
(c) Protoplasm (d) Plasma
(e) Lymph

14. Which of the following is not present in the nucleus of a cell?
(a) Chromosomes (b) Nucleolus
(c) Mitochondrion (d) Genes
(e) Chromatin

15. The network of double membrane that helps to convey materials through the cytoplasm is the
(a) Plasma membrane
(b) Vacuolar membrane
(c) Nuclear membrane
(d) Endoplasmic reticulum
(e) Mitochondrion

16. In aerobic respiration, oxidative phosphorylation takes places in
(a) Cytoplasm (b) Mitochorion
(c) Lysosome (d) Vacuoles
(e) Ribosomes

Study the diagram below and use it to answer **Questions 17 and 18.**



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(a) Movement of the chromosomes to the poles
(b) Arrangement of chromosomes on the equatorial plate
(c) Centromeres moving apart along the spindle
(d) Interaction of the spindles with the chromosomes
(e) Condensation of chromosomes and formation of spindle

19. Auxins are produced in the
(a) Petiole of leaves
(b) Parenchyma of roots and shoots
(c) Epidermis of roots and shoots
(d) Apical regions of roots and shoots
(e) Sclerenchyma tissues of roots and shoots

20. At which of the following stages of cell division can the cell be said to be resting?
(a) Anaphase (b) Telophase
(c) Prophase (d) Interphase
(e) Metaphase

21. Which of the following methods of reproduction is common to *Paramecium* and *Amoeba*?
(a) Conjugation
(b) Sexual reproduction
(c) Binary fission
(d) Budding
(e) Fragmentation

22. At the end of mitosis the number of cells produced from a parent cell is
(a) Four (b) Six (c) Eight
(d) Three (e) Two

23. The stems of young herbaceous plants are kept upright mainly by
(a) Osmotic pressure
(b) Turgor pressure
(c) Transpiration pull
(d) Suction pressure
(e) Root pressure

24. Which of the following tissues is not found in the stem and root of monocotyledons?
(a) Xylem (b) Cambium
(c) Pith (d) Cortex
(e) Pericycle

(d) Water molecules from a dilute solution to a concentrated solution through a semi-permeable membrane
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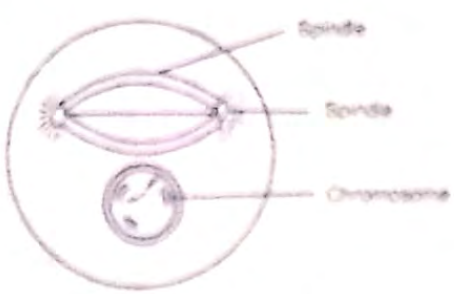

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(e) Pericycle

SSCE/UME–Type Objective Questions

1. In unicellular organisms, essential nutrients can be transported directly to all parts of their body by the process of diffusion only because unicellular organisms have
   (a) a large surface area to volume ratio
   (b) a large volume to surface area ratio
   (c) permeable cell membrane
   (d) outer membrane made of cellulose
   (e) their body always immersed in the nutrient material

2. The ability organisms to maintain a constant internal environment is known as
   (a) diuresis
   (b) endosmosis
   (c) plasmolysis
   (d) dialysis
   (e) homeostasis

3. Autotrophic nutrition may be defined in terms of food obtained
   (a) from other organisms in exchange for some products
   (b) by breakdown of complex substances
   (c) by an organism utilizing its own store of energy
   (d) by synthesizing simple substances using energy from an external source
   (e) in a synthesized form from other living organisms

4. Four cells with osmotic potential equivalent to that of 3% salt solution were immersed respectively in solutions of different concentrations labelled as follows:
   I = 4% salt solution
   II = 6% salt solution
   III = 1% salt solution
   IV = 10% salt solution
   V = 5% salt solution
   Which of the solutions will cause an increase in the osmostic pressure within the cell?
   (a) I only (b) II only (c) III only
   (d) IV only (e) V only

5. Which of the following statements is not true of osmotic process?
   (a) There must be a selectively permeable membrane
   (b) The two solutions must be of different concentrations initially
   (c) It involves only movement of water molecules
   (d) Equilibrium is reached when there is equal distribution of water molecules
   (e) The two solutions are of equal concentration at the beginning of the experiment

6. An association between two living organisms in which one lives on and feeds at the expense of the other organism is known as
   (a) Parasitism (b) Commensalism
   (c) Mutualism (d) Symbiosis
   (e) Predation

7. Which of the following is an autotrophic mode of nutrition?
   (a) Chemosynthesis (b) Saprophytism
   (c) Parasitism (d) Symbiosis
   (e) Commensalism

8. The mechanism of gaseous exchange in living organisms is essentially by
   (a) Osmosis (b) Inhalation
   (c) Diffusion (d) Exhalation
   (e) Breathing

9. Which of these statements is correct diffusion?
   (a) It involves the movement of w only
   (b) Molecules move from a region concentration to a region of lower cc tion
   (c) Differentially permeable membran.. present for diffusion to occur
   (d) It involves the movement of only sc molecules into the Bowman's capsule
   (e) It occurs when a solute can no longer dissolve in a solvent

10. The process of anaerobic respiration of yeast in sugar solution is known as
    (a) Oxidation
    (b) Fermentation
    (c) Tissue respiration
    (d) Alcohol production
    (e) Decomposition

11. Osmosis can be defined as diffusion of
    (a) Atoms and molecules through a membrane to an area of higher concentration
    (b) Water molecules from a dilute solution to a concentrated solution across a permeable membrane
    (c) Water molecules from an area of high concentration to an area of low concentration

(d) Water molecules from a dilute solution to a concentrated solution through a semi-permeable membrane
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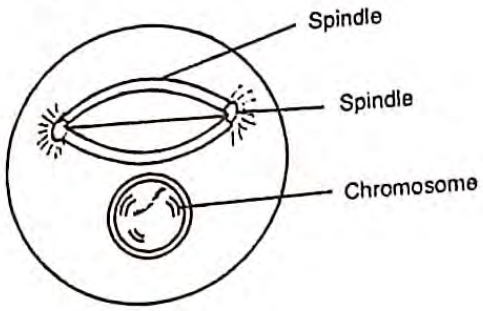

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(e) Pericycle

25. Growing radicles of seedlings are
    (a) Negatively phototropic
    (b) Positively phototropic
    (c) Negatively geotropic
    (d) Negatively hydrotopic
    (e) Positively phototactic

26. If a germinating seed is attached horizontally on a revolving Klinostat, what will be the effect of the seedling after three days? The
    (a) plumule will curve vertically upwards while the radicle will curve towards the ground
    (b) plumule and radicle will not show any curvature
    (c) radicle will curve towards the ground while the plumule will not show any curvature
    (d) plumule will curve vertically upwards while the radicle will not show any curvasture
    (e) plumule and radicle will curve towards the ground

27. Which of the following processes is responsible for the increase in length and dry mass in a root tip?
    (a) Meiosis
    (b) Absorption
    (c) Mitosis
    (d) Conjugation
    (e) Fertilization

28. Which of the following explains the need for transport system in large mammals?
    (a) The area to volume ratio in large animals is high
    (b) The area to volume ratio diminishes with large size
    (c) Diffusion as an adequate means of transport is lacking
    (d) Large animals have internal fluid which contain different types of cells
    (e) Large animals have four-chambered heart.

Answer Key

1.	A	7.	A	13.	C	19.	D	25.	A
2.	E	8.	C	14.	C	20.	D	26.	B
3.	D	9.	B	15.	D	21.	C	27.	C
4.	C	10.	B	16.	B	22.	E	28.	B
5.	E	11.	D	17.	A	23.	C		
6.	A	12.	C	18.	E	24.	B		

Explanations to Objective Questions

1. (A) A large surface area of unicellular organisms like *Amoeba* relative to small volume of the body makes diffusion the only option.

2. (E) Homeostas is clearly defined as the maintenance of a steady state in living organisms by control of the internal environment.

3. (D) Autotrophic nutrition occurs when an organism synthesizes its own food from simple inorganic raw materials using an external energy source, e.g. photosynthesis and chemosynthesis.

4. (C) Since the cells have osmotic potential, equivalent to 3% salt solution, only solution with 1% salt solution will cause an increase in the osmotic pressure within the cell since it is lower than 3%.

5. (E) There must be a concentration differential, i.e. water molecules will move across a semi-permeable membrane from a weaker solution to a stronger solution if osmosis is to take place.

6. (A) Parasitism is the association between two living organisms in which one lives on and feeds at the expense of the other.

7. (A) See answer to question 3.

8. (C) In diffusion there is a concentration differential in the external and internal environments of living organisms.

9. (B) The presence of concentration gradient brings about diffusion of molecules from a region of higher concentration to a region of lower concentration.

10. (B) Anaerobic respiration does not require oxygen and when it occurs in yeast in the presence of sugar solution, alcohol is produced. Hence it is called alcoholic fermentation.

11. (D) Osmosis can be defined as the diffusion of water molecules from a dilute solution to a concentrated solution through a semi-permeable membrane.

12. (C) Energy is stored in the form of ATP in cellular respiration.

13. (C) The protoplasm is used as a medium of transportation of nutrients in unicellular organisms.

14. (C) All the structures are found in the nucleus of a cell except mitochondrion which is found in the cytoplasm of a cell.

15. (D) Endoplasmic reticulum is the network of double membranes responsible for conveyance of materials through the cytoplasm.

16. (B) Mitochondrion is the seat of oxidative phosphorylation resulting in the release of energy.

17. (A) During prophase the centrioles move to opposite sides of the cell — with asters radiating from each. The chromosomes thicken so that their double nature becomes clear.

18. (E) Condensation or thickening of chromosomes and formation of spindle results.

19. (D) Apical regions of roots and shoots are the sites of auxin production.

20. (D) Interphase constitutes the resting phase in cell division because the cell is not involved in division.

21. (C) Asexual reproduction in *Parmecium* and *Amoeba* takes place by binary fission.

22. (E) At the end of mitosis two daughter cells are formed.

23. (B) Turgor pressure resulting from the entry of water into the cells by osmosis, makes the cell wall to be hard, hence young herbaceous plants remain upright.

24. (B) Cambium is absent from the stem and roots of monocotyledons.

25. (A) Growing radicles of seedlings grow away from light, hence they are negatively phototropic.

26. (B) A seedling horizontally attached on a revolving klinostat will demonstrate that the plumule and radicle will not show any curvature.

27. (C) Mitotic division takes place in the root tip where it increases the number of cells and the length of the root.

28. (B) As an animal increases in size (volume) the surface area ratio decreases, hence the area to volume ratio diminishes.

- All arthropods have a segmented body enclosed in a tough, jointed exoskeleton made of **chitin** which gives structural rigidity to the body and points of muscle attachment.
- Cartilages consist of cells embedded in hard organic matrix known as **chondrin**. The embedded cells are the chondroblasts. Complex cartilages, like those of intervertebral discs and pinna of the ear, have their matrix strengthened with connective tissue fibres.
- Bone is much harder than cartilage in that its organic matrix is impregnated with calcium salts of mainly calcium phosphate.
- The organic matrix of bone is secreted by osteoblasts which arrange themselves in concentric rings around nerves and blood vessels giving rise to what is known as **haversian system.**
- This arrangement gives a compactness to the bone (i.e. compact bone) and its found in the shafts of long bones (humerus, femur, tibia and fibula, ulna and radius) while spongy bone which is less compact and harder than the haversian bone is found at the ends (epiphyses) of such bones.

3.1 Types of Supporting Systems

- **Hydrostatic skeleton:** This is found in soft-bodied invertebrates like sea anemones and earthworms. This is not a skeleton in the classical sense, but its place is taken over by a fluid under pressure. This fluid functions as a hydrostatic skeleton, being surrounded by muscles which act against it. In the earthworm circular and longitudinal muscles in the body wall contract against the coelomic fluid in tbe body cavity.
- **Exoskeleton:** This is the hard cuticle found in arthropods (insects, lobsters, etc.) and here it performs the function of a skeleton which, as it lies outside the muscles, is called an exoskeleton. The exoskeleton which is tough and jointed is made by chitin. Growth or increase in size in such animals is by periodical shedding of the exoskeleton by the process of ecdysis or **moulting.**
- **Endoskeleton:** This is so-called because the skeletal elements, bones or cartilage, are internal to the muscles which are attached to them. Examples are found in all vertebrate animals. Endoskeletal animals grow by continuous increase in size and not by ecdysis.

3.2 Functions of the Skeleton

- **Support:** The rigid skeletal support raises the body from the ground, and support some vital organs and maintain the shape of the body.
- **Protection:** The skull protects the brain and houses inner structures of the ear and eyeball.
- The vertebral column protects the spinal cord and the rib cage protects the heart and lungs.
- The pelvic girdle also protects the abdominal organs (Figs. 3.1.1 and 3.1.2).
- **Movement:** The several pieces of bones making up the skeleton allow free movement of the body and limbs.
- Skeleton also serves as points of attachment to muscles responsible for movement through tendons.
- Contraction and relexation of muscles exert pressure on bones to effect movement.
- Respiration: The rib cage moves during inspiration and expiration.
- **Production of blood cells:** The long bones produce blood cells in their marrow before sending them into the circulatory system.

3.3 Joints

Features of a Joint

- A **joint** is where two or more bones meet.
- The bones are connected to one another by tough connective tissue ligaments which are elastic.
- The bones are moved relative to one another by an elaborate system of muscles.
- The muscles are attached to the bones by tendons.
- All muscles have their origin on one bone and their insertion on another situated further away from the centre of the body.

3.4 Types of Joints

- **Ball** and socket joints are found in the shoulder and hip (head of humerus fits into glenoid cavity of pectoral girdle and the head of femur fits into the acetabulum of the pelvic girdle).
- Ball and socket joint allows movement in several directions (Fig. 3.2).

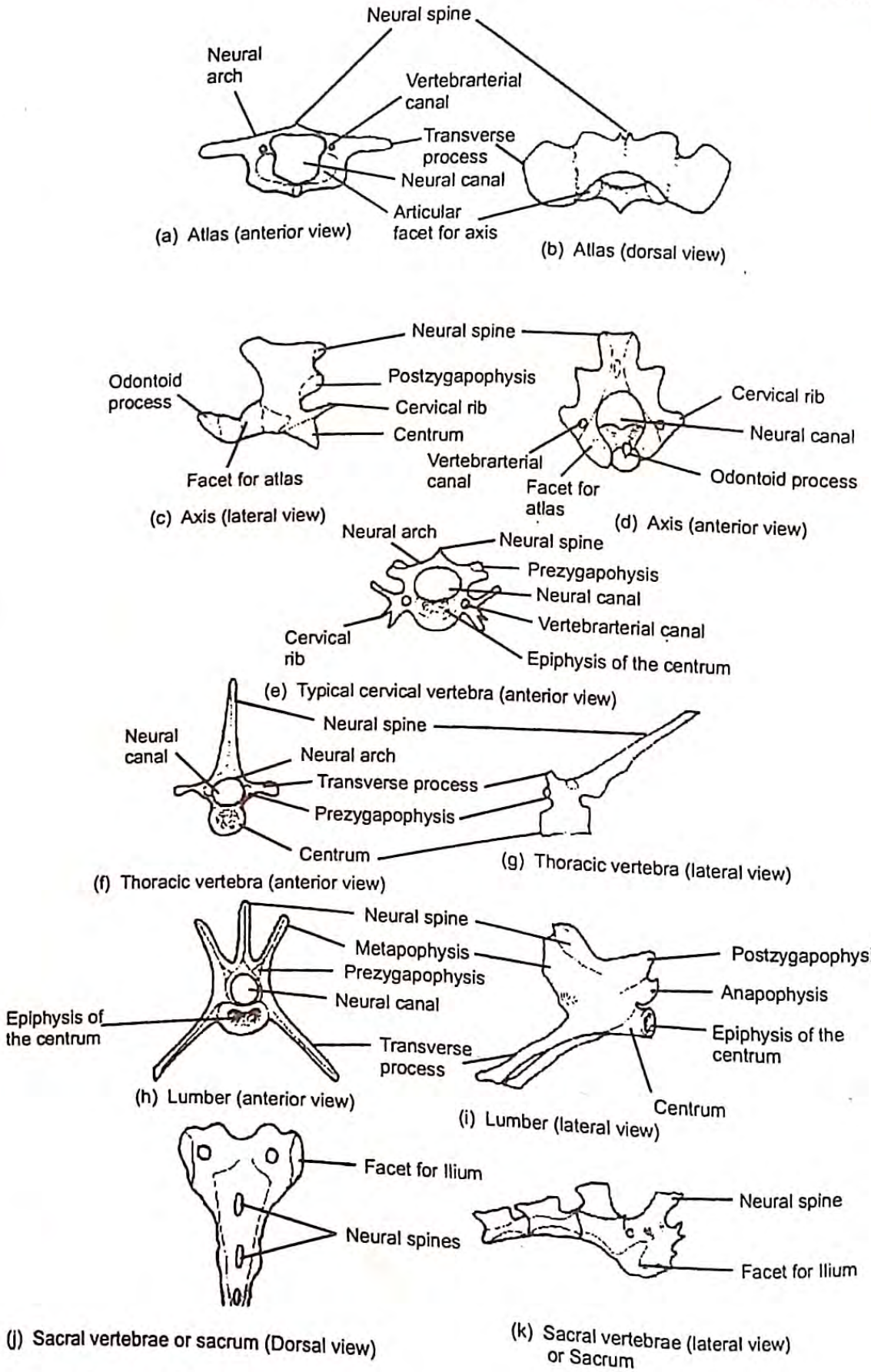


Fig. 3.1.1: Parts of the axial skeleton (a–k)

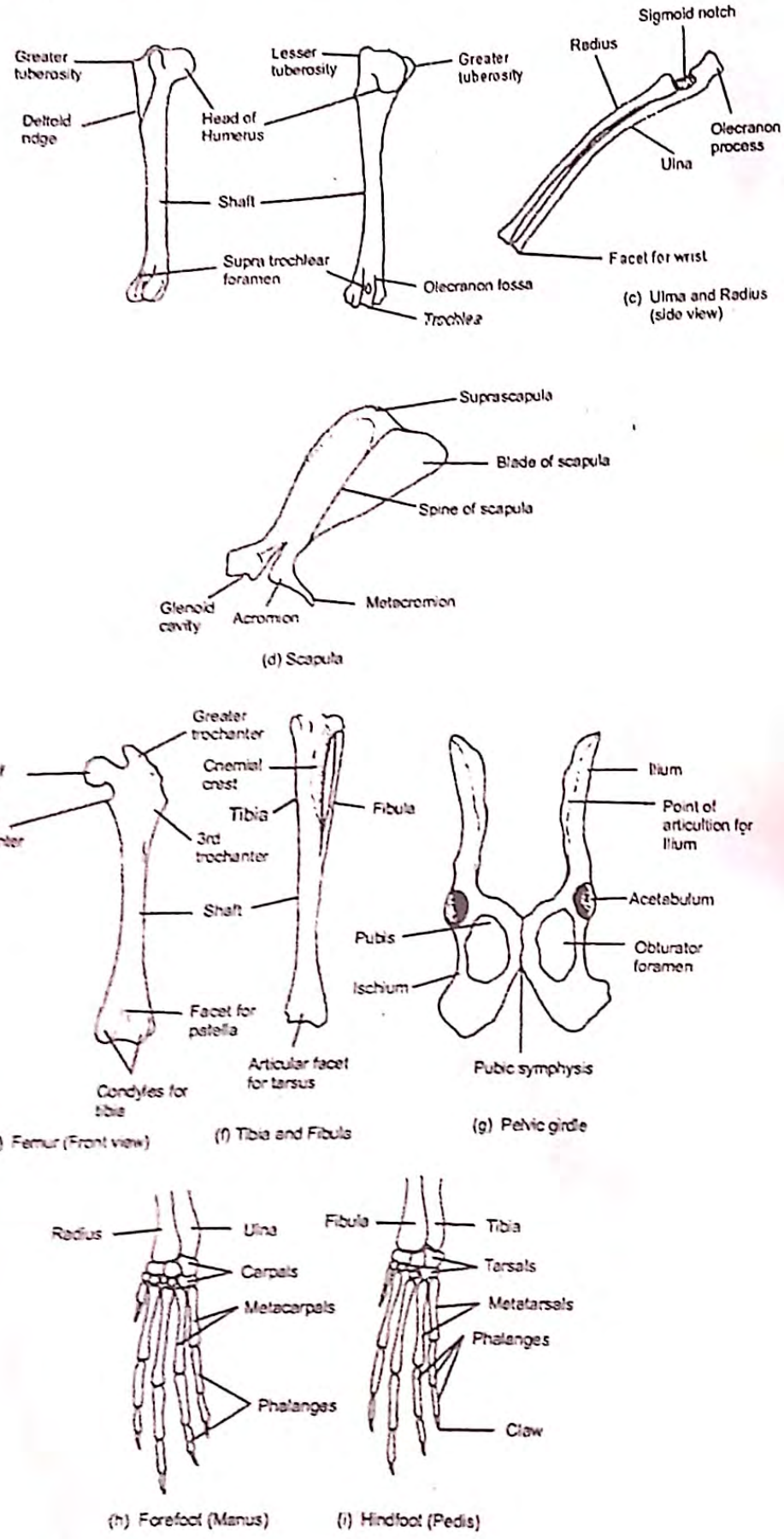


Fig. 3.1.2: Parts of Appendicular Skeleton (a–i)

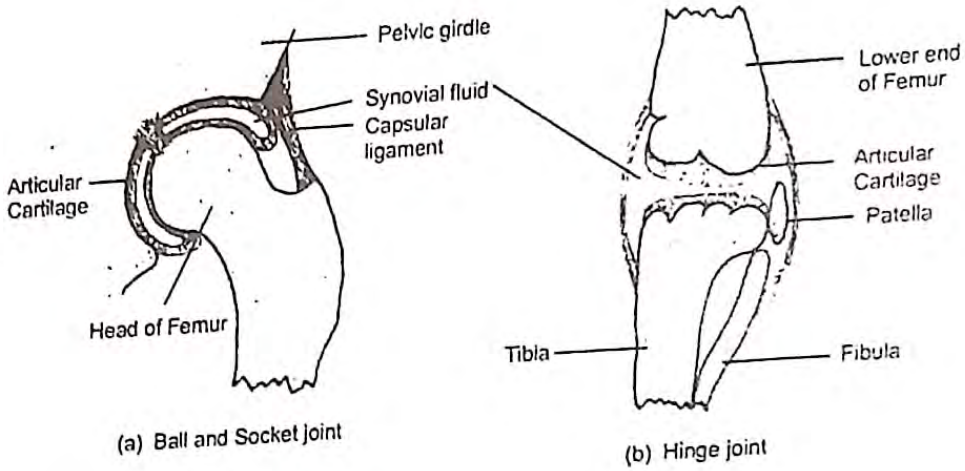


Fig. 3.2: Two types of joints found in mammals

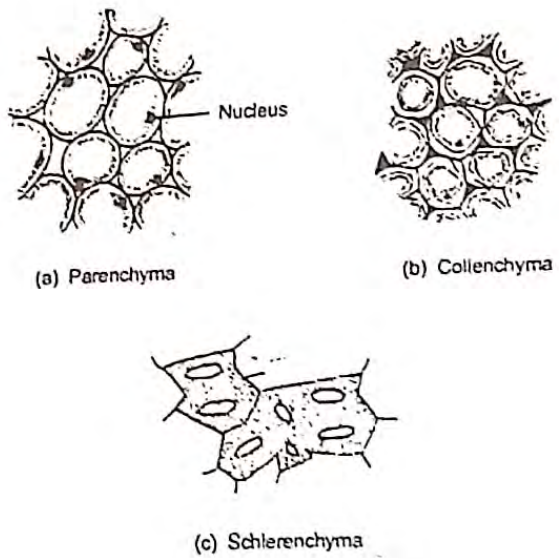


Fig. 3.3.1: Supporting tissues in plants

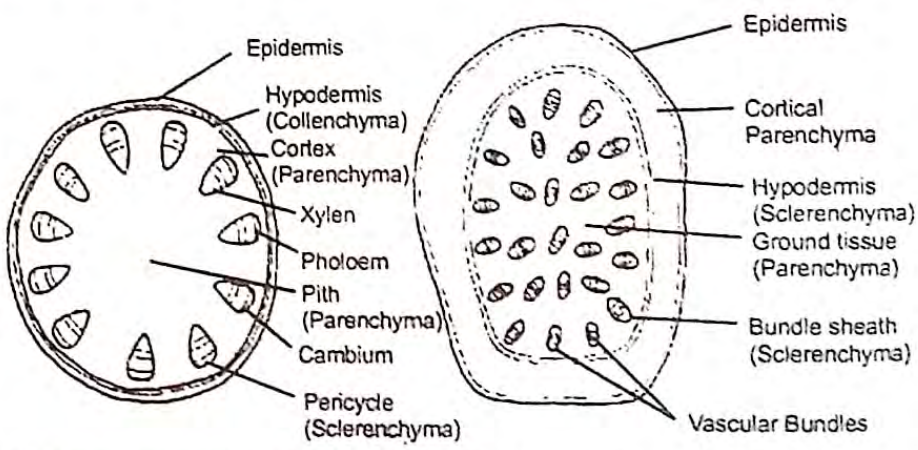


Fig. 3.3.2:Diagrammatic sketch of the transverse sections of young angiosperm stems showing the distribution of supporting tissues

- **Hinge** joint allows movement in only one plane or direction as found at the elbow and knee.
- **Gliding** or **Sliding** joint allows bones to slide over one another as found at the wrist and ankle.
- **Pivot** joint allows rotation of one part of the body on another as found between atlas and axis where the odontoid process of the axis acts as a pivot that allows rotation of the head on the vertebral column.

3.5 Supporting Tissues in Plants

- **Parenchyma** — A packing tissue made up of thin walled cells with a central vacuole each. The cells have intercellular air spaces. Provides flexibility.
- **Collenchyma** — The cells have thick walls that are pronounced at the corners. They provide mechanical support for the plant.
- **Sclerenchyma** — The walls of these cells are very thick and are built up in a uniform layer around the entire margin of the cell. (Figs. 3.3.1 and 3.3.2). Hence they provide rigidity, resilience and hardness.

3.6 Transport System

- The movement of materials from one part of the body to another in a living organism is known as **transport, translocation** or **conduction.**
- Transport system connects different parts of the body with one another, transports food materials to all cells, tissues and storage organs.
- It also serves to distribute water, transport waste products to excretory organs and transport raw materials to food synthesizing tissues.

3.7 Mammalian Circulatory System (Transportation in Higher Animals)

- The heart which is located roughly at the centre of the chest (thorax) cavity is surrounded by a protective membrane called **pericardium** (Fig. 3.4). The thoracic and abdominal cavities are separated by diaphragm.
- The heart is divided into four chambers with the two upper chambers known as the atria or auricles being relatively thin walled.
- The atria receive blood from the veins.
- The two lower chambers known as ventricles are more muscular than the atria and the left ventricle is more muscular than the right ventricle because it sends blood to all parts of the body except the lungs.
- Major branches of the circulatory system are shown in Fig. 3.5.

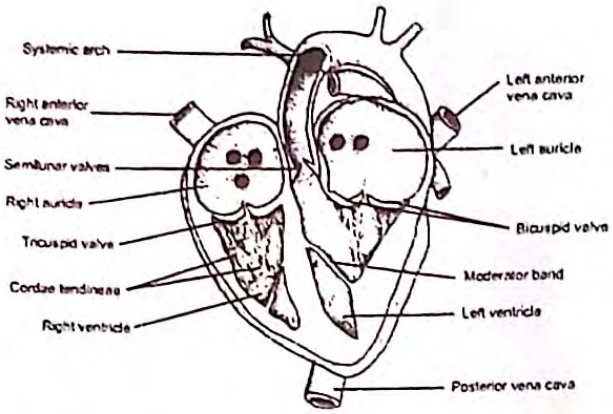


Fig. 3.4: Ventral section through the mammalian heart

- The **tricuspid** valve separates the right auricle from the right ventricle while the **bicuspid** valve separates the left auricle from the left ventricle.

3.8 Arteries, Veins and Capillaries

- Arteries are the blood vessels that carry blood away from the heart.
- They are thick-walled, muscular and elastic so as to withstand high pressure caused by heart beat.
- Arteries divide into smaller arterioles which further subdivide several times to form a dense network of highly microscopic vessels permeating every cell of the body known as capillaries.
- **Capillaries** are tiny vessels whose walls are one cell thick so that water and dissolved substances diffuse freely in and out.
- The capillaries unite into larger vessels which join to form veins which return blood to the heart.
- **Veins** are the blood vessels that take blood back to the heart.
- They are wider and have thinner walls than the arteries.
- They have valves along their length which prevent back flow of the blood.
- All veins carry deoxygenated blood except the pulmonary vein and all arteries carry oxygenated blood except pulmonary artery.

3.9 Blood

- It is a liquid tissue containing cells suspended freely in a watery medium called **plasma** and it is the means of transporting dissolved nutrients.

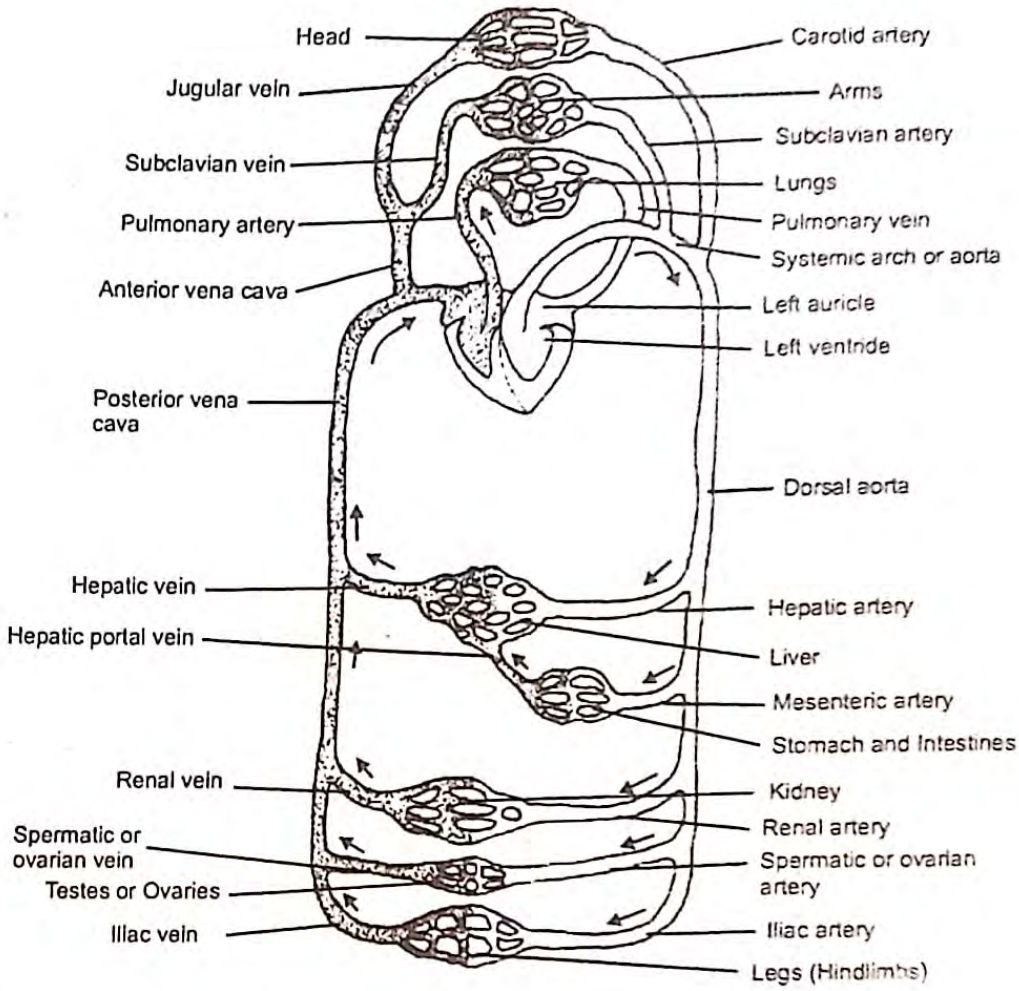


Fig. 3.5: The Mammalian Circulatory System

- The cells and cell fragments make up the "formed" elements of the blood. **Erythrocytes** are the most numerous of the cells which are disc-shaped, biconcave and non nucleated when mature.
- In an adult they are produced in the marrow of bones especially in the ribs, sternum and vertebrae.
- The red colour of the blood is due to the presence of iron-containing pigment called haemoglobin which helps in the transportation of oxygen round the body.
- Aged erythrocytes are destroyed in the liver and spleen.
- When the rate of erythrocyte loss from the body exceeds to the rate of erythrocyte formation anaemia results.

3.10 Leucocytes

- They are much less numerous than the erythrocytes, amoeboid in shape and nucleated.
- There are five distinct types i.e. **neutrophils, eosinophils, basophils, monocytes and lymphocytes.**
- Neutrophils and monocytes are phagocytic in action through ingestion of foreign particles by endocytosis.
- Lymphocytes function in the production of antibodies which combat antigens.
- Eosinophils and Basophils whose numbers are very low in the blood also play a role in combating diseases.
- Leucocytes are synthesized in the bone marrow.
- When a precursor cell in the bone marrow becomes cancerous, the number of leucocyte cells becomes far greater than normal leading to a condition known as leucaemia.

3.11 Platelets

- They are cell fragments produced by large cells in the bone marrow. They are also known as **thrombocytes.**
- Platelets are disc-shaped and much smaller than the

erythrocytes and much less numerous.

- They play an important role in the clotting of the blood (Fig. 3.6).

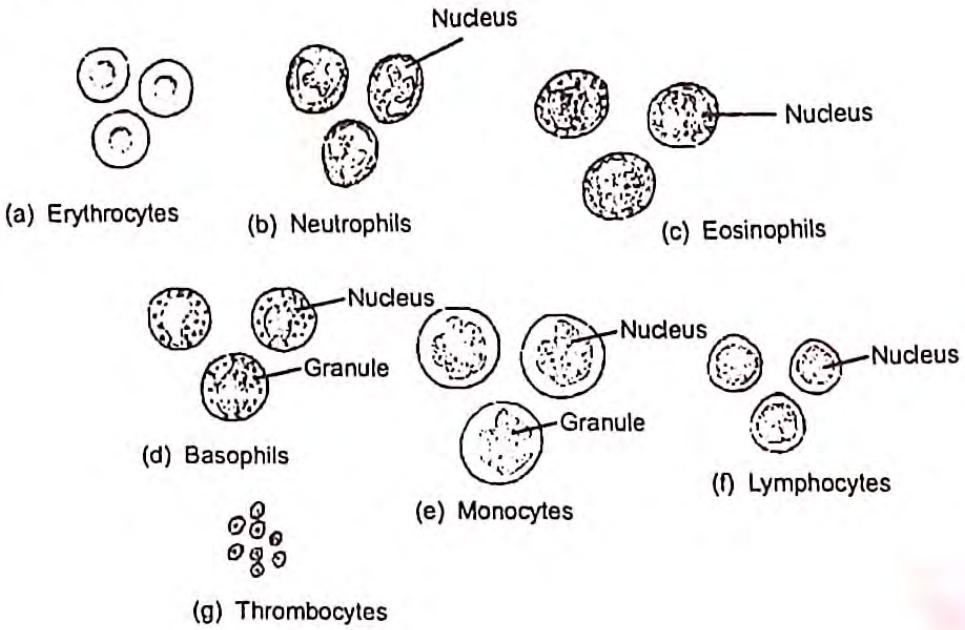


Fig. 3.6: Cells of the mammalian blood in surface view

3.12 Plasma

- The straw-coloured fluid in which the blood cells are suspended. The major component of the plasma is water in which are dissolved glucose, amino acids, waste products of cell metabolism, hormones, vitamins, fat droplets, with sodium and chloride ions being the most abundant.
- 7% of the plasma consists of plasma proteins like **fibrinogen, albumin** and **globulins.**
- Lymph: This is the water and dissolved substances that diffuse from blood capillaries to bathe the tissues of the body as a result of blood pressure and osmotic pressure of the tissues.
- Living cells absorb food and oxygen from plasma and their waste products diffuse into it.
- Lymph is carried away by the lymphatic vessels.

3.13 Transportation in Higher Plants

- This is achieved by the process of transpiration because transpiration accompanies photosynthesis.
- Continuous loss of water from the leaves causes suction or transpiration pull or stream by which more water is drawn from the veins in a leaf, from stem and finally from the roots and soil.
- Mineral Salts (dilute solution) used in the synthesis of carbohydrates,protein etc. are drawn up along with water.
- Transpiration stream transports dissolved salts to the leaves from the roots through xylem vessels.
- The set up in Fig. 3.7 is to demonstrate suction pressure. The leafy shoot is cut under water to prevent air bubbles from entering the xylem vessels and blocking it.

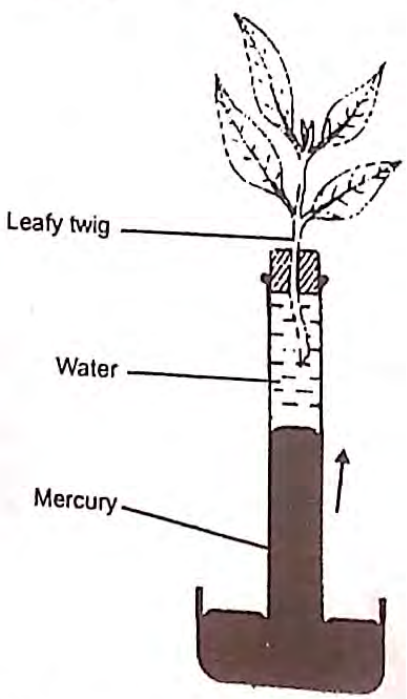


Fig. 3.7: Transpiration pull

- When placed near a window the mercury will rise up as water evaporates from the leaves showing that a pull is being exerted.
- The xylem vessels of a freshly dug-up seedling will be strained red when the roots are placed in water containing **eosin** (which is red colour) for about an hour.
- Sections of the leaf stalk, stem and parts of the roots not in the solutions when cut and examined under the microscope will show that the red parts correspond to the xylem vessels.
- Hence xylem vessels transport water from the roots to the leaves.
- The set up in Fig. 3.8 is to show that water vapour is given off during transpiration.

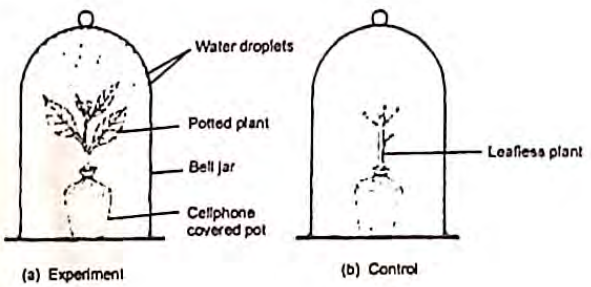


Fig. 3.8: Experiment to demonstrate that water is given off during transpiration

- The base of the potted plant is covered with polythene sheet to prevent evaporation of water from the pot into the bell jar.
- The rim of the dry bell jar is sealed up with vaseline after the potted plant has been placed under it, to prevent escape of water vapour.
- A control experiment in which the plant is devoid of leaves and flowers is also set up.
- After some time, drops of liquid will be found on the inside of the first bell jar but not in the control experiment.
- The liquid will turn anhydrous copper sulphate blue or turn blue cobalt chloride paper pink.

3.14 Respiratory System

(i) Gas exchange in aquatic organisms

- In fishes, gas exchange organs are the gills which present a large surface to water in which they are in contact so that oxygen present in the water diffuses into the cells of the gills.
- Gills are richly supplied with blood (vascularised) which transports oxygen to the inner tissues of the body.
- Blood again transports carbon dioxide from the inner tissues to the gills, and from the gills carbon dioxide passes out by diffusion.

(ii) Gas exchange in roots and stems

- Gaseous exchange in woody stems and mature roots, whose barks are protected by layers of dead cork cells which are impregnated with suberine, takes place through **lenticels.**
- Lenticels are nonsuberised pores that perforate mature roots and wood stems, thus enabling oxygen to reach the intercellular spaces of the interior tissues and similarly allowing CO_2 to be released to the atmos- phere.

(iii) Gas exchange in the leaf

- Exchange of gases in the leaves and young green stems takes place through pores called **stomata.**
- During photosynthesis CO_2 diffuses from the outside air through the stomata into the air spaces of the spongy layer and from here into the air spaces between the palisade layer cells, and then into the cells.
- O_2 produced as a by-product of photosynthesis diffuses out of the stomata in the same way.
- Stomata open when light strikes the leaf so that the guard cells become turgid, and at night when the guard cells lose turgor the stomata close as a result of the cells returning to their original shape (Fig 3.9).

(iv) Gas exchange in Terrestial animals

- Organs of gaseous exchange are located within the body cavity where they are protected from mechanical damage and excessive loss of moisture.
- Among the insects and some other terrestial invertebrates gaseous exchange takes place through an extensive network of air filled tubes called **Trachea** which open outside through small holes in the exoskeleton called **spiracles** (Fig. 3.10).
- **Lungs** are the organs of gaseous exchange among the air breathing land vertebrates.
- The surface area of the lungs is increased by alveoli which are highly vascularised.
- In the amphibians the skin and buccal cavity form supplementary organs of gaseous exchange.
- An efficient respiration surface must be well protected from damage, exposed to a source air, have a large surface area, richly supplied with blood, very thin and moist and highly ventilated.

- In mammals, the efficiency of the lung ventilation is improved by the presence of a muscular diaphragm.
- During inspiration (inhaling) the external intercostal muscles contract, lifting the ribs upwards and outwards. This motion increases the volume of the thoracic cavity and air moves into the lungs as a result of low pressure.

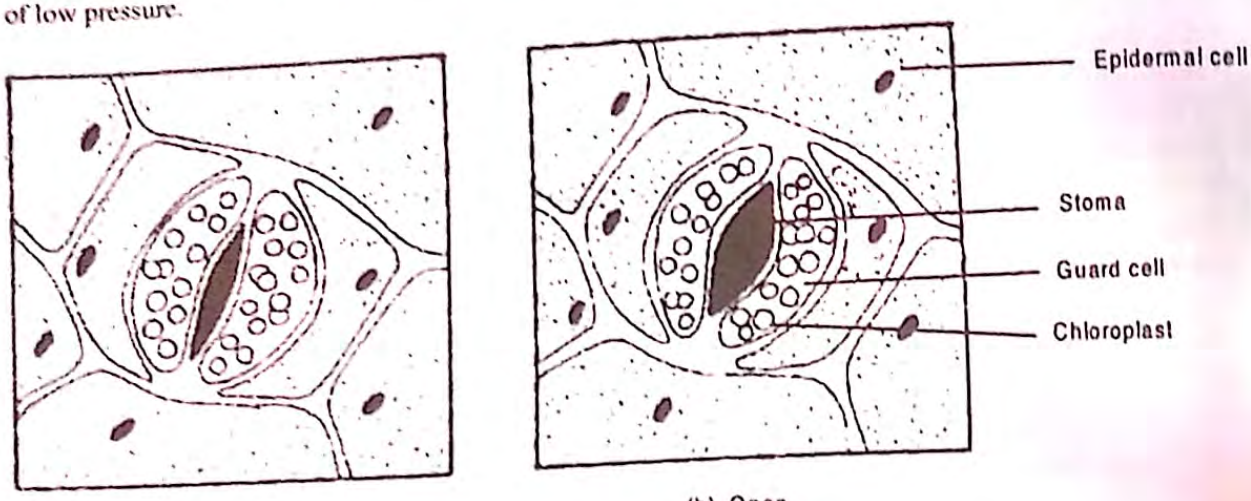


Fig. 3.9: Stomata (Closed

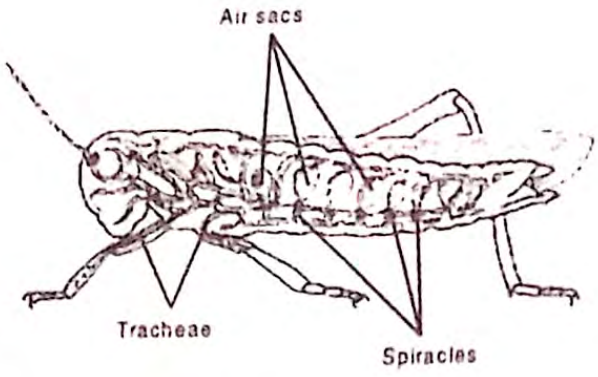


Fig. 3.10: Respiratory system of grasshopper

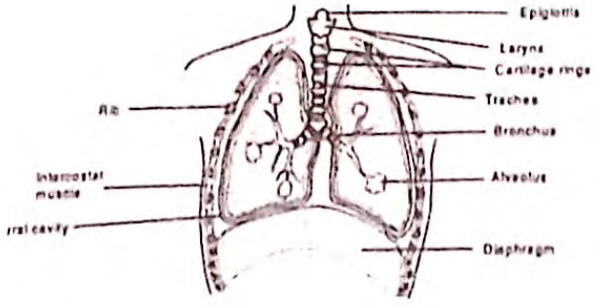


Fig. 3.11: Respiratory system of man

- During expiration (exhaling), the external intercostal muscles draw the ribs down and inwards, thus reducing the volume of thoracic cavity and hence air is forced out of the lungs (Fig. 3.11).

3.15 Excretory Systems

- Among terrestrial plants, waste products of metabolism such as salts and organic acids, are simply stored in the plant either in the form of crystals or as in perennial plants the wastes are deposited in non-living heartwood and are also eliminated when the leaves are shed.
- In platyhelminths, like *Fasciola* and tapeworms, the excretory organs are the flame cells.
- In the earthworms and many other invertebrates, the excretory structures are called **nephridia.**
- In terrestrial arthropods, like the insect, the excretory system is made up of malpighian tubules which lie in the haemocoel and are bathed by blood. The waste product is uric acid which passes outside along with faeces.

3.15.1 The Kidney

- The most important excretory organs of vertebrates are the kidneys which receive blood from the dorsal

aorta through the right and left renal arteries and sends out blood through the right and left renal veins.

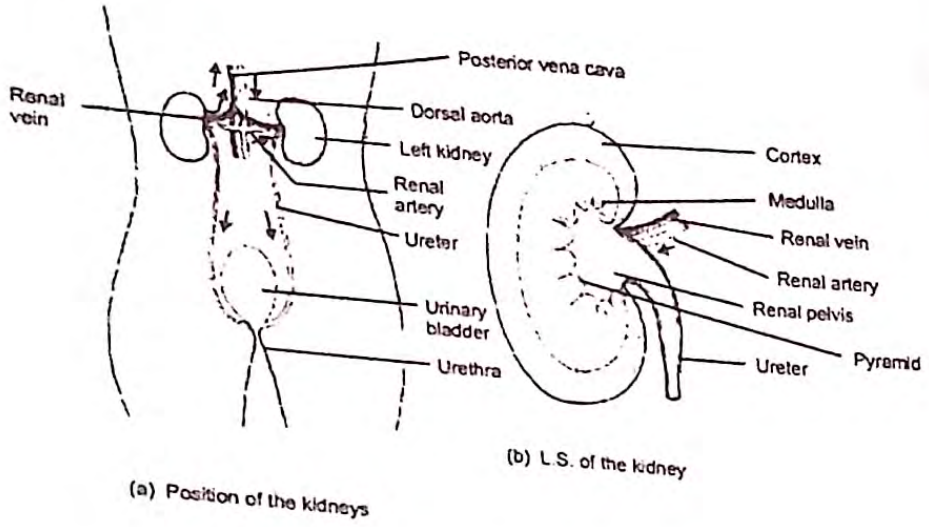
- Transversection of the kidney shows an outer cortex, beneath which is the medulla and beneath the medulla, the hollow pelvis of the kidney.
- The cortex and the medulla consist of thousands of nephrons which are the structural and functional units of the kidney.
- Each nephron consists of a Bowman's capsule, also called renal corpuscle or malpighian corpuscle, with its network of blood capillaries called glomerulus within it.
- The remaining part of a nephron is a tubule consisting of three distinct portions, i.e. the proximal convoluted tubule, the U-shaped loop of Henle, and a distal convoluted tubule.
- The tubule then leads into the pelvis of the kidney via the collecting tubules.
- The cortex contains only the renal corpuscle and the convoluted tubules while the medulla contains the loop of Henle and the collecting ducts.
- Ultrafilteration of the blood takes place in the Bowman's capsules and glucose, amino acids and large amounts of inorganic ions are re-absorbed at the proximal convoluted end from the glomerular filterate.
- Sodium salt is actively transported out of the fluid as it passes through the loop of Henle and finally water is reabsorbed from the fluid at the distal convoluted tubule and the collecting ducts.
- From here the filterate passes through the pelvis into the urinary bladder via the ureter.
- When the bladder becomes full, the urethral sphincter guarding the outlet relaxes so that urine flows out through the urethra.
- Urine contains urea, water and sodium chloride in addition to other compounds.
- In the presence of antidiuretic hormone, dilute urine is produced.
- Kidney stones are solid particles formed in the kidneys which can block kidney tubules.
- Diuresis is a condition of the kidney which causes an increased secretion of urine (Fig. 3.12).

3.15.2 The Liver

- The liver regulates blood sugar by the conversion of excess blood sugar into glycogen.
- It produces bile from bile salts and pigments after decomposition of aged erythrocytes and stores excess iron from such decomposition.
- It carries out deamination of excess amino acids that leads to the formation of urea, ammonia etc. which are excreted by the kidneys.
- It carries out detoxification of poisonous compounds produced in the large intestine by bacterial action and render them harmless.
- It produces fibrinogen which is essential for blood clotting.
- Infective hepatitis is a viral infection of the liver causing inflammation and jaundice.
- Gall stones are solid particles formed within the gall bladder and can block the bile duct.

3.15.3 The Skin

- It is made up of an outer epidermis and an inner dermis (Fig. 3.13). The epidermis has an outer cornified layer (stratum corneum) of dead flat cells, a middle layer called granular layer (stratum granulosum) which contains living cells and an innermost malpighian layer (stratum malpighi) which is a continuous layer of cells that are actively dividing and so produce new epidermis.
- The epidermis contains the pigment melanin which determines the skin colour and protection against harmful ultraviolet rays.
- The dermis is a layer of connective tissue with many elastic fibres, blood capillaries, nerve endings, sweat glands, hair follicles and lymphatics.
- Below the dermis is a layer of subcutaneous fat.
- Increase in the width of blood vessels at high temperatures is known as vasodilation and a decrease in the width is known as vasoconstriction.
- The pigment, melanin, protects the body from ultraviolet rays of the sun.
- Sebum produced by the sebaceous glands of the skin acts as a mild antiseptic apart from keeping the skin supple and water-proof.
- It is a sense organ because of the presence of the nerve endings.
- Sweat produced by sweat glands contains water, urea and sodium chloride which makes the skin an excretory organ.
- In birds and mammals, the skin helps in the maintenance of a constant body temperature (homoithermy) and insulation of the body.



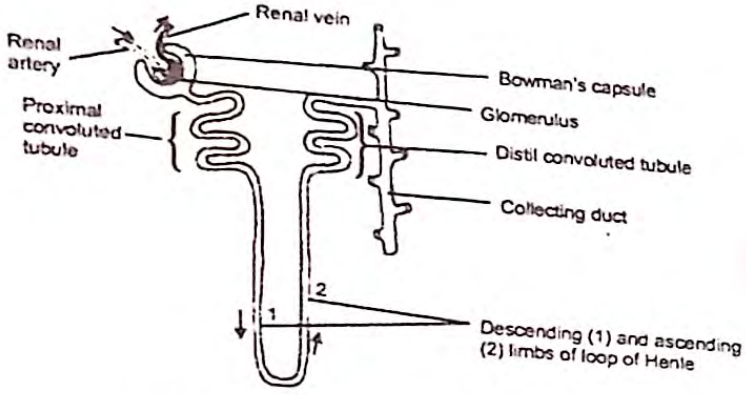


Fig. 3.12: Structure of the mammalian kidney

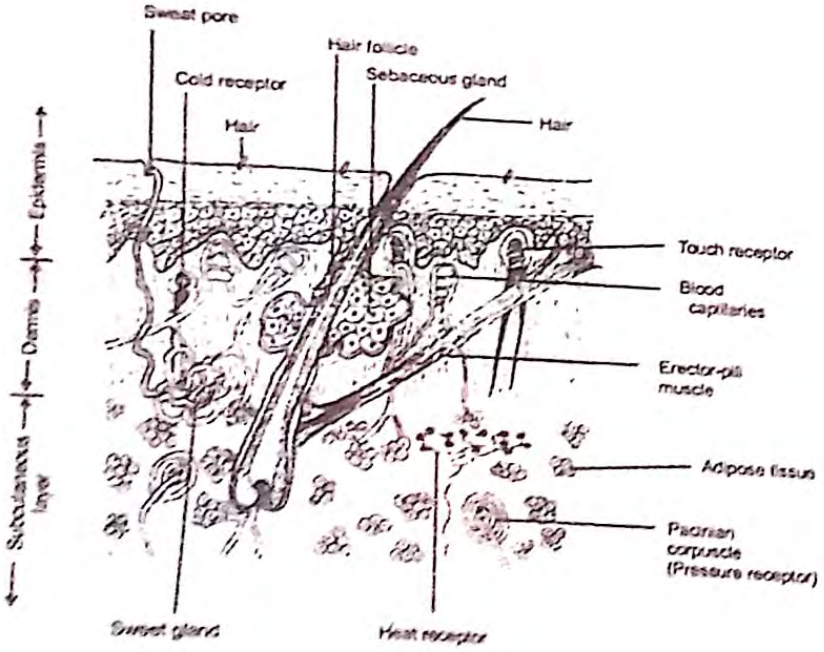


Fig. 3.13: Vertical section through human skin and its sense organs

SSCE/UME–Type Objective Questions

1. The biscuspid valve is located between the
   A. left auricle and left ventricle
   B. aorta and the left ventricle.
   C. Superior vena cana and the inferior vena cava.
   D. right auricle and the superior vena cava.
   E. right auricle and right ventricle.

2. The pulmonary artery carries
   A. deoxygenated blood from the right ventricle to the lungs
   B. oxygenated blood from the right ventricle to the lungs
   C. oxygenated blood from the left ventricle to the left auricle
   D. deoxygenated blood from the left ventricle to the lungs
   E. oxygenated blood from the lungs to the right auricle

3. Which of the following blood components has the greatest affinity for oxygen and carbondioxide?
   A. Blood plasma
   B. Leucocytes
   C. Thrombocytes
   D. Erythrocytes
   E. Lymphocytes

4. The thoracic cavity is separated from the abdominal cavity by a sheet of muscle called
   A. pleural membrane
   B. intercostal muscle
   C. diaphragm
   D. pericardium
   E. epiglottis

5. Deamination is the process during which
   A. excess amino acid is broken down into ammonia and a keto acid
   B. fats and oils are converted into an emulsion for absorption
   C. excess glucose is converted and stored as glucogen in the liver.
   E. excess amino acid is converted into proteins

6. Which of the following statement on urine production is *not* correct?
   A. Concentrated urine is produced under the control of antidiuretic hormone
   B. The total volume of urine produced depends on the need for water excretion
   C. If there is an excess intake of water, dilute urine is excreted.
   D. If there is shortage of water in the body, the volume of urine produced decreases
   E. In the presence of anti diuretic hormone, dilute urine is produced.

7. The pigment in the malpighian layer responsible for skin colouration is known as
   A. haemoglobin
   B. haemocyanin
   C. chlorocruorine
   D. haemoerythrin
   E. melanin

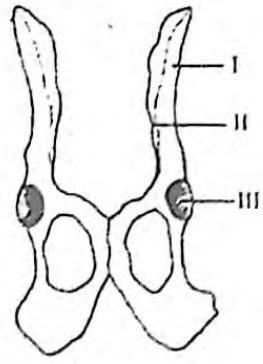
8. Which of the following is the correct route for the movement of material through the kidney?
   A. Glomerulus → Bowman's capsule → loop of Henle–collecting tubule → pelvis.
   B. Glomerulus → loop of Henle → collecting tubule → Bowman's capsule → pelvis
   C. Pelvis → Bowman's capsule → glomerulus → loop of Henle → Collecting tubule
   D. Bowman's capsule → glomerulus → loop of Henle → collecting tubule → pelvis
   E. Bowman's capsule → loop of Henle → glomerulus → colleting tubule → pelvis

9. Blood in mammals transports oxygen because it contains the pigment called
   A. haemoglobin
   B. chlorocruorine
   C. melanin
   D. haemocyanin
   E. haemoerythrin.

10. The loss of water vapour through the aerial parts of the plants to the atmosphere is called
    A. respiration
    B. guttation
    C. Osmoregulation
    D. transpiration
    E. diffusion

11. The upward movement of the sap in the xylem vessels is brought about by
    A. transpiration stream
    B. guttation
    C. capillarity
    D. osmosis
    E. cohesion

Study the diagram below and use it to answer Questions 12 to 14.



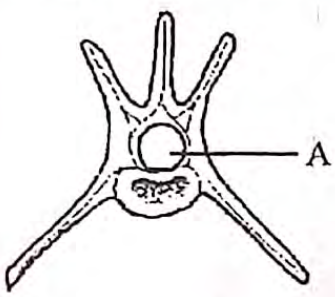
12. The piece of bone represented in the diagram is found in the
    A. pelvic region
    B. pectoral girdle
    C. vertebral column
    D. skull
    E. thoracic region

13. The part labelled I is the
    A. lumber
    B. sacrum
    C. ilium
    D. pubis
    E. coccyx

14. The bone that articulates with the part labelled III is the
    A. humerus
    B. femur
    C. tibia
    D. sacrum
    E. fibula

15. Which of the following is not a function of the mammalian skeleton? It
    A. gives the body its shape
    B. provides a framework on which internal organs are suspended
    C. protects soft and delicate parts of the body
    D. provides attachment for muscles
    E. contracts and relaxes to bring about movement

16. What type of vertebra is represented in the diagram?
    A. Atlas
    B. Axis
    C. Cervical
    D. Sacrum
    E. Lumber



17. The structure labelled A represents the
    A. transverse process
    B. neural spine
    C. neural canal
    D. centrum
    E. metapophysis

18. Which of the following statement is not correct about the function of each group of mammalian vertebrae?
    A. Cervical vertebra support the neck
    B. Thoracic vertebrae articulate with the ribs
    C. Lumbar vertebrae provide attachment for abdominal muscles
    D. Sacral vertebrae support the skull and allow nodding and rotating movements
    E. Caudal vertebrae support the tail and provide attachment for tail muscle

19. The stems of young herbaceous plants are kept upright mainly by
    A. osmotic pressure
    B. turgor pressure
    C. transpiration pull
    D. suction pressure
    E. root pressure

20. Different tissues in plants contribute to the support of the parts as a result of the following characteristics except
    A. malleability
    B. rigidity
    C. flexibility
    D. resilience
    E. hardness

21. Gaseous exchange takes place in the adult toad through the
    A. buccal cavity, bladder and lungs
    B. buccal cavity, skin and lungs
    C. tympanic membrane, lungs and gills
    D. buccal cavity, skin and spiracle
    E. gills, skin and buccal cavity

22. Breathing movement in mammals is accompanied by regular movement of the diaphram and the
    A. intercoastal muscles
    B. vertebral column
    C. Clavicle
    D. pleural cavity
    E. floating ribs

23. Which of the following are components of the glomerular filterate?
    A. All the components of the blood except mineral salts
    B. Water and salts only
    C. All the components of the blood except plasma proteins and cells
    D. Plasma proteins, red and white blood cells
    E. Blood cells, amino acids, salts and nitrogenous waste.

24. Which of the following takes place during the process of detoxication in the liver?
    A. poisonous substances in the body are converted into harmless substances.
    B. Fats and oils are broken into finer particles through emulsification
    C. Excess glucose in the blood are converted into glycogen and stored
    D. Worn-out red blood cells are broken down.
    E. Enough heat is generated to sustain the activities of the liver.

25. The cup-shaped part of the Bowman's capsule contains the
    A. nephridium
    B. glomerulus
    C. loop of Henle
    D. nephron
    E. convoluted tubule

26. The increase in the width of blood vessels in mammalian skin at high temperature is known as
    A. Vasodilation
    B. Sweating
    C. Vasoconstriction
    D. Shivering
    E. Evaporation

27. Muscles are attached to bones by means of
    A. ligaments
    B. cartilage
    C. synovial fluid
    D. tendons
    E. connective tissue

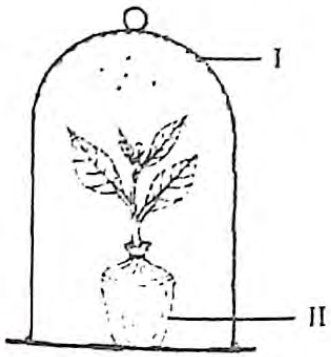
28. Which of the following parts of the mammalian skin is directly involved in excretion?
    A. Sebacious gland
    B. Sweat gland
    C. Horny layer
    D. Hair follicle
    E. Malpighan layer

29. Kidney stone is a disease of the kidney that results in
    A. blockage of the tubules by solid particles
    B. passing out useful materials of the body fluid with urine
    C. presence of large quantity of water in urine
    D. passing out of hormones in urine
    E. prevention of urine production

30. Infective hepatitis is a disease of the
    A. heart
    B. kidney
    C. liver
    D. stomach
    E. pancreas

31. The human skin is regarded as a sense organ because it
    A. separates the body from the outside world
    B. protects the body from cold and heat
    C. regulates the water content of the body
    D. has an outer layer of dead cells
    E. has nerve endings

The diagram below illustrates an experimental set up. Use it to answer Questions 32 and 33.



32. The part labelled I is
    A. a micro organism
    B. a developing insect
    C. a water droplet
    D. an oil droplet
    E. a soil particle

33. The function of the part labelled II is to
    A. prevent the entry of insects into the soil
    B. prevent the entry of light into the soil
    C. prevent light from shining on the delicate roots
    D. prevent evaporation from the soil
    E. make the container air tight

34. Which of the following can be used to detect the loss of water vapour from the leaves of plants?
    A. Hydrated copper sulphate
    B. Blue litmus paper
    C. Red litmus paper
    D. Blue cobalt chloride paper
    E. Lead acetate paper

35. Which of the following structures of the leaf contains air?
    A. Guard cell
    B. Palisade layer
    C. Intercellular space
    D. Vascular bundle
    E. upper epidermis

36. The respiratory organ found in the cockroach is the
    A. air sac
    B. trachea
    C. lung book
    D. lung
    E. gill

37. Which of the following structures functions as an excretory system in flat worms?
    A. contractile vacuole
    B. Nephridium
    C. Flame cell
    D. Malpighian tubule
    E. Kidney

38. Ultrafiltration in the kidney takes place in the
    A. loop of Henle
    B. renal vein
    C. Bowman's capsule
    D. pelvis
    E. pyramid

38. The function of the lenticels is
    A. to remove excess water in the plant
    B. to absorb water from the atmosphere
    C. for gaseous exchange
    D. to absorb light
    E. to store food

40. One of the functions of the xylem is
    A. to strengthen the stem
    B. manufacturing food
    C. reducing loss of water
    D. conducting manufactured food
    E. storing unused sugar

Answer Key

1.	A	9.	A	17.	C	25.	B	33.	D
2.	A	10.	D	18.	D	26.	A	34.	D
3.	D	11.	A	19.	B	27.	D	35.	C
4.	C	12.	A	20.	A	28.	B	36.	B
5.	A	13.	C	21.	B	29.	A	37.	C
6.	E	14.	B	22.	A	30.	C	38.	C
7.	E	15.	E	23.	C	31.	E	39.	C
8.	A	16.	E	24.	A	32.	C	40.	A

Explanations to Objective Questions

1. (A) The valve separating the left auricle from the left ventricle is the bicuspid valve.

2. (A) All arteries carry oxygenated blood except pulmonary artery which carries deoxygenated blood from the right ventricle to the lungs.

3. (D) Erythrocytes have haemoglobin which combines with oxygen to form oxyhaemoglobin that is transported to tissue where they dissociate releasing oxygen. Carbondioxide, from tissues is then carried by erythrocytes in the form of $NaHCO_3$ to where they are released outwards.

4. (C) The diaphragm is the muscular sheet separating the thoracic from the abdominal cavities.

5. (A) Deamination is the removal of amino group (NH_2) from unwanted amino acids.

6. (E) Dilute urine is produced when the blood water level drops, i.e. under the control of antidiuretic hormone.

7. (E) Melanin is responsible for skin colouration.

8. (A) The correct sequence is Glomerulus → Bowman's Capsule → Loop of Henle → Collecting tubule → Pelvis.

9. (A) Mammalian blood has haemoglobin in its erythrocytes which carry oxygen in the form of oxyhaemoglobin.

10. (D) Transpiration brings about the upward movement of water and mineral salts from the roots to the leaves of plants and then to the atmosphere.

11. (A) Transpirational stream brings about the upward movement of sap in the xylem vessels.

12. (A) The bone comes from the pelvic region because it is the pelvic girdle.

13. (C) The part labelled I is the ilium.

14. (B) The bone that articulates with the part labelled III is the femur.

15. (E) Mammalian skeleton is very rigid, it does not contract and relax.

16. (E) The diagram represents the anterior view of lumbar vertebra because it is star-shaped and has a large thick centrum.

17. (C) Structure labelled A represents neural canal through which the spinal cord passes.

18. (D) Sacral vertebrae does not support the skull nor allow nodding and rotation but it articulates with the pelvic girdle.

19. (B) Turgor pressure brings about the stiffness of plant due to intake of water, hence young plant stem is kept upright.

20. (A) All the tissues contribute to the support of the plant parts except malleability which is a characteristic feature of metals.

21. (B) Adult toad uses the buccal cavity, skin and lungs for gaseous exchange.

22. (A) Regular movements of the diaphragm and intercostal muscles accompany breathing.

23. (C) Glomerular filtrate contains water, salts, urea, glucose and amino acids and not plasma proteins and cells.

24. (A) Detoxication is the destruction of various toxins and poisons which enter the body, so that they are rendered harmless.

25. (B) The glomerulus is contained in the cup-shaped part of the Bowman's capsule.

26. (A) Vasodilation is expansion in the width of blood vessels at high temperatures.

27. (D) Tendons connect muscles to bones.

28. (B) Sweat gland excrete water, small quantities of urea, mineral salts and heat in the form of sweat.

29. (A) Blockage of urinary tubules by solid particles in the kidneys is known as kidney stones. Hence urinary function is impaired.

30. (C) Infective hepatitis affects only the liver.

31. (E) The presence of nerve endings in the skin makes it a sense organ because it is sensitive to touch, cold, heat, pain, etc.

32. (C) The part labelled I is a water droplet.

33. (D) The part labelled II is a polythene cover which prevents evaporation of water from the soil.

34. (D) Blue cobalt chloride paper changes to pink in the presence of water.

35. (C) Intercellular space, found within the spongy mesophyll, contains air.

36. (B) All insects respire by means of trachea.

37. (C) All flat worms excrete by means of flame cells.

38. (C) Ultrafiltration in the kidney takes place in the Bowman's capsule.

39. (C) The lenticels play an important role in gaseous exchange , i.e. respiration.

40. (A) Xylem plays an important role in strengthening the stem.

4 Nervous and Hormonal Coordination

4.1 Introduction

- Coordination is brought about by hormones which are secreted by ductless glands also known as **endocrine glands**. Their secretions diffuse directly into the blood where they are circulated all over the body where they exert their effect.
- **Pituitary gland** is a master gland located at the base of the brain because it produces several hormones which stimulate or inhibit the other endocrine glands.
- It controls the rate of growth and excessive secretion produces gigantism and inadequate pituitrin results in dwarfism.
- Thyroid gland located in the neck secretes thyroxine which is an iodine-containing amino acid.
- Thyroxine controls the rate of metabolism, particularly respiration.
- In amphibians thyroxine brings about metamorphosis.
- Deficiency of throxine causes hypothyroidism in which the individual fails to attain normal physical or mental development [cretinism] and inadequate amount of iodine also causes goitre.
- Adrenal glands located just above the kidneys produce two hormones, i.e. **adrenalin** and **noradrenalin.**
- Adrenalin quickens the heart-beat in times of danger such as fright, injury or anger. It increases the rate of oxidation of carbohydrates so that there is an increase in the animals efficiency and it also promotes anxiety.
- Noradrenalin stimulates the contraction of arterioles, thereby increasing blood pressure.
- Pancreas (Islets of Langerhans) secrete **insulin** which determines how much sugar is converted to glycogen and how much is oxidized for energy (regulation of the amount of sugar in the blood).
- Deficiency of insulin results in the disease **diabetes** but formulated regular insulin injection can now control the disease.
- Gonads: The testes and the ovaries produce hormones which control sexual maturity and development, secondary sexual characteristics and activity.
- The testes secrete **testosterone** into the blood which triggers the secondary sexual characteristics found in adult males and it is essential for spermatogenesis.
- The follicles of the ovaries on the other hand produce several hormones called estrogens which promote the development of secondary sexual characteristics in adolescent female and also participate in the monthly preparation of the body for possible pregnancy.
- The corpus luteum formed after ovulation in the ovaries secrete progesterone into the blood stream for the maintenance of pregnancy.
- Relaxin is equally produced as birth approaches.

4.2 Modern Applications of the Plant Growth Hormone: Auxin and Other Plant Hormones

- Fruit development is brought about by the application of auxin to the flowers that are not pollinated. Tomato growers deliberately initiate fruit development in this way.
- Inhibition of Apical Dominance is brought about by the removal of the terminal shoots of plants so that lateral buds can develop. So the downward transport of auxin is inhibited. This principle is regularly exploited by gardeners for ornamental shrubs.
- The ability of auxin to delay abscission has been exploited by apple and orange growers, so careful sprays of auxin will cut down loss of leaves by premature dropping.
- Auxin stimulates the formation of adventitious roots in many species of plants. Adventitious root development is hastened by pretreating the cutting in solutions or powders containing synthetic auxins.
- Other plant hormones are gibberellins and cytokinins.

4.3 Nervous Coordination

- **Central nervous system** consists of the brain and the spinal cord.
- The brain and the spinal cord are covered with three protective membranes (inner pia mater, outer dura mater and a middle arachnoid mater) collectively called **meninges.**

- The **Brain** is divided into three regions: forebrain, midbrain and hindbrain.
- The forebrain is mainly the cerebrum which is the largest part of the brain.
- A median fissure divides the cerebrum into two halves but they are still connected by the corpus callosum. The left cerebral hemisphere controls the right side of the body and vice versa.
- Each cerebral hemisphere is divided into four parts namely; the **frontal, temporal, parietal and occipital lobes** (Fig. 4.1).

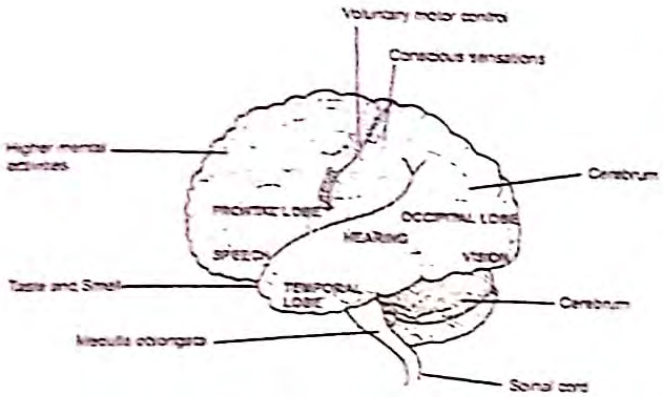


Fig. 4.1: The human brain

- The surface of the cerebrum is grey in colour because of the superficial grey matter and it is thrown into a series of convolutions.
- The cerebrum controls all voluntary actions like learning, intelligence, thinking, imagination, memory, etc. which are aspects of intelligent behaviour.
- Olfactory lobes and hypothalamus are located here.
- The midbrain is quite small and inconspicuous, and relays impulses between the forebrain and the hind brain.
- The hindbrain is made up of cerebellum and medulla oblongata.
- Cerebellum regulates and coordinates muscular movements or balance.
- The medulla oblongata controls involuntary movement of the body like respiration, heartbeat and digestion. It also controls vasoconstriction and vasodilatation of blood vessels, and body temperature.
- The spinal cord in cross-section shows an outer portion which is the white matter and an inner portion which is H-shaped, the grey matter.
- The white matter contains long myelinated nerve fibres and the grey matter the cell bodies of neurones.
- A central canal filled with cerebrospinal fluid runs vertically through the grey matter.
- The spinal cord is concerned with spinal reflex actions and the conduction of nerve impulses to and from the brain.
- **Reflex action** is a rapid and automatic response to stimulus without the involvement of the brain like sneezing, knee-jerk, dilation and constriction of the pupil in the eye, blinking and moving away when you step on something sharp or hot.
- The neurones involved in a reflex action constitute reflex arc.
- A reflex arc involves the reception of a stimulus by a receptor, e.g. skin, which now sends an impulse along a sensory fibre in a nerve that terminates in the spinal cord, after entering through the dorsal root.
- On reaching the spinal cord the impulse passes across a space called synapse, through a transmitter substance called **acetylcholine** to initiate an impulse in an intermediate neurone.
- The intermediate neurone sends the impulse again through a synapse to initiate an impulse in a motor neurone. The axon of the motor neurone leaves the spinal cord through the ventral root to the muscles (effector) which contract when the impulse arrives (Fig. 4.2).
- A **neurone** is the functional unit of the nervous system.
- It transmits electrochemical impulses over a substantial distance.
- A neurone consists of a cell body in which there is a single nucleus.
- The cell body extends to form a number of branches, one of which is produced into a long slender structure called **axon.**
- The axon which carries impulses from the cell terminates in dendrites (Fig. 4.3a).
- Many axons are covered with a fatty sheath called **myelin sheath** which consists of cell membrane of surrounding Schwann cell.
- **Transmission of nerve impulse** along a nerve fibre is achieved by a change in electrical potential, i.e. the passage of an impulse along any part of a nerve fibre makes such a part negative relative to other parts of the nerve fibre.
- Sodium ions flow into the nerve fibre and potassium ions flow out thereby creating an active phase and in

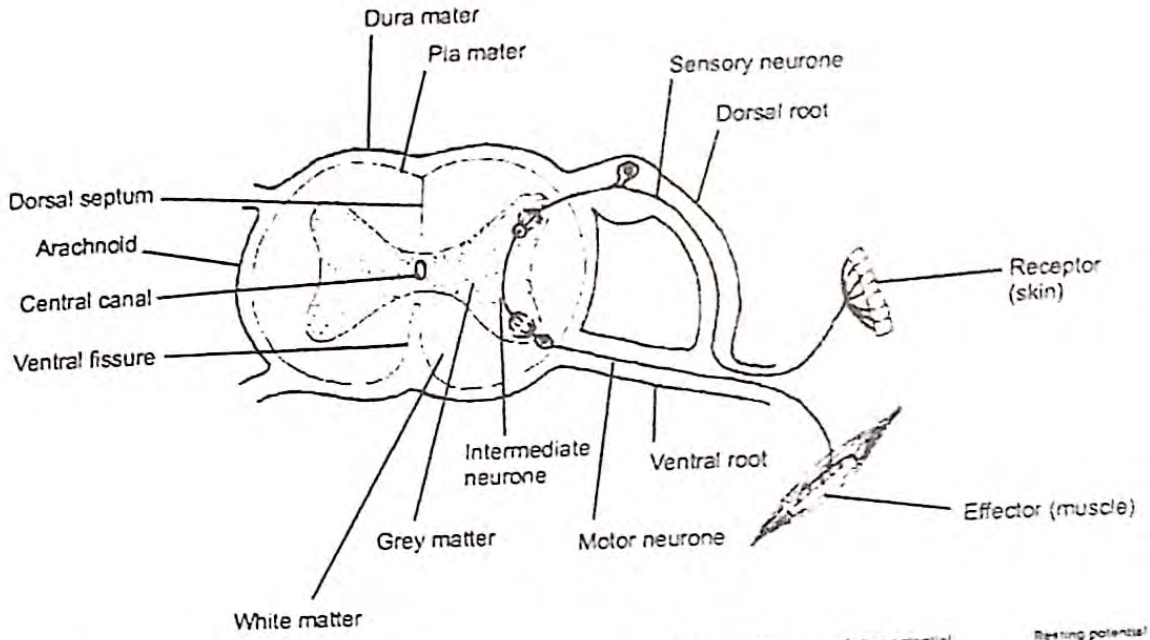


Fig. 4.2: The spinal cord and reflex arc

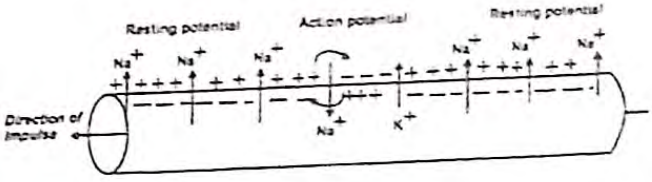


Fig. 4.3b: The nature of nerve impulse

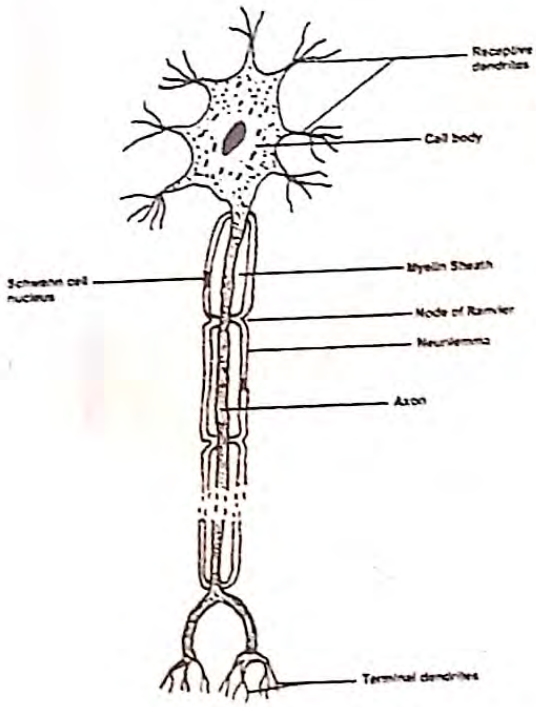


Fig. 4.3a: Neurone structure

a bid to restore the affected part to its resting potential, impulse is transmitted along the fibre in front by the movement of sodium ions inwards and potassium ions outwards and behind the affected part potassium ions move inwards and sodium ions move outwards (Fig. 4.3b).

- **Peripheral nervous system** is made up of sensory and motor nerve fibres which run to and from the central nervous system and the rest of the body, and can be subdivided into sensory-somatic nervous system and automatic nervous system.
- Sensory-somatic nervous system consists of cranial and spinal nerves, and they transmit impulses from the central nervous system to skeletal muscles of the body.
- Autonomic nervous system consists of sensory and motor neurones running between the central nervous system, especially the hypothalamus, and the various internal organs.
- Autonomic nervous system is voluntary and it has two subdivisions, the sympathetic and para-sympathetic nervous systems.
- Conditioned reflex is a learned behaviour and examples include walking, swimming, car driving and bicycle-riding.

4.4 Sense Organs

(a) **The skin** has nerve endings and is sensitive to touch, pressure, pain, heat and cold because of the presence of tactile organs called mechano-**receptors.**

- Touch is detected by receptors located close to the

skin, near the follicles and various internal organs called pacinian corpuscles (pressure receptor).

- Massive mechanical stimulation of the skin produces the sensation of pain.

(b) **Olfactory organ** is the organ of smell which has receptor cells located in the two olfactory epithelia in the nasal cavity.

- As air passes over the olfactory epithelia water- and fat-soluble molecules present in it are dissolved in the mucus layer covering the epithelia.
- This gives rise to the sensation of smell by the stimulation of sensory receptors that send impulses to the brain.
- The organ of smell is located in the antennae of insects.

(c) The **tongue** is the sense organ of taste which has a rough upper surface due to the presence of numerous papillae.

- Groups of sensory cells called taste buds occur on the upper surface of the tongue (Fig. 4.4).

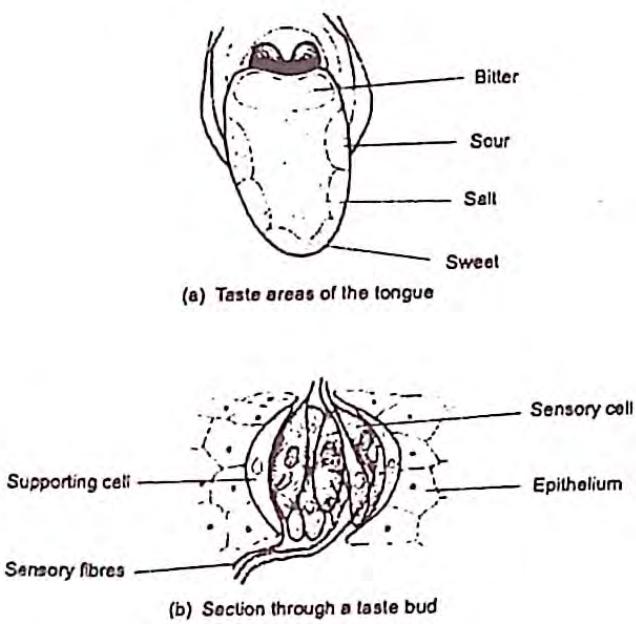
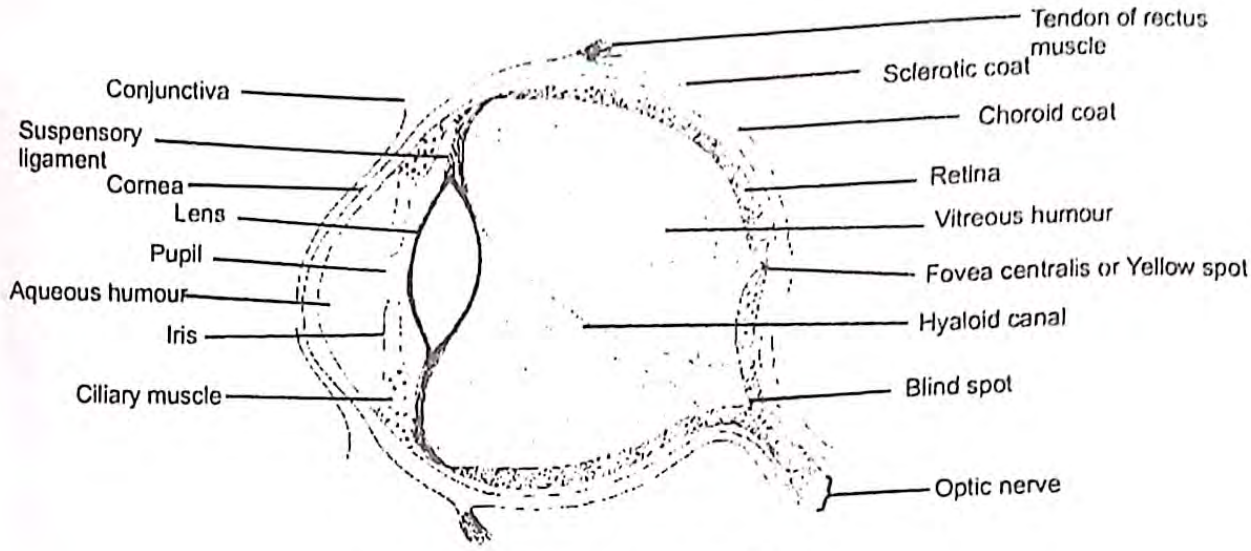


Fig. 4.4: The tongue and taste bud

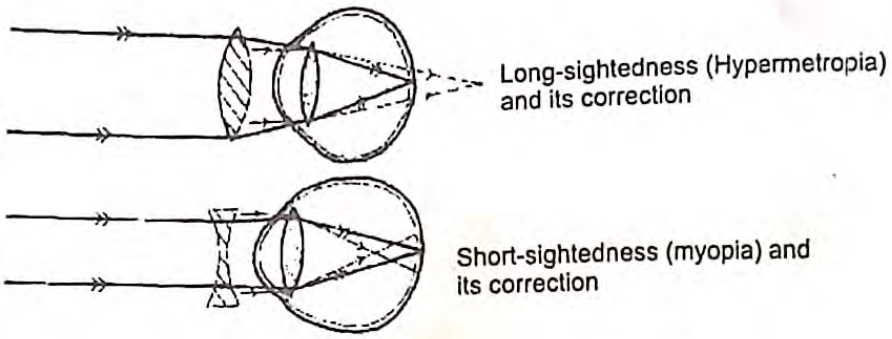
- The sensory cells are long and narrow, and their free surfaces bear short, hair-like processes while there are nerve fibres around their basal portions.
- In man most taste buds are grouped on the tongue such that only four primary sensations of taste are detected namely: sweet sensation at the tip, salty and sour sensation at the sides and centre of the tongue and the back of the tongue to bitter sensation.

(d) The eye is the organ of sight which has a movable upper and lower eyelids protecting its surface (Fig. 4.5a).

- Secretions from the lachrymal (tear) gland lubricates the eye.
- The outer wall of the eyeball is the white and opaque **sclerotic coat** which is transparent in front to form **cornea.**
- The sclerotic coat is lined internally by a vascular layer called **chorold coat** which is modified in front of the eye to form iris.
- The iris is centrally perforated by an aperture called **pupil.**
- The lens which is biconvex in shape lies behind the iris.
- The margin of the lens is attached to ciliary muscles by suspensory ligament.
- The chamber between the lens and cornea is filled with a watery fluid called **aqueous humour** while the chamber behind the lens contains a jelly-like substance called **vitreous humour.**
- The innermost coat of the eyeball is the retina which is concerned with the reception of light (light sensitive).
- The retina contains evenly distributed two types of sensory cells called rods which have visual purple or rhodopsin derived from vitamin A and cones which have iodopsin.
- The cones are more concentrated in a small depressed area of the retina called **yellow spot** or **fovea centralis** which is the most sensitive spot.
- Below the fovea centralis is the blind spot devoid of sensitive cells but is the point at which the optic nerve leaves the eye.
- Accommodation is the ability of the eye to bring about the necessary alterations in the focal length of the lens so that far or near objects are brought to a sharp focus on the retina.
- The size of the pupil controls the amount of light that enters the eye.
- **Longsightedness (Hypermetropia),** i.e. short eyeball or flat lens with the image formed at a point behind the retina; can be corrected by the use of glasses fitted with convex lenses.
- **Shortsightedness (Myopia),** i.e. long eyeball or too spherical eye lens with the image of distant objects brought to a focus in front of the retina; can be corrected by the use of glasses fitted with concave lenses (Fig. 4.5b).
- **Astigmatism** results when the cornea or lens has



(a) Vertical section through the human eye



(b) Common eye defects. The path of light rays without glasses is shown in single arrow; with glasses, in double arrows

Fig. 4.5: The human eye

many irregularities in its curvature hence no focal point, can be corrected by glasses fitted with cylindrical lenses.

(e) The ear is the organ of hearing capable of detecting the vibration of sound (Fig. 4.6). It also functions in balancing.

- The outer or external ear consists of trumpet-shaped pinna which helps in gathering and directing sound waves, and also in judging the distance of sound waves.
- External auditory meatus connects the pinna to the middle ear.
- the middle ear consists of the eardrum (tympanum) and three small bones called **ossicles** which are **Malleus (hammer), Incus (anvil) and stapes** (stirrup) in that order, that function in transmitting vibrations. Stapes then link up with fenestral ovalis.
- The middle ear is air-filled and is connected to the back of the throat by eustachian tube for equalisation of pressure.
- The middle ear makes connection with the inner ear through oval window (fenestra ovalis) and round-window (fenestra rotunda).
- The inner ear which is fluid-filled has a complicated structure. The structure itself is filled with endo-lymph and it has an upper end of the three semi-circular canals at right angles to one another (organs of balance).
- The middle part of the structure is the sacculus and utriculus and the lower part the cochlea.
- The cochlea houses the organ of corti responsible for sound reception.
- Branches of auditory nerve coming from the three semi-circular canals and cochlea send impulses to the auditory centre of the brain.

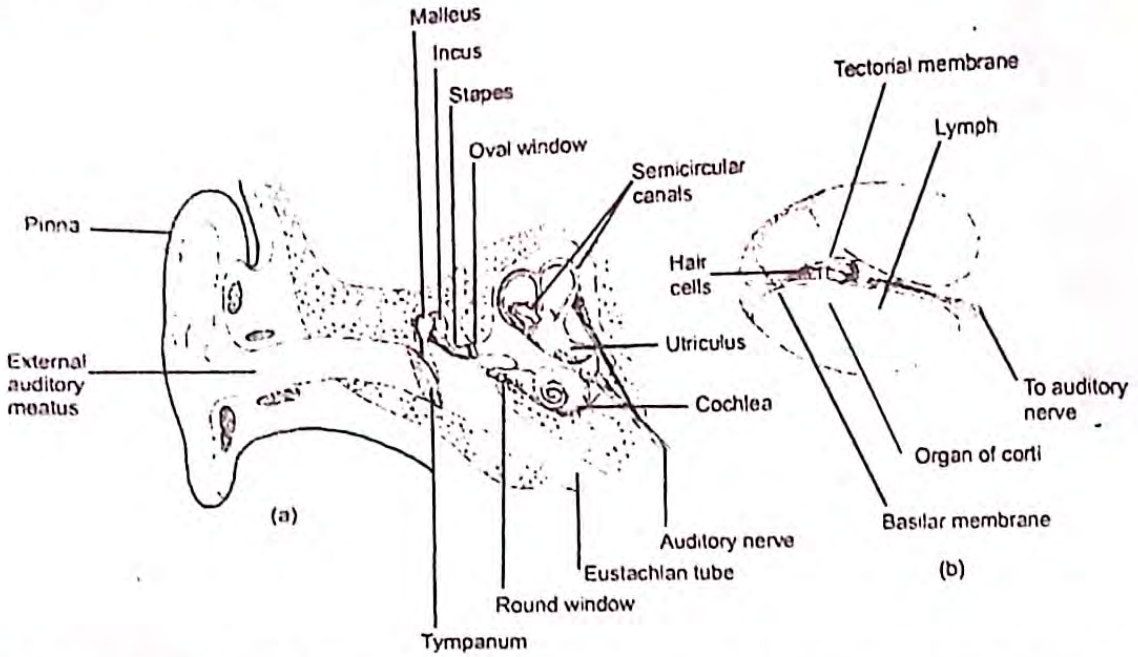


Fig. 4.6: The human ear (a) Vertical section (b) Section through cochlea showing organ of corti

SSCE/UME–Type Objective Questions

1. Which of the following statements is correct of hormones? Hormones are
   - A. secreted into the blood through ducts
   - B. secreted directly into the blood stream
   - C. inactive chemical substances in the blood stream
   - D. non-specific in their mode of action
   - E. secreted at the site of the target organs.

2. The pituitary is called 'Mastergland' because
   - A. it is located in the brain
   - B. its secretions are more numerous than those of any other gland
   - C. its secretion controls other endocrine glands
   - D. it is the only organ that produces hormones
   - E. it regulates muscular movements.

3. The brain and the spinal cord make up the
   - A. peripheral nervous system
   - B. autonomic nervous system
   - C. central nervous system
   - D. somatic nervous system
   - E. sympathetic nervous system.

4. Which of the following specialised structures are stimulated by touch?
   - A. Receptors
   - B. Synapse
   - C. Cell bodies
   - D. Myelin sheath
   - E. Relay neurone

5. Which of the following is the function of the hormone secreted in the Islet of Langerhans? It
   - A. is used in protein metabolism
   - B. is involved in sugar synthesis
   - C. aid the rate at which glucose is converted to glycogen
   - D. controls the fluid content of the body
   - E. plays a role in the pH balance of the body fluid.

6. The spaces between neurones through which impulses cross by means of chemical reaction are called
   - A. axons
   - B. ganglia
   - C. synapse
   - D. dendrites
   - E. pia matter

7. The autonomic nervous system comprises
   - A. Sympathetic and parasympathetic systems
   - B. Cranial and sacral nerves
   - C. Brain and cranial nerves
   - D. Spinal cord and spinal nerves
   - E. Brain and spinal nerves.

8. The condition known as cretinism is caused by the deficiency of
   - A. Adrenalin
   - B. Vitamin A
   - C. Insulin
   - D. Thyroxin
   - E. Vitamin C

9. Which of the following parts of the mammalian brain is involved in taking the decision to run rather than walk?
   A. Cerebellum
   B. Medulla oblongata
   C. Cranial nerves
   D. Midbrain
   E. Cerebrum

10. The hormone which regulates the amount of glucose in the blood is called
   A. adrenalin
   B. auxin
   C. insulin
   D. throxine
   E. ptyalin

11. Which of the following stimuli are not perceived through the skin of mammals?
   A. Light
   B. Pressure
   C. Pain
   D. Heat
   E. Cold

12. Which of the following parts of the tongue does not correspond to taste indicated against it?
   A. Tip-sweet
   B. Centre-salt
   C. Back-sour
   D. Side-sour
   E. Back-bitter

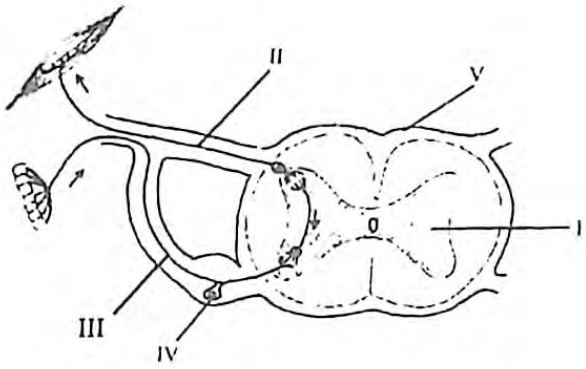
13. Over-secretion of thyroxin is likely to lead to
   A. thinnes of body
   B. Sluggishness
   C. Cretinism in infants
   D. reduced metabolism
   E. dwarfism

14. The human skin is regarded as a sense organ because it
   A. separates the body from the outside world
   B. protects the body from cold and heat
   C. regulates the water content of the body
   D. has an outer layer of dead cells
   E. has nerve endings

15. Which part of the ear is responsible for the maintenance of balance?
   A. Ossicles
   B. Tympanic membrane
   C. Eustachian tube
   D. Semi-circular canals
   E. Cochlea

16. The part of the brain responsible for the control of voluntary movement and interpretation of sensations is
   A. Cerebrum
   B. Cerebellum
   C. Optic lobe
   D. Medulla oblongata
   E. Olfactory lobe

Use the diagram below to answer Questions 17 to 19.



17. The part labelled I is the
   A. white matter
   B. grey matter
   C. central canal
   D. synapse
   E. gangtlon

18. What is the function of the part labelled III?
   A. Conducting nerve impulses to the receptor organ
   B. Conducting nerve impulses to the effector organ
   C. Translating the impulses received from the organ
   D. Amplifying the impulses received from the organ
   E. Transmitting the impulses to the brain

19. Which of the following statements is correct of the structure labelled III? It
   A. consists mainly of cell bodies
   B. conducts impulses to the spinal cord
   C. conducts impulses from the spinal cord to the effector organs
   D. conducts impulses to the brain
   E. interprets stimulus in the spinal cord

20. The organ which is sensitive to smell in a cockroach is
   A. nostril
   B. pedipalp
   C. antenna
   D. mandible
   E. anal cercus

21. Which of the following is the most sensitive spot of the retina?
   A. conjuctiva
   B. cornea
   C. lens
   D. yellow spot
   E. blind spot

22. What part of the brain is concerned with the

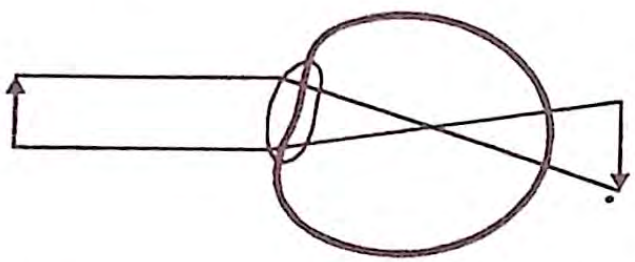
regulation of the heart beat, body temperature and breathing rate?

A. Cerebellum
B. Medulla oblongata
C. Cerebrum
D. Offactory lobe
E. Cortex

23. When an axon is at rest, the concentration of ions on either side of the membrane are different. Which of the following is correct about the concentration of ions on either side of the membrane?
    A. There is an excess of potassium ions inside the axion and an excess of sodium ions outside
    B. The inside of the axon becomes positively charged while the outside is negatively charged.
    C. There is an excess of sodium ions on the inner side of the axon.
    D. Chloride and potassium ions begin to move across the membrane
    E. Calcium ions accumulate on the inner side of the axon.

24. The groups of sensory cells found on the upper surface of the tongue are called
    A. Ampullae
    B. Taste buds
    C. Nerve colls
    D. Somatic cells
    E. Tactile cells

Use the diagram below to answer Questions 25 and 26.



25. What type of eye defect is illustrated in the diagram above?
    A. Hypermetropia
    B. Myopia
    C. Cataract
    D. Astigmatism
    E. Glaucoma

26. This defect can be corrected by the use of
    A. Convex lens
    B. Concave lens
    C. Cylindrical lens
    D. Surgical operation
    E. Biconcave lens

27. The hormone which tones up the muscles of a person in time of danger is from the
    A. Thyroid gland
    B. Pancreas
    C. Adrenal gland
    D. Liver
    E. Spleen

28. What part of the mammalian brain is responsible for maintaining balance?
    A. Medulla oblongata
    B. Cerebellum
    C. Optic lobe
    D. Cerebrum
    E. Olfactory lobe

29. A gland in the human body secretes two hormones, one of which causes a disease called diabetes when there is a deficiency. Which of these organs produces the hormones?
    A. Spleen
    B. Parathyroid
    C. Adrenal body
    D. Thyroid
    E. Pancreas

30. The following are connected with the movement of a reflex action
    I. Central nervous system
    II. Muscle
    III. Skin
    IV. Sensory nerve
    V. Motor nerve

    Which of the following sequences indicate a correct path?
    A. 1–2–3–4–5
    B. 2–1–4–5–3
    C. 3–4–1–2–5
    D. 4–1–5–2–3
    E. 3–4–1–5–2

31. Accommodation of the human eye is best described as
    A. Ability to see distant objects
    B. Movement of upper and lower eyelids
    C. Focussing a near and distant objects
    D. Changing of lens shape due to varying light intensities
    E. Constant adjustment of the iris

32. The function of the ossicles (malleus, incus and stapes) in the mammalian ear is the
    A. transmission of vibrations
    B. regulation of pressures
    C. support of the inner ear
    D. maintenance of balance during motion
    E. secretion of oil

33. Unlike auxins, gibberellins
    A. induce the formation of adventitious roots
    B. do not affect leaf and fruit absicssion
    C. cannot stimulate stem elongation
    D. are quite effective as herbicides
    E. increase growth in the girth of a plant

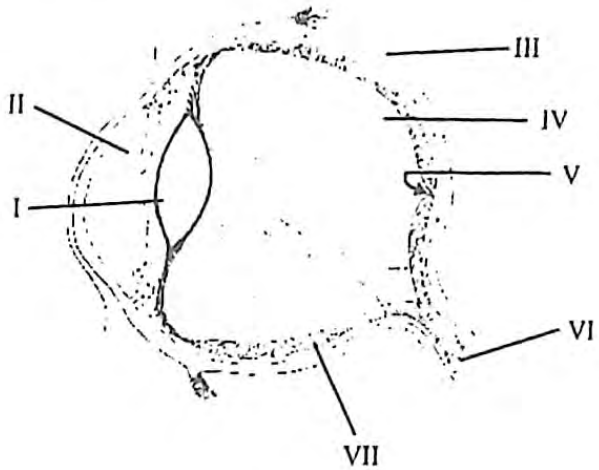
young ovary with

A. cytokinin, abscisic and the enthylene
B. gibberellins ethylene and abscisic acid
C. auxin, abscisic acid and ethylene
D. auxin, cytokinin and gibberellin
E. auxin, kinin and gibberellin

35. A fundamental similarity between nervous and hormonal systems is that both
    A. involve chemical transmission
    B. have widespread effects
    C. shed chemicals into the blood stream
    D. evoke rapid responses
    E. are controlled by the conditional reflex

36. A dwarf plant can be stimulated to grow to normal height by the application of
    A. thyroxin
    B. gibberellin
    C. insulin
    D. kinin
    E. abscisic acid

37. Neurones that receive stimuli from the body or internal organs are called
    A. sensory neurones
    B. efferent neurones
    C. motor neurones
    D. relay neurones
    E. multipolar neurones



38. The greatest concentration of light receptors is found in the part labelled
    A. VI
    B. V
    C. IV
    D. III
    E. I

39. The aqueous humour is represented by the part labelled
    A. I
    B. II
    C. III
    D. VI
    E. VII

40. The light sensitive layer is labelled
    A. I
    B. III
    C. IV
    D. VI
    E. VII

Answer Key

1.	B	9.	E	17.	B	25.	A	33.	C
2.	C	10.	C	18.	D	26.	A	34.	D
3.	C	11.	A	19.	B	27.	C	35.	D
4.	A	12.	C	20.	C	28.	B	36.	B
5.	C	13.	A	21.	D	29.	E	37.	A
6.	C	14.	E	22.	B	30.	E	38.	B
7.	A	15.	D	23.	A	31.	C	39.	B
8.	D	16.	A	24.	B	32.	A	40.	E

Explanations to Objective Questions

1. (B) Hormones are produced from ductless glands and secreted directly into the blood stream.

2. (C) The action of the pituitary gland controls the endocrine system, hence it is a 'master gland'.

3. (C) The brain and the spinal cord make up the central nervous sytem.

4. (A) Receptors are sense organs which are stimulated by touch, pressure, pain, heat and cold.

5. (C) Islets of Langahans present in the pancreas secrete insulin which regulates the amount of glucose in the blood.

6. (C) Synapses are gaps across which neurones pass impulses through chemical reaction.

7. (A) The sympathetic and parasympathetic systems make up the autonomic nervous system.

8. (D) Cretinism is caused by the deficiency of thyroxine, a hormone secreted by the thyroid gland in the neck region.

9. (E) The decision to run rather than walk is a voluntary action and thus is controlled by the cerebrum in the fore-brain.

10. (C) Insulin produced by the Islets of Langahans of the pancreas regulates the amount of glucose in the blood.

11. (A) Light is not perceived through the skin of mammals but through the eyes which are photoreceptive.

12. (C) The back of the tongue is for bitter taste only and not for sour.

13. (A) Thyroxine regulates body metabolism and its overproduction causes overactivity and loss in weight which leads to thinness of the body.

14. (E) The presence of nerve endings in the skin make it sensitive to touch, cold, heat, pressure and pain.

15. (D) The three semi-circular canals lying at right angles to one another maintain balance in mammals.

16. (A) Cerebrum controls voluntary movement and interpretation of sensation.

17. (B) The part labelled I is the grey matter made up of cell bodies.

18. (D) The part labelled III is the sensory neurone amplifying the impulses received from the organ and conducting them to the spinal cord.

19. (B) See question 18.

20. (C) The antenna in a cockroach is sensitive to touch,smell and vibrations.

21. (D) The most sensitive spot of the retina is the yellow spot, i.e. fovea centralis because photosensitive cells called cones are more concentrated here.

22. (B) The medulla oblongata controls the involuntary actions in the body like heart beat, body temperature, breathing rate, construction and dilation of blood vessels.

23. (A) When an axon is at rest, there is an excess of potassium ions inside the axon and an excess of sodium ions outside.

24. (B) The groups of sensory cells found on the upper surface of the tongue are the taste buds.

25. (A) Hypermetropia (Long sightedness) because the image is formed behind the retina.

26. (A) A convex lens will converge light rays so that the image will fall on the retina.

27. (C) Adrenal glands secrets adrenalin which tones up the muscles of a person in time of danger.

28. (B) Cerebellum is concerned with coordination of movement and maintenance of equilibrium.

29. (E) Pancreas contains Islets of Langahans which secretes insulin which regulates the amount of glucose in the blood.

30. (E) The correct sequence is 3→4→1→5→2, i.e. skin → sensory nerve → central nervous system → motor nerve → muscle.

31. (C) Accommodation is the ability of the eye to focus near and distant objects on the retina.

32. (A) The ossicles are responsible for the transmission of vibrations from the tympanum to the oval window (fenestra ovalis).

33. (C) Gibberellins cannot stimulate stem elongation.

34. (D) Auxins, cytokinin and gibberellins bring about fruit enlargement when sprayed on young ovary.

35. (D) Nervous and hormonal systems evoke rapid responses.

36. (B) Gibberellin (a plant hormone), when applied to a dwarf plant will stimulate the plant to grow to normal height.

37. (A) Neurones that receive stimuli from receptors are called sensory neurones.

38. (B) The part labelled V has the greatest concentration of light receptors.

39. (B) The part labelled II is the aqueous humour.

40. (E) The part labelled VII is the light sensitive layer, i.e. retina.

5 Reproduction

5.1 Reproductive Behaviour (Animals)

- Males of many species engage in courtship behaviour which ensures that the male and female reproductive structures will be ready for mating at the same time.
- Males of many species, e.g. lizards and birds establish territories which they defend from other males of their species and into which they attract females of their species for breeding.
- Territories serve as breeding grounds and places of rearing young.

5.2 Male Reproductive System of a Mammal

- The male reproductive structure produces sperm cells and also delivers these cells to the reproductive tract of the female.
- Sperm production takes place in the two testes which is outside the abdominal cavity so that the testes remain at a lower temperature than that of the rest of the body and hence favours sperm production (spermatogenesis) (Fig. 5.1a).
- Each testis consists of a tight mass of seminiferous tubules the walls of which have diploid cells (spermatogonia) which undergo meiotic division resulting in the formation of haploid spermatozoa.
- The seminiferous tubules meet and join to form sperm ducts which lead to the epididymis which also in turn leads into vas deferens.
- The vas deferens of each testes open into the top of the urethra as it leaves the urinary bladder.
- The seminal vessicle, prostate gland and cowpers gland empty into the vas deferens, and urethra respectively.
- The secretions of the seminal vessicle, prostate and cowper's glands and the sperm cells form semen.

5.3 Female Reproductive System of a Mammal

- Female reproductive organs are a pair of ovaries.
- Oviducts lie very close to each ovary and it is through these that the egg (ovum) passes down into the womb (uterus) lower down in the abdomen (Fig. 5.1b).
- The uterus communicates with the outside through the vagina.
- The neck of the womb is the cervix situated at the junction between the vagina and the womb.
- The female reproductive system function in the production of eggs (ova), reception of male sperm, provision of right conditions for fertilization to occur and be able to nourish the developing baby not only before birth but also after.

5.4 Metamorphosis in a Toad

- Metamorphosis is the stages which certain animals like insects, toads and frogs undergo before they attain adulthood.
- Twelve days after fertilization the embryo wriggles free of its membranes to begin life as a tadpole, swims for a short time and then attaches itself to a water-weed or stone by means of the sticky secretion of glands in the oral sucker or cement gland (Fig. 5.2).
- The tadpole is without mouth and it subsists on yolk in the endoderm cells.
- External gills appear, v-shaped somites are clearly visible in the tail region, stomodaeum (mouth) perforates through the foregut in front of the oral sucker and develop horny outgrowths with which the tadpole rasps algae.
- External gills are replaced by internal gills, tadpole then feeds on vegetation and a long coiled intestine becomes visible.
- By the end of the seventh week, two buds from which hind-limbs will develop emerge and by the eighth week they are fully developed and soon after the fore-limbs emerge through the operculum.
- The tadpole comes to the surface frequently to take gulps of air and as a result the lungs gradually replace the gills as respiratory structure.
- The tongue becomes longer in preparation for food capture, the fore- and the hind-limbs are well developed with the hind-limbs becoming considerably elongated.
- The tail shortens and by the third month, the little toad leaves the water with a simple tail that slowly disappears.

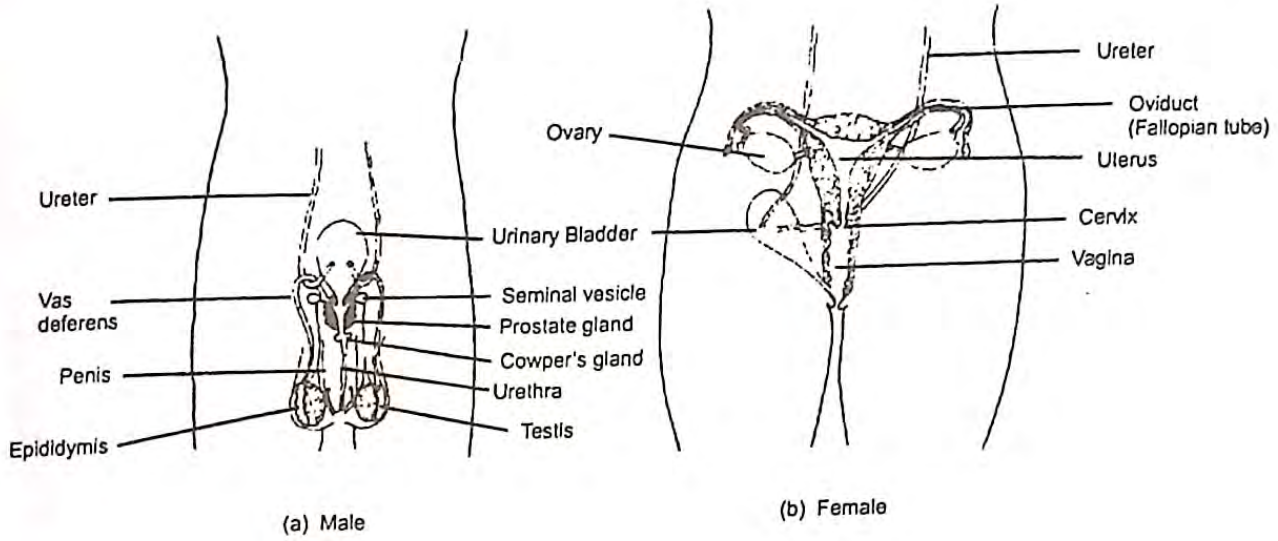


Fig. 5.1: Human reproductive organ

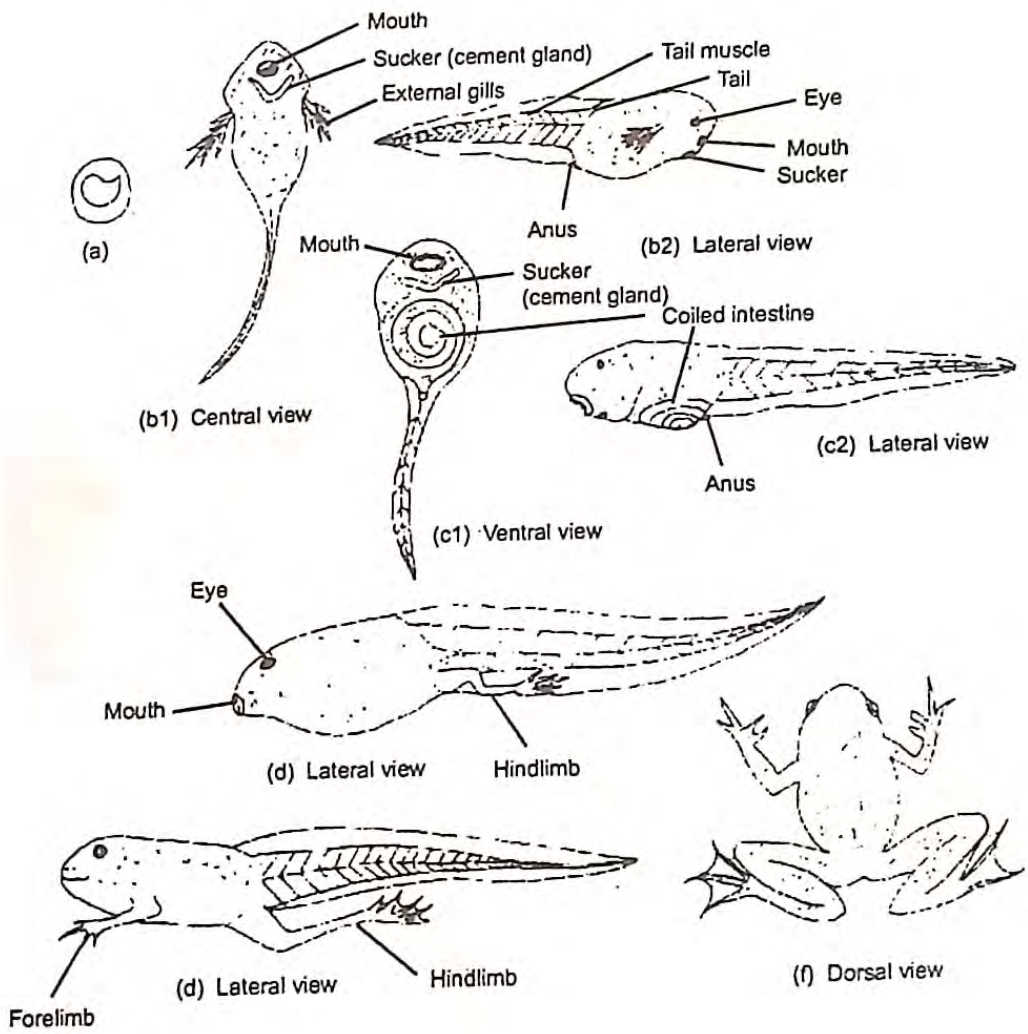


Fig. 5.2 Development and metamorphosis in the toad
(a) Egg (b) External gill stage (c) Internal gill stage (d) Hindlimb stage
(e) Hind- and Fore-limb stage (f) Immature adult

5.5 Metamorphosis in Insects (Life History)

a. Housefly

- Goes through complete metamorphosis, i.e. egg —> larva —> pupa —> adult.
- A female lays 100–150 eggs in 2–7 batches on moist decaying animal or plant matter including faeces.
- Eggs hatch in 2–3 days into tiny white larvae called maggots, each of which has twelve body segments, pointed anterior and a very broad posterior end.
- The head bears a pair of hooked mandibles used for tearing up food and locomotion is achieved by the use of spiny pads on the ventral surface of sixth to twelfth segments.
- A pair of spiracles lie on the dorsal surface of the second segment and another pair on the last segment.
- The maggot moults two or three times, then pupates about five days after hatching.
- The pupa is encased in puparium, does not feed, and metamorphosis is completed in six to twelve days.
- The adult emerges by dialating a sac-like organ called ptilium in front of its head so that the pressure exerted breaks open the puparium.
- The life cycle lasts for about four weeks.

b. Cockroach

- Undergoes complete metamorphosis, i.e. egg → larva → pupa → adult.
- After mating the female cockroach lays its eggs in a honey egg-case called ootheca formed in the body.
- Ten to sixteen eggs are arranged in two rows in the

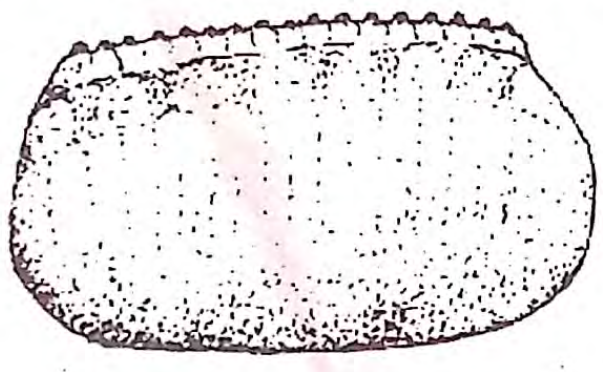
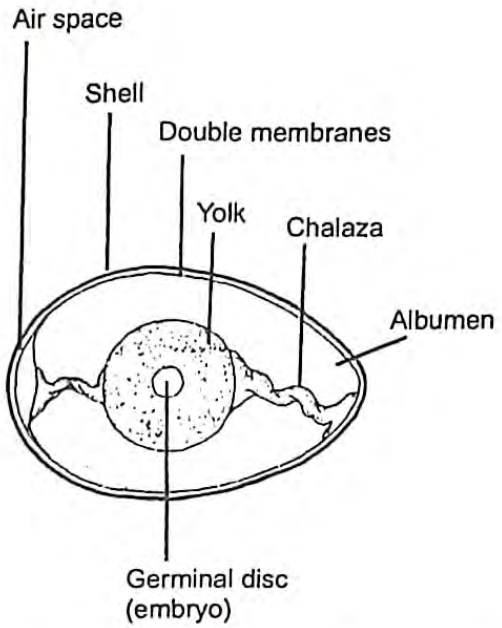


Fig. 5.3: **Ootheca (Egg case)**

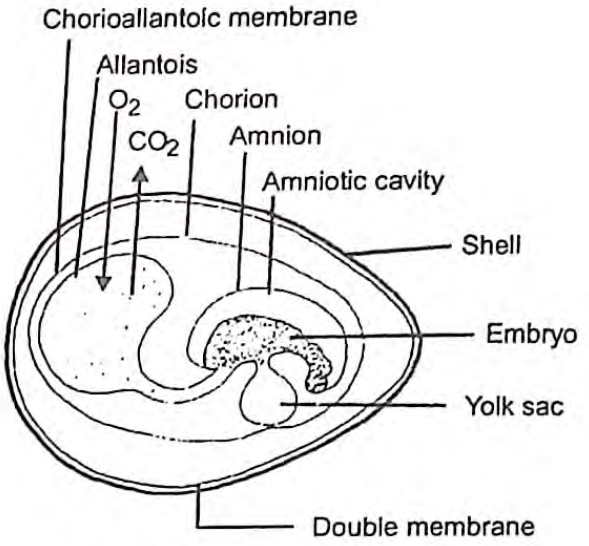
- The eggs are deposited in a dark, warm and humid place after being carried around for some time in the abdominal pouch.
- Eggs hatch into tiny, wingless white nymphs which increase in size and then moults, and after the last moult the wings become fully formed.
- Development from egg to adult lasts for about 11 to 20 months and the adult life lasts for 3 to 12 months.
- The nymphs are wingless, small in size and sexually immature compared to the adult.

5.6 Adaptive Features in a Developing Animal

- The eggs of fish, amphibians, reptiles and birds are telolecithal in that the yolk is aggregated chiefly towards one pole of the cell, i.e. the vegetative pole (V.P.) whilst the egg nucleus lies near the other pole, i.e. the animal pole (A.P.).
- In birds so much yolk is present with the result that non-yolky portion is restricted to a tiny patch called blastodisc, at the animal pole.
- Most amphibians do not take care of their eggs because the eggs are simply left in water to develop after fertilization but the gelatinous layer of albumin surrounding the egg absorb water, swell and provide some physical protection for the egg, keep the eggs warmer compared to surrounding water. The dark upper surface absorbs sun's rays and provides protective colouration when viewed from above. Similarly, the light underside of the egg blends with the sky when viewed from underneath.
- Reptile eggs are protected by waterproof shell and are laid and buried in a warm location abandoned by the mother. There is a plentiful supply of yolk and when the young emerge, they resemble their parents.
- Birds lay shelled eggs fully tended by the parents. The eggs are richly supplied with yolk, and the young hatch with most of the adult features. Parents continue to care for the newly hatched young until they are ready to take care of themselves.
- A shelled egg (Fig. 5.4) is an important adaptation permitting bird and reptile embryos to develop away from water. The egg shell is permeable to gases and impermeable to water. As the embryo develops from the zygote after fertilization, four special membranes called extraembryonic membranes are also formed, i.e. the **yolk sac**, which surrounds the yolk and connects the embryo with its major source of good; the **amnion** which grows around the embryo, enclosing it in a fluid-filled cavity; i.e. amniotic cavity; the **chorion** which lines the inner surface of the egg shell, aids exchange of gases (O_2 and CO_2) between the embryo and the outside air, and the **allantois**, which serves as a reservoir for metabolic wastes created by the embryo during development.



(a) Shelled egg (unincubated)



(b) The extraembryonic membranes of the developing chick (incubated)

Fig. 5.4: The chicken egg

- In higher mammals, the extraembryonic membranes form a placenta and umbilical cord, which connects the embryo to the mother's uterus by the process of implantation. The placenta extracts not only food but also oxygen from the uterus. Carbon dioxide and other excretory wastes (urea) are similarly transferred to the mother's circulatory system for eventual disposal by excretion. Gestation is the period between formation of zygote and birth. After birth, in mammals the young is nourished on milk supplied by the mother's mammary gland. Training in behavioural pattern characteristic of the species forms part of parental care.
- Menstration occurs on monthly basis when ovum is not fertilized resulting in the removal of the lining of the uterus which may cause pains.

5.7 Reproduction in Flowering Plant (Insect-pollinated Flower)

- A flower is the reproductive part of a flowering plant and its basal part is made up of a receptacle which is found on the end of a stalk called **pedicel.** On the receptacle are found four types of floral leaves arranged in rings and starting from the outside the whorls are **sepals** which are collectively called **calyx; petals** which are collectively called **corolla; stamen** also known as **androecium** (male part of a flower), and **carpels** collectively called **gynoecium** or **pitsil** (the female part of a flower) (Fig. 5.5).
- Sepals are usually green in colour but in some plants they may have same colour as the petals, i.e. they become **petaloid.** The calyx protect the flower when in bud.
- The corolla in many cases are scented and their function is to attract insects for pollination. The calyx and corolla are known as **perianth.**
- The androecium has the male organs or stamens. Each stamen consists of two parts, i.e. the anther which contains the **pollen grains** and the **filament** which is the stalk of the anther.
- The gynoecium is the female organ of a flower which contains ovules. The female part of a flower is made up of three parts namely:
  - the ovary which contains the ovules;
  - the style which connects the ovary to stigma;
  - the stigma which receives the pollen grains during pollination.
- When there is only one carpel in one pistil, the pistil is known as **monocarpous pistil.** When it is more than one carpel that are free, it is known as **apocarpous pistil** but when they are fused together, it is known as **syncarpous pistil.**
- **Placentation** is the arrangement of ovules on the placenta inside the ovary. There are four basic types:
  - **Marginal placentation** where the ovary is formed from a single carpel joined along its edges and the ovules situated along these edges, e.g. Cassia.

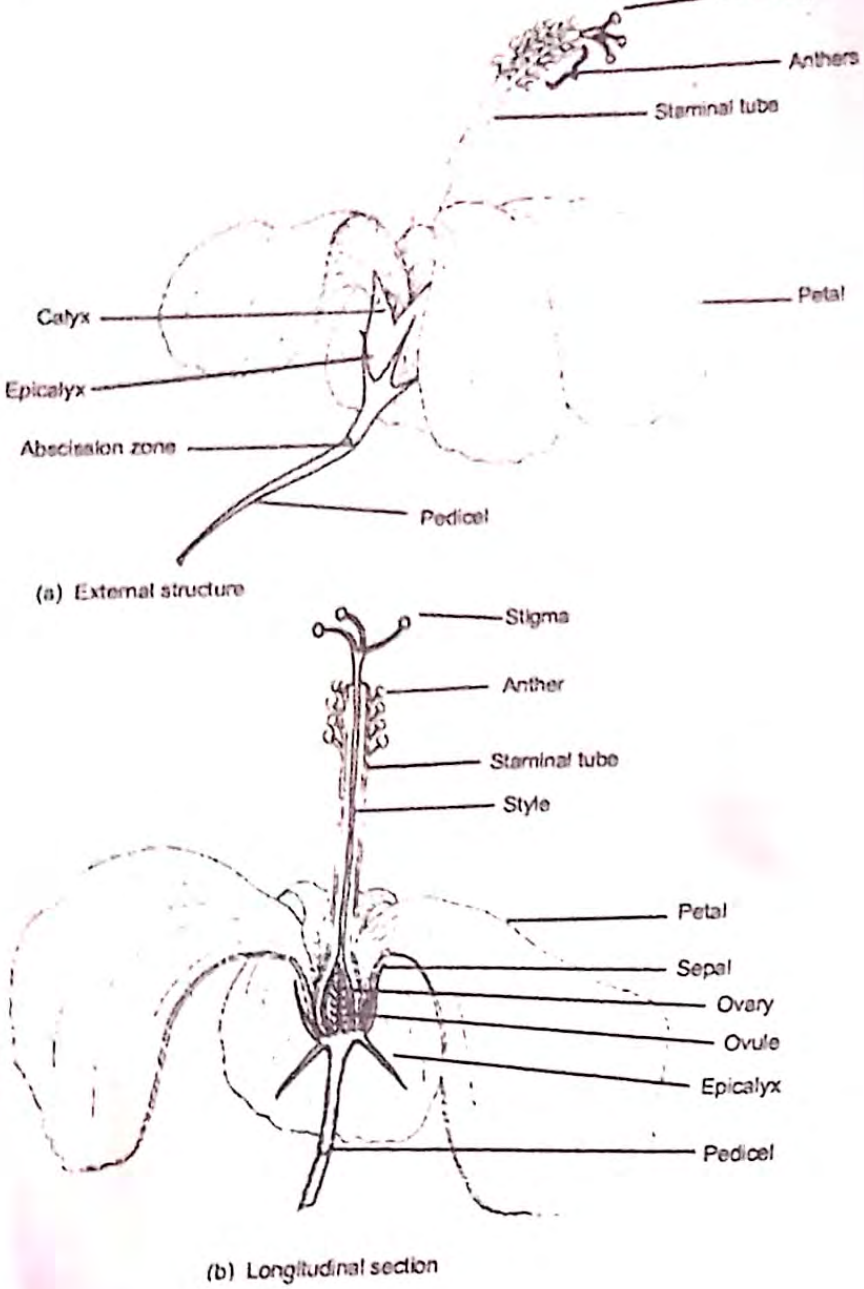


Fig. 5.5: **The Hibiscus flower**

— ***Parietal placentation*** where the ovary is made up of many carpels fused along their edges (syncarpous ovary) and the ovules located at the edges of these carpels, e.g. pawpaw.

— ***Axile placentation*** where the edges of the syncarpous ovary are pushed inwards so that they meet in the centre of the ovary and the ovules are equally located centrally on the axis.

— ***Free central placentation*** where the ovules are located on the knob which projects from the base of the ovary.

5.8 Arrangement of Floral Parts in a Wind-Pollinated Flower

- The flowers are borne on inflorescences.
- The flowers are in pairs along the branches and a pair is enclosed in bracts to form a spikelet.
- Inside each spikelet the lower of the two flowers consists of only three yellowish stamens but the upper flower has two tiny structures called lodicules which represent perianth.
- The feathery stigmas of the flower ripen before the stamens and with the opening of the bracts, the

anthers protrude and burst, releasing several light pollen grains that are dispersed by the wind.

- When the anthers mature before the stigma the condition is known as **protandry**.

5.9 Pollination in plants

- **Pollination** is the transfer of pollen grains from the anther to the sigma of a flower.
- If the transfer is from the anther to the stigma of the same flower or another of the same plant, then self pollination results.
- If the transfer of pollen grains is from the anther to the stigma of another flower on a different plant of the same species, then cross-pollination results.
- **Cross-pollinated flowers**
  - Cross-pollination is only possible on unisexual flowers or dioecious plants, i.e. plants that have male organs and female organs on different flowers of the same species, e.g. pawpaw.
  - In some hermaphroditic flowers (bisexual) the stamens or stigma ripen at different times and so only cross-pollination can satisfy effective fertilization.
- **Self-pollinated flowers**
  - Both stamens and pistils are found within the same flower, i.e. monoecious.
  - The corolla of some completely encloses the stamens and pistil so that insects cannot easily reach the structures.
- **Agents of Pollination**
  - Animal agents include beetles, flies, butterflies, moth, bees, humming birds, sunbirds, and bats.
  - Non-biological agents are the wind and water.
- **Characteristics of Insect-pollinated flowers**
  - Petals are large and brightly coloured
  - they have a sweet scent,
  - they secrete nectar,
  - they have sticky stigma, and
  - they have rough and heavy pollen grains.
- **Characteristics of wind-pollinated flowers**
  - the flowers are not brightly coloured because they have inconspicuous perianth,
  - they have no scent,
  - they do not secrete nectar,
  - they have large feathery stigma,
  - they produce light and smooth pollen grains.

5.10 Fertilization and Zygote Development in Flowering Plants

- After pollination a pollen grain sends out a pollen tube through the stigma and style into the ovary, finally reaching an ovule.
- Within the pollen tube are three nuclei, i.e. two male nuclei and a pollen tube nucleus which is responsible for the development of the pollen tube.
- The moment the pollen tube reaches the ovule through the micropyle, the pollen tube nucleus disappears.
- Each ovule consists of a large oval-shaped cell called **embryo sac** which is surrounded by two protective coats called **integuments**. Integuments later give rise to seed coat.
- The mature embryo sac contains eight nuclei, two of which later fuse to form one called **secondary nucleus**. The arrangement of the nuclei is shown in Fig. 5.6.

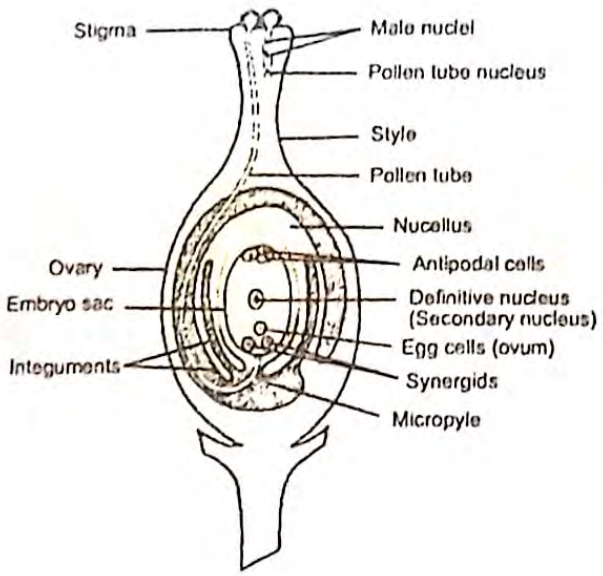


Fig. 5.6: Embryo sac and its fertilization

- One of the two male nuclei fuses with the female nucleus (ovum) to form the zygote and the second male nucleus fuses with the secondary nucleus to form **endosperm.**
- The zygote later forms the embryo, the ovules form the seeds and the ovary forms the fruit.
- A fruit which develops without fertilization is described as pathenocarpic.
- A seed is a dormant embryo sporophyte with stored food and protective coats.
- A seed consists of:
  - A plumule, made up of two embryonic leaves, which will become the first true leaves of the seedling, and a terminal (apical) bud.
  - The hypocotyle and radicle, which will grow

into the stem and primary root.
  - One or two cotyledons, which store food that will be used by the germinating seed.

- Angiosperms that produce seeds with two cotyledons are called **dicotyledons**, e.g. bean and angiosperms that have a single cotyledon are called **monocotyledons**, e.g. maize grain and other grass seeds.

5.11 Fruits

- A fruit is a mature ovary and it is different from a seed in that:
  - a fruit has two scars while a seed has one;
  - a fruit is mature ovary while a seed is a mature embryo.
- A true fruit has the following parts:
  - a **pericarp**, i.e. the ovary wall,
  - the seed or seeds,
  - the remains of style and stigma or scar where these have been detached.
- **Simple fruits** are produced from a single flower in which the pistil is made up of one carpel or several fused carpels and they are divided into three types, i.e., dry **indehiscent (non-splitting) fruit, dry dehiscent (splitting fruits)**, and **succulent fruits**.
- **Aggregate fruits** are produced from a single flower with an apocarpous pistil, i.e. several free carpels and hence called a collection of simple fruits, e.g. Cola fruit and rose fruit.
- **Multiple fruits** are produced from many flowers the ovaries of which are fused together, e.g. pineapple and fig.
- Dry indehiscent fruits (Fig. 5.7a) include:
  - An **achene** which is a dry fruit formed from a single carpel enclosing only one seed. It is the smallest possible fruit, e.g. sunflower fruits, *Clematis* and *Anemone*.
  - A *nut* is a dry fruit with a very hard or woody or tough pericarp, e.g. the hard part of cashew fruit (Fig. 5.7b).
  - A **caryopsis** has the pericarp and the seed coat fused together, e.g. grains of cereals (maize, guinea corn, millet) and grasses (Fig. 5.7c).
  - A **samara** has the pericarp extended to form one or more wings, e.g. combretum and African rosewood (Fig. 5.7d).
  - A **cypsela** is a one-seeded indehiscent dry fruit that originates from a bicarpellary ovary. It has a persistent calyx that has developed into a parachute of hairs called **pappus**, e.g. *Tridax* and *Emilia* (Fig. 5.7e).
- Dry dehiscent fruits (Fig. 5.8).
  - These are fruits in which the pericarp breaks open to release the seeds and they are classified according to the number of splits that occur in the pericarp.
  - A **follicle** is a dehiscent dry fruit formed from a single carpel bearing one or more seeds and splitting along one suture (ventral) only (i.e. one side only), e.g. Crestus, Magnolia, Marsh marigold (Fig. 5.8a).
  - A **legume** is formed from a single carpel having one or more seeds and splitting along both sides longitudinally, e.g. Pride of Barbados, bean fruit and flamboyant (Fig. 5.8b).
  - A **capsule** is formed from a syncarpous ovary which contains several seeds and it opens by longitudinal slits along many sides to release the seeds, e.g. castoroil fruit, cotton fruit and *Abelmoschus esculentus* (Okro) fruit (Fig. 5.8c).
  - A **schizocarp** is formed from a syncarpous ovary in which the carpels separate from one another as one-seeded mericarps, e.g. Desmodium and some cassia (Fig. 5.8d).
- Succulent fruits (Fig. 5.9)
  - They have some of their parts or the whole fruit being fleshy and can be eaten.
  - A **drupe** has a pericarp made up of three layers, an outer epicarp, middle mesocarp and an inner endocarp which forms the hard part covering the seed, e.g. mango, coconut and palm fruit.
  - a **berry** has the epicarp, mesocarp and the endocarp fleshy, and sometimes the whole fruit is eaten, e.g. orange, guava, banana, tomato, lemon, grape, cucumber, date and tangerine.

5.12 Dispersal of Fruits and Seeds by Wind

- Plants like orchid and begonia produce **small** and **light** seeds that may be air-borne as easily as spores. In numerous cases the seed or fruit wall is expanded into a wing-like structure which acts as a sail or propeller to catch the wind, e.g. winged seed of *Tecoma* and jacaranda, and the winged fruit of Combretum.
- The cypsela of *Tridax* has a parachute-like tuft of hairs called **pappus**.
- The seeds of cotton and silk cotton plants are surrounded by a mass of silky threads called **floss**, so that they are easily carried about by the wind.

5.13 Dispersal by Animals

- Animals like birds, fruitbats and monkeys may be agents of dispersal of fruits and seeds.

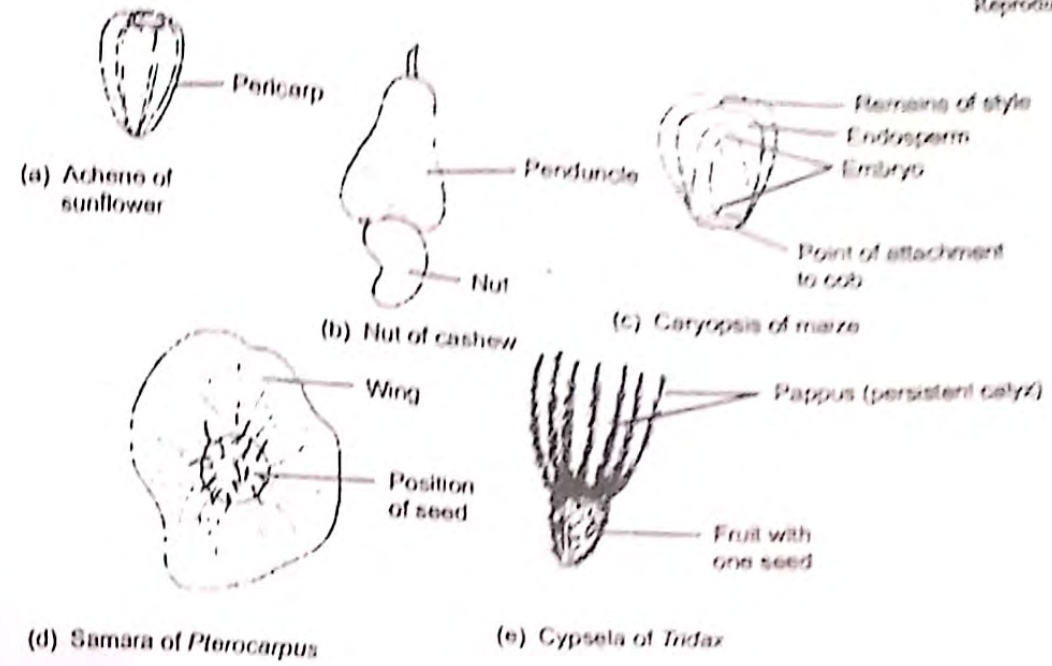


Fig. 5.7: Dry indehiscent fruits

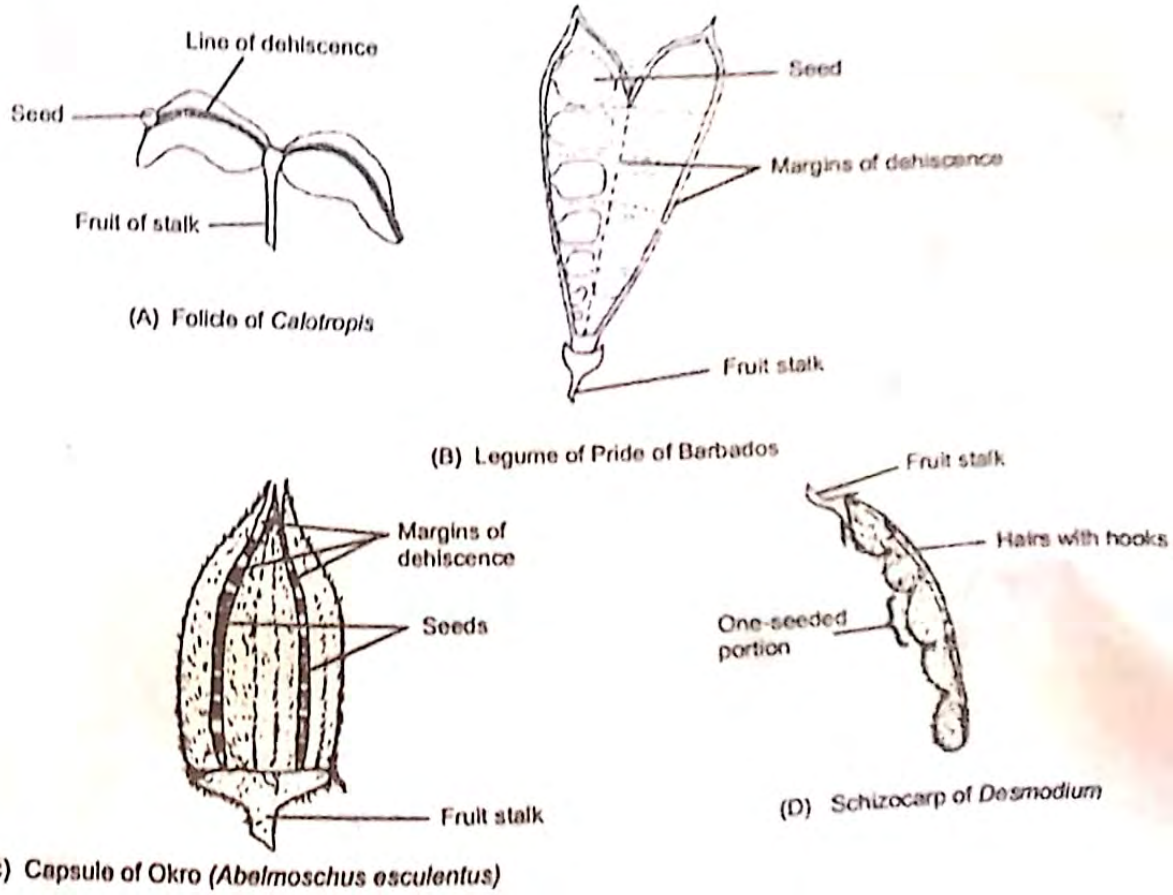
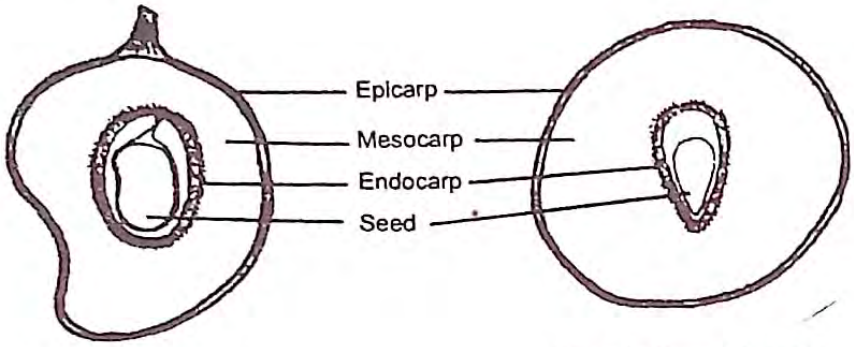
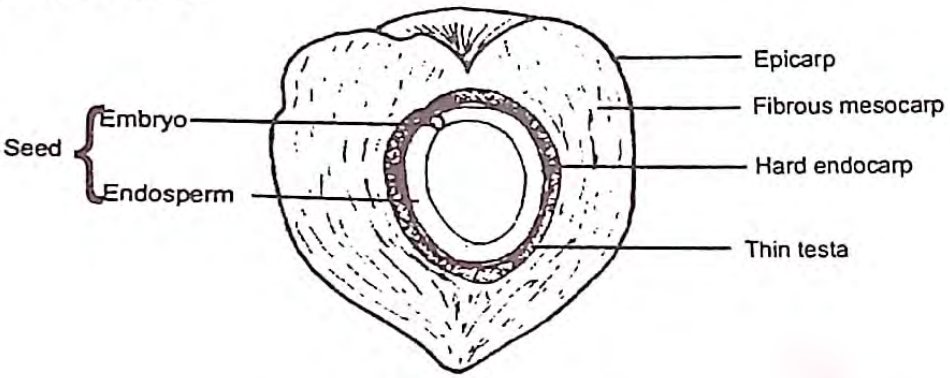


Fig. 5.8: Dry dehiscent fruits

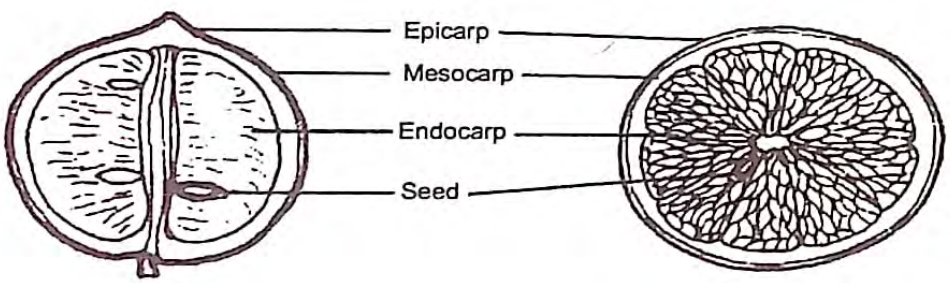


(A) L.S. Drupe of Mango

(B) T.S. Drupe of Mango

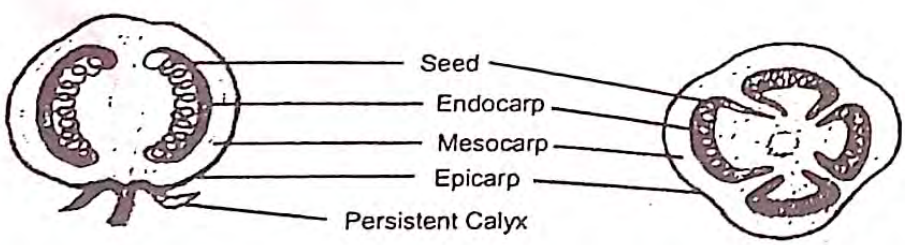


(c) T.S. Drupe of Coconut



(d) L.S. Berry of Orange

(e) T.S. Berry of Orange *(Hesperidium)*



(f) L.S. Berry of Tomato

(g) T.S. Berry of Tomato

***Fig. 5.9:* Succulent fruits**

- Many animals feed on the soft succulent fruits but they either discard the strongly protected seeds of berries, the stony endocarp of drupes and achenes or they may swallow them. In the latter case many of the hardened parts are resistant to enzymes that they pass through the gut without harm, and when later dropped, can germinate successfully, e.g. guava, tomato.
- Fruits and seeds may be carried externally on an animal's body by reason of the fact that they develop special hooks or spines which catch in the fur or wool (hairy coats of animals), e.g. triumfetta.
- Some fruits have sticky hairs which get attached to the hairy coat of animals, e.g. *Desmodium.*

5.14 Dispersal by Man

- Dispersal of several fruits and seeds have been aided by the agricultural practices of man, e.g. guava, sweet potato and cassava which are native to tropical America now found in all tropical areas of the world.
- Seeds of weeds are accidentally transported from one part of the world to another by man's activities.

5.15 Dispersal by Water

- The seeds of certain water lilies have air spaces in the pericarp which give the buoyancy required in water.
- The coconut fruit achieves dispersal of its single seed by floating to new locations.

5.16 Dispersal by an Explosive Mechanism (Self-Dispersal)

- These are dependent on some rapid movement of the fruit wall which causes the sudden jerking outwards of the seeds when dry, e.g. fruits of legumes (Pride of Barbados) and capsules (fruit of okro — *Hibiscus esculentus*).

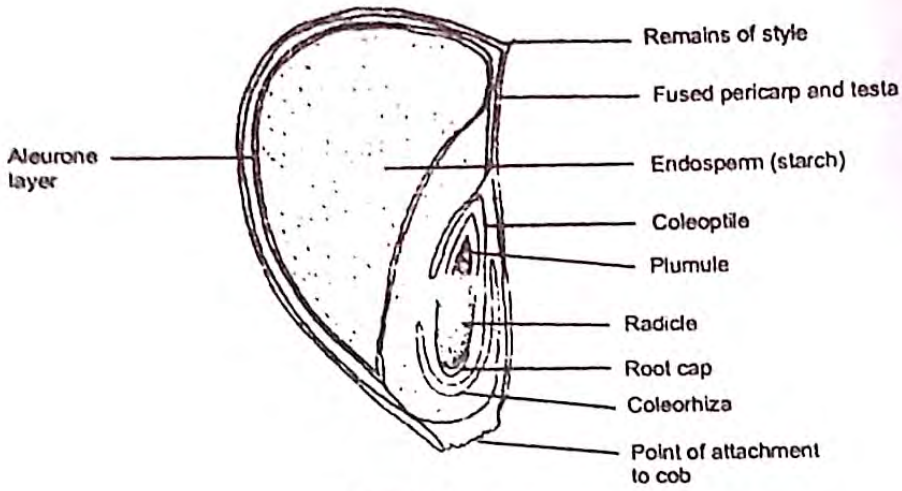
5.17 Types of Seeds

- **Monocotyledonous seed** (e.g. Maize grain) (Fig. 5.10a)
  - It is an endospermous seed in which the pericarp and the testa are fused together (food is stored in the endosperm).
  - There is only one cotyledon.
  - The fruit has a stalk at the narrow end and at the opposite end is the remains of style.
  - In longitudinal section, the endosperm takes up more than half of the seed and the cotyledon occupies a space less than half of the seed.
  - The axis of plumule and radicle lies beside the cotyledon.
  - The plumule is protected by plumule sheath or coleoptile and the radicle is protected by radicle sheath or coleorhiza.
- Castor oil seed (Fig. 5.10d-f)
  - It is an endospermous dicotyledonous seed.
  - The testa is hard with the colour varying from brown to black.
  - A spongy knob called **caruncle** is located at one end of the seed.
  - The endosperm is a white mass inside which is found the embryo.
  - The embryo consists of two weak cotyledons, which are flat, and a short radicle projecting from its base.
  - The plumule is very short and it is continuous with the radicle.
- Cowpea (bean) seed (Fig. 5.10b-c)
  - It is a non-endospermous dicotyledonous seed.

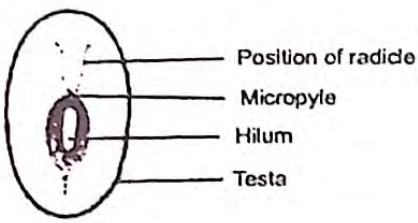
5.18 Germination

- This is the resumption of growth of the embryo plant inside the seed.
- Proper temperature, adequate amounts of water and an adequate supply of oxygen are essential for germination

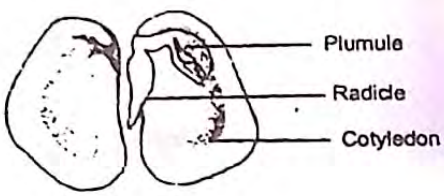
- Germination starts with the absorption of water and the first part of the embryo to emerge is the radicle. Further growth is ensured by the food reserve inside the endosperm or cotyledons.
- Germination in which the cotyledons remain in the soil is known as **hypogeal germination** but when carried above the soil into the air is known as **epigeal germination.**
- Most monocotyledons show hypogeal germination and most dicotyledons show epigeal germination.
- During epigeal germination the hypocotyl (part of the embryo just below the cotyledons) elongates and bends much more than epicotyl (part of the embryo above the cotyledons). The epicotyl then emerges from the soil carrying the plumule and cotyledons into the air (Fig. 5.11).
- During hypogeal germination the epicotyle elongates much faster than the hypocotyl with the result that the cotyledons are never carried above the ground. The radicle and plumule force their way through their sheaths to emerge (Fig. 5.12).
- Alkaline pyrogallol is used in experiments on germination to show that oxygen is essential for germination, i.e. energy production during germination.



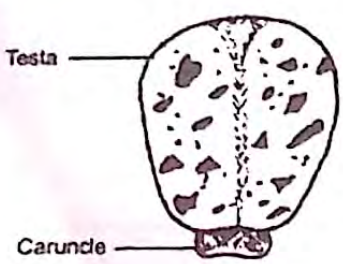
(a) L.S. Maize grain



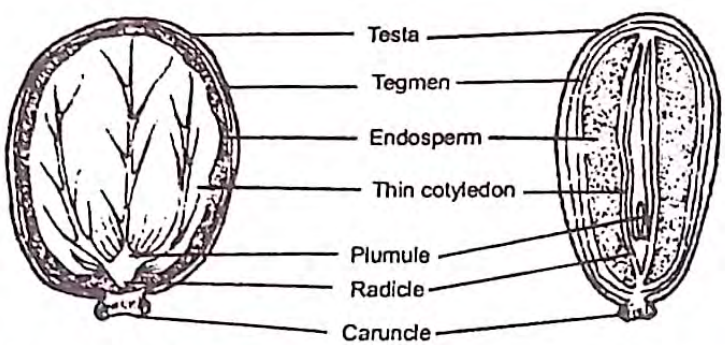
(b) External structure of bean seed



(c) Been seed with testa removed



(d) External structure of Castor Oil seed



(e) Internal structure of Castor Oil seed

(f) L.S. Castor Oil seed

Fig. 5.10: Structure of seeds (a) Monocotyledon seed (b) and (c) non-endospermous dicotyledonous seed; (e) (f) (g) endospermous dicotyledonous seed

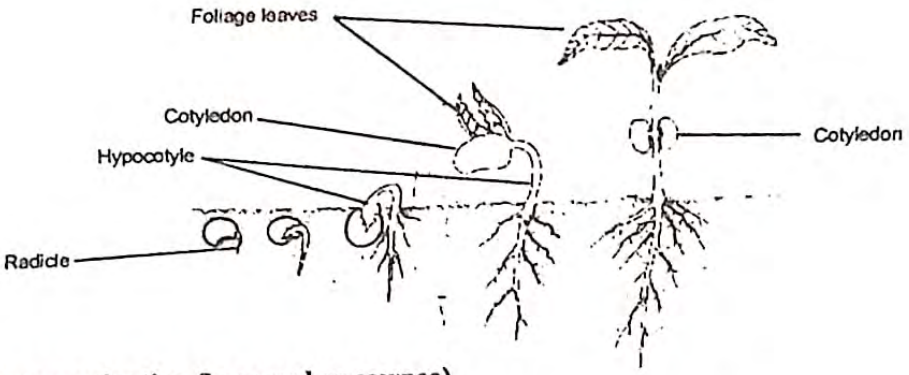


Fig. 5.11: Epigeal germination (beanseed or cowpea)

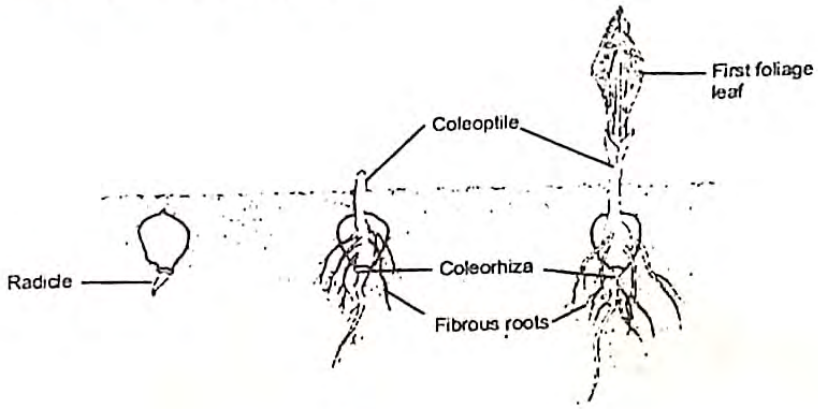


Fig. 5.12: Hypogeal germination in maize grain

SSCE/UME–Type Objective Questions

1. Which of these is not a characteristic of wind pollinated flowers?
   A. Smaller quantities of sticky pollen produced.
   B. Feathery styles and stigmas pendulously hang outside the flower.
   C. Small, inconspicuous, scentless flowers.
   D. Large anthers which are loosely attached to filaments.
   E. Light pollen grains produced in large quantities from anthers.

2. A germinating seed requires oxygen which is essential for
   A. converting carbohydrate into glucose
   B. transporting energy from one part of the plant to another
   C. the production of energy by oxidising essential carbohydrates
   D. hydrolysis of proteins
   E. the formation of water molecules within the germinating seed.

3. Alkaline pyrogallol was used in an experiment. That experiment must have been connected with
   A. excretion
   B. germination
   C. transpiration
   D. digestion
   E. photosynthesis

4. The plumule is best described as
   A. part of the cotyledon
   B. part of the endosperm
   C. a miniature shoot
   D. structure near the micropyle
   E. structure opposite the hilum

5. A dry, indehiscent, winged fruit formed from one carpel is known as
   A. schizocarp
   B. caryopsis
   C. samara
   D. nut
   E. follicle

6. The maize grain is regarded as a fruit and not a seed because
   A. it is covered by a sheath of leaves
   B. the testa and the fruit wall fuse after fertilization
   C. it has both endosperm and cotyledon
   D. it has coleorhiza and coleoptile
   E. the pericarp and seed coat are separate.

7.

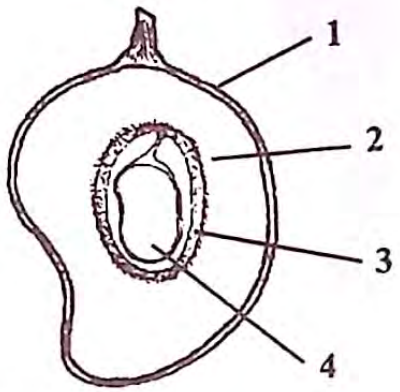


Fig. 5.13

The figure above is a section of a drupe. The part numbered 2 is
   A. epicarp
   B. endocarp
   C. mesocarp
   D. seed
   E. pericarp

8. The one-seeded fruit in which the pericarp and seed coat have become fused together is known as
   A. achene
   B. samara
   C. caryopsis
   D. legume
   E. drupe

9. A true fruit is formed from
   A. a fertilized ovary and other floral parts
   B. a fertilized ovary
   C. a fertilized ovary and calyx.
   D. a fertilized ovary and fleshy receptacle.
   E. an unfertilized ovary and other floral parts

10. For pollination and fruit formation, the essential part(s) of the flower should be the
   A. corolla
   B. ovary
   C. pistil (gynaecium)
   D. ovules
   E. receptacle

Use the figure below to answer Questions 11 and 12.

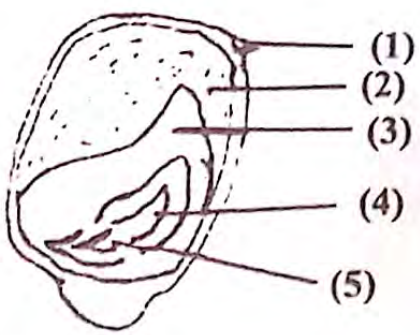


Fig. 5.14

11. The testa
    A. is normally absent
    B. is included in the part numbered (1)
    C. is fused with the part numbered (3)
    D. is the part numbered (1)
    E. surrounds the parts numbered (4) and (5)

12. Reserved food material is
    A. stored in the parts numbered (2) and (3)
    B. stored in the part numbered (3) only
    C. stored in the parts numbered (4) and (5)
    D. not concentrated in any particular part
    E. found in all parts except (1)

Questions 13 and 14 refer to the figure below.

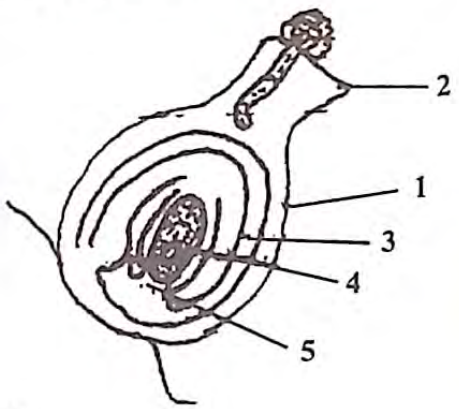


Fig. 5.15

13. The correct figured sequence for the following labelled parts (stigma, ovary wall, embryo sac, integument and micropyle) is
    A. 12453 B. 24135
    C. 21435 D. 14325
    E. 31425

14. Which structure becomes the seed coat after fertilization?
    A. 1 B. 2
    C. 3 D. 4
    E. 5

15. Which of the following statements is NOT true of menstration?
    A. It occurs monthly
    B. The discharge contains amniotic fluid
    C. It signifies no conception
    D. It involves the lining of uterus
    E. It may cause pains

16. A fruit formed from a single flower having several free carpels, is
    A. a multiple fruit
    B. a dry dehiscent
    C. a dry indehiscent fruit
    D. a simple fruit
    E. an aggregate fruit

17. If the anthers of a flower mature before the stigma, the condition is termed
    A. epigyny B. hypogyny
    C. protogyny D. protandry
    E. perigyny

18. In human reproduction, the zygote formed as a result of fertilization gets attached to the wall of the uterus. The process is called
    A. sexual reproduction B. cohesion
    C. pregnancy D. implantation
    E. viviparity

19. A dry fruit which can break into several parts each containing one seed is a
    A. capsule B. aggregate fruit
    C. legume D. schizocarp
    E. follicle

20. In what order do the following structures develop during the metamorphosis of the toad?
    1. External gills 2. Internal gills
    3. Forelimbs 4. Hindlimbs
    5. Mouth

    A. 12345 B. 15243
    C. 21345 D. 53412
    E. 54321

21. The mammalian organ through which nourishment and oxygen diffuse into a developing embryo is called
    A. amnion B. chorion
    C. umbilical cord D. oviduct
    E. placenta

Use the figure following to answer Questions 22 and 23.

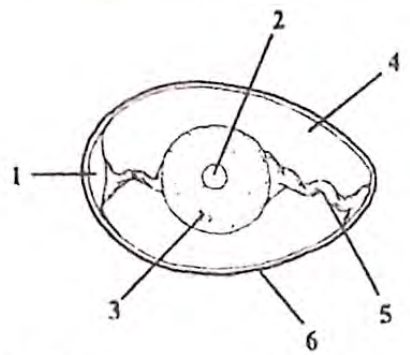


Fig. 5.16: Transverse section of a fowl's egg

22. The young chick is formed from
A. 3
B. 3 and 4
C. 1, 3 and 5
D. 2
E. 4

23. Which parts provide food for the developing chick?
A. 2 and 3
B. 4 and 5
C. 3 and 4
D. 2 and 5
E. 1 and 2

24. A flowering plant is monoecious if
A. the androecium is found on one plant
B. the gynoecium is monocarpous
C. it produces essential organs
D. the gynoecium and androecium are on the same plant
E. the flowers are unisexual

25. How many nuclei are found in a pollen tube during fertilization?
A. 2
B. 3
C. 5
D. 6
E. 7

26. Gestation in mammals is the period
A. required for growth after birth
B. between the formation of the foetus and birth
C. of development from zygote to birth
D. before the formation of the zygote
E. of implantation of embryo after fertilization

27. A dehiscent fruit formed from several fused carpels with many seeds is classified as
A. an achene
B. a capsule
C. a follicle
D. a legume
E. a nut

28. What is the difference between viviparous and oviparous animals?
A. Possession of yolked eggs
B. Laying of unfertilized eggs
C. Possession of yolkless eggs
D. Protection of eggs from predator
E. Laying and brooding of eggs

29. A dry indehiscent fruit containing one seed and possessing a hairy pappus is described as
A. a nut
B. a drupe
C. a cypsela
D. a follicle
E. an achene

30. Which of the following is not a part of the female reproductive system?
A. Ovary
B. Seminal vesicle
C. Uterus
D. Vagina
E. Fallopian tube

31. Which of the following substances from the mother's blood diffuse through the placenta into the blood vessels of the foetus?
A. Carbon dioxide and oxygen
B. Glycogen and starch
C. Urea and carbon dioxide
D. Glucose and oxygen
E. Digested food and carbondioxide

32. The correct sequence of structures from the ovary to the vagina is
A. ovary —> fallopian funnel —> uterus —> cervix —> vagina
B. ovary —> fallopian tube —> cervix —> uterus —> vagina
C. ovary —> cervix —> fallopian tube —> fallopinan fumel —> vagina
D. ovary —> cervix —> uterus —> fallopian tube —> vagina
E. ovary —> fallopian tube —> fallopian funnel —> cervix —> vagina

33. Airspaces are characteristic of seeds or fruits dispersed by
A. birds
B. water
C. wind
D. explosive mechanism
E. man

34. The components of castor oil seed and the maize grain are similar *except* for the
A. number of cotyledons
B. location of the embryo
C. number of radicles
D. number of plumules
E. number of micropyles

35.

Fig. 5.17: Transverse section of an ovary

The type of placenta shown in the figure above is

A. parietal
B. marginal
C. axile
D. free central
E. triple marginal

36. A fruit which develops without fertilization is described as

A. simple
B. pathenocarpic
C. aggregate
D. compound
E. multiple

37. Epigeal germination can be found in

A. Sorghum
B. Maize
C. Millet
D. Groundnut
E. Wheat

38. Double fertilization in higher plants is significant because it ensures the

A. formation of a fertile embryo
B. formation of a fertile embryo and the endosperm
C. development of the seed
D. development of the fruit
E. production of double seeds

39. Hypogeal germination is characterised by the

A. emergence of plumule out of the ground
B. provision of nourishment by the endosperm
C. elongation of the hypocotyl
D. elongation of the epicotyl
E. elongation of both epicotyl and hypocotyl

40. The main function of the petal of a flower is to

A. attract pollination agents
B. protect the flower while still in bud
C. serve as landing stage for insects
D. protect the inner parts from dessication
E. produce nectar

Answer Key

1.	A	9.	B	17.	D	25.	B	33.	B
2.	C	10.	C	18.	D	26.	C	34.	A
3.	B	11.	B	19.	D	27.	B	35.	A
4.	C	12.	A	20.	B	28.	B	36.	B
5.	C	13.	C	21.	E	29.	C	37.	D
6.	B	14.	C	22.	D	30.	B	38.	B
7.	C	15.	B	23.	C	31.	D	39.	D
8.	C	16.	E	24.	D	32.	A	40.	A

Explanations to Objective Questions

1. **(A)** Smaller quantities of sticky pollen is not a characteristic of wind polinated flower but of insect pollinated flower.

2. **(C)** Germinating seed requires oxygen for the production of energy by oxidising carbohydrates, i.e. respiration.

3. **(B)** Alkaline pyrogallol is used to demonstrate that oxygen is essential for germination.

4. **(C)** Plumule is a miniature shoot.

5. **(C)** Samara is a dry, indehiscent winged fruit formed from one carpel.

6. **(B)** A maize grain is a fruit and not a seed because the testa and the fruit wall fuse after fertilization.

7. **(C)** The part numbered 2 is the mesocarp.

8. **(C)** Caryopsis is a one-seeded fruit with fused testa (seed coat) and pericarp (fruit wall).

9. **(B)** A true fruit is formed from a fertilized energy.

10. **(C)** The essential part of a flower for pollination and fruit formation is the pisil, i.e. gynaecium.

11. **(B)** The testa is included in the part numbered (1).

12. **(A)** Reserved food material is stored in the parts numbered (2) and (3).

13. **(C)** The correct figured sequence is 21435.

14. **(C)** The structure numbered 3, i.e. integument becomes the seed coat after fertilization.

15. **(B)** The discharge during menstruation does not contain amniotic fluid because no pregnancy/birth involved.

16. **(E)** An aggregate fruit is formed from a single flower with several free carpels.

17. **(D)** Protandry is a condition in which the anthers of a flower mature before the stigma.

18. **(D)** The process by which a zygote gets attached to the wall of the uterus is called implantation.

19. **(D)** A schizocarp is a dry fruit formed from a syncarpus ovary (fused) in which the carpels break into several parts each containing one seed, i.e. mericarps.

20. **(B)** The correct sequence is 15243.

21. **(E)** Nourishment and oxygen diffuse into a developing embryo through the placenta.

22. **(D)** The young chick is formed from the part numbered 2, i.e. blastodisc.

23. **(C)** The parts numbered 3 and 4, i.e. yolk and albumin, provide food for the developing chick.

24. **(D)** A flowering plant is monoecious if the gynoecium (female part) and androecium (male part) are on the same plant.

25. **(B)** Pollen tube which develops from a pollen grain during fertilization has three nuclei.

26. **(C)** Gestation in mammals is the period of development from zygote to birth.

27. **(B)** A capsule is a dehiscent fruit formed from several fused carpels with many seeds and splits by longitudinal slits.

28. **(B)** Eggs are fertilized within viviparous animals and are not laid, e.g. mammal, while oviparous animals lay unfertilized eggs, e.g. fish.

29. **(C)** An achene is a dry indehiscent fruit containing one seed but when an achene has a pappus of hairs, it is a cypsela.

30. **(B)** Seminal vescile which secrete seminal fluid is found only in males.

31. **(D)** Glucose and oxygen will always diffuse through the placenta into the foetal blood vessels while carbondioxide and urea diffuse into mother's blood for removal.

32. **(A)** The correct route through which ovum passes from the ovary to the vagina is Ovary → fallopian funnel → uterus → cervix → vagina.

33. **(B)** Air spaces are characteristic of seeds or fruits dispersed by water so that they can float easily without sinking.

34. **(A)** The components of castor oil seed and the maize grain are similar except for the number of cotyledons, i.e. castor oil seed has two cotyledons and maize grain has one cotyledon.

35. **(A)** The type of placentation shown in the figure is parietal.

36. **(B)** A fruit which develops without fertilization is described as pathenocarpic.

37. **(D)** Epigeal germination can be found in groundnut (dicotyledon) and not in the others which are monocotyledons.

38. **(B)** Double fertilization in higher plants ensures the formation of fertile embryo and the endosperm.

39. **(D)** In hypogeal germination the epicotyle elongates.

40. **(A)** The petals of a flower helps to attract pollination agents.

6 Nutrition

6.1 Photosynthesis

- This is the process by which plants manufacture simple carbohydrates such as sugar, from carbon-dioxide and water in the presence of sunlight and chlorophyll so that oxygen is set free as a by-product.

6.2 Mechanism of Photosynthesis

- It is essentially an oxidation-reduction process by which hydrogen is transferred from water to carbon dioxide through a carrier substance.
- The volume of carbon dioxide absorbed is almost equal to that of oxygen liberated in most green plants.
- Glucose appears to be the first stable carbohydrate formed during photosynthesis with the overall reactions represented as

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6H_2O + 6O_2\uparrow$$

- Photosynthesis takes place in a series of complex steps which is divided into two broad reactions i.e., light reactions and dark reactions.
- During *light reactions* chlorophyll absorbs light energy from the blue and red regions of the visible spectrum so that the chlorophyll becomes activated.
- Activated chlorophyll (CL) which is highly unstable uses the trapped energy to split water molecule into hydrogen (H^+) and (OH^-) ions. This is the photolysis of water

$$4H_2O + CL \xrightarrow{\text{Light}} 4(OH^-) + 4(H^+)$$

- H^+ is now available for reduction and fixation of carbon dioxide in dark reactions.
- OH^- eventually forms water and oxygen when four molecules combine together, i.e.

$$4(OH^-) \longrightarrow 2H_2O + O_2\uparrow$$

The oxygen produced in photosynthesis comes from water.

- *Dark reactions* also called Calvin cycle involves the uptake of CO_2 by the plants and the reduction of CO_2 by hydrogen atoms produced during photolysis.
- The energy required for the process is the potential chemical energy stored in green cells and water is the specific hydrogen-donor for the reduction of CO_2.
- The first formed intermediate product of photo synthesis is phosphoglyceric acid before being converted to sugar; i.e. glucose.
- The glucose formed by photosynthesis can also be used in the synthesis of starch, cellulose and ultimately all other molecules (i.e., lipids, proteins and nucleic acids) in the cell.
- Excess food manufactured during photosynthesis is translocated to other parts of the plant or downwards to storage organs from the leaves through the phloem during the night.
- Leaf test for starch — Detached leaf is exposed to boiling water for 30 seconds to kill and soften the tissues, boiled in methylated spirit to remove chlorophyll, and a few drops of iodine solution is added. A blue-black colouration shows the presence of starch. (Detach the leaf during the day.)
- A variegated leaf of ice-plant when plucked during the day and tested for starch will show that only the green parts will turn blue-black with iodine solution. The parts that are white will be stained brown. Hence starch has been produced in the green parts, i.e. chlorophyll is necessary for photosynthesis (Fig. 6.1).
- The leaves of a plant kept in the dark and another exposed to sunlight when tested for starch will show that only the leaf exposed to sunlight will turn blue-black with iodine solution. Hence, sunlight is necessary for photosynthesis (Fig. 6.2).
- Carbon dioxide is also essential for photosynthesis (Fig. 6.3).
- Oxygen is set free as a by-product of photosynthesis (Fig. 6.4).

6.3 Mineral Requirements of Plants

- Minerals which are derived from rocks are made available for plant use as a result of weathering.
- The minerals, i.e. nutrients are present in the soil in solution and plants absorb them as ions.
- Nutrients must be present in balanced quantities, for a plant may be unable to use one element because it lacks another element.

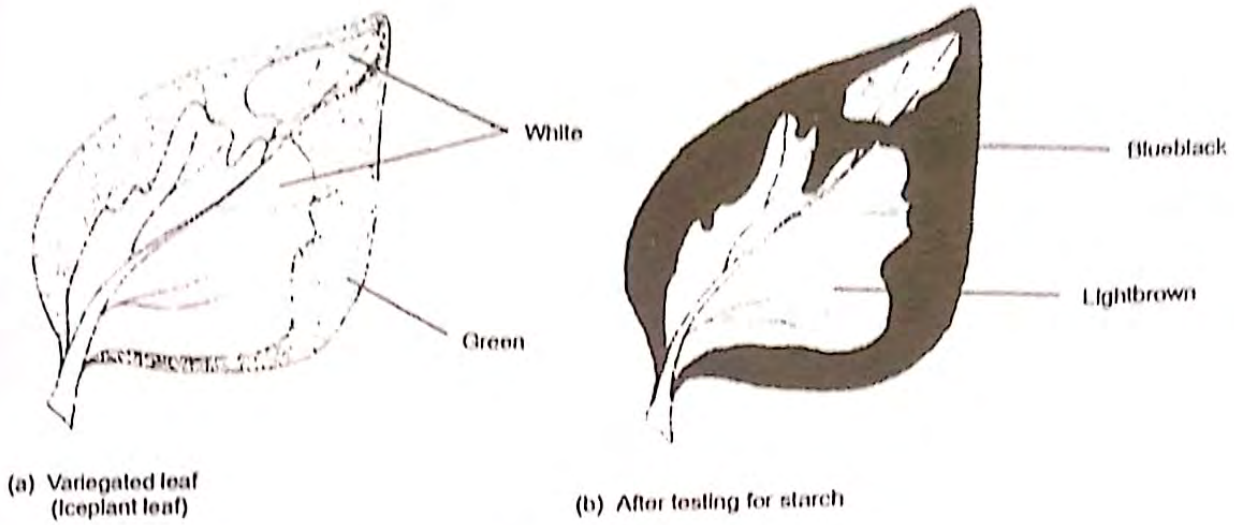


Fig. 6.1: To show that chlorophyll is essential for photosynthesis

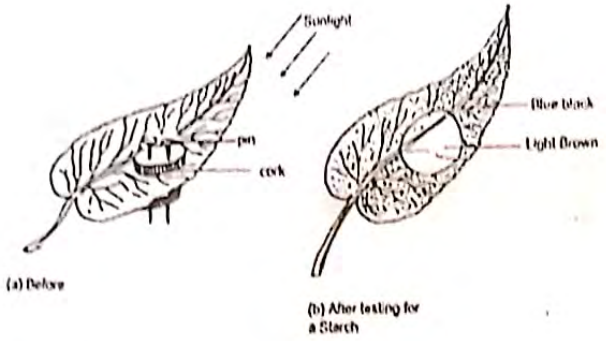


Fig. 6.2: To show that sunlight is necessary for photosynthesis

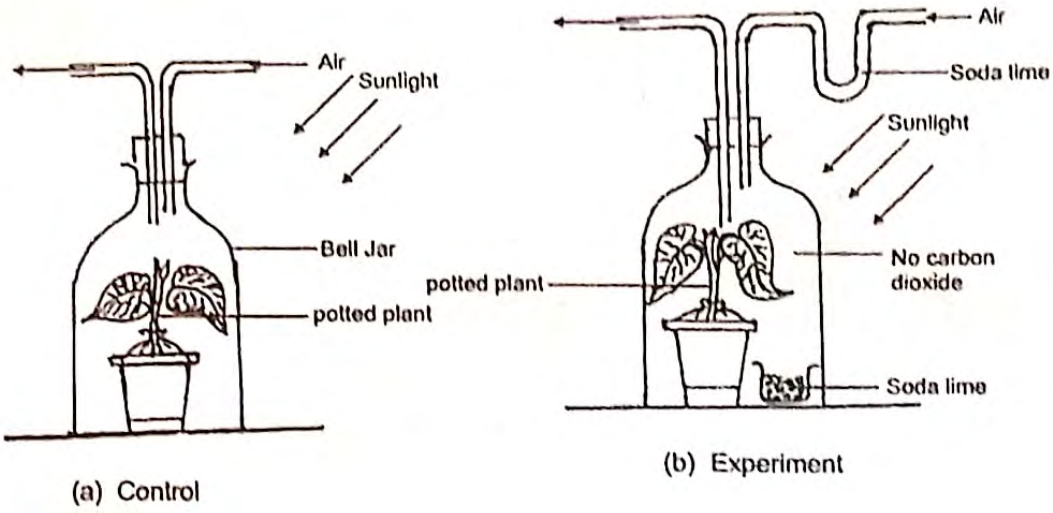


Fig. 6.3: To show that carbon dioxide is essential for photosynthesis

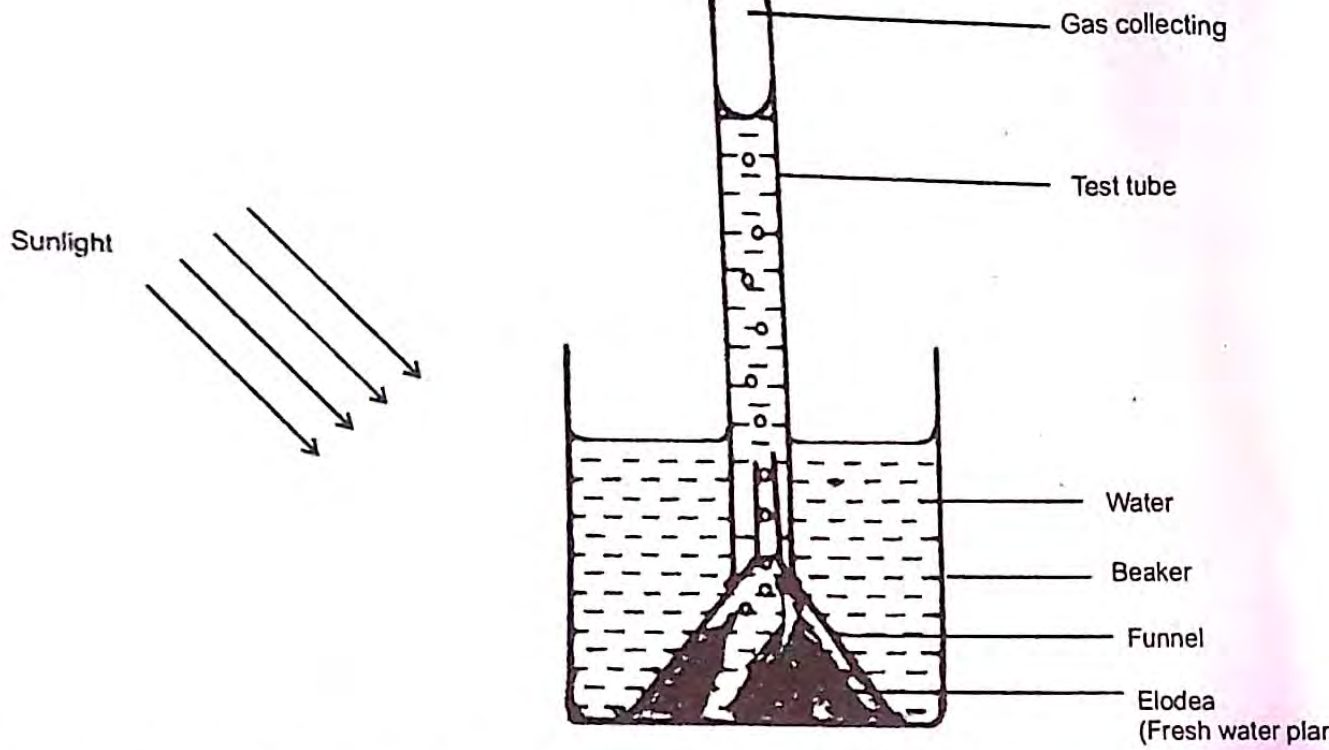


Fig. 6.4: To show that oxygen is set free during photosynthesis

- Those elements that are required by plants in large quantities are called macro-elements (i.e. carbon, oxygen, hydrogen, sulphur, nitrogen, phosphorus, potassium, calcium and magnesium) and those required in small quantities are termed micro-elements (i.e. iron, manganese, copper, zinc, boron and molybdenum).
- A nutrient solution is known as culture solution and a solution which contains all the required elements in their correct proportions is known as a comple culture solution.
- Bottles containing culture solutions are alwa darkened to prevent the rays of the sun (light) fro breaking down the mineral elements.
- The deficiency or lack of any one of the elemen can affect plant growth and hence plants sho symptoms of such deficiency.

The deficiencies and their importance are:

Element	Importance	Deficiency	Source
Nitrogen	Production of chlorophyll, protein and nitrogenous bases.	Stunted growth, dormant lateral buds, Yellowing of leaves [chlorosis] and small leaves.	From soil in the form of nitrate ion and ammoniu cation.
Potassium	Formation of cells and regulation of cellular activities.	Yellow to brown discolo-ration of the leaves. Deformed leaves and very poor growth.	From soil in the form of potassium cation.
Iron	Cholorophyll formation	Poor growth and chlorosis of leaves.	From soil in the form of ferrous and ferric cation.
Magnesium	Chlorophyll formation, and cell wall formation.	Poor growth and chlorosis of leaves.	From soil in the form of magnesium cation.
Calcium	Cell wall formation, activation of certain enzymes.	Weak and stunted growth coupled with poor root development.	Calcium cation from the soil.

Phosphorus	Nucleic proteins formation, and nuclear division and acts as buffer in cell sap.	Thin lanky growth with reddish leaves and stems, poor root development.	Phosphate anion from the soil.
Sulphur	Synthesis of some proteins in the cell.	Thin and slender stems coupled with chlorosis of the leaves.	Sulphate anion from the soil and sulphur dioxide through leaves.

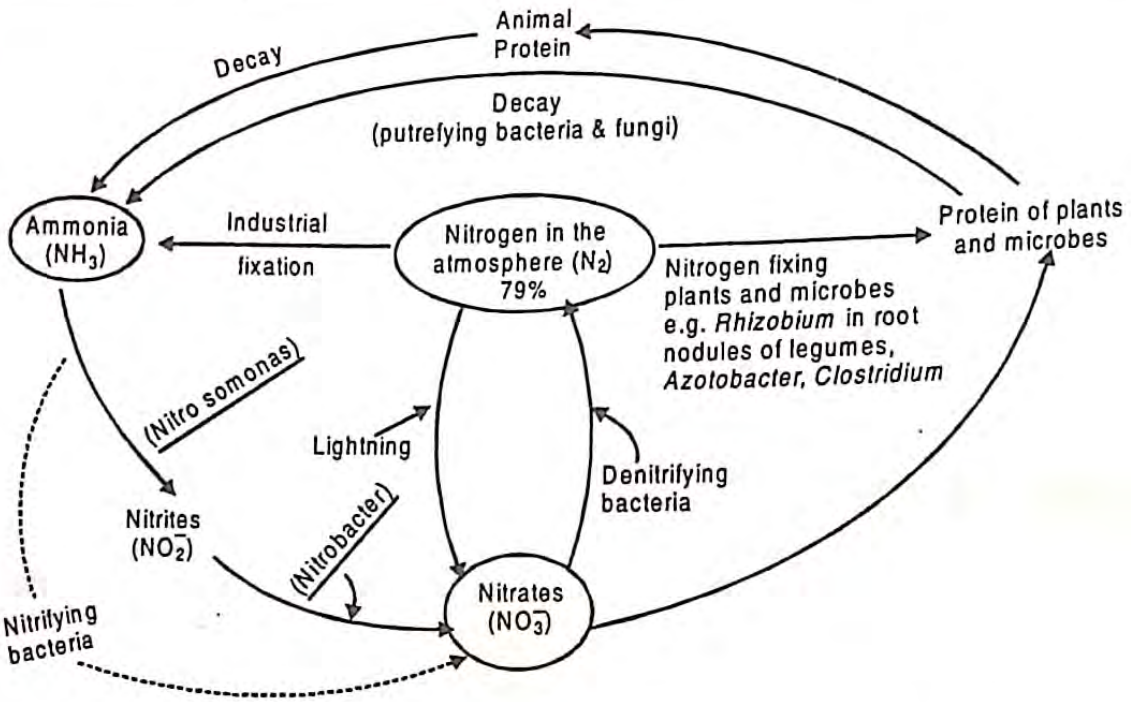


Fig. 6.5: Nitrogen-cycle (showing the role played by several micro-organisms in the cycling of nitrogen through the biosphere

Nitrogen Cycle

- All living things require nitrogen atoms for the building of proteins and a variety of other essential organic molecules.
- Air, which is 79% nitrogen, serves as a reservoir of nitrogen.
- In order to manufacture proteins, plants must secure nitrogen in "fixed" form, i.e. incorporated in compounds.
- The most commonly used form is nitrate ions (NO_3). Other substances such as ammonia (NH_3), and urea [$(NH_2)_2(CO_2)$] are used successfully in both natural systems and as fertilizers in agriculture.
- The enormous energy of lightning breaks nitrogen molecules and enables them to combine with oxygen in the air. The nitrogen oxides formed dissolve in rain-water, forming nitrates. In this form they are carried to the earth (Fig. 6.5).
- The growing needs of agriculture have led to an ever-increasing production of industrially fixed nitrogen such as nitrogen fertilizers.
- The widespread cultivation of legumes especially alfalfa and soybeans has greatly increased the world-wide rate of nitrogen fixation. The roots of legumes harbour gram-negative bacilli of the genus **Rhizobium** which are able to fix atmospheric nitrogen, for their host.
- Other nitrogen-fixing bacteria, e.g. **Azotobacter, Clostridium**, live free in the soil.
- Microorganisms of decay such as putrefying bacteria and fungi, though their activities break down organic nitrogen compounds (proteins) in corpses, etc. into ammonia.
- Most ammonia produced by decay is converted into nitrates by bacteria of the genus **Nitrosomonas** and **Nitrobacter**. Through their activities nitrogen is made easily available to the roots of plants.
- Denitrification reduces nitrates to nirtogen, thus

replenishing the atmosphere. Once again bacteria are the agents involved and they live deep in the soil and in aquatic sediments where oxygen is in short supply.

6.4 Animal Nutrition

- Food eaten by an animal is divided into seven groups, i.e. carbohydrates, proteins, fats, water, mineral salts, vitamins and roughages, which together constitute a balanced diet.
- Carbohydrates which are the primary sources of energy to the body contain the elements of carbon, hydrogen and oxygen with the ratio of hydrogen to oxygen being 2:1 in a molecule.
- Carbohydrates occur in three major forms, i.e., monosaccharides (simple sugars) Oligosaccharides (complex sugars) and polysaccharides (starches, cellulose and glycogen).
  - Monosaccharides are found in plants and animals as hexose sugars with the chemical formula $C_6H_{12}O_6$ e.g. glucose (from grapes and onions) and fructose (from fruits).
  - Oligosaccharides are formed by condensation of two or more monosaccharide molecules. When two monosaccharide molecules are involved it is called disaccharide with the chemical formula $C_6H_{22}O_{11}$, e.g. sucrose (from sugarcane, banana, pineapple), lactose (milk sugar) and maltose (malt sugar).
  - Polysaccharides which are formed by the condensation of very large number of simple sugar molecules have the chemical formula $[C_6H_{10}O_5]n$ where n is very large, e.g. starch, cellulose and glycogen.
  - Mono- and oligo-saccharides are soluble in water but polysaccharides are insoluble and the hydrolysis (breaking down) of oligo and polysaccharides will yield monosaccharides.
  - Test for monosaccharides (simple reducing sugars) involves the addition of equal volumes of Fehlings solutions I and II to a solution of glucose or fructose which when boiled gives an orange or brick-red precipitate — or when an equal volume of Benedicts solution is added to monosaccharides a red, orange or yellow precipitate will be formed showing the presence of simple sugar.
  - Test for sucrose (Disaccharide) involves the heating of a solution of sucrose with dilute hydrochloric acid for about 5 minutes in a water bath after which drops of sodium hydroxide are added until it is alkaline. Subsequent addition of Benedicts solution will produce a red, orange or yellow precipitate confirming the presence of sucrose before hydrolysis.
  - Test for starch involves the addition of iodine solution to boiled starch which will give a blue-black colour.
  - Test for glycogen involves the addition of iodine solution to glycogen which will produce a reddish-brown colour
- Proteins are body building substances that are essential for building new cells, repair work and replacement of old cells.
  - They are soluble in dilute acids and alkali but not in water.
  - They coagulate when heated because they are easily denatured.
  - They contain the elements of carbon, oxygen and nitrogen and some may have sulphur or phosphorus in addition
  - Proteins found in muscles are called myosin, milk and cheese have casein, and albumin is found in eggs.
  - The building blocks of proteins are amino acids of which about 25 are known.
  - Proteins are broken down to proteoses, peptones and polypeptides in a decreasing order of complexity.
  - Biuret test for proteins involves the addition of dilute sodium hydroxide to protein solution, with vigorous shaking and finally adding 1% copper (II) sulphate drop-wise. A violet colour shows the presence of proteins in solution and a pink colour denotes the presence of peptones in the solution
  - Millons test for proteins involves the addition of Millon's reagent to the test solution which after heating will produce a deep red colour or precipitate that is indicative of the presence of a protein.
- Fats and oils which are also known as lipids. They contain a high proportion of carbon and hydrogen but little oxygen, i.e. the ratio of hydrogen to oxygen is not 2 to 1.
  - They are stored in the body as reserve energy source and are also good insulators
  - They are insoluble in water but they form stable emulsion with dilute alkali
  - Fats and oils are broken down into fatty acids and glycerol before being absorbed into the body.
  - When Sudan III is added to oil or fat, the oil picks up the red colouration.
  - When oil or fat is boiled with osmic acid a black precipitate is formed.
  - A drop of oil or fat placed on a white paper will leave a transluscent spot on the white paper

without evaporating.

- Vitamins
  They influence many important chemical processes in the body of organisms.

Vitamin	Sources	Soluble in	Destroyed by	Deficiency cases
A	Liver, eggs, animal fat, milk, fish, liver oil, red palm oil, green vegetables	Fat	Sunlight & Oxidation	Eye diseases, loss of weight, reduced resistance to diseases
B	Yeast, palm wine, eggs, tomatoes, groundnuts, germ of cereals, liver, beans, brown rice	Water	Alkalis	Beriberi
C	Lemons, limes, oranges, guava, tomatoes, grapefruit, pawpaw, green vegetables	Water	Cooking, canning, bottling, oxidation	Scurvy
D	Sea fish, fish liver oil, milk, eggs	Fat	—	Rickets
E	Liver, eggs, vegetable oil	Fat	—	Sterility
K	Green vegetables	Fat	—	Blood unable to clot

6.5 Digestive Enzymes

- Enzymes are organic (biological) catalysts secreted by living cells to bring about many biochemical reactions in various metabolic processes.
- Digestive enzymes are divided into four major groups, those that act on carbohydrates are called amylases, those acting on proteins are proteases, those acting on fats and oils are lipases and those acting on cellulose in herbivorous animals are known as cellulases.
- Enzymes need co-enzymes to activate them.
- Enzymes are proteins, soluble, highly specific and sensitive to temperature (i.e. range between 35-40°C) but are denatured at temperatures above 40°C. The optimum temperature is 37°C when reaction rate is fastest for most enzymes.
- They are sensitive to the degree of acidity or alkalinity and many bring about reversible reactions.
- Table 6.1 shows different digestive enzymes and their activities.

6.6 The Digestive System

- Alimentary systems of an earthworm, cockroach, bird and rabbit are shown in Figs. 6.6, 6.7, 6.8 and 6.9.
- Villi of the small intestine of higher vertebrates increases the surface area of absorption of digested food materials.
- The villi are numerous and thin-walled to make it easier for the diffusion of digested food across them.
- The villi are well supplied with blood and lymph (lacteal) vessels for the transportation of absorbed digested food.
- The small intestine is long and coiled so as to delay the food inside it and hence ensures maximum absorption of digested food.

6.7 Feeding Habits

- Heterotrophic organisms are classified in terms of the type of food they eat.
- Herbivores highlight animals which feed on plants, carnivores which feed on animals, fluid-feeders which feed on a variety of animal and plant juices and microphagous feeders (filter-feeders) which feed on small particles suspended in water.
- **Herbivorous mammals**, e.g. cattle, use premolars and molars to grind up plant tissues before exposing them to digestive enzymes.
- Herbivorous molluces such as snail have radula with which they tear plant food.
- Herbivorous insects like grasshopper have a pair mandibles with jagged edges for cutting through leaves and blades of grass. These, together with a set of manipulating devices, make up the mouth parts.
- **Carnivores:** Highlight animals that are adapted for catching prey.
- These adaptations take various forms: fast movement, canine teeth in great cats, suckers nematocysts bearing tentacles in *Hydra*.
- In mammals the sharp incisor teeth are used for biting pieces of flesh and tearing it off bone.
- In many mammals the last pair of premolars in the upper jaw and the first pair of molars in the lower jaw are enlarged and have sharp ridges for shearing flesh. These are the carnassial teeth.
- Carnivorous plants inhabit areas deficient of nitrates, e.g. the venus fly-trap and pitcher plants.
- They obtain carbohydrates by photosynthesis but nitrogen is obtained from the bodies their victims

Table 6.1: Activities of Digestive Enzymes

Secretion	Source	Site of Action	Enzymes	Substrate	Products
Saliva (slightly alkaline)	Salivary glands	Mouth cavity	Ptyalin	Starch	Maltose
Gastric juice (acidic)	Stomach wall (gastric glands)	Stomach	HCl (not an enzyme) Pepsin Renin	Pepsinogen Protein Caseinogen (Soluble)	Pepsin Polypeptides Casein (insoluble)
Bile (alkaline)	Liver	Duodenum	Bile Salts (not enzymes	Fats	Fat droplets (chylomicrons)
Pancreatic Juice (alkaline	Pancreas	Duodenum	Amylopsin Trypsin Peptidases Steapsin (Lipase) Nucleases	Starch Protein Polypeptides Fats Nucleic acids	Maltose Polypeptides Amino acids* glycerol Nucleotides
Intestinal Juice or Succus entericus	Wall of the small intestine	Small intestine	Maltase Invertase Enterokinase Erepsin Nucleotidases	Maltose Sucrose Trypsinogen Peptones + Polypeptides Nucleotides	Glucose* Glucose + Fructose* Trypsin Amino acids* Pentose sugars + phosphoric acid + organic base *
Caecal Juice in Herbivores	Wall of caecum	Caecum	Cellulase	Cellulose	Glucose*

* The end products of food digestion which are eventually absorbed by the villi of the small intestine.

(insects).

- The insect is attracted by colour, scent or sugary bait, then trapped, killed and digested by a fluid rich in proteolytic enzymes.
- Amino acids are absorbed into the plant.
- In the venous fly-trap there are folding leaves with spikes along the edges and a hinge-like midrib.
- When an insect alights on a leaf, the two halves spring together, the spikes interlocking so that the unfortunate animal cannot escape.
- In the pitcher plants, the pitcher-shaped leaves contain digestive fluid, the nectar at the lips of the pitcher attracts insects which slip and slide into fluid. The insect is then killed and digested.
- **Fluid-feeders:** Highlighted animals are divided into two groups, wallowers and suckers.
- Wallowers include gut parasites such as the tapeworm which inhabit the small intestine.
- They feed on the digested food of the host, absorbin it direct through the integument. *Ascaris lum bricoides* found in the small intestine of man, muscular pharynx helps in sucking the hosts foo into their own gut.
- A saprophyte is another 'wallower' which penetrate into the material, secreting a variety of enzyme which break down solid components of food int soluble products.
- Enzymes are secreted across the body surface t digest the surrounding food so that the saprophyt becomes surrounded by a solution from which it ca absorb all the chemicals it requires directly across i surface.
- Sucking insects feed on blood or plant juices an have their mouth parts modified into a shar proboscis which can pierce the integument, throug which fluid is drawn by the suctorial action of th pharynx.

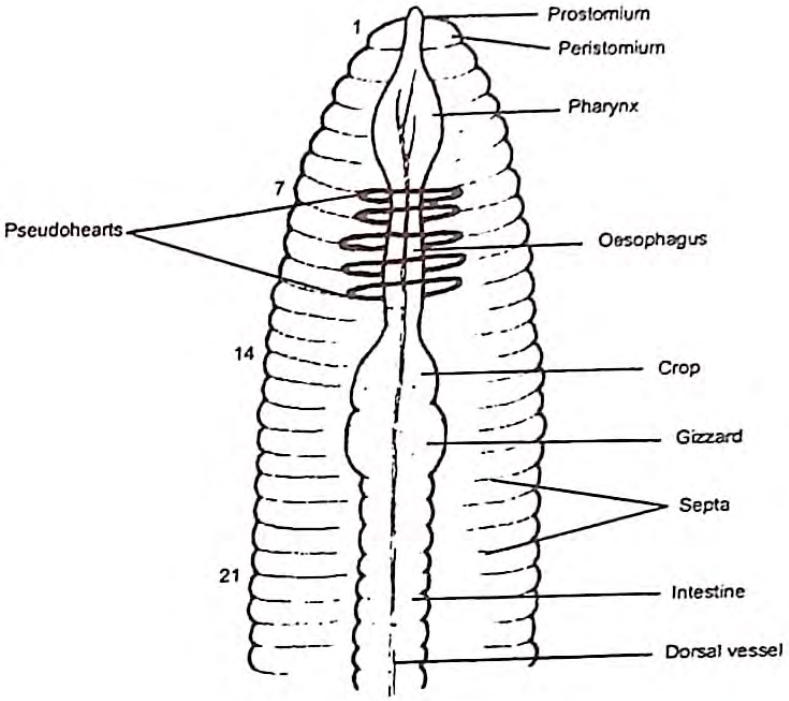


Fig. 6.6: The alimentary canal and blood vessels of earthworm as seen when dissected from the dorsal surface

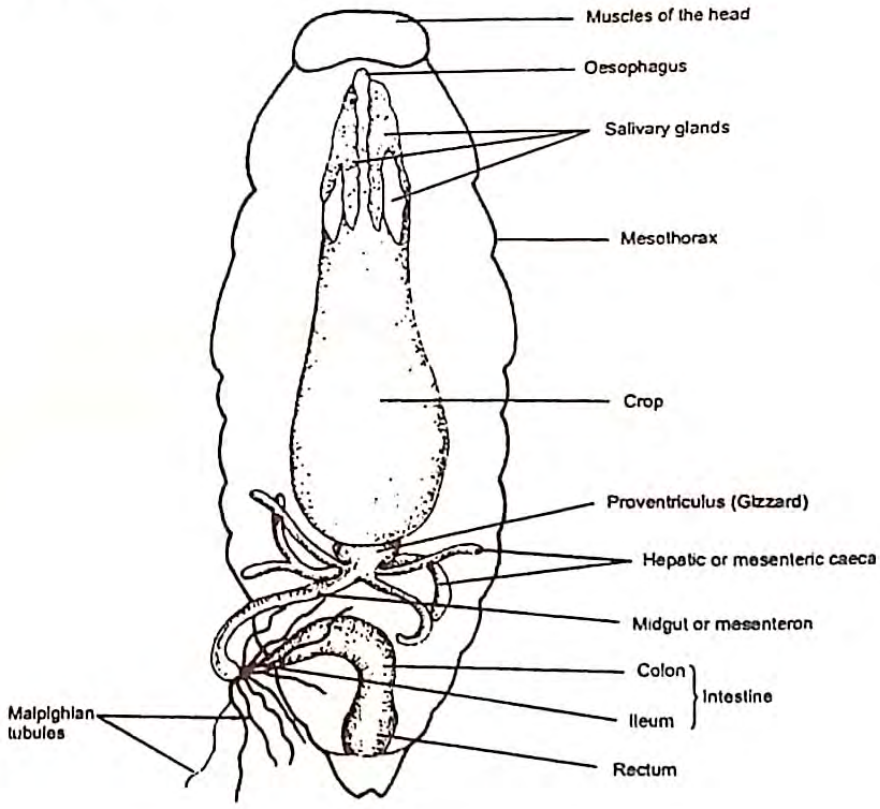


Fig. 6.7: The alimentary canal of cockroach, with the terga (Dorsal) removed

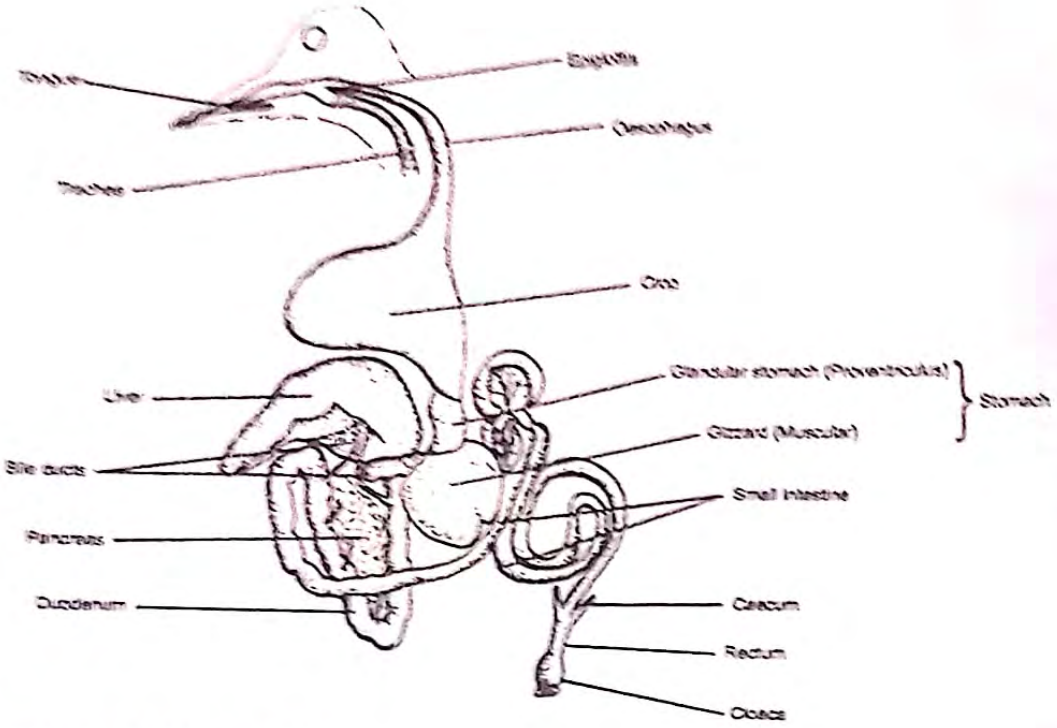


Fig. 6.8: Alimentary canal of a bird

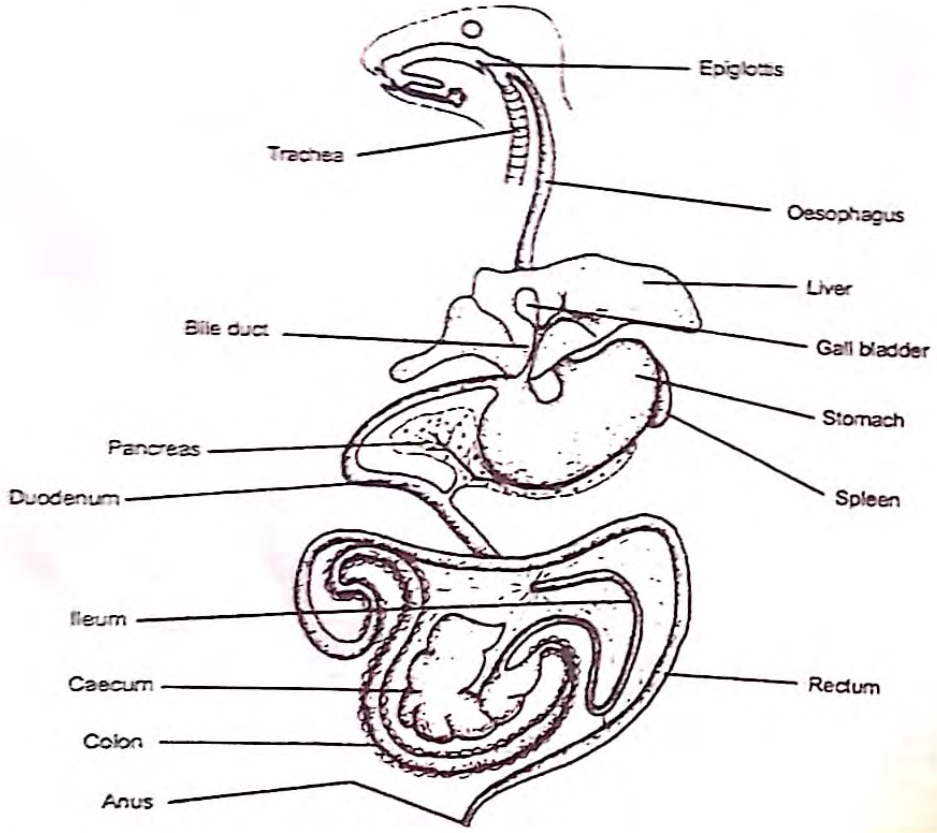


Fig. 6.9: The alimentary canal of a rabbit

- In the mosquito the proboscis consists of a sucking tube surrounded by a group of sharp stylets with which they pierce the skin to suck blood.
- The butterflies have long flexible proboscis with which they reach the nectar.

6.8 Microphagous or Filter Feeders

- They are always aquatic and feed on tiny particles suspended in water.
- In the *Anodonta* (aquatic mollusc), the gills are used for sorting and straining food particles as water is drawn towards the body.
- Four sheet-like gills, two on each side of the muscular foot, hang into the mantle cavity.
- The cilia on the gills draw water into the inhalent siphon and through the pores into the vertical tubes.
- Particles in water then get trapped in mucus on the surface of the gill and the water eventually passes out through the exhalent siphon.
- Only small particles of nutritional value are concentrated and sent towards the mouth where further sorting takes place before being swallowed.
- In the crustaceans, water is drawn over the body by the movement of the appendages.

SSCE/UME–Type Objective Questions

1. Which of the following does not occur during photosynthesis?
   A. Energy from sunlight is absorbed
   B. Carbon dioxide is evolved
   C. Glucose is synthesized
   D. Oxygen is given off
   E. Water is split

2. Which of the following is not regarded as a micro-nutrient or trace element essential for plant growth?
   A. Phosphorus B. Zinc
   C. Boron D. Silicon
   E. Molybdenun

3. In a water culture experiment, a plant showed poor growth and yellowing of the leaves. These symptoms were probably due to the absence of
   A. calcium B. phosphorus C. iron
   D. zinc E. copper

4. Autotrophic nutrition may be defined in terms of food obtained
   A. from other organisms in exchange for some products
   B. by the breakdown of complex substances
   C. by an organism utilizing its own store of energy
   D. by synthesizing simple substances using energy from an external source
   E. in a synthesized form from other living organism

5. Autotrophs are also described as
   A. consumers B. decomposers
   C. carnivores D. producers
   E. herbivores

6. In photosynthesis, the following processes are part of the light reaction except
   A. transfer of radiant energy which reduces a co-factor
   B. absorption of radiant energy by chlorophyll
   C. utilization of the energy in the electron transfer chain to form ATP
   D. formation of glucose using energy from NADPH
   E. splitting water into its component parts.

7. Which of the following is not an adaptation of the small intestine for absorption of digested food. The
   A. Villi are well supplied with blood and lymph vessels
   B. Villi are numerous and thin-walled
   C. Small intestine has large surface area
   D. Small intestine is long and coiled
   E. Small intestine leads into the large intestine

8. Thunderstorm can be beneficial to plants because it.
   A. kills the pests that attacks crops
   B. destroys some of the major crops
   C. adds nitrates to the soil
   D. adds line to the soil
   E. make rain water available to plants.

9. Which of the following is a nitrifying bacterium?
   A. Nitrobacter B. Rhizobium
   C. Azotobacter D. Clostridium
   E. Nitrosomonas

10. Which of the following reagents is used for the test for starch?
    A. Millon's reagent B. Fehling's solution
    C. Sudan III D. Iodine solution
    E. Benedict's solution

The diagram below represents a section through the earthworm showing its alimentary system. Use it to answer questions 11 to 13.

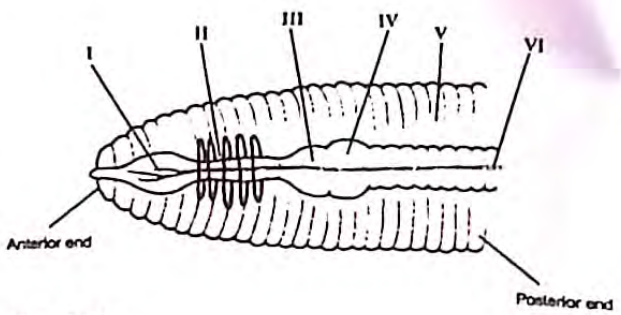


Fig. 6.10

11. What is the name of the part labelled I?
    A. Oesophagus B. Nephridium
    C. Pharynx D. Intestine
    E. Crop

12. What is the name of the structure labelled III?
    A. Gizzard B. Crop
    C. Heart D. Seminal vesicle
    E. Coelom

13. What is the function of the part labelled IV?
    A. Production of sperm
    B. Temporary storage of food
    C. Mechanical grinding of food
    D. Pumping blood all over the body
    E. Producing cocoon for the eggs

14. In testing for a reducing sugar, the food substance is usually warmed with

A. Sulphuric acid
B. Million's reagent
C. Sudan III
D. Benedict's solution
E. Sodium hydroxide

15. The part of the alimentary system of a bird where grinding of maize occurs is the
A. crop
B. stomach
C. gizzard
D. cloaca
E. rectum

16. The following are necessary for photosynthesis to take place except
A. chlorophyll
B. water
C. carbon dioxide
D. light
E. oxygen

17. Which of the following organisms reduces nitrates in the soil to gaseous nitrogen?
A. Euglene
B. Protozoan
C. Denitrifying bacterium
D. Parasitic mould
E. Nitrifying bacterium

18. $6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$

The process represented by the above equation is:
A. proteinsynthesis
B. respiration
C. photosynthesis
D. transpiration
E. translocation

19. The oxygen given off during the process is derived from
A. sunlight
B. chlorophyll
C. carbon dioxide
D. atmosphere
E. water

20. Which of these is a direct photosynthetic product?
A. Glucose
B. Starch
C. Protein
D. Fats
E. Latex

21. Which of these is the terminal portion of the alimentary canal of a mammal?
A. Oesophagus
B. Stomach
C. Rectum
D. Colon
E. Appendix

22. Bacteria inhabiting legume root nodules and which add atmospheric nitrogen to the soil are referred to as
A. denitrifying bacteria
B. nitrifying bacteria
C. nitrogen fixing bacteria
D. nitrogen freeing bacteria
E. nitrogen putrefying bacteria

23. The digestive enzyme that coagulates proteins in milk is
A. ptyalin
B. pepsin
C. rennin
D. trypsin
E. amylase

24. The element nitrogen is utilized in
A. formation of ATP
B. formation of glucose
C. formation of amino acids
D. photosynthesis
E. none of the above

25. Human beings require vitamins in their diet because vitamins
A. contains carbohydrates and fats
B. prevent kwashiorkor
C. stimulate the alimentary canal
D. digest proteins in the body
E. influence many important chemical processes in the body

26. Which of the following food substances is digested in the stomach?
A. Carbohydrates
B. Fats and oils
C. Fats and proteins
D. Proteins
E. Carbohydrates and fats

27 In testing for glucose the necessary reagent and the condition under which the reagent reacts best are
A. Fehling's reagent, in the cold
B. Millon's reagent, boiled
C. Fehling's reagent in acid medium, heated
D. Iodine solution, boiled
E. Fehling's reagent in neutral or alkaline medium boiled

28. All of the following are digestive enzymes except
A. bile
B. lipase
C. maltase
D. pepsin
E. ptyalin

29. The deficiency of vitamin D leads to
A. scurvy
B. pellagra
C. rickets
D. beriberi
E. polio

30. Which of the following pairs are the products of sucrose hydrolysis?
A. Glucose + glucose
B. Fructose + fructose
C. Glucose + fructose

D. Galactose + glucose
E. Fructose + galactose

31. Which of the following statements is NOT true of enzymes? They
A. are proteins
B. need cofactors to activate them
C. are sensitive to hydrogen ion concentration
D. are specific in their action
E. can withstand high temperatures

32. Which of the following represents the sequence of protein hydrolysis?
1. Polypeptides, 2. Amino acids,
3. Proteins, 4. Peptones
A. $3 \rightarrow 1 \rightarrow 2 \rightarrow 4$
B. $3 \rightarrow 2 \rightarrow 4 \rightarrow 1$
C. $3 \rightarrow 4 \rightarrow 2 \rightarrow 1$
D. $3 \rightarrow 4 \rightarrow 1 \rightarrow 2$
E. $3 \rightarrow 1 \rightarrow 4 \rightarrow 2$

33. A food substance which produces red colouration with Sudan III contains
A. Protein
B. Sugar
C. Starch
D. Cellulose
E. Fat

34. The main organic substances found in the human body are
A. carbohydrates, proteins and salts
B. salts, fats and proteins
C. fats, carbohydrates and proteins
D. salts, fats and carbohydrates
E. proteins, fats, carbohydrates, salts and vitamins

35. The severe deficiency of vitamin C leads to
A. kwashiokor
B. beriberi
C. pellagra
D. scurvy
E. sterility

36. In demonstrating the importance of mineral elements in plants, the culture bottle must be darkened to
A. prevent algal growth in culture solution
B. allow root growth
C. prevent breakdown of mineral elements
D. prevent photosynthesis in the roots
E. prevent insects from getting inside the culture solution

37. If the gall bladder of a mammal is damaged, which of the following will be most seriously affected?
A. Glycolysis
B. Digestion of starch
C. Digestion of fats and oils
D. Digestion of proteins
E. Digestion of mineral salts in food

38. Which of the following will be first digested if ingested at the same time?
A. Cooked beans
B. Cooked rice
C. Cod liver oil
D. Roasted beef
E. Boiled green vegetables

39. The pancreatic juice contains the enzymes amylopsin
A. pepsin and trypsin
B. rennin and streapsin
C. streapsin and trypsin
D. streapsin and ptyalin
E. maltase and erepsin

40. The element common to protein, carbohydrate and lipid is
A. hydrogen
B. sulphur
C. nitrogen
D. phosphorus
E. chlorine

Answer Key

1.	B	9.	A	17.	C	25.	E	33.	E
2.	A	10.	D	18.	C	26.	D	34.	C
3.	C	11.	C	19.	E	27.	E	35.	D
4.	D	12.	B	20.	A	28.	A	36.	A
5.	D	13.	C	21.	C	29.	C	37.	C
6.	D	14.	D	22.	C	30.	C	38.	B
7.	E	15.	C	23.	C	31.	E	39.	C
8.	C	16.	E	24.	C	32.	D	40.	A

Explanations to Objective Questions

1. **(B)** Carbon dioxide is used up rather than being evolved during photosynthesis.

2. **(A)** Phosphorus is a macronutrient available in the soil, though found limiting in soils.

3. **(C)** Yellowing of leaves (etiolation) and poor growth are deficiency symptoms of iron used for the synthesis of chlorophyll.

4. **(D)** Autotrophic nutrition occurs when an organism synthesizes its organic nutrients from simple inorganic raw materials using an external energy source in the process.

5. **(D)** Autotrophs are producers because they synthesize organic food from inorganic raw materials using external energy source.

6. **(D)** Light reaction in photosynthesis does not involve glucose formation using energy from NADPH.

7. **(E)** The small intestine leading into the large intestine is not in any way an adaptation of the small intestine for the absorption of digested food.

8. **(C)** Thunderstorms provides energy from lightning discharges to combine nitrogen and oxygen forming nitrous oxide which in the presence of rain water forms nitric acid. This acid then reacts with other chemicals in the soil to form nitrates which are absorbed by plant roots.

9. **(A)** Nitrobacter is a nitrifying bacterium that converts nitrites (from ammonia) to nitrates.

10. **(D)** Iodine solution is used for testing for starch, which is turned blue-black.

11. **(C)** The part labelled I is Pharynx.

12. **(B)** The structure labelled III is crop.

13. **(C)** The part labelled IV (gizzard) is for mechanical grinding of food.

14. **(D)** In testing for a reducing sugar, the food substance is usually warmed with Benedict's solution so that a red or orange precipitate is formed.

15. **(C)** The gizzard is used for grinding maize in the alimentary canal of a bird.

16. **(E)** The conditions necessary for photosynthesis are chlorophyl, water, carbon dioxide and light. Oxygen is a by-product of photosynthesis.

17. **(C)** Denitrifying bacteria reduces nitrates in the soil to gaseous nitrogen.

18. **(C)** The process involving

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$$

is photosynthesis.

19. **(E)** Oxygen given off during the process is derived from water.

20. **(A)** Glucose is a direct photosynthetic product.

21. **(C)** The rectum is the terminal portion of the alimentary canal which opens outward through the anus.

22. **(C)** Nitrogen fixing bacteria are found in the root nodules of legumes.

23. **(C)** Renin produced in the stomach coagulates protein found in milk.

24. **(C)** Nitrogen is used in the synthesis of amino acids which are the building blocks of protein.

25. **(E)** Vitamins influence many important chemical processes in the body of organisms.

26. **(D)** The enzymes produced in the stomach are pepsin and renin which act on proteins only.

27. **(E)** In testing for glucose, the reagent and the condition under which the reagent reacts best are that the reagent must be in a neutral or alkaline medium and then boiled.

28. **(A)** Bile is not a digestive enzyme, it only emulsifys fats and oils during digestion.

29. **(C)** Deficiency of vitamin D causes rickets.

30. **(C)** The products of sucrose hydrolysis are glucose and fructose.

31. **(E)** Enzymes cannot withstand high temperature because they are denatured at high temperatures.

32. **(D)** The sequence of hydrolysis of protein is from proteins → peptones → polypeptides → amino acids (3 → 4 → 1 → 2).

33. **(E)** Fat produces red colouration with Sudan III.

34. **(C)** The main organic substances found in the human body are fats, carbohydrates and proteins.

35. **(D)** Severe deficiency of vitamin C leads to scurvy.

36. **(A)** Culture bottles used in culture experiments must be darkened to prevent growth of algae which will damage the roots.

37. **(C)** If the gall bladder of a mammal is damaged th digestion of fats and oils will be most seriously affected.

38. **(B)** Digestion of carbohydrates (cooked starch only starts from the mouth, so cooked rice (carbohydrate) will be first digested.

39. **(C)** Pancreatic juice contains the enzymes amylop sin, streapsin and trypsin.

40. **(A)** The element common to protein, carbohydrate and lipid is hydrogen.

7 Basic Ecological Concepts

7.1 Ecosystem

- The ecosystem is an ecological unit made up of a community of all the plants and animals living in a habitat and the non-living part of the environment.
- The living or biotic component and the non-living (abiotic) environment interact to produce a stable system e.g. a lake is an ecosystem which consists of the plant and animal components, water, minerals, dissolved oxygen, soil and sunlight on which they depend. In a forest ecosystem, the plants absorb light and water for their photosynthesis while animals feed on plants and on each other. The dead remains of animals and plants return the nutrients to the soil. An ecosystem is thus self-supporting.
- The whole part of the Earth's surface which contains living organisms is called the biosphere. All the energy released on Earth is derived from sunlight. The energy released by animals is derived ultimately from green plants that they or their family eat and the green plants depend on sunlight for making their food.

7.2 Components of the Ecosystem and Sizes

(a) Ecological Components

- (i) **Environment:** This is made up of everything in the surroundings of an organism that could possibly influence it. The tadpole environment consists of water whose temperature will influence the growth and activity of the tadpole. The watery environment contains plants and animals on which the tadpole feeds, but it also contains fish and other vertebrates that may feed on the tadpole. The water contains dissolved oxygen used by the tadpole for respiration by means of its gills. Thus the water, the oxygen, the food and the predators are all part of the tadpole's environment.
- (ii) **Biosphere:** This is that part of the Earth's surface which contains living organisms. It is composed of the solid crust or substratum on which living organisms live, as well as the large bodies of water of the earth and the air surrounding the living things on earth.
- (iii) **Lithosphere:** This is the solid crust or substratum on which organisms live.
- (iv) **Hydrosphere:** This comprises the large bodies of water on earth.
- (v) **Atmosphere:** Envelope of air surrounding the earth, which consists of gases and water vapour.
- (vi) **Habitat:** It is a place where an organism lives. The habitat of the tapeworm is the intestine of a mammal. The habitat of a bean weevil is a bean seed or plant.
- (vii) **Niche:** A place which is suited to the way of life of an organism within a community. A herbivore and a carnivore may share the same habitat but their different feeding methods mean that they occupy different niches.
- (viii) **Population:** A group of organisms, all of the same species living together in a particular habitat or within a community e.g. the population of weaver birds in a village or the population of *Tilapia* in a lake. The total number of weaver birds or *Tilapia* in a stated area is the population. A population forms part of the community.
- (ix) **Biotic community:** A community is made up of either plant or animals living in a habitat. In the soil there is a community of organisms which includes earthworms, termites and other insects. In a lake, the animal community includes fish, insects, crustacea, molluscs, protozoa. The plant community consists of rooted plants with submerged leaves, rooted plants with floating leaves, filamentous algae like *Spirogyra* and single-celled plants like diatoms in the surface waters. The collection of populations makes up a community.
- (x) **Ecosystem:** The community of living organisms in a habitat and the non-living part of the environment.

(b) Components of the ecosystem

- An ecosystem is composed of two main divisions:
  (i) the living or biotic component, and
  (ii) the non-living or abiotic components.
- The biotic components are plants, animals and micro-organisms while the abiotic components comprise the soil, mineral nutrients, gases and the sun's energy.

- Biotic components: These are made up of the producer, consumer and decomposer organisms. The producer organisms include all chlorophyll containing plants that can produce organic matter in the presence of sunlight through photosynthesis. They are known as autotrophs. The main producer organisms on land are the grasses, herbs, shrubs and trees while there are algae in aquatic habitats. The bacteria are also producers who obtain their energy from chemical compounds like sulphur and iron. Oxygen production from photosynthesis is a second important by-product of producer organisms.
- The consumer organisms are the animal life of the ecosystem that feed on plants and other animals. These are known as heterotrophs. Consumer organisms which feed directly on plants are called primary consumers or herbivores e.g. sheep, rabbit, insects. The secondary consumers are those that feed on herbivores, e.g. hawk and lion. They are also called carnivores or predators. Sharks that feed on carnivorous fishes are regarded as tertiary consumers.
- Decomposer organisms are bacteria that can break down dead bodies of organisms in order to obtain their nutrients in form of inorganic materials. The decomposers which make up only a small percentage of the total mass of living matter play an important part in completing the cycling of materials in the ecosystem.
- **Abiotic Components:** These are made up of the non-living components of the ecosystems such as sun's energy, water, minerals from soil and gases. The sun is the major source of heat and light and as such are important in the life of organisms. The heat energy from the sun warms the earth so as to make it habitable to organisms. The light from the sun is necessary for photosynthesis as well as plant and animal activities. Temperature influences a lot of activities in living organisms.
- Water is very important for all life processes. It is a universal solvent in which life chemical reactions take place in all organisms. Water is available in streams, rivers, lakes, seas and oceans. Water can also be present in form of moisture, rain and snow. Humidity is affected by water and this influences activities of organisms.
- Mineral salts in soil and water are essential for plant life where they are used at various stages in photosynthesis (metabolism).
- Oxygen and carbon dioxide are the most important gases found in the living environments of plants and animals.

7.3 Local Biotic Communities

- Local biotic communities or biomes in Nigeria ca be divided into two major types: forest and savanna The important factors that determine the distributio of forest and savanna are climatic. They are
  (i) the mean annual rainfall;
  (ii) the length and severity of the dry season;
  (iii) the minimum relative humidity.
- The annual rainfall decreases progressively from th south to the north and the vegetation change accordingly.
- In tropical Africa, the biomes are rain forest, guine savanna, sahelian savanna, desert, swamp, fresh water, estuarine and marine types.

(a) Rain Forest

- The rain forests are near the equator and as such characterized by plenty of sunshine and heavy annual rainfall of about 1500 mm spread over nine months. The temperature is high and the relative humidity is at least 70%.
- The vegetation is dominated by tall trees with their widespread canopies often touching one another. Below the upper layer of tree tops, lighting is poor and does not permit the growth of grasses at the ground level.
- The plant community falls into five layers and can be identified with the three top layers being trees, the fourth a shrub layer while the fifth is the ground vegetation. Tree trunks and branches may be woven together by twisting wooden stems which provide support and exposure for plants like ferns, orchids and mosses called epiphytes. The ground vegetation comprise non-woody plants like lichens, liverworts and fungi.
- The trees of the rain forests are all evergreen with massive buttress roots, thick barks and broad dark green leathery leaves. The economic trees include *Milicia excelsa* (*iroko*), *Khaya ivorensis* (mahogany), *Terminalia* (Afara), *Triplochiton* (obeche), and *Elaeis guineensis* (oil palm).
- The animal fauna include antelopes, elephants and wild pigs found in forest or along banks of streams and rivers. Others are reptiles such as lizards, turtles, and snakes, arthropods such as insects, millipedes and centipedes which grow to large sizes, and birds and mammals.

(b) Southern Guinea Savanna

- The mean annual rainfall is above 500 mm but less than 1250 mm. There are not more than six months in the year with less than 25 mm of rainfall. The

lowest monthly relative humidity is above 28%.

- The vegetation is an open savanna woodland type, with tall grasses up to 3 m high. The trees have short trunks or boles, with broad leaves. They are usually up to 17 m but rarely up to 30 m high. The area is a grassland vegetation with scattered trees and shrubs.
- The vegetation is green and luxuriant in the rainy season with many of the herbs and grasses flowering. However in the dry season, the trees shed their leaves and the vegetation turn dry and brown. This enhances fire occurrence which tend to burn off the grasses, leaving behind standing clumps of burnt grass stems and black ash on the ground. Several of the trees are adapted to fires by possessing thick, fire-resistant barks. They also produce new sucker growths soon after each fire. Many of the trees have crooked trunks as a result of their exposure to fire. The grasses also have remarkable powers of producing new suckers from the bases of old shoots, soon after fires.
- The savanna tree species include *Daniellia oliveri* (false balsam), *Terminalia* (afara), *Lophira* and *Khaya senegalensis* (mahogany), *Butyrospermum* (shea butter tree) *Parkia* (locust bean). The grass species include *Andropogon, Panicum, Pennisetum.*
- The common forms of animals found in this biome are guinea fowls, deer, rats, grass cutters and snakes. Birds and termites are abundant.

(c) Northern Guinea Savanna

- The mean annual rainfall in this zone is above 500 mm, as in southern guinea savanna but the dry season is longer.
- The vegetation is dominated by grasses with scattered trees and shrubs. The tree species are shorter with thorny bushes. The common trees in this biome include *Isoberlina doka, Vapaca somone* and *Monotes kerstingii.* The grasses in this zone are shorter than those in southern guinea

(d) Sahelian Savanna

- This zone in West Africa spreads from Senegal to Chad and from Sahara to Northern Guinea Savanna but in Nigeria it occurs only at the North-east corner in Borno State, near Lake Chad.
- Rainfall is variable in this zone, with an annual rainfall of between 250-500 mm concentrated in 3 to 4 months of the year. The dry season is severe and long, resulting in drought.
- The typical vegetation is an open thorn savanna, with trees 5 m tall, having small leaves and thorns. The grasses are short and sparse. The common plant species include *Acacia* (gum arabic), *Lepthdenia spartium* and date palms. The common grass in the zone is *Aristida stipioides.* Seasonal changes in vegetation are pronounced but annual fires are less fierce because of the sparse vegetation.

(e) Desert

- The natural desert belt of the sahara does not extend into Nigeria. Desert encroachment has however occurred in the extreme northern boundaries of the country including Sokoto and Borno States.
- Sunshine is intense in the desert zone with very high temperatures in daytime but low at night. Rainfall is scanty and generally below 250 mm annually.

(f) Swamp/Estuarine

- A swamp is a lowland flooded most of the time, with shrubs and trees grown within the zone. Two main types of swamp communities recognized in Nigeria are the salt water (estuary) type and the freshwater type.
- The *estuarine type* forms a belt along the west of Nigeria, from Lagos State across Ondo, Delta, Rivers, Cross River and Akwa Ibom States. In the estuary, marine tide causes the salt water of the sea and freshwater from inflowing rivers into the sea to mix. The estuary along the coast occurs as marshes, deltas or lagoons densely grown with mangrove plants.
- The mean annual rainfall in the belt is over 2000 mm. During the rainy season, the river water brings down large amounts of sand and silt which are deposited at the delta. However, in the dry season, the flow of the rivers subsides and seawater brought in by tides makes the water in the estuaries and creeks brackish. The vegetation is dense with plants having prop-roots and breathing roots *(Pneumatophores).* The prominent plant of the mangrove community include the red mangrove, *Rhizophora racemosa* and the white mangrove, *Avicennia mitida.* Coconuts are on sandy beaches along the sea shores.
- Animals found in the estuaries include fishes: the *Periophthalmus,* tilapia species and angel fish; shelled animals: bloody clam *(Arca),* barnacles, mangrove crab, hermit crab and lagoon crab; mosquitoes and birds such as heron.
- The *freshwater swamp* belt occurs beyond the range of the tides. The rainfall in the biome is very heavy, relative humidity is very high and the evaporation from the rivers is low. Rivers overflow their banks during the rainy season and flood the land, making the soil marshy and waterlogged.

- The vegetation is rich with many types of trees such as palms and other fibre plants, e.g. *Raphia* sp, *Pandamus* sp, *Synphonia* sp and *Alstonia* sp which are used for building canoes. The herb community is dominated by sedges, bushes, reeds, grasses and many floating plants.
- Due to the low oxygen content of the water, most animals found in the zone are with accessory respiratory apparatus like snails (*Biomphalaria*, lungfishes (*Polypterus*) and mud fishes (*Clarias*).

7.4 Major Biomes of the World

- (a) **Tropical Forest:** This occurs in the equatorial region of the world where rainfall is heavy. It consists of luxuriant forest vegetation with tall trees and woody climbers with stems that climb up and hang down from trees, called lianes. The fauna include specialized birds such as hornbills, and woodpeckers, small antelopes, monkeys, chimpanzees and forest elephants.
- (b) **Savanna:** This is a tropical region dominated by grasses with scattered trees and fire-resisting thorny shrubs. The fauna include a great diversity of grazers and browsers such as antelopes, buffalo, zebra, elephant and rhinoceros, the carnivores include cheetah, hyena and lion.
- (c) **Desert:** Deserts occur between latitudes 15° and 30° north and south of the equator. The mean annual rainfall in deserts in less than 25 mm. Daily temperature fluctuation is great.

  Only few species of plants thrive in the dry zone. These include annuals which flower during short rainy period, produce seeds and then die while the succulent plants store water which enables them to survive dry periods. The woody species have tiny, reduced leaves or thorns in place of leaves. Common plants include cactus and acacia while date palms occur around oasis.

  The desert animals have the ability to conserve water and they tend to excrete solid urea instead of urine. The camel which is common has feet adapted for walking on loose sand.
- (d) **Shrub:** This can also be referred to as deciduous forest especially in the temperate regions of Europe, Britain, Asia, North America and Eastern USA. Most of the areas originally occupied by this vegetation have been altered by man's activities through urbanisation and agriculture.

  The trees are broad-leaved and include the beech, birch, walnut, and many herbaceous plants with sparse grass population. They shed their leaves in autumn and grow new leaves in spring. The annual rainfall is low.

  Animals in this zone include black bear, deer, red fox, badger and various rodents as well as many insects, reptiles and birds.
- (e) **Afro Alpine:** This vegetation occurs on highlands and mountains in Africa and is brought about by altitude and rainfall. The vegetation changes as one ascends the mountain, e.g. Kenya highlands with savanna between 600 m and 1600 m, temperate forest between 1600 m and 2000 m, bamboo forest between 2000 m and 3000 m and Alpine pasture above 3000 m. Mountain vegetation occurs on Guinea and Cameroun mountains between 900 and 1800 m while temperate forest occurs above 1800 m and beyond this, is grass.
- (f) **Swamp:** This is made up of the saltwater and fresh-water swamp forest with very heavy rainfall and high relative humidity. In the saltwater swamp, river water brings down large amounts of sand and silt which are deposited at the delta with many creeks and lagoons. During the dry season, the flow of the rivers subsides and seawater is brought in by tides thus making the water in the estuaries and creeks brackish. Saltwater also floods the aluvial deposits, making them salty. Mangrove swamp forests develop on the bare, water-logged salt mine deposits. The typical tree is the red mangrove (*Rhizophora racemora*), white mangrove (*Avicenia mitida*), and coconut and reeds which grow in sandy beaches along sea shores. The animals include angel-fish, clam, oysters, barnacles and crab.
- The freshwater swamp forest occurs beyond the range of tides. Rivers overflow their banks during the rainy season and flood the land, making the soil marshy and waterlogged. The forest vegetation is irregular and broken. Shrubs and climbing plants form chains of thick bush. Patches of open water are present in dry season. In standing water, plant species include water lettuce, *Lemna* and *Salvinia*. At the edges of the water bodies are grasses, ferns and raphia palm.

7.5 Population Studies by Sampling

- Population simply means the number of organisms living in an area: specifically it is a group of organisms of the same species living in a particular area at the same time.

a. Population Size

- Population size can be measured in several ways:
  - * abundance: absolute numbers in a population;
  - * numerical density: number of individuals per unit area or volume;

* biomass density: biomass per unit area or volume.

b. Dominance

- In any community, any species of organism that are more numerous than others is said to be the dominant species. The dominant species often exert more influence on the habitat and on the species in the area.

c. Density

- The density of an organism in a habitat is the average number of that individual per unit area of the habitat. Population density is thus the number of individuals (N) divided by the unit area (A).

$$\text{Density} = \frac{\text{Population size}}{\text{Area of habitat}}$$

- In an ecosystem with multiple species any species with the highest density is said to be dominant over the other species.
- Population studies cannot be carried out on a whole habitat. Such investigations can only be carried out on specific species or random small sections of the habitat. Estimates of population sizes are made through process known as sampling. There are various types of sampling methods:

- **i. Quadrat Method**
  A quadrat is a rectangular or square frame made from thick wire or wood. The quadrat is thrown at random several times on specific areas and the area thus covered is studied. The number of species of organisms and their types within the quadrat are recorded. The average number of times each species appears is calculated and the dominant species is then determined.

- **ii. Transect Method**
  In this situation, a tape or rope worked at convenient intervals is stretched across the area. The organisms found at each interval are recorded. This is repeated a few times and as such an estimate of the number of organisms in the area can be recorded.

- **iii. Capture-Recapture Method**
  This method is specific for estimating animal populations. The method involves catching a particular species of animal, then counted (C_1) and marked with dye or specific mark and released into the population. The process is repeated after a period of time when the marked animals would have mixed properly with the unmarked animals. The number of animals/later caught are then recorded (C_2). The number of marked recaptures in the second sampling are then recorded (r). The population of the animal (N) in the area can then be calculated using the formula:

$$\text{Population in the area (N)} = \frac{C_1 \times C_2}{r}$$

d. Factors that affect population

- The number of individuals in a population changes and it is affected by the following factors: weather, food availability, disease, reproductive rate, death rate, emigration, immigration, self regulation of numbers, and unfavourable natural events such as fires, droughts and floods.

7.6 Ecological Factors

- Ecological factors which are common to all habitats include temperature, rainfall, light intensity, hydrogen ion concentration, wind and pressure. Temperature and rainfall are the main factors that determine the major biomes of the world.
- The major ecological factors of terrestial environments are temperature, sunlight, rainfall, humidity, wind, pressure and the soil. In the aquatic environment however, the important ecological factors are temperature, sunlight, rainfall, wind and pressure, density, dissolved salts and dissolved gases.

Temperature

- Terrestrial habitats show a greater temperature range than aquatic ones. In temperate countries, the temperature varies from below 0°C in winter to above 20°C in summer but in the tropics lower temperatures are recorded during the rainy season than in the dry season. However lower temperatures are recorded in the tropics when the cold wind blows during the harmattan period of the dry season.
- In aquatic habitats, temperatures decrease with depth. In tropical marine waters, the surface temperature may be as high as 30°C while at a depth of 1500 m it may be as low as 4°C. On the other hand, temperatures of terrestrial habitats decrease with height, with a drop of about 1°C for every 150 m rise in height above sea level.
- Majority of living organisms are killed by high temperatures but they are inactive at low temperatures. Many of the living organisms grow and reproduce rapidly when they have increased metabolic activities at optimum temperatures.

Light

- Light has a more direct effect on the growth of plant populations than on that of animals. Plants are dependent on light for photosynthesis and as such their distribution.

- Light also affects the activities of animals in various ways. Majority of animals are active during the day but animals such as cockroaches appear at night.

Rainfall

- Rainfall is an important ecological factor that affects the terrestrial habitat. Abundant rainfall of over a mean of 1800 mm is typical of the rain forest while low rainfall of less than 250 mm is characteristic of deserts.
- Rainfall affects freshwater aquatic habitats especially streams and ponds which depend on rainfall to exist. Lack of rainfall may lead to the death of the plants and animals living in the water bodies except for animals like lungfish which can survive dry periods.

7.7 Relationship Between Soil Types and Water-Holding Effects of Soil on Vegetation

- The major components of soil are inorganic particles from rocks, organic materials from decomposed dead organisms, water and air from the atmosphere. Every soil is composed of three main horizons:
  - (i) litter and top soil which is a few centimetres thick, darkened by organic matter from the thin litter layer and is responsible for plant growth and activities of soil organisms;
  - (ii) subsoil which is several meters deep, sandy clay, light in colour and lacks humus; and
  - (iii) weathered material developing into subsoil on top of the parent rock which occur 30 m or more below the surface.

Soil Types

- Soils are classified world-wide according to a complex related to their modes of formation: podzolization, laterization and calcification.
- **Podzolic Soil:** This is a soil type typical of cold temperate climates. The soluble mineral nutrients of the topsoil are leached by rainfall to subsoil while slow decay of litter produces acid topsoil.
- Soils can be classified in many different ways. It can be classified as alluvial or sedentary, depending on how it was formed; as topsoil or subsoil, depending on the depth at which it is situated; and as loam, sand, or clay, depending on its composition.

Sedentary and Alluvial Soils

- The sedentary soil is one situated on top of the rock from which it developed by weathering. The alluvial soil is one composed of mineral particles carried over long distances from their place of origin by rivers and then deposited on flat areas.
- Topsoil and Subsoil: Both sedentary and alluvial particles develop into fertile soil as they are colonized by plants and animals, and as they gather humus from the activities of bacteria and fungi. Then processes take place in the upper regions of the soil where they form topsoil. Below the topsoil is subsoil, which is lighter in colour owing to the absence of humus, and contains very little life and as such less fertile than topsoil. The best topsoil is called *loam.*
- Sedentary and alluvial soils can be classified into different types according to the substances of which they are composed; loam, sand and clay.
- **Loam Soils:** They are the most fertile of all soils and they promote vigorous plant growth. They are well-aerated, drain freely and yet retain plenty of moisture and dissolved minerals.
- **Sandy Soils:** They contain high proportion of large mineral particles combined with little humus and large spaces between the particles which give the soil good drainage and aeration. This enables plant roots to penetrate quickly during germination; sandy soil does not retain water for long after rainfall, and this causes plants to wilt and die very quickly during drought.
- **Clay Soils:** These contain a high proportion of minute mineral particles. Unlike sand, clay retains moisture during drought and it loses minerals at a slow rate owing to its poor drainage. Clay particles are small and they are sticky and heavy when wet, making the soil extremely difficult to cultivate. Wet clay is almost completely devoid of air, and in prolonged drought it dries and splits into rock-hard lumps.

7.8 Simple Measurement of Ecological Factors

a. Physical Factors

- **Temperature:** It measures the degree of coldness or hotness of a body. A mercury thermometer is used to measure temperature in degrees celsius (°C). At least two readings in a particular area of the habitat are taken to ensure accurate results. Air temperatures are taken in the shade. Soil thermometer is used to measure soil temperature. A maximum-minimum thermometer is used to record the highest and lowest temperature of the day. A waxed bulb thermometer tied to a string which is knotted at different depths, is used to measure the temperature of water at different depths in a pond or river.

- **Rainfall:** It is measured with a rain gauge. The height of the water in the cylinder indicates the rainfall after each rain storm.
- **Light:** The intensity of light is measured by a photometer.
- **Relative Humidity:** It is measured with a hygrometer using either the pocket type or the wet-and-dry bulb hygrometer.
- Humidity is read off the pocket hygrometer after exposing it in the air and allowing the lever arm to stabilize.
- The wet-and-dry bulb hygrometer is swung in the air for about half a minute. Then the readings on both the thermometers are taken and then converted to relative humidity using a conversion table.
- **Wind:** The direction of the wind is indicated by a wind-vane, while the wind speed is measured by an anemometer on which the wind speed is shown in meters per second.
- **Pressure:** The pressure of air in any habitat is measured by a barometer in millimeters of mercury (mmHg).
- **Water Depth:** This is measured with a meter rule to the end of which is attached a weighted line to ensure that the rule is vertical. However for deep waters, a weighted line which is knotted at one metre intervals is used.
- **Water Flow:** The rate of water flow is measured by throwing a floated object into the water and the time taken to cover a known distance is recorded in metres per second. A water-speed meter can also be used to compare the speed of water flow at different positions in a river.
- **Turbidity:** The degree of water turbidity affects the water transparency. This can be measured by lowering a secchi disc into the water and noting the depth at which the disc is no longer visible. This is a method for comparing the turbidity of different waterbodies.

b. Edaphic Factors

Soil or edaphic factors are related to:

— the type of soil particles,
— the humus, mineral, water and air content,
— the pH, and
— the depth of the water table

- **i. Relative Amounts of Different Soil Particles**
  This can be determined by filling the soil sample into two-thirds of a measuring cylinder and adding water. It is then shaken thoroughly and allowed to stand and settle. The types of soil and the amounts of soil particles can be seen in layers under the water.

- **ii. The Percentage of Water in a Soil Sample**
  The soil sample is weighed in a crucible of known weight. The soil sample is then heated in an oven maintained at 100°C. After half an hour, the crucible and its contents are removed, cooled in a desiccator and reweighed. The heating, cooling and weighing procedure is repeated until a constant weight is obtained.

  The percentage of water in the soil sample can be determined from the weight differences.

- **iii. The pH of the Soil Sample**
  The soil sample is filled to a depth of 25 mm in a test-tube and about 12 mm barium sulphate solution is added. The test-tube is then filled with distilled water and shaken thoroughly. A pH paper is then dipped into the solution to determine the acidity or alkalinity of the soil sample.

SSCE/UME–Type Objective Questions

1. The study of the organisms and the environment of an abandoned farmland is the ecology of:
   A. a community B. a population
   C. a species D. a habitat
   E. an ecosystem

2. The complex relationship between the members of a community and between the community as a whole and its physical environment is
   A. ecosystem B. habitat
   C. environment D. niche
   E. microhabitat

3. A secchi disc is used in the determination of
   A. rainfall B. tides
   C. waves D. turbidity
   E. current velocity

4. If three 30cm lengths of glass tubing are tightly packed with clay , sand and loamy soils respectively, and then stood in a beaker of water for one week, the level of water will be
   A. lowest in the tube with clay
   B. the same in all the tubes
   C. lowest in the tube with loamy soil
   D. highest in the tube with sandy soil
   E. lowest in the tube with sandy soil

5. A few grams of dried soil were first heated until red hot and then further heated until no more smoke was released. This experiment was to determine the
   A. amount of water in soil
   B. percentage of water in soil
   C. presence of humus in soil
   D. resistance of laterite to heat
   E. release of smoke form soil

6. Soil is said to be fertile if
   A. it is black
   B. it can supply usable nutrients to plants
   C. the oxygen content is very high
   D. artificial manure is added to it
   E. it is waterlogged.

7. A population is defined as the collection of
   A. similar organisms that are found in the same habitat
   B. similar organisms that breed in the same habitat
   C. similar organisms that interbreed freely in the same habitat
   D. similar organism that eat the same feed
   E. different organisms in the same habitat

8. 28g of soil sample was heated to constant weight of 24g. When further heated to red hot and cooled, it weighed 18g. What is the percentage of humus in the soil?
   A. 21.4 B. 55.6
   C. 75.0 D. 25.0
   E. 35.7

9. A anemometer is an instrument for measuring
   A. relative humidity B. altitude
   C. wind speed D. turbidity
   E. salinity

10. Which of the following groups of factors is completely abiotic?
   A. salinity, tide, plankton, turbidity
   B. temperature, pH, soil, insect
   C. wind, altitude, humidity, light
   D. conifers, pH, rainfall
   E. soil, water, bacteria, salinity

11. If a handful of soil is shaken with water and left to settle, the soil particles will settle from light to heavy particles as follows:
   A. humus, clay, silt, sand, stones
   B. humus, silt, clay, sand, stones
   C. humus, clay, silt, stones, sand
   D. clay, humus, silt, sand, stones
   E. humus, sand, silt, clay, stones

12. Which of the following ecological factors are common to both terrestrial and aquatic habitats?
   A. rainfall, temperature, light and wind
   B. salinity, rainfall, temperature and light
   C. tides, wind, rainfall and altitude
   D. pH, salinity, rainfall and humidity
   E. wind, pH, temperature and light

13. Which of the following has the greatest influence on the distribution of animals in marine and fresh water habitats?
   A. pH B. salinity
   C. water current D. turbidity
   E. altitude

14. In a savanna ecosystem, the abiotic factors include
   A. legumes, temperature and sandy soil
   B. water, temperature and soil

C. minerals, oxygen and reptiles
D. waters, oil and grasses
E. soil, minerals and bacteria

15. Mangrove swamp can be found
A. on a seashore with a flat terrain
B. where a river meets the sea
C. on a mud flat near the sea
D. where two rivers meet
E. on a piece of land surrounded by water

16. Soil with the finest particles is called
A. silt B. clay
C. sand D. gravel
E. loam

17. Which of the following is not a characteristic feature of tropical rainforest habitats?
A. trees with longer boles than those in savanna
B. fewer canopy strata than the savanna
C. many more trees and climbers than savanna
D. plenty of leaf litter on the ground
E. more diverse communities than the savanna

18. Which of the following factors is not associated with aquatic habitat?
A. temperature B. light intensity
C. turbidity D. wave action
E. humidity

19. Which of the following types of soil has the highest water-retaining capacity?
A. clay B. laterite
C. loam D. gravel
E. sand

20. Water rises most rapidly in
A. sandy soil B. clayey soil
C. sandy-loam soil D. loamy soil
E. humus soil

21. Puddles, ponds, rivers, lakes, seas and oceans are grouped together as
A. ecosystem B. biosphere
C. terrestrial habitat D. marine habitat
E. aquatic habitat

22. The number of individuals in a habitat in relation to the unit space available to each organism is referred to as the
A. birth rate B. density
C. mortality D. frequency
E. dispersion

23. Predation is an example of a factor in an ecological system referred to as
A. edaphic B. typographic
C. abiotic D. biome
E. biotic

24. Which of the following world biomes is found in Nigeria?
A. Sahel savanna
B. Tundra
C. Desert
D. Afro-alphine
E. Temperate deciduous forest

25. A hygrometer is used for measuring
A. relative humidity B. specific gravity
C. light intensity D. turbidity
E. wind speed

26. How are savanna plants protected from bush fires? They
A. have sunken stomata
B. roll their leaves during hot days
C. have thick bark
D. always produce flowers
E. shed their leaves gradually throughout the year

27. Which of the following best describes a marine habitat? A large body of water
A. which has no distinctive colour or taste
B. with high concentration of salt
C. with little suspended materials
D. with no water weeds
E. which sustains no animal life

28. The following are features of Northern Guinea Savanna except
A. presence of tall trees with thick bark
B. bare soil with very few trees
C. presence of fire-resistant trees
D. abundant herbivores
E. predominance of woody trees

29. The following are features of the tropical rain forest except
A. abundant water supply
B. loose and moist soil
C. short trees growing beneath tall trees
D. scanty trees with small leaves
E. presence of many animals

30. Which of the following is not a characteristic of overcrowding in plant and animal community?

A. Population outstripping available space
B. Population exceeding available food
C. Competition within the population
D. Increase in primary production
E. Population increasing at the same rate as the birth rate

Answer Key

1.	E	7.	A	13.	A	19.	A	25.	A
2.	A	8.	A	14.	B	20.	C	26.	C
3.	D	9.	C	15.	C	21.	E	27.	B
4.	E	10.	C	16.	B	22.	B	28.	B
5.	C	11.	A	17.	B	23.	E	29.	D
6.	B	12.	A	18.	E	24.	A	30.	D

Explanations to Objective Questions

1. (E) An ecosystem is the study of the organism and the environment.

2. (A) The ecosystem deals with the members of the community and their relationship with the environment.

3. (D) A secchi disc is used to determine the turbidity of water.

4. (E) The level of water will be lowest in the tube with sandy soil because sandy soil does not retain water for long.

5. (C) It is only the presence of humus in the soil that can release smoke because of the organic matter content.

6. (B) Soil can be regarded as fertile because it can supply usable nutrient to plants.

7. (A) A population can be defined as a collection of similar organisms that are found in the same habitat at the same time.

8. (A) The percentage of humus in the soil is 21.4 because humus can only be released from soil when heated to red hot.

9. (C) An anemometer is an instrument used for measuring wind speed.

10. (C) Winds, altitude, humidity and light are all abiotic factors.

11. (A) When a handful of soil is shaken with water and left to settle, the soil particles will settle with light humus on top, followed by heavier particles in the order: clay, silt, sand and stones.

12. (A) Rainfall, temperature, light and wind are the ecological factors that are common to both terrestrial and aquatic habitats.

13. (A) pH has the greatest influence on the distribution of animals in marine and fresh water habitats.

14. (B) The abiotic factors in a savanna ecosystem include water, temperature and soil.

15. (C) Mangrove swamp can be found on a bare mud flat near the sea.

16. (B) Clay is the soil with the finest particle.

17. (B) Any forest with fewer canopy strata than the savanna has no feature typical of a tropical rain forest.

18. (E) Humidity is not associated with aquatic habitat.

19. (A) The clay soil has the highest water-retaining capacity.

20. (C) Water rises most rapidly in sandy-loam soil because it is well aerated and drain freely.

21. (E) Puddles, ponds, rivers, lakes, seas and oceans are grouped together as aquatic habitat.

22. (B) The number of individuals in a habitat in relation to the unit space available to each organism is referred to as the density.

23. (E) Predation is an ecological system.

24. (A) Sahel savanna is a world biome found in Nigeria.

25. (A) A hygrometer is used for measuring relative humidity.

26. (C) The thick bark on savanna plants protect them from bush fires.

27. (B) A marine habitat is a large body of water with high concentration of salt.

28. (B) Bare soil with very few trees are not features of Northern Guinea Savanna.

29. (D) Scanty trees with small leaves are not features of the tropical rain forest.

30. (D) Increase in primary production is not a characteristic of overcrowding in plant and animal community.

8 Functioning Ecosystem

8.1 Food Webs and Trophic Levels

Autotrophs and Heterotrophs

- Living organisms can be divided into two main groups according to the way they feed: *Autotrophs* and *Heterotrophs*.
- **Autotrophs:** These are organisms that are able to use sunlight or chemical energy to manufacture their own food from simple inorganic substances. The autotrophs are the only organisms that can produce food in an ecosystem and as such they are called *primary food producers*. They form a direct link between the abiotic and the biotic components of an ecosystem because they are able to utilize sunlight directly. Examples of autotrophs are green plants, photosynthetic protists and some bacteria.
- **Heterotrophs:** These are organisms that cannot manufacture food on their own. They depend on ready-made food produced directly or indirectly by autotrophs. They utilize the nutrients and chemical energy stored in the food they feed on for their activities. Examples of heterotrophs are animals, carnivorous plants, fungi, most protists and bacteria. In an ecosystem, heterotrophs are either *consumers* or *decomposers*.

8.2 Trophic Levels

- Energy and nutrients are transferred step by step among organisms along a feeding pathway in an ecosystem.
  Feeding in nature follows the following pattern:
- The starting point is a primary producer (green plant) which is eaten by a primary consumer (animal: sheep) which ultimately is eaten by a secondary consumer (animal: lion) and finally a decomposer which converts the remains of the dead producers and consumers into simple inorganic substances which return to the non-living environment.
- The feeding pattern is similar in all ecosystems except that it may end at the primary consumer level or may proceed on to the secondary consumer, tertiary consumer and beyond before it is decomposed.
- Each step along the feeding pathway is known as a *trophic* level.
- Trophic level 1 is mainly an autotroph or primary producer. Trophic level 2 consists of primary consumers. The trophic level standing from one increases in an ascending order and indicates the path of energy flow.

8.2.1 Food Chain

- A food chain shows the transfer of energy and nutrients from organism to organism in a feeding pathway. Following are common examples:

  (i) Grass (primary producer) ⟶ Sheep (primary consumer) ⟶ Lion (secondary consumer)

  (ii) Grass → Grasshopper → toad → snake → hawk

  (iii) Green algae → shrimp → fish → man

- A food chain involves at least two links and more organisms at the lower trophic level than the higher trophic level. The food energy transfers decrease from the lower to the higher trophic levels. The food chains in an ecosystem can be illustrated in a form of pyramid of numbers, or biomass or energy (Fig. 8.1).

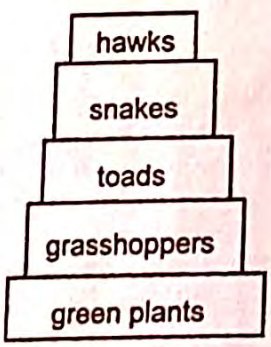


Fig. 8.1: Pyramid of number/biomass

8.2.2 Food Web

- Numerous food chains are present in an ecosystem and these can be linked together in a food web (Fig. 8.2).

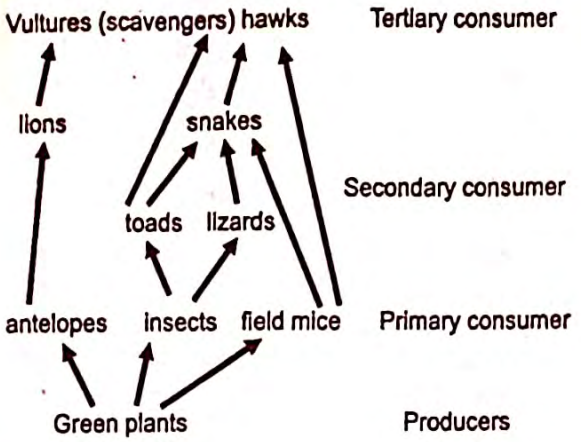


Fig. 8.2: A food web

8.2.3 Energy Flow

- In the food chain illustrated below, the grass is the primary producer. The grass traps solar energy through photosynthesis and produces chemical energy in the form of food. It uses some of the energy in the food for its metabolic activities and the energy is eventually lost as unusable heat energy.
- When the grass is eaten by a primary consumer such as zebra, part of the energy stored in the grass is transferred to the consumer while part of it is lost as heat. The zebra uses part of the energy in the food it consumed for its metabolic activities while part of the energy is lost as heat. When the zebra is eaten by a secondary consumer such as the lion part of the energy is similarly lost as heat while the energy stored as food in the lion is eventually released as heat by decomposers when the lion dies.

Solar energy ⟶ Grass ⟶ Zebra ⟶ Lion

- Thus, the food chain starts with an input of solar energy in the ecosystem and at every trophic level, some of the stored energy is lost as heat energy and as the food chain progresses, the stored energy decreases progressively and eventually all the stored energy is lost from the ecosystem.
- In the aquatic environment, the food chain starts with an input of solar energy which is trapped by phyto-plankton which is eaten up by a zooplankton or shrimp which is then eaten up by a fish and eventually the stored energy is progressively lost through the chain and is released as heat by decomposers.

Solar energy → Food chain ⟶ Phytoplankton ⟶ Zooplankton ⟶ Fish

8.3 Ecological Pyramids

- In communities where there appear to be more plants than herbivores, more herbivores than small carnivores, more small carnivores than large carnivores, these represent *pyramid of numbers* as in Fig. 8.3.
- However, in many instances, a simple pyramid of numbers is not enough to demonstrate the flow of energy through an ecosystem. For example, a tree can support a large population of grasshopper which in turn will be eaten by a small population of preying mantis which, on the other hand, are the prey of a fewer number of birds us shown in Fig. 8.3.

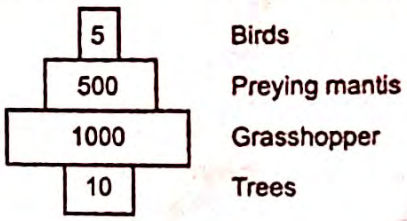


Fig. 8.3: Pyramid of numbers/biomass

- Biomass takes into account both the size of the individual organisms and their numbers. The pyramid of biomass thus gives the accurate picture of the relationships between the organisms at the various trophic levels in a food chain than the pyramid of numbers.
- **Pyramid of Energy:** This represents the rate of flow of food energy through each trophic level of a food chain.
- A pyramid of energy is always upright. This is because there is always a great loss of useful energy as food is transferred from one trophic level to the next higher level. Since only a small fraction of energy is passed on to successive trophic levels, the number of links in a food chain is limited, usually to three or four.

SSCE/UME–Type Objective Questions

1. Which of the following does not contribute to the biomass in an ecosystem?
   A. Producers B. Food chains
   C. Consumers D. Microorganisms
   E. Saprophytes

2. Autotrophs are also described as:
   A. consumers B. decomposers
   C. carnivores D. producers
   E. herbivores

3. Which of the following statements is not correct about food chains?
   A. All food chains start with a green plant
   B. Food chains involve feeding relationships among organisms
   C. Shorter food chains indicate more effective utilization of energy than longer ones
   D. There Is no energy loss in a food chain
   E. All food chains have primary consumers

4. At which trophic level would the highest accumulation of a non-biodegradable substance occur?
   A. Primary consumers
   B. Tertiary consumers
   C. Primary producers
   D. Secondary consumers
   E. Secondary producers

5. Which of the following groups of organisms feeds directly on green plants?
   A. producer
   B. decomposer
   C. primary consumer
   D. secondary consumer
   E. tertiary consumer

6. Changes in energy flow between organisms in a habitat can be represented by a
   A. pyramid of biomass
   B. pyramid of numbers
   C. pyramid of energy
   D. flow chain
   E. food web

7. Which of the following would be the primary producer in a food chain?
   A. saprophytes
   B. herbivores
   C. carnivores
   D. green plants
   E. parasites

8. A food web can be described as a
   A. complex feeding relationship in a community
   B. transfer of food energy from one community to another
   C. complex pattern of food production in a community
   D. group of many populations competing for the same type of food
   E. group of organisms belonging to the same trophic level

9. Which of the following explains the term pyramid of numbers?
   A. The number or organisms in a trophic level
   B. The relationship between plants in different trophic levels
   C. The number of saprophytes and parasites in a habitat
   D. The number of predators in a habitat
   E. Progressive decrease in the number of individuals from lower to higher trophic level

10. Which of the following represents the correct order in a possible food chain?
    A. Crustacean → diatom→ fish → man
    B. Fish → crustacean → man → diatom
    C. Man → fish → crustacean → diatom
    D. Diatom → crustacean → fish → man
    E. Man → diatom → fish → crustacean

11. In complex food relationships in a community, the primary, secondary and tertiary consumers are referred to as:
    A. symbionts B. omnivores
    C. heterotrophs D. autotrophs
    E. carnivores

12. Which of the following is the correct chain for the organism found in an aquatic community?
    A. Paramecium, Phytoplankton, Prawn, Water flea, Tilapia, Shark
    B. Waterflea, Paramecium, Phytoplankton, Prawn, Tilapia, Shark
    C. Phytoplankton, Waterflea, Paramecium, Prawn, Shark, Tilapia
    D. Phytoplankton, Paramecium, Waterflea, Prawn, Tilapia, Shark
    E. Paramecium, Phytoplankton, Waterflea, Prawn, Shark, Tilapia

13. In a food chain, the position occupied by an organism is called
A. the trophic level
B. energy level
C. the feeding level
D. the habitat level
E. niche

14. Which of these statements about food chain is not correct?
A. Animals in the chain are consumers
B. A food chain usually begins with a green plant
C. All organisms in a food chain are animals
D. Living things are dependent on one another
E. The food chain involves energy transfer in an ecosystem

15. Which of the following organisms will have the highest biomass?
A. Small Fish B. Tadpole
C. Hawk D. Phytoplankton
E. Big fish

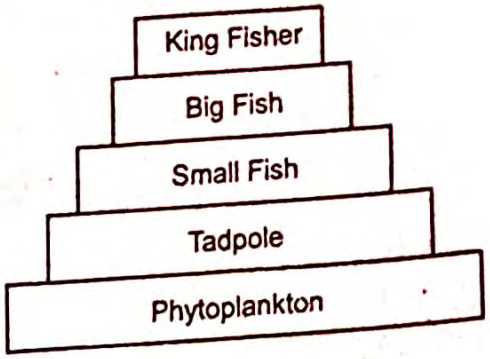
16. Which of the following is the correct sequence for energy transfer and nutrient cycling among living things in an ecosystem?
A. Consumers → producers → decomposers
B. Producers → decomposers → consumers
C. Decomposers → producers → consumers
D. Producers → consumers → decomposers
E. Consumers → decomposers → producers

17. In any food chain, the first member must be a(n)
A. carnivore B. herbivore
C. zooplankton D. autotroph
E. predator

18. The total mount of energy entering a food chain is that which is present in the
A. consumer B. ecosystem
C. producer D. decomposer
E. habitat

19. In a food chain, each stage in the chain is a
A. chain level B. web level
C. consumption level D. trophic level
E. production level

Use the diagram below to answer questions 20–22.



20. The diagram above illustrates
A. the size of various organisms
B. types of organisms in a habitat
C. the flow of energy in a habitat
D. the hierarchy of organisms in a habitat
E. the food chain

21. Which of these organisms occupies the 4th heterotrophic level?
A. King Fisher B. Small fish
C. Phytoplankton D. Big fish
E. Tadpole

22. Which of these organisms is a herbivore?
A. Tadpole B. Phytoplankton
C. King Fisher D. Small fish
E. Big fish

23. Which of the following organisms is a producer?
A. Spirogyra B. Minor
C. Rhizopus D. Yeast
D. Fungus

24. Which of the following groups consists of heterotrophs?
A. Mucor , Chlamydomonas and Euglena
B. Spirogyra, Mucor and mushroom
C. Man, Mucor and mushroom
D. Man, maize and cowpea
E. Chlamydomonas, man and maize

Answer Key

1.	B	6.	C	11.	C	16.	D	21.	A
2.	D	7.	D	12.	D	17.	D	22.	A
3.	D	8.	A	13.	A	18.	C	23.	A
4.	C	9.	E	14.	C	19.	D	24.	C
5.	C	10.	D	15.	D	20.	C		

Explanations to Objective Questions

1. **(B)** Food chain does not contribute to the biomass in an ecosystem.

2. **(D)** Autotrophs are also described as producers.

3. **(D)** There is energy loss in a food chain.

4. **(C)** Primary producers have the highest accumulation of non-biodegradable substances.

5. **(C)** Primary consumer feeds directly on green plants.

6. **(C)** Pyramid of energy represents changes in energy flow in a habitat.

7. **(D)** Green plants represent primary producer in a food chain.

8. **(A)** A food web can be described as a complex feeding relationship in a community.

9. **(E)** Pyramid of numbers is the progressive decrease in the number of individuals from lower to higher trophic level.

10. **(D)** Diatom → crustacean → fish → man is the correct order in a possible food chain.

11. **(C)** The primary, secondary and tertiary consumers are referred to as heterotrophs in complex food relationships in a community.

12. **(D)** Phytoplankton, Paramecium, Waterflea, Prawn, Tilapia and Shark is the correct chain for the organism found in an aquatic community.

13. **(A)** The position occupied by an organism in a food chain is called trophic level.

14. **(C)** All the organisms in a food chain are not necessarily animals.

15. **(D)** Phytoplankton is the organism which has the highest biomass in a food chain.

16. **(D)** The correct sequence for energy transfer and nutrient cycling among living things in an ecosystem is as follows: producers → consumers → decomposers.

17. **(D)** In any food chain, the first member must be an autotroph.

18. **(C)** The total amount of energy entering a food chain is that which is present in the producer.

19. **(D)** In a food chain, each stage in the chain is a trophic level.

20. **(C)** The flow of energy in a habitat can best be represented in a pyramid of energy.

21. **(A)** In the food chain: Phytoplankton → Tadpole → Small fish → Big fish → King fisher, the organism that occupies the fourth heterotrophic level is the King fisher.

22. **(A)** Tadpole is a herbivore.

23. **(A)** Spirogyra is a producer.

24. **(C)** Man, mucor and mushroom is a group consisting of heterotrophs because they cannot synthesize their own food.

9 Energy Transformations in Nature

9.1 Introduction

- Energy exists in various forms that are interconvertible and as such one form of energy can be transformed into another form. In nature, energy transformations are brought about by living organisms. Their activities cause energy to flow through ecosystems unidirectionally.

Energy Losses in the Ecosystem

- Solar Energy: The sun is the external source of energy for ecosystem on earth. More than half of the solar energy that reaches the earth's atmosphere is reflected or absorbed. The amount of solar energy that finally reaches the earth's surface in one year is about two to eight million kilojoules per square metre, depending on the latitude. When this energy falls on a grassland, about 20% of the energy is reflected by the vegetation, 40% warms the vegetation, soil and air, and 39% evaporates water. Only about 1% of the solar energy that reaches the earth surface is available for use by the grass.
- Primary Producers: The photosynthetic organisms that make up the primary producers manufacture food using light energy from the sun and carbon dioxide and water. In the process, the photosynthetic organisms transform the light energy into chemical energy. In green plants, the process occurs in the chloroplasts in leaves.
- About half of the amount of solar energy that falls on the plant leaf can be absorbed by chlorophyll and other photosynthetic pigments. About one quarter of the energy is converted to chemical energy such as starch while the rest is lost as heat energy.
- **Consumers:** Primary consumers feed on producers. The primary consumers are herbivores such as cow. The secondary consumers are usually carnivores.
- **Energy Laws:** Energy transformations in nature are governed by the laws of thermodynamics.
- The first law of thermodynamics states that each form of energy is readily converted into other forms but that energy is never created or destroyed. Hence, in the process of burning, chemical energy of wood changes into heat and light. In a motor vehicle, energy in form of fuel changes into mechanical energy.
- The second law of thermodynamics implies that when one form of energy is converted into another, a proportion of it is turned into heat and it is lost to the system.
- Then energy conversions in an ecosystem obey the first and second laws of thermodynamics. Hence, in a food chain starting from the grass to goat and to man, even if the goat eats up the grass with its roots, not all the energy in the grass is passed on to the goat. Similarly, only a fraction of the energy received by the grass can get to man through the goat.
- Every time heat is lost from a food chain, highly complex organised food molecules are broken down or disorganized. A measure of the degree of disorganization is called *entropy.*
- Every time chemical energy is transferred from one trophic level to a higher trophic level, the entropy of the ecosystem increases. Since photosynthetic producers are continuously harnessing the sun's energy which is eventually lost as heat, the entropy of the universe as a whole can be said to be constantly increasing.

9.2 Nutrient Cycling in Nature

- A functioning ecosystem uses both energy and inorganic nutrients. As energy flow is unidirectional, it is constantly supplied by the sun from outside the ecosystem. Nutrient substances like mineral salts and gases from some parts of the environment (air, soil, water) enter into living things at some stage only to be returned to the environment at another stage. These movements involving the organisms, the earth and chemicals, take place in cycles and are therefore termed *biochemical* cycles.

- (a) **Carbon Cycle:** Carbon is the basis of all life. In all living organisms, carbon, hydrogen, oxygen and nitrogen are elements that make up the molecules. Carbon occurs naturally in the atmosphere as carbon dioxide, in tissues of living and dead organisms, in rocks and as fossil fuels such as coal.
- Carbon in the form of carbon dioxide is removed from the atmosphere by photosynthesis while it gets back through respiration of plants, animals and soil

organisms, as well as combustion of fossil fuels and decomposition. The carbon cycle shown in Fig. 9.1 below is mainly about the processes which bring about the circulation of carbon compounds in nature.

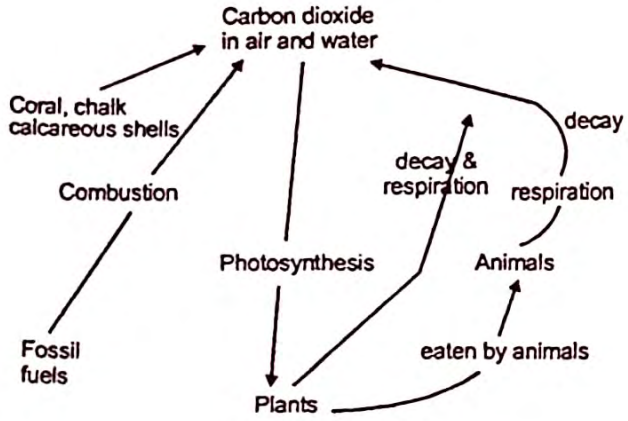


Fig. 9.1: Carbon cycle

- (b) **Water Cycle:** All living organisms contain a high percentage of water. The external source of water to all ecosystem is precipitation in form of dew or rainfall or melted snow all of which start from the atmosphere (Fig. 9.2).

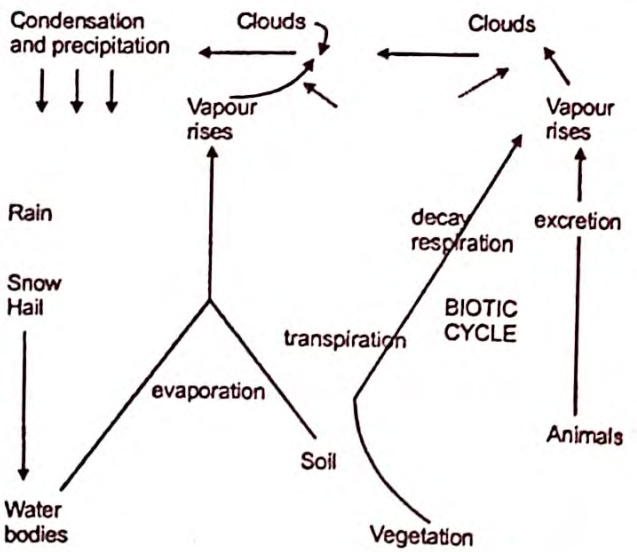


Fig. 9.2: The water cycle

- Plants take up water from the soil in large amounts and lose most of it by transpiration. Only a small amount of the absorbed water is used in photosynthesis and other metabolic reactions. Animals take in water constantly from their food and drink and lose the water in the air they breathe out, and in faeces and urine. All living organisms release water during cellular respiration. Decomposers also release water during most decay processes. Combustion of most fuels also releases water.
- Water circulates in nature within the abiotic environment and only a little through living organisms as shown in the figure.
- Water that enters the biotic component of the ecosystem returns to the abiotic environment via respiration, decay, excretion and transpiration.
- Water bound up in the protoplasm of organisms is also returned to the environment whenever the organisms get decomposed.
- (c) **Decomposition in Nature:** Decomposers play an essential role in the cycling of nutrients in ecosystems. They form the link between the producers and consumers and the abiotic environment in all food chains.
- Decomposers are saprophytes. They feed on dead and decaying organisms and their wastes and bring about their decay or decomposition.
- There are two types of decomposers: micro-decomposers and macrodecomposers. Micro-decomposers are the microorganisms like bacteria, fungi and protozoa that are capable of using components of dead organisms as food. The macro-decomposers are the bigger organisms such as millipedes, mushrooms, mites and certain insects that use fragments of dead organisms and microdecomposers as food.
- The inorganic compounds released during decay include gases like carbon dioxide, ammonia, hydrogen sulphide and water vapour, and salts like nitrates, sulphates, phosphates and potassium ions.
- The activity of decomposers releases the nutrients in simple forms. Green plants absorb such nutrients through their roots and leaves, and use them to manufacture energy-rich food. The nutrients are again returned to the abiotic environment by decomposers to be used by green plants. This use and re-use of nutrients is called recycling.

SSCE/UME–Type Objective Questions

1. Which of the following processes is not involved in the carbon cycle?
   A. Burning B. Decay
   C. Transpiration D. Respiration
   E. Photosynthesis

2. Which of the following processes are not involved in a water cycle?
   A. Evaporation, condensation and precipitation
   B. Evaporation, percolation and transpiration
   C. Evaporation and precipitation
   D. Evaporation and transpiration
   E. Precipitation and condensation

3. Water in the human body is derived from the following sources except through
   A. drinking water
   B. food substances
   C. cytoplasm or worn-out cells
   D. respiratory activity
   E. absorption by the skin

4. The pathway through which nutrient elements are replenished and circulated in an ecosystem is called a
   A. cycle B. food chain
   C. food web D. Kreb's cycle
   E. pyramid of numbers

5. All of the following can illustrate the dynamic nature of the ecosystem except
   A. nitrogen cycle B. carbon cycle
   C. water cycle D. nutrient cycle
   E. locomotion in organism

6. Which of these processes removes carbon from the atmosphere?
   A. putrefaction of animals
   B. Burning of fuels
   C. Photosynthesis
   D. Volcanic eruption
   E. Respiration in plants

7. Which of these is not applicable to the circulation of water in nature?
   A. Water removed from oceans are permanently lost
   B. Water constantly evaporates from the seas, rivers, lakes and soil
   C. Water vapour rises, cools, condenses and precipipates as rain and dew.
   D. Water is absorbed by plants and animals from their external environment.
   E. Plants lose water by transpiration and decay.

8. Which of the following processes removes water from the water cycle?
   A. Condensation B. Perspiration
   C. Photosynthesis D. Transpiration
   E. Respiration

9. Carbon is added to the atmosphere by the following processes except
   A. respiration B. burning
   C. photosynthesis D. volcanic eruption
   E. oxidation

10. Which of the following is a means of measuring primary production?
    A. Rate of reproduction of animals
    B. Life span of an organism
    C. Number of predators in an environment
    D. Harvested plants in a unit area
    E. Rate of consumption by herbiviores

Answer Key

1.	C	3.	E	5.	E	7.	A	9.	C
2.	B	4.	A	6.	C	8.	C	10.	D

Explanations to Objective Questions

1. **(C)** Transpiration is not a process involved in the carbon cycle.

2. **(B)** Evaporation, percolation and transpiration are processes not involved in water cycle because percolation is out of place.

3. **(E)** Water in human body is not derived from absorption by the slum.

4. **(A)** The pathway through which nutrient elements are replenished and circulated in an ecosystem is called a cycle.

5. **(E)** Locomotion in organisms cannot illustrate the dynamic nature of the ecosystem.

6. **(C)** Photosynthesis is a process which removes carbon from the atmosphere.

7. **(A)** Water which is removed from oceans are permanently lost and does not take part in the circulation in nature.

8. **(C)** Photosynthesis removes water from the water cycle for the process of food manufacture.

9. **(C)** Photosynthesis does not add carbon to the atmosphere.

10. **(D)** Harvested plants in a unit area can be used to measure primary production.

10 Ecological Management

10.1 Association

- There are many ways in which organisms interact or associate with one another in nature. Associations may occur between individuals of the same kind or species, i.e. intraspecific or between different species, i.e. interspecific.

- (i) **Symbiosis:** This is a close and prolonged living together and association of two or more organisms of different species. The relationship in a symbiosis may be temporary or permanent; harmful, beneficial or neutral to one or more or all the partners.

  The three basic types of symbiosis are parasitism, commensalism and mutualism.

- (ii) **Parasitism:** This is an association in which one organism, the parasite, lives on another, the host, deriving benefit from it and causing harm to it. Animals are often hosts to a range of parasitic organisms such as bacteria, fungi, protozoa and invertebrates like flukes, roundworms, ticks, mites, etc. Plants too are hosts to parasites such as viruses, bacteria, fungi and nematodes. Parasites like ticks, fleas, lice and leeches that live on body surfaces of their hosts are known as ectoparasites. Endoparasites on the other hand live inside the bodies of their hosts, e.g. roundworms and hookworms.

- (iii) **Commensalism:** This is an association between unrelated organisms where one partner, the commensal, benefits while the other, the host is unaffected. Examples include the fish called remora (shark-sucker) and the shark. The remora benefits by being carried freely from place to place and gathering with little effort, pieces of prey left by the shark. The remoras dorsal fin is modified into a hold fast-like structure by which it attaches itself temporarily to the underside of the shark. Many bacteria inhabiting the intestine of the vertebrates feed on undigested food particles in the gut and do no harm to their host. The association between an epiphyte and a tree may be regarded as commensalism. The epiphyte gets a site where it can get enough sunlight to carry out photosynthesis, e.g. ferns on trunks of oil palm trees.

- (iv) **Mutualism:** This is an association between unrelated organisms which is beneficial to both. In a loose association, the two partners are capable of leading independent lives. In a very close association, both partners cannot live without the other. Often one partner lives inside the other in such close associations. Examples are as follows:

  — **Root nodule of legumes:** Nitrogen-fixing bacteria receive nourishment and shelter from host legume. The legume benefits from nitrogen salts manufactured from the nitrogen of the air fixed by bacteria.

  — **Mycorrhiza:** Many trees live mutually with certain fungi in the soil. The fungus forms a covering (mycorrhiza) around the roots of the tree. The covering helps the tree to absorb water and mineral from soil while the fungus receives nourishment from its host tree.

  — **"Aphid-Ant" Association:** Some species of ants and aphids benefit from living together. Most aphids live on plants and do not move about a lot. They suck juices from tender parts of their hosts. Aphids are tended by ants which also help to move them to more favourable feeding sites. Tending ants obtain nourishment in the form of special secretion from their aphids.

- (v) **Saprophytism:** This is a feeding association between living organisms (saprophytes) and dead remains of once-living organisms. Saprophytes absorb food from organic remains of plants and animals after digesting it outside their bodies (extracellular digestion). Common saprophytes include plants, mainly fungi (mushrooms or mould on bread) and bacteria saprophytic. Fungus that live on dead epithelial cells of animals cause ringworm and other skin diseases. Saprophytes also help in the decay and removal of dead remains of organisms on earth.

10.2 Adaptation to Environmental Factors

- Living organisms are found in different types of habitats. These organisms show features that enable them to live successfully in these environments. These features may be structural, functional or behavioural. Such features are known as *adaptations*. These adaptations can be seen in organisms as they relate to their aquatic or terrestrial or arboreal

habitat and to particular modes of life whether they are burrowing or parasitic and to other activities such as feeding whether they are carnivorous or herbivorous.

Aquatic and Terrestrial Adaptations

- The aquatic and terrestrial environments are the two most contrasting. Organisms living in each of these environments show adaptations that are suited to their particular environment.
- **Structural Adaptations:** Terrestrial plants tend to grow tall and erect because of their supporting woody stems. The terrestrial plants have greater tissue differentiation, structure for water absorption and conduction and water-proofing coatings against drying up.
- Aquatic plants on the other hand are smaller, less differentiated and lack special absorbing and conducting tissues and prominent coatings to prevent dessication.
- Small petals and sepals and abundant pollen are adaptations for wind pollination in plants. Insect pollinated flowers have relatively large coloured petals and sepals, and nectary.
- In deserts, cactus survives because its leaves are reduced to spines and its photosynthetic and storage tissues are in the thin-skinned stem.
- Mammalian hair takes the form of quills in porcupine, horns in rhinoceros, flat plates (scale-like) in pangolins and wool in sheep.
- Beaks and feet of birds are suited to specific feeding habits. These structural adaptations change gradually as the environmental conditions change.

Physiological Adaptations

- Physiologically, many aquatic animals get rid of their nitrogenous wastes in the form of ammonia; birds and insects as solid uric acid and mammals as urea. These modes of excretion are linked up with conservation in their different habitats. Human beings tend to urinate more on a cold day than on a hot day.
- **Behavioural Adaptations:** These include those special actions or activities of the organisms which are peculiar for avoiding enemy by resembling other objects such as mimicry in butterflies, colour change in chameleon and adoption of night life (nocturnal) in some animals. Desert animals avoid dessication by being nocturnal.
- Different groups of organisms may develop similar structures as they adapt to similar ways of life in their history. This is known as *adaptive convergence* e.g. Tadpoles (amphibia) and fishes are adapted to life in water.

  The shark, which is a fish and the whale, which is a mammal swim in water.

  The spiny leafless *Euphorbia* of African desert also resemble cactus of mexican desert.

10.3 Tolerance

- Abiotic factors play an important role in the distribution of living organisms in the various terrestrial and aquatic habitats of the world. Some of these factors include temperature, rainfall, light intensity, salinity and edaphic factors. Each of these factors occurs over a range in the various habitats.
- Organisms can only live within certain minimum and maximum limits for each abiotic factor. The range between the upper and lower limits is known as the *tolerance range* for the factor.
- Tolerance ranges may differ for each abiotic factor. For example, an organism may have a broad range of tolerance for temperature and a narrow one for pH. They may change as the organism passes through various stages in its life. Usually the young organism has a narrower tolerance range than the adult. The minimum and maximum limits of tolerance for all abiotic factors may vary *geogra- phically* and *seasonally.* One factor may affect the tolerance ranges of other factors. For example, when the oxygen level of water is low, lobsters can only tolerate temperatures up to 20°C but at higher oxygen levels, they can tolerate temperatures up to 30°C.
- The distribution of most organisms over the earth follows definite patterns. In West Africa, for instance, oil palms grow in fairly distinct climatic belts or zones. Reptiles like snakes and lizards are found mainly in the tropical region. Similarly insects are more abundant in the tropical zones of the world. Organisms thus occur within roughly defined climatic belts called their *geographical range.*

10.4 Pollution of the Atmosphere

- Pollution can be defined as the addition of unwanted substance into the environment that can cause damage. The unwanted materials or energy that cause pollution or produce undesirable effect is called a *pollutant.* Most pollutants are caused by industrial or agricultural activities.
- Many pollutants accumulate in the environment because they are *non-biodegradable* i.e. they cannot be broken down into simple harmless substances.

  There are many types of pollution: air pollution, water pollution and pollution by pesticides.

- **(i) Air Pollution:** The major causes of air pollution are the gaseous and solid particle pollutants. The gaseous-pollutants are mainly oxides of carbon, i.e. carbon monoxide and carbon dioxide and the hydrocarbons. Oxides of nitrogen and sulphur compounds are gaseous pollutants. Most of the gaseous pollutants arise from incomplete burning of coal, oils and natural gas as fuels of motor vehicles and engines in factories and homes. The hydrocarbons come from evaporation of oils and paints.
- Smog which appears over cities and industrial areas arise from a combination of smoke, chemical pollutants and water droplets.
- Solid pollutants which include smoke, soot and dust are emitted by motor vehicles, industrial plants, electric power stations and incinerators.
- (ii) **Effect of air pollutants:** Carbon monoxide is a poisonous gas which combines readily with haemoglobin to form a stable compound and as such the blood carries less oxygen to the body cells. This causes dizziness, headache, nausea and tiredness.
- High concentrations of carbon dioxide in air may give rise to greenhouse effect.
- Sulphur dioxide is an acidic poisonous gas which causes irritation and damage to the sensitive lining of the eyes. Prolonged exposure to this pollutant may lead to lung diseases, cancer and heart attacks.
- Sulphur dioxide causes *acid rain* when it dissolves in rainwater, it can reduce the growth of vegetations and damage their leaves. Acid rain also dissolves metallic structures.
- Hydrocarbon pollutants can cause cancer.
- Solid pollutants emitted by motor vehicles cause the irritation of the nose, throat and air passages. Lead dust can cause anaemia and brain damage in children.
- Chlorofluorocarbons which are emitted from the coolants of refrigerators and air conditioners can accumulate in the upper atmosphere and react with ozone layer that prevents ultra violet radiation in sunlight from reaching earth surface, i.e. depletion of ozone layer.
- **(iii) Noise Pollution:** This is the unpleasant and irritating sound that are caused by motor vehicles, aeroplanes, machines, television, radios and human beings.
- The loudness of sound is measured in decibels (dB). Prolonged noise can cause headache or loss of hearing. Workers in factories are advised to wear ear-muffs to prevent loss of hearing.

10.5 Pollution of water and soil

- Water and soil are readily polluted by similar agents. Water and soil pollutants fall into the following categories:
  - (i) Organic remains of decaying plants and animals as well as food wastes, domestic sewage and animal wastes.
  - (ii) Non-living compounds, especially those containing dissolved metals such as lead, mercury and copper; industrial wastes, detergents, artificial fertilizers and hospital wastes.
  - (iii) Non-degradable organic materials such as oils, insecticides and herbicides.

Effects of Water and Soil Pollutants

- Organic wastes favour the spread of human diseases such as dysentery, typhoid fever, cholera, and other diseases of the gut.
- Inorganic remains such as fertilizers and detergents lead to excessive growth of algae and other water weeds on water bodies.
- Industrial wastes contain poisonous metals and chemicals which endanger the lives of many water and soil organisms.
- Crude oil pollution can lead to widespread death of many plants and animals.
- Pollution of water and soil can destroy food chains in the ecosystems.

SSCE/UME–Type Objective Questions

1. The relationship between a herbivore and the bacteria which live in its caecum is known as:
   A. parasitism B. symbiosis
   C. commensalism D. saprophytism
   E. predation

2. A constituent of the exhaust gases of motor vehicles which causes serious air pollution is
   A. water vapour B. carbon dioxide
   C. carbon monoxide D. oxygen
   E. ozone

3. When large numbers of organisms share limited space and resources the result is
   A. immigration B. commensalism
   C. symbiosis D. extinction
   E. competition

4. The change in colour of the chameleon serves as a means of
   A. attraction to the opposite sex
   B. repulsion of the enemy
   C. a camouflage from a predator
   D. regulation of body temperature
   E. production of a variety of skin pigments

5. An association between living organisms in which one lives and feeds at the expense of the other organism is known as
   A. parasitism B. commensalism
   C. mutualism D. symbiosis
   E. predation

6. Which of the following is not a way of reducing air pollution?
   A. siting factories and power stations away from residential areas
   B. building tall factory chimneys
   C. passing waste gases through filters and absorbers
   D. using lead-free petrol in cars
   E. recycling tins, cans and bottles

7. Natality is a factor affecting population density because it involves
   A. increase in population due to immigration
   B. increase in population due to emigration
   C. the number of mature females in the population
   D. the number of females in the population
   E. the number of all the young ones produced in the population

8. Which of the following may not contribute to environmental pollution?
   A. solar energy radiation
   B. industrial wastes
   C. waste disposal
   D. smoke
   E. nuclear power plant

9. Which of the following is a pollutant of air?
   A. oxygen B. water vapour
   C. carbon monoxide D. nitrogen
   E. hydrogen

10. Which of the following water pollutants may contain organisms that cause dysentery?
    A. Pesticides B. Sewage
    C. Industrial waste D. Fertilizers
    E. Crude oil

11. Which of the following does not illustrate adaptation to the environment?
    A. colour changes by chameleon
    B. streamline shape of fishes
    C. light bones in birds
    D. development of big muscles by a weight lifter
    E. possession of fins by fishes

12. The changing of colour by a chameleon to that of the environment is an example of
    A. adaptive radiation
    B. protective colouration
    C. courtship display
    D. display of body colour
    E. territorial behaviour

13. Which of the following is not a pollution control measure?
    A. Penalising those who dump refuse in water or land.
    B. Controlling the emission of smoke from automobiles and industries.
    C. Ensuring that there is no spillage of oil.
    D. Establishing government agencies like Federal Environmental Protection Agency (FEPA).
    E. Channelling sewage into streams.

14. An association between two organisms in which both benefit is called
    A. parasitism B. commensalism
    C. symbiosis D. predation
    E. competition

15. The association between protozoa and termites is an example of
    A. symbiosis
    B. parasitism
    C. predation
    D. saprophytism
    E. epiphytism

16. Which of the following associations can upset the balance of nature?
    A. Epiphytism
    B. Parasitism
    C. Symbiosis
    D. Saprophytism
    E. Scavengers

17. Which of the following examples best describes camouflagic colouration in organisms?
    A. A butterfly which has a red colouration on the wing.
    B. A green snake in a green grass.
    C. A zebra with vertical stripes on the body.
    D. Certain tree frogs with bright colour patterns.
    E. A flower whose petals and sepals are of the same colour.

18. In commensalism
    A. one of the organisms gains and the other loses
    B. both organisms benefit from the association
    C. one of the organisms gains while the other neither gains nor loses
    D. one of the organisms produces extra cellular digestive guices
    E. one of the organisms gains from the association

19. Parasitism can best be defined as an association between two organisms in which
    A. both organisms benefit
    B. one organism gains and the other neither gains nor loses
    C. both organisms suffer some disadvantages
    D. one organism leads to the death of the other
    E. both organisms neither benefit nor suffer any disadvantages.

20. Two organisms of different species, living in close association but not dependent on each other are referred to as
    A. parasites
    B. commensals
    C. symbiosis
    D. autotrophs
    E. saprophytes

21. Carbon monoxide poisons tissues by
    A. constricting the blood vessels
    B. killing the cells
    C. combining irreversibly with haemoglobin
    D. rapturing the blood vessels
    E. combining with tissue cells

Answer Key

1.	B	6.	E	10.	B	14.	C
2.	C	7.	E	11.	D	15.	A
3.	E	8.	C	12.	B	16.	B
4.	C	9.	C	13.	E	17.	C
5.	A					18.	C
						19.	D
						20.	B
						21.	C

Explanations to Objective Questions

1. **(B)** The relationship between a herbivore and the bacteria which live in its caecum is known as symbiosis.

2. **(C)** A constituent of the exhaust gases of motor vehicles which causes serious air pollution is carbon monoxide.

3. **(E)** Competition results when large numbers of organisms share limited space and resources.

4. **(C)** The change in colour of the chameleon serves as a means of camouflage from a predator.

5. **(A)** An association between living organisms in which one lives on and feeds at the expense of the other organism is known as parasitism.

6. **(E)** Recycling of tins, cans and bottles is not a way of reducing air pollution.

7. **(E)** Natality is a factor affecting population density because it involves the number of all the young ones produced in the population.

8. **(C)** Waste disposal does not contribute to environmental pollution.

9. **(C)** Carbon monoxide is a pollutant of the air.

10. **(B)** Sewage is a water pollutant that may contain organisms that cause dysentery.

11. **(D)** The development of big muscles by a weight lifter does not illustrate adaptation to the environment.

12. **(B)** The changing of colour by a chameleon to that of the environment is an example of protective colouration.

13. **(E)** Channelling sewage into streams is not a pollution control measure.

14. **(C)** An association between two organisms in which both benefit is called symbiosis.

15. **(A)** The association between protozoa and termites is an example of symbiosis.

16. **(B)** Parasitism is an association that can upset the balance of nature.

17. **(C)** A zebra with vertical stripes on the body best describes camouflagic colouration in organisms.

18. **(C)** In commensalism, one of the organisms gains while the other neither gains nor loses.

19. **(D)** Parasitism can best be defined as an association between two organisms in which one organism gains and the other loses.

20. **(B)** Two organisms of different species, living in close association but not dependent on which other are referred to as commensals.

21. **(C)** Carbon monoxide poisons tissues by combining with haemoglobin.

11 Habitats

11.1 Introduction

- **Habitat:** is a suitable place where an organism lives. Following are examples of major habitats: aquatic and terrestrial.

11.2 Aquatic Habitat (Water bodies)

- The aquatic habitat can be sub-divided into marine, estuarine and fresh habitats.

(i) Marine Habitat — Characterised by very high salinity

- The sea or ocean make up the marine habitat. It can be divided into *littoral* and *benthic* zones.
- The **littoral** zone is the region that extends over the continental shelf to a depth of about 200 metres. Beyond the littoral zone is the benthic zone which extends to a depth of 10,000 metres. Light can only penetrate the littoral zone while the benthic zone is dark and cold. The flora and fauna which make up the rocky shore comprise barnacles, oysters and mussels where they are exposed to severe wave action. However, anemones, sponges and seaweeds are found on the sheltered shores.
- The surface waters of the ocean support *plankton* and *nekton*.
- The *plankton* consists of small to microscopic organisms which float, drift passively or swim slowly in the surface waters. The *nektons* consist of actively moving or swimming animals like fishes, squids, crabs, prawns and whales. The fishes have streamlined muscular bodies. The bony fishes have gas-filled swim bladders. Examples of the tropical open sea fishes include shark, mackerel, sea catfish, bonga fish and croaker.
- The oceans have high concentration of mineral salts and greater uniformity of their physical factors and biotic contents. Oxygen is abundant in the littoral zone and the substratum could be rocky, sandy or muddy. Oxygen concentration is minimum at ocean bed and salinity is fairly uniform, averagely 35%.

(ii) Estuarine Habitat

- An estuary is an ecological zone where river and sea water meet, thus to establish brackish conditions. Brackish water has a salinity which fluctuates with the tides and the wet and dry seasons.
- Powerful currents in the estuaries prevent a number of organisms from settling down or establishing themselves successfully. Temperature changes are fairly small and last for short periods because there is constant inflow and outflow of water.
- The sediments of the estuary are soft and muddy. When excessive, sediments slow down filter-feeding in animals and photosynthesis in submerged plants.
- Estuarine organisms cope with fluctuating salinity in their habitat in the following ways.
- Organisms such as *Arenicola*, a burrowing worm, have cells which have a wide tolerance range for salinity.
- Some organisms have body surfaces that are fairly impermeable.
- Some organisms such as the mitten crab have osmoregulatory mechanisms that maintain the osmotic concentration of their body fluids slightly above that of the surrounding water.
- The estuarine habitat may have marshes, swamps, deltas and lagoons.
- The plankton in the estuarine ecosystem include producer plankton protists such as diatoms and flamentous algae, as well as consumer plankton such as copepods and larval forms.
- Mangrove forests are common in the swamps of Nigeria. The white mangrove has salt secreting glands in the upper epidermis of its leaves.
- The mudskipper is a typical fish found in lagoons and mangrove forests. It can swim like fishes and use its modified pectoral fins to walk on land and climb trees. Its gills and gill slits are modified for breathing in water as well as on land. Many animals such as the Nigerian lancelet flourish in the lagoons during the dry season and perish when the rains come.

(iii) Freshwater Habitats — Characterised by very low salinity

- Freshwater habitats are made up of two main types:
  - standing waterbodies such as ponds and lakes; and

— running waterbodies, such as streams and rivers.

- Both habitats have littoral and benthic zones. The organisms that make up the surface waters are producers and consumers while the deeper waters comprise consumers and decomposers, especially bacteria.
- Many streams, rivers, ponds and lakes dry up during the dry season. The animals that live in such habitats either move away during the dry season or produce resistant eggs that hatch out at the onset of the rains or exist as resistant cysts during the period of drought, e.g. the African lungfish exists in a cocoon during drought. It utilizes its lung for breathing air during drought and produces urea a waste product instead of ammonia produced during the wet period.
- Lakes harbour a wide variety of organisms such as plankton, nekton, green plants and decomposers. Some lake animals spend their larval stages in water while the adults live mainly on land. Some reptiles and birds visit lakes to feed on aquatic organisms.
- The plant community in lakes include rooted flowering plants like grasses and sedges, rooted flowering plants with submerged leaves like *Ceratophyllum,* and with floating leaves like water lily, *Nymphaea* and free-floating plants such as water lettuce, water hyacinth and water fern, *Salvinia.*
- Other plant community members include photosynthetic plankton, *Chlamydomonas,* blue-green algae and filamentous algae *Spirogyra.* The animal community of lakes include *cyclops,* rotifers, insect larvae, flatworms, watersnails, tadpoles, turtles, water snakes, fishes such as *Tilapia* and water birds.
- The river harbours the following organisms: snails, flatworms, fishes and prawns. Snails and flatworms use mucous secretions to cling to the substratum while insect larvae and oysters have suckers and hooks for attaching themselves to the substratum. Many fishes and prawn can jump up rapids and swim against the current.
- **Marshes and swamps:** These are wetlands with vegetation. The land is usually covered with shallow standing water or is waterlogged. The vegetation in marshes is mainly grasses while that in swamps is mainly trees.
- In freshwater marshes and swamps, the important abiotic factor that organisms have to adapt to include low oxygen level and a soft substration that offers poor support or anchorage.
- Marsh plants include grasses, sedges and weeds. Papyrus and rice are important tropical freshwater marsh plants.
- A variety of invertebrates such as worms, crustaceans, insects and molluscs live in marshes.

Swamp

- Swamp vegetation may be grouped as — tropical freshwater swamp forests, temperate freshwater swamp forests, and mangrove swamp forests found in tropical coastal and estuarine regions which receive high annual rainfall.
- The freshwater swamps comprise evergreen trees, ferns, shrubs, lichens and algae while the animals include insects, fishes, frogs, crocodiles, snakes, turtles, birds and certain land mammals.
- The mangrove swamps comprise mangrove trees such as white mangrove, *Avicennia;* red mangrove, *Rhizophora* and *Raphia.* The animal life in mangrove swamps include polychaete worms, oysters, mussels, barnacles, shrimps and mudskippers. Animals found among mangrove trees above water include insects, snakes and birds.
- Typical food chains in marshes and swamps wetlands are as follows:

  — Detritus → Worm → Bird
  — Plant → Insect → Frog → Alligator

  Economic products of wetlands include timber, peat, shellfish and fish.

11.3 Terrestrial Habitat

- Terrestrial habitats are characterised by their vegetation which is influenced by rainfall and temperature. Abundant rainfall and warm temperatures throughout the year produce luxuriant forests, while low rainfall and cold temperatures result in scrubland with poor vegetation.
- **Forests:** The dominant plant communities in a forest are trees and they require abundant rainfall.
- **Tropical Rain Forests:** The three important abiotic factors that are necessary for growth and reproduction of living organisms are temperature, rainfall and humidity.
- The five main vegetation layers in a rain forest are: ground layers of non-woody herbs, mosses, liverworts, lichens and fungi; a lower layer of shade loving dwarf trees, palms and shrubs; a middle layer of smaller trees with elongated narrow crown; an upper layer of tall trees whose crowns form a continuous canopy; and the emergent trees which come up above the canopy at intervals.
- The rain forest has many types of trees including timber trees like mahogany, *Kyaya ivoriensis,* abura, *Mitragyria;* cola and iroko, *Chlorophora.*
- The forest ground-dwellers in the rain forest include

invertebrates such as earthworms, termites, nematodes, soil vertebrates such as burrowing snakes, pouched rat and shrew.

- The forest arboreal dwellers include butterflies and grasshoppers, chameleons, snakes, birds, bats and monkeys.

Grasslands

- Grasslands lie between forests and deserts. In these habitats, the rainfall is not enough to support tree growth.
- Temperate grasslands experience low rainfall, a hot wet summer and a cold dry winter. Tropical grasslands experience a high temperature throughout the year and a distinct wet season and a dry season.
- The vegetation of grasslands has the ability to withstand long period of drought, to recover from fires and to recover after being eaten up by grazing herbivores.

Savanna

- The savanna experiences low and uneven distribution of rain and as such support the growth of tall, medium and short grasses and a few short trees.
- The plant community in Guinea Savannah include tall and thick grasses during the rainy season and trees such as mahogany, *Khaya senegalensis*; timber tree, *Afzelia africana*; locust bean, *Parkia;* shea butter, *Butyrospermum paradoxum;* African balsam, *Daniellia oliveri* and baobab.
- In the drier sudan savanna, the grass is shorter and the woody trees are fewer and smaller. These include *Combretum, Terminalia, Acacia* and baobab.
- In the dry sahel savanna bordering the sahara desert, the land is bare with clumps of short grass and a few isolated shrubs and trees such as *Euphorbia* and *Acacia.* Narrow forest zones occur along rivers in the savanna belt.
- The animal community include a wide variety of herbivores such as antelopes, gazelles, bush cows, warthogs, giraffes, zebras, rhinoceroses and elephants. There are also a few running and hopping mammals such as rats, squirrels, hares, porcupines and cutting grass. These animals are preyed upon by carnivores like lions, cheetah, leopards, lynx, hyenas and jackals.
- Other savanna mammals include monkeys, bats, wild cat, civets, hedgehogs, shrews and pangolins. The birds in the savanna include seed-eaters such as weavers, as well as ostrich, vultures and marabou. The reptiles include lizards, geckos, chameleons, skinks and snakes while the amphibians include the frog, *Rana galamensis* and the toad, *Bufo regularis.* The invertebrates include locusts, grasshoppers, flies, termites and ants. Other invertebrates include earthworms, spiders and scorpions. Following are examples of savanna food chains:

  — Grass → Zebra → Lion
  — Seed → Weaver bird → Civet
  — Leaves → Elephant → Vulture

Arid Lands

- Arid lands are the driest habitats, receiving less than 200 mm annual precipitation e.g. tropical deserts.
- Tropical deserts lie between latitudes 15° and 30° N and S of the equator with temperature ranges between 27 and 33°C in summer and about 15°C in winter. The absence of clouds in desert creates very hot days and very cold nights. Rainfall is scanty and the dry winds that blow erode the land.

Sahara Desert

- The sahara is the largest of the world's deserts with an area of over seven million square metres. It is a sandy desert. The abiotic factor in sahara are: low rainfall, high temperature, low relative humidity at noon which rises at night, and strong dry winds.

SSCE/UME–Type Objective Questions

1. Which of the following is not classified as a terrestrial habitat?
   A. Forest
   B. Guinea Savannah
   C. Afro Alpine
   D. Littoral zone
   E. Desert

2. Which of the following factors is not associated with aquatic habitat?
   A. Temperature B. Light intensity
   C. Humidity D. Turbidity
   E. Wave action

3. The number of individuals in a habitat in relation to the unit space available to each organism is referred to as the
   A. birth rate B. density
   C. mortality D. frequency
   E. dispersion

4. The following conditions form the basis of competition in a habitat except
   A. food B. space
   C. reproductive mate D. light
   E. body structure

5. Which of the following is not an aquatic habitat?
   A. Estuaries B. Ponds
   C. Rain forest D. Swamp
   E. Salt Lake

6. Which of the following factors does not affect the distribution of organisms in an aquatic habitat?
   A. Turbidity B. Depth of water
   C. Humidity D. Temperature
   E. Availability of nutrient

7. Which of the following organisms can be found in fresh water habitat?
   A. Sargassum and dog fish
   B. Red mangrove and pond skater
   C. Sargassum and red mangrove
   D. Spirogyra and pond skater
   E. Red mangrove and dog fish

8. Which of the following factors may not affect living organisms in an aquatic habitat?
   A. Turbidity B. Temperature
   C. Humidity D. Light
   E. pH

9. Which of the following instruments is not used in measuring abiotic factors in any habitat?
   A. Microscope B. Thermometer
   C. Hydrometer D. Wind vane
   E. pH meter

10. What ecological condition favours the breeding of blackflies?
   A. Fresh water habitat
   B. Water in ponds and swamps
   C. Water in small containers
   D. Fast flowing streams
   E. Stagnant lakes

11. The most important physical factors which affects an organism living in the inter-tidal zone of the seashore is
   A. pH B. salinity
   C. wave action D. temperature
   E. turbidity

12. Important abiotic factors which affect all plants and animals in the habitat are
   A. temperature and turbidity
   B. rainfall and relative humidity
   C. salinity and wind direction
   D. temperature and rainfall
   E. salinity and relative humidity

Answer Key

1.	D	4.	E	7.	D	10.	D
2.	C	5.	C	8.	C	11.	C
3.	B	6.	C	9.	A	12.	D

Explanations to Objective Questions

1. **(D)** Littoral zone is a region of the sea or ocean that extends over the continental shelf while the others are terrestrial habitats.

2. **(C)** Humidity is the amount of water vapour in air while the others are associated with aquatic habitats.

3. **(B)** Density is the amount of individuals in a unit space.

4. **(E)** Body structure is fixed and does not form a basis for competition in a habitat.

5. **(C)** Rain forest is a terrestrial habitat while the others are aquatic habitats.

6. **(C)** Humidity is the amount of water vapour in air and does not affect the distribution of organisms in an aquatic habitat.

7. **(D)** Spirogyra and pond skater are found in fresh water habitats while the others can be found in marine habitats.

8. **(C)** Humidity is the amount of water vapour in air and does not affect living organisms in an aquatic habitat.

9. **(A)** Microscope cannot be used to measure abiotic factors in any habitat while the others are relevant.

10. **(D)** Fast flowing streams which are highly oxygenated favour the breeding of blackflies while the others are not hgihly oxygenated.

11. **(C)** Wave action is an important factor that affects an organism living in the inter-tidal zone of the seashore while the others are not effective.

12. **(D)** Temperature and rainfall are abiotic factors which can affect all plants and animals in the habitat while the others are not effective.

12 Ecology of Populations

12.1 Ecological Succession

- Succession refers to the series of gradual changes being carried out by plant and animal communities in an area, in which, with time, lower organisms are replaced by higher ones.
- Soil formation is an important process in a new habitat such as a bare rocky area. Lichens and mosses are the first organisms to grow and colonize the bare rock. Thus lichens and mosses are known as *pioneer organisms*. Their presence together with abiotic factors like rain, diurnal temperature changes and wind crumble the rock surface leading to soil formation. This process takes thousands of years. This succession which starts from primary bare rock and ends up in the formation of a stable forest is called *primary succession*. The final or stable stage ever attained through the series of changes into a forest is called the *climax*.
- Succession begins as soon as land area such as bare rock is first colonized by lichens and mosses, then followed by annual grasses and herbs; and then are succeeded by biennial and perennial grasses and herbs which are succeeded by shrubby growth and later hardy shrubs and fast growing trees and finally by forest trees.
- A pond is similarly colonized by submerged rooted aquatic plants, then followed by floating and rooted aquatic plants and then by marsh or wet soil with partially submerged plants like sedges and reeds and later dry soil with shrubby growth and finally succeeded by forest.
- Ecologists recognize two types of succession:
  - primary succession, and
  - secondary succession
- Primary succession occurs in an area where no community existed before and involves the formation of a new soil.
- Secondary succession occurs when an existing community on a terrestrial habitat is disturbed and leaving some soil and probably some certain organisms. The secondary succession can be caused by fires or droughts or floods.
- While primary succession takes thousands of years to occur, secondary succession is much faster because of the existing soil and the surrounding and suppressed plants that can recolonize the area.

12.2 Population and Population Density

- A population is the total number of individuals of a given species in a habitat at a given time.
- Population density is the average number of individuals of a species per unit area of the habitat.

Population density of people in a state

$$= \frac{\text{Total number of people}}{\text{Area of the state}}$$

Factors Affecting a Population

- The following factors cause the population in a given area to increase or decrease:
- **Natality:** Production of young which increases population by adding new individuals to the population.
- **Mortality:** Death which decreases population by removing individuals from a population.
- **Immigration:** increase in population as individuals from other areas move permanently into the area.
- **Emigration:** decrease in population as individuals move out permanently from the area to other areas.

Overcrowding

- Overcrowding occurs when a population in a given habitat increases beyond a point where the resources in the habitat are not enough to support all the individuals in the population.
- Overcrowding may be caused by either or both of the following reasons:
  - (i) Failure of organisms to move out or disperse from their places of birth, or
  - (ii) Addition of individuals of the same species (immigration) to an existing population.

Factors that cause Overcrowding

- Population increases beyond the carrying capacity of a habitat cause overcrowding because the resources in the habitat, such as food and space, are

not enough to support all the individuals in the population.

- Natality and immigration are two important factors that add individuals to a population and so lead to overcrowding. High natality causes a population to increase in size rapidly.
- High survival rate causes an increase in population size and as such most individuals will reach sexual maturity and reproduce before they die, thus causing an increase in population size.
- A young population with a high natality rate and high survival rate is therefore capable of increasing in size rapidly. Such a population will invariably overshoot the carrying capacity of the habitat, thus, overcrowding it.

12.3 Food Shortage

- When organisms become overcrowded, living space and food are the first essential requirements that become inadequate.
- In any habitat, producers provide the food that supports all consumer species, either directly or indirectly. The population sizes of various species are adjusted to the amount of food that is available in the habitat. Certain environmental factors can decrease the food supplies of some of these populations and cause food shortage.

Causes of Food Shortage

- Plants which are the main source of food in many habitats may be affected drastically by drought. Drought causes water scarcity and the death of plants and eventually death of primary consumers when subjected to food shortage.
- Death or emigration of primary consumers cause food shortage for secondary consumers and eventually death of consumers at the various trophic levels.
- Disease can bring about food shortage by killing off large numbers of herbivores and thus food shortage for consumers that prey on the herbivores.

Effects of Food Shortage

- Food shortage causes population sizes to decrease. The decrease is brought about by several factors such as:

  (i) **Competition:** There is competition for available food during food shortage and as such only the vigorous individuals in the population succeed in getting food while the others starve or ultimately die.

  (ii) **Emigration:** As a result of food shortage, members of the population move away or emigrate to areas where they can find food.

  (iii) **Rate of reproduction:** Food shortage reduces the rate of reproduction and delays sexual maturity in animal populations. Growth is retarded and this delays sexual maturity. Plants are also affected by food shortage by a reduction in the rate of reproduction.

  (iv) **Mortality:** Food shortage increases the rate of death in animal populations especially among the younger age groups.

Balance in Nature

- In every ecosystem, all the organisms in the population grow and become stabilized in a new habitat. The populations increase or decrease according to the overall effects of some factors. Such a population is said to be in a balance.
- The factors that affect the populations and contribute to the balance in nature can be divided into abiotic and biotic factors.

Factors Affecting a Population

- Abiotic factors include temperature, water, light, oxygen, food, soil and availability of space.
- Biotic factors include predation, competition, parasitism, diseases, natality, mortality, territorial behaviour and dispersal.
- Some of these factors are density-dependent and some are density-independent.

12.4 Response of Population to Factors Affecting Them

- Some abiotic factors affect the growth of a population irrespective of its population density. Such factors include flooding which may kill large number of individuals, and many catastrophic events like bush burning, droughts and storms which often reduce population sizes drastically.
- Many biotic factors that affect population growth, increase or decrease in intensity as the population density changes. Such biotic factors include availability of food which decreases as the population density increases.
- Seasonal fluctuations affect population densities. Plant population increases during the rainy season in the savanna region and this results in a corresponding increase in population densities of the herbivores that feed on them. However, the plant population decreases during the dry season and as such a decline in the population density of the herbivore.

- Predator-prey fluctuations affect the population size. When predators eat prey, the population size of the prey reduces. This leads to increased competition among the predators for the remaining prey.

12.5 Dynamic Equilibrium in Ecosystem

- The biosphere is composed of many ecosystems. The biotic community in each ecosystem is composed of populations of many species.
- In an ecosystem with a stable climax community, the population sizes of the various species are constant. The various populations in a climax community are said to be in balance or in *dynamic equilibrium.*
- In an ecosystem, dynamic equilibrium is maintained by density-dependent and density-independent factors that increase or decrease the reproductive rate, death rate, emigration, and immigration of each population in the community.

Family Planning

- Population growth especially in developing countries has to be controlled because the earth will not be able to support the additional growth of human population with time. Unless birth is limited, human misery is likely to increase with increased population but diminishing food supply. This undue increase in human population beyond food production can be prevented by practicing birth control and family planning.
- In Nigeria, the family planning council has established clinics in all parts of the country to advise people on the advantages of planning their family by:
  - keeping births spaced out, and
  - having a few children whom they can look after and support properly.

The techniques of family planning are as follows:

(i) **Rhythm Method:** In this situation, partners go into sexual act only during certain periods in the menstrual cycle of the female when she is most unlikely to have a ripe and receptive ovum in her tract.

(ii) **Coitus Interruption:** This is the method whereby the male withdraws his organ from that of the female just before release of spermatozoa during intercourse.

(iii) **Use of Condom:** This is the method whereby the male wears a protective rubber device on his organ, to trap spermatozoa released during sexual act, instead of allowing sperm to get into the female's body.

(iv) **Diaphragm:** This is the method whereby the female wears a protective rubber device at the lower opening of the womb. The diaphragm stops spermatozoa released into the vagina, from entering the upper parts of the female tract during mating.

(v) **Pills:** These are hormone preparations which females usually take by mouth in order to prevent fertilization in their ova.

(vi) **Intra-uterine device (IUD):** These are metal or plastic objects designed in many forms and placed in the womb to prevent pregnancy.

(vii) **Sterilization:** This is done by tying or ligating the sperm ducts or the oviducts in the man and woman respectively.

SSCE/UME–Type Objective Questions

1. Which of the following is not an effect of over-crowding on a population?
   A. Death B. Competition
   C. Commensalism D. Aggression
   E. Cannibalism.

2. A climax community is characterised by
   A. a stable composition of species.
   B. rapid changes in the composition of species
   C. constant changes in appearance of the habitat
   D. different species occurring at different times
   E. gradual change in animal population.

3. Which of the following is not a characteristic of overcrowding in plant and animal community?
   A. Population outstripping available space.
   B. Population exceeding available food.
   C. Competition within the population.
   D. Increase in primary production.
   E. Population increasing at the same rate as the birth rate.

4. Which of the following factors does not control population growth?
   A. food shortage B. emigration
   C. abundance of food D. predation
   E. natural disaster

5. Which of the following is the least safe method of birth control? Using:
   A. spermicide cream
   B. contraceptile pills
   C. barriers such as condoms
   D. intrauterine device
   E. the withdrawal method

6. Which of the following statement is not correct?
   A. Shortage of food may be caused by pestilence, rinderpest and neglect of farming.
   B. Shortage of food may lead to competition, emigration, death and decline in the rate of reproduction.
   C. Overcrowding may lead to scarcity of food.
   D. Dispersal of seeds or fruits prevent over-crowding.
   E. Territorial defence encourages food scarcity in a habitat.

7. Which of the following is not an outcome of ecological succession?
   A. There is a progressive change in species, structure, organic matter and energy flow.
   B. The climax stage consists of a comparatively long lasting species composition.
   C. There is no dynamic equilibrium between the community and habitat in the climax stage.
   D. Communities occurring later in the sequence of succession are more complex.
   E. There are fewer dominant species in the late stages of succession.

8. Which of the these organisms will have the lowest population in an ecosystem?
   A. Hawk B. Praying Mantis
   C. Mouse D. Grasshopper
   E. Bees

9. Density, mortality rate, birth rate are the factors that affect
   A. population B. dominance
   C. cover D. habitat
   E. abiotic community

10. Which of the following statements is not true of a population?
    A. Members of a population may be of the same species.
    B. Members of a population may be of the same species but do not interbreed.
    C. Size of a population may be limited by space.
    D. Some species can form another population elsewhere.
    E. Members of a population must co-exist.

11. Which of the following is an abiotic which affects a population?
    A. Predator B. Parasite
    C. Consumer D. Producer
    E. Temperature

12. Which of the following is an abiotic factor?
    A. Predation B. Rainfall
    C. Cropping D. Grazing
    E. Migration

13. Which of the following factors may not cause over-crowding?
    A. Limited space
    B. Scarcity of food
    C. Reduced birth rate

D. Loose immigration
E. Increased birth rate

14. Which of the following habitats cannot be used for the study of succession?
A. Savanna grassland
B. Abandoned farmland
C. A pond
D. Well cultivated land
E. Lawn

15. Which of the following statements is not true of a climax vegetation? It:
A. is an ecological phenomenon.
B. is a stable community.
C. eliminates competition.
D. results from succession
E. involves the colonization of a habitat

16. Which group of plants would be the first colonizers in an ecological succession changing rock to soil?
A. Mosses
B. Ferns
C. Lichens
D. Grasses
E. Shrubs

17. Which of the following **cannot** be associated with over-crowding?
A. Epidemic
B. Competition
C. Food shortage
D. Succession
E. Cannibalism

Answer Key

1.	C	5.	E	9.	A	13.	C	17.	D
2.	A	6.	E	10.	B	14.	D		
3.	D	7.	A	11.	E	15.	C		
4.	C	8.	A	12.	B	16.	C		

Explanations to Objective Questions

1. (C) Commensalism does not result from overcrowding on a population.

2. (A) A climax community is characterised By a stable composition of plants and animal species.

3. (D) Increase in primary production cannot arise from overcrowding in plant and animal community.

4. (C) Abundance does not control population growth.

5. (E) The withdrawal method is the least safe method of birth control.

6. (E) Territorial defence does not encourage food scarcity in a habitat.

7. (A) The progressive change in species structure, organic matter and energy flow is not an outcome of ecological succession.

8. (A) Hawk is a carnivore which is a tertiary consumer in the list provided.

9. (A) Density, mortality rate, birth rate are the factors that affect population.

10. (B) Members of a population may be of the same species and have to interbreed.

11. (E) Temperature is an abiotic factor which affects a population.

12. (B) Rainfall is an abiotic factor.

13. (C) Reduced birth rate may not cause over-crowding.

14. (D) A well cultivated land cannot be used for the study of succession.

15. (C) A climax vegetation eliminates competition.

16. (C) Lichens are the first group of plants that would colonize an ecological succession changing from rocks to soil.

17. (D) Succession cannot be associated with over-crowding.

13 Relevance of Biology to Agriculture

13.1 Introduction

- Biology is the study of living things. Agriculture is the growing of plants and the rearing of farm animals to provide our needs such as food. A basic knowledge of biology is necessary to practice agricultural activities. Such knowledge includes the following:
  - (i) the structure, function and nutrition of plants and animals;
  - (ii) diseases which affect plants and animals, and the organisms which cause them;
  - (iii) ecological systems;
  - (iv) soil; and
  - (v) genetics.

13.2 Classification of Plants

- Plants can be classified botanically according to their structure, presence of flowers and their mode of reproduction. Based on these criteria they are grouped into: non-flowering plants (lower plants); and flowering plants (higher plants).
- The following plants known as spermatophytes or seed plants are further classified into:
  - (i) Angiosperms with seeds enclosed in protective structures.
  - (ii) Gymnosperms with unprotected seeds.
- Flowering plants have four main organs: root, stem, leaf and flower. The flower in most cases is reproductive and forms the seeds.
- The two groups of angiosperms are characterised as follows:
- (a) **Dicotyledons:**
  - (i) Possess two seed leaves or cotyledons.
  - (ii) Leaves are net-veined or reticulate.
  - (iii) Flower parts in fours or fives or multiples of fours or fives.
  - (iv) Have tap root system.
- (b) **Monocotyledons:**
  - (i) Possess single seed leaf or cotyledon.
  - (ii) Leaves are mostly parallel — veined.
  - (iii) Flower parts are in threes or multiples of threes.
  - (iv) Have fibrous root systems.

The non-flowering or seedless plants are classified into three main groups:

- (a) **Thallophyta:** They lack roots, stems, leaves or flowers. This group consists of the algae, the fungi and bacteria.
- (b) **Bryophyta:**
  - (i) They have stem-like and leaf-like structures but lack roots or flowers.
  - (ii) They reproduce by spores and absorb nutrients from the soil by means of rhizoids, e.g. mosses.
- (c) **Pteridophyta:**
  - (i) They possess roots, stems and leaves but lack flowers.
  - (ii) They reproduce by spores.
  - (iii) They have primitive vascular system, e.g. ferns like *Dryopteris* and *Nephrolepis*.

13.3 Agricultural Classification of Plants

- Plants are classified agriculturally according to the products for which they are grown.
- (i) **Cereals:** These plants belong to the grass family. They include maize, rice, millet, guinea corn, and wheat; these are grown mainly for food.
- (ii) **Legumes:** These are important sources of dietary protein and they also supply protein-rich fodder for livestock. The legumes replenish soil nitrogen because nodules on their roots contain nitrogen-fixing bacteria and as such are important in crop rotation. The legumes include bean, groundnut, cowpea, soya bean and lima bean.
- (iii) **Tubers:** These are grown mainly for starch and form staple food. They include cassava, yam and sweet potato.
- (iv) **Vegetables:** These are grown to supply dietary vitamins and minerals, e.g. tomato, okra, onion and pepper.
- (v) **Fruits:** These are plants grown for their fruits which are rich sources of vitamins, minerals and sugars e.g. orange, pineapple, banana, plantain, mango, pawpaw and native pear.

- (vi) **Beverages:** The crops which yield these products include cocoa, coffee.
- (vii) **Drugs:** These include tobacco and quinine.
- (viii) **Spices:** These include pepper, ginger, cinnamon and cloves.
- (ix) **Oils:** The fruits and seeds of certain plants are rich in oil. Such crops include oil palm, shea butter, coconut and sunflower.
- (x) **Forage Crops:** These are grasses and legumes that are grown for animal food.
- (xi) **Latex:** When a cut is made on the trunk of certain trees, a milky fluid called latex flows out. The rubber tree is grown for its latex that is used to make natural rubber.
- (xii) **Fibres:** Plants such as jute, hemp, and cotton are grown for their fibres which are used for making ropes and cloth.
- (xiii) **Economic Trees:** The following trees are grown for economic purposes:
  (a) Timber producing trees, e.g. iroko (*Chlorophora excelsa*) *Ceiba,* and *Tripilochiton*
  (b) Wood pulp used for paper e.g. *Gmelina arborea.*
- (xiv) **Forest regeneration Trees:** The following trees are grown because of their ability to stabilize the soil through rapid growth, e.g. *Cassia,* Dogonyaro (*Azadirachia indica*) and Agunmaniye (*Gilsicidia*).

13.4 Classification based on Life Cycle

- Flowering plants can be classified into herbs, shrubs and trees.
- (i) **Herbs:** These are plants that do not grow tall and are also not woody. Herbs can be further classified based on their life cycle into annuals, biennials, and perennials:
  (a) **Annuals:** These herbs complete their life cycle in one year e.g. maize and yam. Some annuals such as goat weed, *Ageratum* have very short life cycle of only a few weeks.
  (b) **Biennials:** These are herbaceous plants that complete their life cycle in two years e.g. onion which uses the first year to form bulb and the second year to produce flowers, seeds and eventually die.
  (c) **Perennials:** There are herbaceous plants that live for many years or have many growing seasons in which the life cycle is completed, e.g. ginger.
- (ii) **Shrubs:** These are plants which are larger than herbs but are smaller than trees. They have woody stems which form many branches close to the ground. Shrubs are perennials, e.g. *Allamanda, Hibiscus.*
- (iii) **Trees:** These are plants taller than shrubs. They are tall woody plants and they are perennials.

13.5 Effect of Agricultural Activities on Ecological System

- Agricultural activity or farming involves both the growing of crops as well as rearing of livestock such as poultry, cattle, goats and pigs.
- (i) **Bush clearing:** Bush cleaning exposes the soil and its rich organic matter and subjects it to leaching. The exposed soil is also subjected to erosion and desiccation.
- (ii) **Bush Burning:** It destroys the soil organic matter and useful organisms. Soil fertility is lost but water holding capacity of the soil improves. The crumb structure of the soil improves and food is provided for beneficial soil micro-organisms.
- (iii) **Tillage:** It increases the water penetrating capacity of the soil. It mixes the soil and aerates it. Tillage can also introduce weed into the ecosystem.
- (iv) **Fertilizer:** Fertilizers maintain soil fertility or correct shortages in mineral salts. The undesirable side effects of fertilisers include the washing of excess phosphates into ponds and lakes which may lead to the development of algal blooms that may pollute the water.
- **Excess nitrogen and phosphates** from fertilizers may deplete oxygen contents of bodies of water and the fishes and other aquatic organisms in them may get suffocated.
- **Herbicides** are used to remove unwanted weeds from the farms. When herbicides are non-selective, both wanted and unwanted plants are eliminated. However the herbicides can accumulate toxic compounds in soil and water when washed down by running water thereby constituting pollutants to the biosphere.

13.6 Different types of farming

- **Shifting Cultivation:** This involves the cultivation of a land and then abandoned for another place so as to allow the used land to recover its lost fertility. This type of farming is only possible where the population is small and land is abundant. Over-use of soil can lead to total loss of soil fertility.

- **Crop Rotation:** This is a method of farming adopted to help maintain soil fertility while cropping repeatedly. This involves dividing a piece of farm land into plots, each being planted with a particular type of crop since different crops make different demands on the different elements of the soil. Different crops are rotated in the plots each growing season with one of the plots being left fallow or planted with leguminous cover crops. This method ensures better growth for the respective crops.
- **Mixed Farming:** This type of farming combines livestock rearing with crop production. This practice involve the use of farmyard manure to enrich the soil, while the crop by-products such as straw, groundnut haulms, and cowpea and soya bean top serve as livestock feed.

13.7 Pests and Diseases of Agricultural Importance

- Pests are organisms that cause considerable damage to man, his domestic animals and crops in the course of living their normal lives. The common pests of agriculture include fungi, nematodes, insects, some mammals (especially rodents) and birds.
- Disease conditions render crops, man and his domestic animals weak and insufficient in their activities. Diseases of crops and animals are caused mainly by micro-organisms and a few parasitic worms. Disease causing organisms include bacteria, viruses and some fungi.

Pests of Crops

- Pests attack crops and decrease their production. The pests may be plants or animals. Plant pests are known as weeds. The most important animal pests are insects. Other animal pests include mites, nematodes, snails, birds and rodents, ticks. The following are some crop pests:

- **Stem Borers:** Larvae of certain moths bore the stems of cereal crops such as maize. These larvae feed on the stem and cause stem weakness. The stem breaks even in slight wind.
- **Root Feeders:** Insect larvae or adults feed on roots of plants like yam. The mature yam beetles breed and lay egg in damp places.
- **Leaf Feeders:** Leaves of crops are eaten by snails, leaf-feeding beetles, caterpillars of various moths and butterflies, grasshoppers and locusts.
- **Young shoot feeders:** Insects like aphids, mealy bugs and scale insects pierce and suck juices from young shoots of crop plants.
- **Fruit and seed feeders:** These include fruit-piercing moths, fruit flies, cotton stainers and certain beetles.

Pest Control

- This is aimed at reducing the pest population and protecting crop plants. Pest control can be effectively carried out with a knowledge of the pest's habits and life cycles. Following are some control measures.
- **Physical methods:** involve the collection of pests by using traps and then destroying them. Snails and caterpillars are often picked, while light traps and sticky traps are used to trap adult insects. Other methods include burning, flooding and using deterrents such as bird scares to keep away birds.
- **Cultural methods:** involve the modification of the farming practice so that environmental factors do not favour the development of the pest.
- **Crop rotation,** closed-season practice, fallowing and changing the time of planting make sure that susceptible crops are not around during the development of the pest so that the pest starves to death. Cleaning or burning fields destroys the pests. Tilling land brings soil pests to the surface and kills them.
- **Biological methods:** Natural enemies of pests are used to destroy pests. Such enemies may be predators, parasites or diseases of pests.
- **Chemical methods:** Chemicals such as pesticides are used to kill or reduce the pest population, or to protect the crop from attack by pests. e.g. yam beetles are controlled by dusting yam setts with Aldrin 2.5% dust or Gammalin A at planting.
- **Modern trends in pest control:** In the sterile male technique, male insect pests are sterilized by chemicals or radiation, and then released into the pest population. This reduces the pest population.

Diseases of Crops

- Most diseases of crops are caused by fungi, bacteria and viruses.
- **Fungal diseases:** Most disease-causing fungi are microscopic and parasitic smut, rust, blight, root-rot and mildew, e.g. black pod disease of cocoa and rust of maize.
- **Viral diseases:** Viruses are parasites. Mosaic mottling causes dark and light patches on leaves like cassava, lesions, chlorosis, stunting and necrosis. Many plant virus infections are spread by insect vectors e.g. mosaic-causing virus is carried by aphids.
- **Bacterial diseases:** These include bacteria blight of

cassava, bacteria wilt of tomato, leaf spots, bacterial galls and soft rot.

Control of Diseases

- This is done by preventing crops from becoming infected by disease-causing agents and destroying the disease-causing agents and the vectors that transmit them.
- **Physical methods:** These include the removal of diseased plants, removal of weed hosts and bush burning and soil flooding.
- **Cultural method:** Modification of unusual farming practices to destroy disease-causing agent. Planting resistant varieties also helps to control the disease. *Chemical methods:* Chemicals are used to eradicate disease — causing agents or protect the crops from such agents.

Pests and Parasites of Farm Animals

- The common farm animals are poultry, swine, horses, rabbits, goats, sheep and cattle.
- Most pests of farm animals are external parasites like lice, fleas, bugs, tsetse-flies, ticks and mites. They have biting or piercing and sucking mouth parts. These allow them to feed mainly on the blood of the farm animal. Their bites make the farm animals irritable and nervous leading to low productivity e.g. low milk production in cows and low meat production in fowls.
- Internal parasites such as flukes, trypanosomes, tapeworms lower the resistance of farm animals, making them unhealthy or eventually killing them. Some diseases of animals are:
- **Foot rot** of sheep is a fungal disease spread through contact of the animals' foot with contaminated marshy pastures. It is controlled by dropping the feet of the sheep into dilute copper sulphate solution.
- **Fowl typhoid** is spread by bacteria in animals coming into contact with faeces of infected birds. It can be controlled by removing infected birds or innoculation of the birds when about five weeks old.
- **Anthrax of cattle** is also a bacterial infection spread by ingestion of contaminated food or water. It can be controlled by vaccination and disinfection of the immediate environment.
- **Fowl pox** is a viral infection spread by mosquitoes causing scabs all over the head, eyes, nostrils and comb with nasal discharge and tear production. This can be controlled by removing the infected birds when about 1½ weeks old.
- **New castle disease of chickens** is also a viral disease spread by contact with contaminated water, food or faeces — causing respiratory and nervous disorders. It can be controlled by proper sanitation of the immediate environment, disinfection of poultry equipment, removal of infected birds and vaccination of birds twice at one week and five and half weeks old.

13.8. Food Production and Storage

- **Food production:** Food for human beings comes from both crops and livestock. Food production needs arable land. It is increased by improving agricultural yields.

  Food production can be met by considering the following:
- **Land for cultivation:** Available land should be preserved for cultivation and not for other purpose.
  - The soil fertility of the land under cultivation should be maintained by manuring, adding artificial fertilizer, practicing crop rotation and preventing soil erosion.
  - Land not fit for crop cultivation should be converted into fertile land.
  - Unproductive desert land can be converted to fertile fields by irrigation, hill and mountain slopes can be used for cultivation by terracing.
- Improvement of agricultural yields can be achieved as follows:
  - Maintaining the fertility of the land by ensuring high level of crops production through the use of organic manure, artificial fertilizers and rotation of crops.
  - Use of pesticides, vaccines and other protective chemicals to improve crop yields and livestock productivity.
  - Improvement of crop and livestock breeds through selective breeding and production of new strains of crop plants and highly productive livestock breeds.
  - Mechanization in order to obtain high yields from large farms.
- **Preserving and storing food:** These can be carried out in the following ways:
  - Drying of cereals by spreading under the sun.
  - Low-temperature preservation methods involve refrigeration and freezing of fresh fruits, vegetables and seeds.
  - Refrigeration of meat and fish
  - Freezing of food as to keep them for long periods.

— Freeze – drying of food to obtain high quality dried foods.
— Canning of food through sophisticated process.
— Pasteurization of milk by heating so as to destroy micro-organisms and enhance keeping for a longer time.
— Salting of meat diminishes bacterial activity and helps in preservation.
— Smoking of fish and meat helps to destroy bacteria and assist in preservation.
— Chemical preservation so as to preserve and enhance the flavour of food.
— Irradiation of micro-organisms and parasites in food to prolong its keeping quality.
— Fermentation of fruits and vegetables so as to improve the flavour.

Causes of Wastage

- The causes of wastage of food are as follows:
  - Poor harvesting methods which create wounds on farm produce.
  - High temperatures and relative humidities of the tropical region which hasten spoilage by micro-organisms.
  - Premature harvesting of crops which give low-quality products.
  - Poor handling and inefficient transportation system for harvested food.
  - Wastage arising from processing
  - Poor storage conditions which expose crops to pests and spoilage organisms.

Preventing Wastage of Food

- This can be prevented through the following methods:
  - Good arable land can be prevented from wastage through proper farm management and good farming practices.
  - Well-planned irrigation with anti-pollution measures can help to prevent water wastage and reduce pollution.
  - Proper education on the use of farmyard manure and artificial fertilizers enhance crop yields.
  - Careful farm management prevents loss of crops to weeds, pests and diseases.
  - Losses due to disease in livestock rearing can be prevented by good nutrition, good sanitation, and proper use of vaccines and prophylactic medicines.
  - Careless harvesting of root crops and vegetables causes damage and can be prevented by improved harvesting methods.
  - Proper storage methods can help to prevent food wastage.

Population Growth and Food Supply

- Population growth is very rapid. The world's population growth is estimated at an average of 1.7% but in Nigeria, growth is faster at an estimate of 3.2% per year.
- The increase in human population creates problems such as decrease in land available for farming and increase in food demand in a situation of diminishing food supply.

Factors that affect availability of food

- The major factor that affects availability of food is food production.
- Rapid increase in population which leads to increase in food demand.
- Diminishing agricultural lands.
- Low level of agricultural mechanization
- Neglect of agriculture as a profession by youths.
- Lack of incentives for farmers.

Relationship between availability of food and human population

- The availability of food supply cannot support the human population at a certain level regarded as its carrying capacity. The factors operating to keep the population of an area at a level commensurate with the available food are as follows:
  - Competition for the available food
  - Diseases resulting from under-nutrition and malnutrition
  - Famine and starvation
  - Mortality.

SSCE/UME–Type Objective Questions

1. Which of the following statements is not correct about the ecological effects of agricultural activities.
   A. The use of herbicides and pesticides on farms can cause pollution.
   B. Bush-burning encourages dessertification.
   C. Excessive use of fertilizers in farms may have harmful effects.
   D. Tillage of tropical soils with ploughs and harrows may encourage erosion.
   E. Monocropping is one of the cultural methods of controlling pests.

2. Pests can be controlled without the subsequent danger of polluting the ecosystem by
   A. introducing a population of species similar to that of the pest.
   B. increasing the population of the natural enemies of the pest.
   C. eliminating the natural enemies of the pest.
   D. keeping the population of natural enemies of the pest constant.
   E. spraying the ecosystem with pesticide.

3. Predators are beneficial in pest control because they
   A. are natural enemies of small mammals
   B. devour farm animals
   C. feed on pest of crops
   D. prey on game animals
   E. compete for food with the pests

4. Which of the following bacteria reduces nitrates in the soil to gaseous nitrogen?
   A. putrefying bacteria
   B. nutrifying bacteria
   C. saprophytic bacteria
   D. denitrifying bacteria
   E. parasitic bacteria

5. The burning of farmlands should be discouraged because it
   A. destroys some plant pests
   B. reduces the population of wild animals on the farm
   C. makes bush clearing more difficult
   D. destroys the organic part of the soil
   E. reduces the dormancy period of some weeds

6. Which of the following organisms reduces nitrates in the soil to gaseous nitrogen?
   A. Euglena
   B. Protozoan
   C. Denitrifying bacteria
   D. Parasitic mould
   E. NItrifying bacterium

7. The use of predators or parasites to control pests in the farm is known as
   A. predator control
   B. chemical control
   C. biological control
   D. animal control
   E. parasitic control

8. Which of the following is the underlining principle in the adoption of biological control of pests?
   A. Knowledge of agricultural practices by the farmer.
   B. Relationship between plants and animals.
   C. The presence of poisonous chemicals in the farm.
   D. The relative population of plants and animals.
   E. The predator-prey relationship in the ecological community.

9. Which of the following plant diseases is indicated when grains of cereals are covered with a mass of spores?
   A. smut
   B. mosaic
   C. maize streak
   D. fungal blast
   E. rust

10. The farming practice by which an exhausted farm land is left for a number of years before cultivation is known as
   A. crop rotation
   B. continuous cropping
   C. monocropping
   D. bush fallowing
   E. mixed cropping

11. Which of the following farm practices can cause loss of soil fertility.
   A. mulching
   B. compost application
   C. use of fertilizers
   D. terracing
   E. intensive cropping

12. The importance of practising crop-rotation in agriculture is to
   A. control soil erosion
   B. maintain the fertility of the soil

C. improve the nutritional value of crops
D. ensure growth of cash crops
E. conserve use of soil

13. Which of the following measures is most likely to cause a damage to the ecosystem in the control of pests and weeds?
A. Use of ladybird insects to feed on aphids
B. Introduction of bacterium to kill larvae of butterfly.
C. Introduction of *Tilapia* fish in ponds to feed on mosquito larvae.
D. Spraying insecticides to destroy cotton-strainers in a cotton field.
E. Clearing the weed by slashing.

14. A vegetable farmer will benefit from a cattle rearer in
A. the faecal remains of the cattle are made available to the farmer.
B. the farmer makes available fresh green vegetables for consumption by the cattle daily.
C. the farmer provides shelter for the cattle and the cattle rearer.
D. the cattle rearer makes available meat to the farmer daily.
E. the cattle feeds on the weed surrounding the vegetable.

15. Soil fertility can be conserved and renewed by
A. yearly monocropping
B. crop rotation and cover crops
C. bush burning to remove unwanted debris
D. avoiding artificial manures
E. irrigation of land

16. Which of the following methods is used in food preservation for a long period?
A. adding oil to food
B. exposing food to fresh air
C. keeping the food in a store
D. warming the food daily
E. salting and drying the food

17. The practice of warming soup regularly as it concerns food preservation is essentially to
A. retain the taste
B. help to break down the cellulose
C. remove the excess water
D. kill the bacteria in it
E. raise the temperature

18. In which of the following are food grains best preserved?
A. Silos
B. Sacks
C. Refrigerators
D. Ovens
E. Baskets

Answer Key

1.	E	5.	D	9.	A	13.	B	17.	D
2.	B	6.	C	10.	D	14.	A	18.	A
3.	C	7.	C	11.	E	15.	B		
4.	D	8.	E	12.	B	16.	E		

Explanations to Objective Questions

1. **(E)** Monocropping encourages same group of pests to attack the crop plants.

2. **(B)** Pests can be controlled without the subsequent danger of polluting the ecosystem by increasing the population of the natural enemies of the pest.

3. **(C)** Predators are beneficial in pest control because they feed on pest of crops.

4. **(D)** Denitrifying bacteria reduces nitrates in the soil to gaseous nitrogen.

5. **(D)** The burning of farmlands should be discouraged because it destroys the organic part of the soil.

6. **(C)** Denitrifying bacterium reduces nitrates in the soil to gaseous nitrogen.

7. **(C)** The use of predators or parasites to control pests in the farm is known as biological control.

8. **(E)** The predator-prey relationship in ecological community is the underlining principle in the adoption of biological control of pests.

9. **(A)** Smut is the plant disease that leads to the covering of grains of cereals with a mass of spores.

10. **(D)** The farming practice by which an exhausted farm land is left for a number of years before cultivation is known as bush fallowing.

11. **(E)** Intensive cropping can lead to a loss of soil fertility.

12. **(B)** The importance of practising crop-rotation in agriculture is to maintain the fertility of the soil.

13. **(B)** Introduction of bacterium to kill larvae of butterfly is most likely to cause a damage to the ecosystem.

14. **(A)** A vegetable farmer will benefit from a cattle rearer if the faecal remains of the cattle are made available to the farmer.

15. **(B)** Soil fertility can be conserved and renewed by crop rotation and cover crops.

16. **(E)** Salting and drying the food is a method that can be used to preserve food for a long time.

17. **(D)** The practice of warming soup regularly as it concerns food preservation is essentially to kill the bacteria in it.

18. **(A)** Food grains are best preserved in silos.

14 Micro-Organisms: Man and His Health

14.1 Introduction

- Micro-organisms are very small living organisms which are normally not visible to the naked eye but can be seen with the aid of a microscope. The micro-organisms include bacteria, viruses, protozoa, some algae and some fungi.

14.2 Micro-Organisms Around us

- Micro-organisms are found almost everywhere in nature. They are found in water, soil, air, surfaces of objects and living organisms, within living organisms where they are able to invade living cells.
- Many micro-organisms are disease-causing agents known as pathogens and are mainly the parasitic micro-organisms.
- Some micro-organisms are beneficial to man, especially the saprophytic ones that bring about the decay of organic matter.
- Micro-organisms can be grouped as follows:
  - — Those without cell structure which comprise all viruses.
  - — Those with prokaryotic cell structures comprising all bacteria and blue green algae, and
  - — Those with eukaryotic cell structures comprising all protists, certain fungi and algae.

(i) Micro-Organisms in Air

- Micro-organisms do not grow in air but they occur in dust and water droplets in air. They produce spores which are light and can be easily dispersed by air. When the spores land on suitable substrates, they germinate, multiply and produce more spores. Nutrient agar which is exposed to air grows various types of colonies of bacteria and fungi. The common micro-organisms found in air are as follows:
  - — **Virus:** Common cold virus, influenza virus, polio virus, pox virus and measles virus.
  - — **Bacteria:** Pneumococci, Staphylococci, Streptococci, *Bacillus anthracis* causes anthrax, *Mycobacterium tuberculosis* causes tuberculoses.
  - — **Fungi:** *Penicillium* (blue-green mould), *Aspergillus, Rhizopus nigricans* (black or bread mould), *Saccharomyces* (yeast).

(ii) Micro-Organisms in Water

- Aquatic organisms are rich in both organic and inorganic nutrients. The various aquatic environments such as wells, ponds, lakes, streams, rivers and seas support a wide variety of bacteria, blue-green algae, protists, green algae and fungi. The plankton are common micro-organisms that constitute aquatic water-bodies.
- Micro-organisms found in water can be grouped as follows:
  - — natural water micro-organisms
  - — soil micro-organisms
  - — sewage micro-organisms
- The natural water micro-organisms comprise bacteria: *Coccus, Bacillus, Pseudomonas, Azotobacter, Thiobacillus, Sarcina, Micrococcus, Vibrio, Spirillum* and *Spirochaeta* which are mostly heterotrophic.
- The blue-green algae include *Oscillatoria, Nostoc* and *Anabaena.*
- The protists include autotrophic diatoms, dinoflagellates, *Chlorella, Chlamydomonas* and certain species of *Englena* and heterotrophic *Amoeba* and *Paramecium*
- The Algae include green algae such as *Spirogyra* and *Volvox.* These green algae along with protists such as diatoms are the main primary producers in aquatic environments.
- The fungi comprising aquatic fungi, moulds and mildews.

(iii) Soil Micro-Organisms

- These are made up of nitrifying and nitrogen-fixing bacteria, *Streptomyces* and certain fungi.

(iv) Sewage Micro-Organisms

- These include coliform micro-organisms such as viruses and non-pathogenic bacteria e.g. *Escherichia coli, Streptococcus faecalis,* pathogenic bacteria such as *Vibrio cholerae* and *Salmonella typhi* and pathogenic protozoa such as *Entamoeba histolytica* all of which are found in the intestine of vertebrates,

animal origin.

14.3 Micro-Organisms in our Bodies and Food

Normal Microflora in the Body

- Micro-organisms colonize infant bodies shortly after birth. At the adult stage, numerous bacteria, yeasts and protozoa are established on and in the body.
- These normal microflora prevent or interfere with the invasion of the body by pathogens and they become pathogenic when the hosts resistance becomes low e.g. *E. coli* which is normally found in the colon causes urinary tract infections.

Pathogenic Micro-Organisms in the Body

- Pathogenic micro-organisms usually establish themselves in human bodies and cause diseases.
  - — when the body resistance is low
  - — when the normal microflora in the body is disrupted by antibiotic therapy, and
  - — in infants where the normal microflora is not yet established.
- The inability of the body to resist or fight against pathogens is caused by:
  - — malnutrition
  - — stress and overwork
  - — harmful habits like smoking and drinking, and
  - — environmental pollution.

14.4 Carriers of Micro-Organisms

- The non-living agents that carry micro-organisms from one place to another are air, water and food.
- The living agents that carry micro-organisms from place to place are animals. These animals that carry pathogenic micro-organisms are known as *Vectors.* These vectors include:
  - — insects like mosquitoes, houseflies, cockroaches, fleas and tsetse-flies, and
  - — other animals like rats, dogs and cats.

14.5 Micro-Organisms in Action

- The effects of microbial activity can be seen in the form of organic decay and infectious diseases. The activities of micro-organisms can however be seen in food production, industrial processes, medicine, agriculture; water treatment and waste disposal.

(I) Beneficial effects of micro-organisms

- The beneficial effects of micro-organisms are seen and primary producers. Micro-organisms are also used in various ways to benefit mankind. The use of biological organisms in industries is known as biotechnology. This may involve the use of micro-organisms or the use of enzymes extracted from micro-organisms.

Beneficial effects in nature

- Micro-organisms play important role in
  - — maintaining soil fertility, and
  - — recycling nutrients in the ecosystem

Sewage disposal

- Saprophytic micro-organisms decompose sewage into harmless inorganic compounds. This is effective in sparsely populated areas.
- Aerobic and anaerobic bacteria act upon sewage in sewage treatment plants in cities. The final harmless liquid is discharged into water bodies while the sludge is used as fertilizer.

Food Industries

- Micro-organisms are used in food industries mainly to:
  - — carry out fermentation on certain edible substances to produce foods with properties that help to preserve them; have improved flavours and textures; and make them easily digestible.
  - — produce single cell proteins (scp) from bacteria, yeasts and blue-green algae for cattle feed and human food; and
  - — food additions such as vitamins, amino-acids and nucleotides produced from microbial cultures.

Chemical Industries

- Micro-organisms are used to produce many substances and enzymes of industrial importance such as citric acid, dextrans and cellulose.
- They are also used to produce substances that are important in *medicine* such as antibiotics, steroid hormones and vaccines.
- Bacteria are being used to extract metals such as copper and uranium from low-grade ore and waste residue from mines.

Genetic engineering

- With the use of genetic engineering techniques, scientists are able to produce micro-organisms which act as "chemical factories" and manufacture substances difficult to obtain by the usual means.

(ii) Harmful effects of Micro-organisms

- Micro-organisms are harmful in two ways:
  - by causing food spoilage, and
  - by causing diseases.

Disease-causing micro-organisms

- The types of disease-causing or pathogenic micro-organisms in humans are viruses, bacteria, protozoa and fungi.

Spread of Disease-causing micro-organisms

- Pathogens can spread through
  - air by air-borne pathogens entering through the nose or mouth causing diseases like pneumonia, tuberculosis, influenza and mumps.
  - water by water-borne pathogens, spread by drinking contaminated water, affecting the intestines of the host and causing diarrhoea, typhoid and cholera.
  - food by food-borne pathogens through the eating of contaminated food such as milk and ice cream and causing bacillary dysentary, amoebic dysentary and cholera.
  - direct skin contact especially in crowded places or during kissing or sexual intercourse and may cause leprosy, ringworm, yaws, syphillis and gonorrhoea; and
  - animal vectors such as insects can spread disease-causing organisms from person to person.

14.6 Towards Better Health

- In order to attain better health, there is the need to control disease-causing micro-organisms and their animal vectors and improve public health facilities.

(i) Control of harmful micro-organisms

- In order to prevent the spread of infectious diseases, there is the need to kill or inhibit the growth of disease-causing micro-organisms. This can be done as follows:
  - through *sterilization* methods which kill or remove all micro-organisms from the objects that are being treated, by using high temperature and sterilizing chemicals like chlorine.
  - use of chemicals like disinfectants, antiseptics and antibiotics to kill or prevent the growth of pathogenic micro-organisms.
  - use of preservation techniques to prevent food spoilage and growth of pathogens in medicines.
  - use of high salinity salt solution to wash cuts and wounds to prevent them from becoming infected.
  - removal of water or dehydration to inhibit microbial growth.

(ii) Control of Vectors

- Important vectors of diseases are insects and rodents. The insect vectors include mosquitoes, houseflies, tsetse-flies, lice, and fleas. Through the knowledge of their habits and life cycles, environmental, biological and chemical methods can be used to control a given vector.

(iii) Public Health

- The maintenance of good public health in a nation would require the following:
  - refuse and sewage disposal;
  - protection of water supply and provision of clean water;
  - food hygiene;
  - control of diseases; and
  - healthy organizations.

(iv) Public Health Organisations

- Family Health International (FIH)
- Federal Environmental Protection Agency (FEPA)
- Global Health Council (GHC)
- Global Health Watch (GHW)
- International Planned Parenthood Federation (IPPF)
- National Institute of Public Health (NIPH)
- National Institute of Health (NIH)
- United Nations Children's Fund (UNICEF)
- World Health Organization (WHO).

SSCE/UME–Type Objective Questions

1. Which of the following are not causative organisms of plant diseases?
   A. Fungi B. Viruses
   C. Bacteria D. Nematodes
   E. Cercariae

2. The process whereby micro-organisms can convert atmospheric nitrogen into nitrogenous cmpound is known as
   A. nitrogen cycle B. nitrogen fixation
   C. denitrification D. putrefaction
   E. decomposition

3. Which of the following substances cannot control the growth of harmful microorganisms?
   A. Antibiotics
   B. Hypertonic salt solution
   C. Disinfectants
   D. Isotonic sugar solution
   E. Antiseptics

4. One of the ecological conditions which favour the spread of malaria is
   A. improper disposal of decaying organic matter
   B. clearing bushes around dwelling houses
   C. presence of water weeds
   D. presence of stagnant water in ponds
   E. improper disposal of sewage into the sea.

5. Microorganisms which break down dead organisms and absorb their contents are called
   A. decomposers B. consumers
   C. parasites D. commensals
   E. symbionts

6. The substances recycled in the soil by the activities of micro-organisms during the decay of dead organisms are first utilized by the
   A. carnivores B. herbivores
   C. scavengers D. autotrophs
   E. decomposers

7. Micro-organisms are important in recyling nutrients because
   A. they are capable of multiplying quickly.
   B. they are eaten by large organisms
   C. they break down dead organic materials
   D. many of them are parasites
   E. they can occur anywhere.

8. The following are useful effects of micro-organisms except
   A. production of vaccines
   B. curing of tobacco
   C. tanning of leather
   D. decay of meat
   E.

Answer Key

1.	E	3.	D	5.	A	7.	C
2.	B	4.	D	6.	C	8.	D

Explanations to Objective Questions

1. **(E)** Cercariae are not causative organisms of plant diseases.

2. **(B)** The process whereby micro-organisms can convert atmospheric nitrogen into nitrogenous compound is known as nitrogen fixation.

3. **(D)** Isotonic sugar solution cannot control the growth of harmful micro-organisms.

4. **(D)** One of the ecological conditions which favour the spread of malaria is the presence of stagnant water in ponds.

5. **(A)** Micro-organisms which break down dead organisms and absorb their contents are called decomposers.

6. **(C)** The substances recycled in the soil by the activities of micro-organisms during the decay of dead organisms are first utilized by the scavengers.

7. **(C)** Micro-organisms are important in recycling nutrients because they break down dead organic materials.

8. **(D)** The decay of meat is not a useful effect of micro-organism.

15 Conservation, Genetics and Evolution

15.1 Introduction

- Conservation is the proper maintenance of natural resource base for human survival and development.
- Man depends essentially on the resources of nature for survival — his food, water, clothing, shelter and the energy he uses.
- There is a rapidly increasing degradation and depletion of the country's natural resources, resulting from the heightened drive to satisfy vastly expanded human demands.
- Non-renewable natural resources like petroleum, coal and mineral ores have accumulated over geological time scales and continue to accumulate at rates at least a million times slower than the rates of their extraction.
- Renewable natural resources (living resources) are formed at rates closer to those of their use; therefore, man can control the rate of their use, the rate of formation and their distribution in space.
- The essence of renewable resources conservation is to balance the rates of their use against those of their formation in order to maintain stability and ensure sustained yield.

15.2 Resources to be conserved

— Soil, — water, — wildlife, — forest, and — minerals.

Aims of Conservation

- To manage the ecosystem in such a way that they yield the greatest sustainable benefit to present generations while maintaining the potential to meet the needs and aspirations of future generation.
- To maintain essential ecological processes and life-support systems, such as those responsible for soil regeneration, soil protection, nutrient recycling and water supply.
- To maintain the range of genetic material found in living things, which is essential for supporting breeding programmes for the improvement of cultivated plants and domesticated animals.
- To ensure the security of raw materials for industries and medicines.

Objectives of Renewable Resources Conservation

- To maintain genetic diversity in order to ensure permanence in the supply of materials to satisfy basic human needs and thus improve the well-being of society.
- To regulate environmental balances in such factors as carbon dioxide, radiation levels and biogeochemical cycles.
- To protect the environment, i.e. catchment areas, in order to enhance water resources and check soil erosion, protection of grazing lands against desert encroachment, and the stabilization of coastal environments.
- To maintain the scientific value of natural ecosystems, the study of which is required to enhance conservation itself, to improve the management of man-made systems, and to provide clues to technical innovations in agriculture, medicine and industry.
- To enhance the amenity values of natural resources, including aesthetic, heritage, religious, sentimental, ethical and recreational values on which tourism may be built.

Problems of Conservation

- The lack of awareness of the benefits of conservation and its relevance to everybody, which prevents governments, policy makers, development practitioners, and the general public from seeing the urgent need to achieve conservation objectives.
- Undue emphasis on short-term interests in preference to longer term stability and sustainability.
- The inadequacy of the capacity to conserve because of poor legislation, ineffective enforcement of legislation and poor organizational framework.
- Inadequacy of trained personnel and lack of basic information on priorities, productive and generative capacities of living resources.
- Failure to deliver conservation-based development where it is most needed (rural areas).

Agencies Responsible for Conservation

- Federal and State's forestry departments / Ministries of Agriculture and Natural Resources.

- International Union for Conservation of Nature and Natural Resources (IUCN)
- Nigerian Conservation Fund (NCF).
- World Wildlife Fund (WWF).
- Game Reserve and National Park Authorities.
- Federal Environmental Protection Agency (FEPA).

Conservation Education

- Public enlightenment is inevitable as to educate farmers who ignorantly burn bushes, poachers who decimate wildlife for short-term monetary gains and fishermen who use poisons in order to catch more fish evidently unaware of the harm done to other aquatic resources.
- Cattle is grazed almost wherever suitable pasture is available, not minding the harm done to the environment.
- Soil is cleared for construction or ploughed in a way satisfactory to the user who is unmindful of the harm his action does to others (erosion), Remedy is contour ploughing, strip cultivation, terracing, cover crop cultivation.
- The public should be made to be aware of the loss resulting from lack of adherence to conservation measures, as in practices of deforestation, unregulated land clearing, indiscriminate killing of marine and wildlife resources. The negative effects of such actions on the survival of man should be highlighted.
- The need to spread the use of resources over time.
- All conservation agencies should embark on intensive and carefully organized public education programmes since the acceptance of conservation programmes by decision makers depends on their level of awareness (use extension staff where extension staff/farmer ratio is high).
- Apart from using extension services, news media and educational institutions for effective public enlightenment campaign, the support of relevant voluntary organizations and societies should be enlisted.

Conservation Laws in Nigeria

- The Forestry Ordinance 1937 (Chapter 75).
- The Eastern Region Forest Law, 1955.
- The Forestry Ordinance with Amendments (Northern Region, 1960).
- The Forestry Law (Eastern State), Cap. 38.
- The Forestry (amendment) Edict; Western State, 1969.
- The Forestry (amendment) Edict; Western State, 1973.
- The Wild Animals Preservation Law (Cap. 132).
- The Sea Fisheries Decree 1971.
- Sea Fisheries (Fishing) Regulation 1972.
- Exclusive Economic Zone Decree 1978.
- The Wild Animals Preservation Law (Western Nigeria) 1959.
- The Wild Animals Law (Northern Nigeria) 1963.
- The Wild Animals Law (Eastern Nigeria) 1965.
- The Wild Animals Preservation Law (Lagos State) 1972.
- The Wild Animals Amendment Edict (NE State) 1975.
- The Wild Animals Amendment Edict (Kano State) 1978.
- The Kainji Lake National Park Decree 1979.
- The Endangered Species Decree 1985.
- The Federal Environment Protection Agency Decree 1988.

15.3 Variation in Population

A. Morphological Variations in physical appearance (Continuous Variations)

- Many traits found in a population of plants or animals vary in a smooth, continuous way from one extreme to the other.
- Body weight, body length, colour (skin, eye and coat of animals), and finger prints (Arches, loops are just four traits in which we may expect to find a whole range of variations.
- Typically, most of the individuals fall near the middle of the range, with fewer individuals at the extremes (see Fig. 15.1).
- Calculation of the heights or weights (of one sex) in a classroom will reveal a wide, unbroken range of values with more individuals in the middle of the range than at the extremes.
- Finger print is unique in every individual, difference being recognizable in the patterns of the arches, loops, whorls and compounds.
- This attribute is employed by the police in crime detection because finger prints left by the criminal at the scene of crimes are developed and matched with those of the suspects.

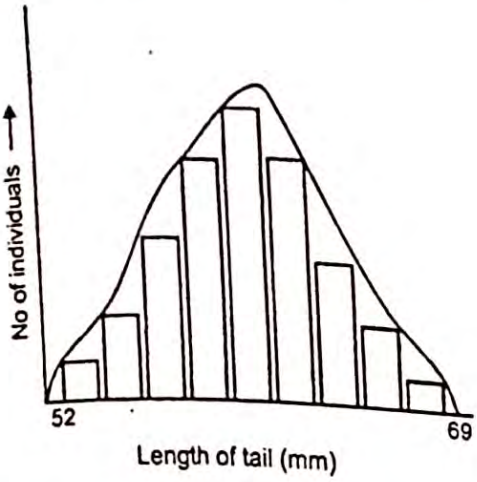


Fig. 15.1: Bar Chart of the distribution of length of tail in a population of mice

- Continuous variations result from combined interaction of many genes whereas discontinuous variations are due to single genes.

B. Physiological Variations (Discontinuous Variations)

- With respect to traits, the individuals of some populations fall into two or more distinct categories with no intergrading between them, i.e. when there are no intermediate or graded forms between two extremes, variation is discontinuous.
- Some physiological variations like the ability to roll the tongue, ability to taste phenyl thiocarbamide (PTC) and the occurrence of blood groups are discontinuous. There are no intermediate stages, you are either able to do tongue rolling / taste PTC or not.
- In humans, four blood types are recognized: A, B, AB and O.
- The erythrocytes in type A blood bear antigen A on their surface, in type B–antigen B, in type O–neither A nor B, in type AB–both A and B.

Antigen-antibody contents of human blood types

Blood Group	Blood Content	
	Cellular antigens	Plasma antibodies
A	A	anti–B
B	B	anti–A
AB	A and B	None
O	None	anti–A and anti–B

Blood transfusion relationships of human blood groups

Blood group	Donor	Recipient
A	A, AB	O, A
B	B, AB	O, B
AB	A and B	None
O	O, A, B, AB	O

- O blood group is a universal donor and AB blood group is a universal recipient.
- Blood typing is commonly used as a source of evidence in court paternity cases where the rightful father of a child is to be determined among claimants. e.g. a man with AB blood could not possibly be the father of a type O child because the child must have received an i allele from his father but an AB man has no such allele.

15.4 Heredity (Genetics)

- Living organisms have characters transmitted to them by their parents and such characters, like variations in height, skin colour, size and weight, are said to be *hereditary* or *inheritable.*
- Variations that can not be passed from parents to offsprings are said to be *non-hereditary.* For instance, modifications brought about by environmental factors alone are said to be acquired and not inherited. The child of a wrestler cannot inherit the bulging and rippling muscles of his father.

Mendelian Genetics

- When Mendel cross-pollinated plants with red flowers and those with white flowers all the offsprings usually called F1 or *first filial generation* were all alike in that they all had red flowers.
- The character, i.e. red flower, that appeared in F1 generation is said to be *dominant* (R) while the white flower trait that did not appear or remain hidden was said to be *recessive* (r).
- When F1 flowers were crossed among themselves, 75% of their offspring (i.e. *F2* or *second filial generation*) were red flowered and 25% white flowered (see figure below).

Parents: Red flower (RR) X White flower (rr)

Gamates: (R) (r)

F_1: All red flowers (Rr)

Self-fertilization of F_1 generation

Parent: All red flowers (Rr) X All red flowers (Rr)

Gamates: (R) (r) (R) (r)

F_2: (RR) (Rr) (Rr) (rr)

75% or ¾ red 25 or ¼ white

Ratio: 3 1

or

Male gamate	Female gamate		
	R	r
R	RR	Rr
r	Rr	rr

- **First Law:** Mendelian law of segregation states that contrasting factors (i.e. genes) are present in pairs in non-reproductive or somatic cells so that when an organism forms gamates the pairs separate so that each gamate contains only one factor.
- **Second Law:** Mendelian law of independent assortment states that when more than one pair of factors are considered each pair segregates independently of every other pair, e.g.when mendel crossed-pollinated pure-breeding round-seeded, (RR) yellow-cotyledon (YY) pea plant variety with a pure-breeding, wrinkled-seeded (rr), green-cotyledon (yy) variety. All the seeds were round and had yellow cotyledons (F_1-generation is a dihybrid because two different traits are involved, i.e. shape and colour). F_1-generation were allowed to self pollinate and the resulting. F_2-generation showed 315 round-yellow peas, 101 wrinkled-yellow peas, 108 round-green peas, and 32 wrinkled-green peas.
- Thus four types of combination of the yellow and green peas and the two kinds of round and wrinkled peas were obtained in the ratio 9:3:3:1 (see figure below).

Parents: RRYY X rryy

Gamates RY ry

F1-generation: Ryry Round seeds and yellow cotyledons

Self-Gross of F1: RYry X RYry

Male gamate	Female gamate			
	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

Round Wrinkled

Yellow Green Yellow Green

↓ ↓ ↓ ↓

315 101 108 32

9 : 3 : 3 : 1

- Individuals in which factors of a given pair are alike e.g. RR and rr are said to be *homozygous;* those with dominant and recessive genes or factors, e.g. Rr are *heterozygous.*

The genetic make-up, i.e. genotype, of some of the red flowered plants is Rr, but their physical characteristics, i.e. phenotype, is red.

Chromosomes: the basis of heredity

- Body cells, i.e somatic cells, of all animals and sporophytes of plants have fixed number of chromosomes for particular species.
- The total number of chromosomes for each species which is found in the nucleus of each somatic cell is the genome and the genome has a diploid number, i.e. 2n, of chromosomes.
- Man has a diploid number (2n) of 46 chromosomes which means that there are two of each kind of 23 chromosomes, hence haploid number is 23.
- Paired chromosome is made up of genes which are the functional units of inheritance.
- A gene for a particular trait exists at the same location or locus on two homologous chromosomes.
- A gene is the functional region of the DNA (deoxyribonucleic acid) of chromosomes and it directs the formation of a particular protein which is often an enzyme.
- The enzyme produced then regulates the production of cell products like a particular colour of flowers etc.
- Each chromosome which has duplicated itself during early cell division, is bound or connected together by a *centromere* and each member of the bound pair is called a *chromatid* (Fig. 15.2).

- DNA molecule consists of a double helix in which two chains are formed by repeating small chemical units called *nucleotides*.

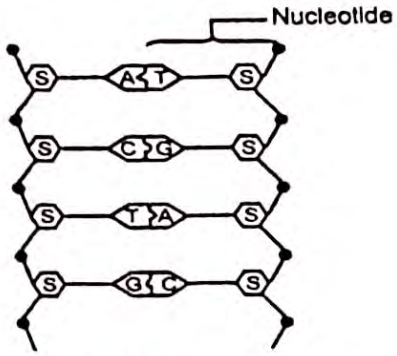


Fig. 15.2: Each nucleotide consists of a deoxyribose sugar (S), a phosphate group (P) and a nitrogenous compound base

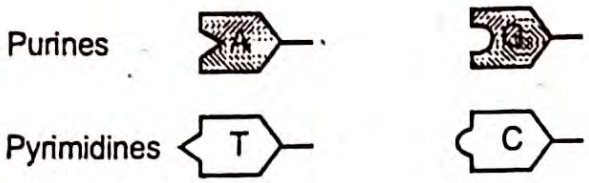


Fig. 15.3: The nitrogenous base may be Adenine (A), Guanine (G), Cytosine (C) and Thymine (T)

Applications of the principles of heredity in:

(a) *Agriculture*

- New crop plants can be generated by artificially crossing (hybridizing) two separate species which will have higher protein content, better yield than either parents and greater resistance to most environmental hazards like diseases, insect pests, harsh climates etc.
- Improvement of the nutritive value and palatability of crops and easier harvesting.
- Better breeds of animals can be produced, e.g. the crossing of humped and humpless cattle gave the keteku cattle breed which produces good beef and well adapted to Nigerian Climatic conditions.
- Animals like horses and dogs with attractive or desired traits have been produced from crosses with close relatives by farmers. Hence the idea of pure breeds.
- In breeding through, sexual acts between e.g. brother and sister should be avoided. The practice increases the percentage of harmful recessive alleles (homozygous) expressing themselves in such offsprings.
- Potential couples are advised to seek medical advice in order to make sure that their offsprings would not have a hereditary disease or be exposed to such a risk e.g. sickle-cell disease.
- It may be necessary to study many pedigrees in terms of hereditary diseases in order to ascertain or predict the genetic future of offspring and the chances of the disease inheritance.
- The Rh factor (Rhesus factor) of individuals must be known so that no antigen creates any immunological problems during blood transfusion, i.e. blood clotting.

15.5 Evolution

Competition

- Organisms frequently compete with one another for commodities like food, light, water, shelter, mate or resting site. Competition has resulted in numerous adaptations, structural and behavioural, which enable individuals to win in a "struggle for existence." Competition between members of the same species is known as intra-specific competition and competition between individuals of different species is known as inter-specific competition.
- The closer the ecological niches of the competing organisms the fiercer is the competition.

Relationship between competition and succession

- Communities (animal group or plant group) do not remain stable indefinitely.
- A slight change in the environment may upset the balance and result in a change in the structure of the community, e.g., on an unexploited rocky terrain, Lichens break up the rock surface so that dust particles can sustain mosses and later ferns and grasses. Eventually shrubs and trees move in, in that order as there is increased build-up of humus on the rock surface.
- Those trees that can thrive well on poor soils will first move in and later the dominant species can be recognized.
- This then influences the environment of competitors, which then replaces the former as the dominant species, and so on. Eventually a more or less stable climax community comes into existence.

Some structural adaptation to protection and defence

- In the course of evolution some animals have come to bear a striking and often detailed resemblance to part of a plant, thereby protecting themselves from attack by predators.
- This is quite obvious in some insects which resemble objects like leaves, sticks, thorns or twigs. In such a case the plant species form a most important part of the insects biotic environment.
- Some animals also closely resemble other species that happen to be unpalatable to a predator, a phenomenon known as mimicry.
- The unpalatable species possess distinctive colours or markings (warning colouration) and predators learn to recognize these signs and avoid attacking the species.
- Armed with similar markings the mimic is also protected from attack, e.g. the swallowtail butterfly, *Papilio dardanus*, mimics another butterfly, *Amaurus albimaculata*, which is highly impalatable to predators.
- In some cases the mimic is also unpalatable.

Behavioural Adaptations in Social Animals: the Honeybee as an Example

- The life of the honeybee colony revolves around the activities of its single queen which spends most of her time laying eggs in the wax cells of the honey comb.
- The queen fertilizes most of these eggs, just before they are deposited in the cells, by releasing sperm from storage sacs, which were filled at the time of her mating flights.
- These eggs latch into larvae (grubs) after three days and are fully taken care of and fed by worker bees for six days.
- The workers seal up the cells with wax and the larvae undergo metamorphosis, so that, eventually, new worker bees emerge three weeks after the eggs are laid.
- The worker is a female but without functional sex organs.
- When a fertilized egg is deposited in each of the several special cells (Queen cells), it develops into a fertile female, one of which will become the future queen of the hive.
- Larvae destined to be workers are fed on protein — rich secretion from the salivary stands of adult bees and those destined to be queens are fed on "royal jelly."
- The queen can also lay unfertilized eggs which will eventually develop into male bees called drones.
- So an active beehive contains one queen, a few hundred drones and many thousands of workers.
- Before a new queen emerges, the old queen leaves the hive, taking a large proportion of the workers with her to avoid competition. This is the phenomenon of swarming.

15.6 Theories of Evolution

- The idea of evolution involves two processes, the first is the gradual change in genotype and phenotype of a population of living organisms, i.e. adaptive changes.
- The second is the formation of new species.

Larmack's Theory

- The theory states that when an organism develops a need for a particular structure, this induces its appearance.
- The idea was based on the observation that structures which are subjected to constant use become well developed, whereas, those that are not used tend to degenerate.
- These beneficial characteristics, acquired during an individuals lifetime, could be handed on to the progeny, i.e. evolutionary change could be achieved by the transmission of acquired characters.

Darwin's Theory

- All organisms produce far more offsprings than the environment can support, and there tends to be a struggle for existence due to over-crowding and competition between individuals.
- Those individuals with favourable variations will stand a better chance of survival in this struggle. This can also be expressed as survival of the fittest.
- Darwin coined the term natural selection to describe the weeding-out process.
- As environmental conditions are constantly changing, natural selection is for ever favouring the emergence of new forms, or, as Darwin put it, the origin of species.
- Wallace conceived the idea of evolution and natural selection independently of Darwin.

Evidences for Evolution

1. *From fossil records*

- Paleontology is the study of fossils and a fossil is any sort of remains of an organism that lived in a past

geological age.

- Under special circumstances, the entire body of an organism may be preserved after death.
- Insects trapped in the sticky pitch of conifers over 30 million years ago, can now be studied entombed in amber as easily as if they had just died.
- Frozen carcasses of the mammoth, an extinct relative of the elephant, are occassionally unearthed in Siberia.
- Even as many as 40,000 years in the nature's deep-freeze, the meat of these animals is sufficiently well preserved to be used for biochemical studies.
- Usually, the soft portions of the body are quickly destroyed after death by scavengers or decomposed by decay bacteria.
- Hard parts, such as bones or shells, are more resistant to such destruction and hence more likely to be fossilized and surrounded by sediments of clay or sand. They may yield easily recognizable fossils hundreds of millions years later, long after the enclosing sediments have turned to rock such as shale or sandstone.
- Another common fossil is the putrefactor which is a copy in stone of some plant or animal part. As the original remains disintegrated, they were replaced bit by bit with mineral deposits.
- This process can proceed so slowly that the original specimen is reproduced in all its detail.
- With rare exceptions, fossils are not the remains of organisms still found living on the earth.
- The idea that all organisms alive today share a common ancestry at some period in history implies that there were fewer kinds of living things in the past and that these were less complex. This describes fossil records very well.
- The presence and distribution of fossils already discovered provide us with some of the most direct evidence of the theory of evolution.

2. *From Comparative Anatomy*

- Organs which have the same basic embryonic development are said to be homologous. For example the forelimbs of man, whale and bat (wing) are homologous organs, even though they are used in variety of ways, they have the same basic structural plan.
- This is hardly surprising in view of our belief that all organisms have shared a common ancestry at some time in their evolutionary history.
- The fused vertebrae which make up the base of the human spine are interpreted as the vestigial remnants of the tail, possessed by our ancestors.
- In fact, human babies occassionally are born with short tails. These are, however, quickly removed.

3. *From Embryology*

- The embryonic development of all vertebrates shows striking uniformities particularly during cleavage, morphogenesis, and all the early stages of differentiation.
- These similarities are often cited as evidence of an evolutionary relationship among vertebrates.
- The month-old human embryo has a series of paired tracheal grooves in the neck region which are matched on the interior by a series of paired gill pourches. This pattern is found in the embryonic development of all vertebrates.
- In the fishes, the pouches and grooves eventually meet and form gill slits, the openings which allow water to pass from the pharynx over the gills and out of the body.
- In higher vertebrates the grooves and pouches disappear and in man the chief trace of their existence is the eustacian tube and auditory canal, which connect the pharynx with the outside of the head.
- The temporary possession of a tail and a two-chambered heart are other examples of developmental stages, through which the human embryo passes.
- The idea that our embryonic development repeats that of our ancestors is called the theory of recapitulation. This theory is not restricted to vertebrates but is also found in insects.
- Physiological as well as anatomical recapitulation occurs.
- Fish excrete a large part of their waste nitrogen as ammonia, while amphibians have the less toxic urea as their chief nitrogenous waste.
- Tadpole excretes ammonia until it undergoes metamorphosis into the adult frog. Only then does the chief nitrogenous waste become urea.
- Birds convert waste nitrogenous compounds into the almost insoluble compound uric acid. Even the developing chick embryo passes through ammonia-excreting stage which is followed by a urea-excreting stage. This suggests that the chick is repeating stages in the physiological or biochemical development of its ancestors.

SSCE/UME–Type Objective Questions

1. In a breeding programme a cross was made between two true-breeding cowpea types: one with round seeds and the other with wrinkled seeds. If roundness is dominant while wrinkleness is recessive, in the first filial generation all the seeds produced will be
   A. 25 percent wrinkled, 75 per cent round
   B. 100 per cent wrinkled
   C. 50 per cent wrinkled, 50 per cent round
   D. 75 per cent wrinkled, 25 per cent round
   E. 100 per cent round

2. A farmer cannot check soil erosion effectively by
   A. constructing his ridges for planting across the slope
   B. cultivating cover crops
   C. practising strip cropping
   D. clearing the land of vegetation
   E. constructing terraces

3. If you cross-breed a tall variety (Tt) of maize with a short variety (tt), the ratio of tall to short plants in the offspring will be
   A. 2 : 1 B. 1 : 2
   C. 1 : 1 D. 1 : 3
   E. 3 : 1

4. What is the genetic ratio of a cross between a homozygous tall plant and a homozygous dwarf plant?
   A. O tall : 4 short B. 3 tall : 1 short
   C. 2 short : 2 tall D. 1 tall : 1 short
   E. 4 tall : O short

5. Normally any character shown by an organism is due to the effects of
   A. hormones and chromosomes
   B. chromosomes
   C. mutations
   D. homones and genes
   E. genes and environment

6. A person whose blood can be donated to all other people, must have the blood group
   A. O B. AB
   C. B D. A
   E. none of the above

7. If a 26 year old blind man married a young one-eyed woman and they had four children, how many of them would be blind like their father?
   A. All B. 3
   C. 2 D. 1
   E. None

8. When a person with blood group O has lost much blood due to an accident, he can be given blood from persons who have blood belonging to
   A. group O only
   B. group A only
   C. group AB only
   D. groups A and O only
   E. groups AB and O

9. An organism having one pair of identical genes is
   A. a heterozygote B. hybrid
   C. an allelomorph D. a homozygote
   E. a diploid

10. The character-producing factors in living organisms are
    A. chromomeres B. alleles
    C. chromatids D. chromosomes
    E. genes

11. If a child can receive blood from a l donors, he belongs to the blood group
    A. O B. A
    C. B D. AB
    E. AS

12. The characters by which an organism s recognized are termed its
    A. phenotype B. genotype
    C. morphology D. anatomy
    E. physiology

13. The hereditary material in a cell is known as
    A. ADP B. CNS
    C. RNA D. ATP
    E. DNA

14. The ratio of carriers to sicklers in the F2 generation derived from a parental cross of two carriers of haemoglobin S gene is
    A. 3 : 1 B. 1 : 3
    C. 2 : 1 D. 1 : 2
    E. 1 : 1

15. There will be agglutination when the
    A. Group A serum mixes with Group A erythrocytes
    B. Group A serum mixes with Group B erythrocytes

C. Group AB serum mixes with Group O erythrocytes
D. Group B serum mixes with Group B erythrocytes
E. Group AB serum mixes with Groups A and B erythrocytes

16. If R and r denote the genes for a character, the offsprings of the cross between RR and Rr are
A. RR, 2Rr, rr
B. 2RR, 2rr
C. 2RR, 2Rr
D. 4Rr,
E. 3RR, rr

17. The theory of natural selection was developed by
A. Lamarck and Darwin
B. Darwin and Wallace
C. Wallace and Mendel
D. Mendel and Lamarck

18. Fossil records found in sedimentary rocks offer some explanation for the theory of evolution because
A. the deposits have remains of organisms characteristic of when they were formed
B. different strata have remains of organisms of the same kind
C. only organisms with strong parts are fossilized
D. most animal and plant fossils bear little resemblance to present day living specimens
E. the deposits have remain of organisms uncharacteristic of when they were formed.

19. Which of the following is an example of discontinuous variation observed in man?
A. Skin colour
B. Tongue rolling
C. Body weight
D. Height
E. Finger prints

20.

Gamete from the other parent	Gamate from one Parent	
	SS	Ss
	Ss	ss

S = smooth
s = wrinkled

In the illustration above, the genotypes of the offsprings are
A. 1 heterozygous smooth : 2 homozygous smooth : 1 wrinkled
B. 1 homozygous smooth : 2 heterozygous smooth : 1 wrinkled
C. 2 homozygous smooth : 2 wrinkled
D. 3 heterozygous smooth : 1 wrinkled
E. 1 homozygous smooth : 3 heterozygous wrinkled

21. Which of the following is not part of the objectives of conservation of natural resources?
A. To eliminate wild animals in order to enable other plants and animals survive
B. To use our resources on sustained yield basis
C. To guard against indiscriminate killing of plants and animals
D. To guard against the extinction and exhaustion of natural resources
E. To protect the environment

22. Which of the following major ideas were proved by Gregor Mendel?
A. There is a tendency among organisms to increase rapidly in number if conditions are favourable
B. There is a struggle for existence among a very large number of offsprings
C. Nature selects organisms which will survive to produce their kind
D. All species exhibit structural and functional variations
E. Characteristics of an organism are controlled by genes occurring in pairs.

23. Which of the following is not a constituent of DNA?
A. Purine
B. Phosphate
C. Cytosine
D. Deoxyribose
E. Uracil

24. The haploid number of chromosomes in man is
A. 48
B. 46
C. 42
D. 24
E. 23

25. Which of the following is not an evidence of evolution?
A. Fossil
B. Anatomy
C. Embryology
D. Genetics
E. Behaviour

26. The term used in describing the effective management of natural resources by man is
A. preservation
B. utilization
C. economics
D. development
E. conservation

27. The theory that new organs or characteristics develop in organisms when there is need for them was postulated by
A. Charles Darwin
B. Jean Lamarck
C. Louis Pasteur
D. Gregor Mendel
E. Wallace

28. Which of the following is NOT a type of finger print?
    A. Whorl
    B. Compound
    C. Simple
    D. Loop
    E. Arch

29. Which of the following is the precise location of the gene?
    A. Chromosome
    B. Centrosome
    C. Centriole
    D. Ribosome
    E. Endoplasmic reticulum

30. The following agencies are responsible for conservation in Nigeria except
    A. Forestry Departments
    B. Nigerian Conservation Society
    C. Game Reserve Authority
    D. Nigerian Red Cross Society
    E. Ministry of Agriculture and Natural Resources

Answer Key

1. E	7. E	13. E	19. B	25. E
2. D	8. A	14. C	20. B	26. E
3. C	9. D	15. B	21. A	27. B
4. E	10. E	16. C	22. E	28. C
5. D	11. D	17. B	23. E	29. A
6. A	12. A	18. A	24. E	30. D

Explanations to Objective Questions

1. **(E)** Since the cowpea types breed through, roundness will be homozygous dominant (i.e. RR) and wrinkled seeds will be homozygous recessive (ze, rr). So when crossed

RR X rr ..... Parent
R r ..... Gamate
Rr ..... F1 generation

or

Female gamates	R	R
r	Rr	Rr
r	Rr	Rr

= 4Rr

Since R is dominant all F1 generation will be 100% round.

2. **(D)** Soil erosion cannot be effectively checked by clearing the land of vegetation because it will promote soil erosion.

3. **(C)** Tt × tt ... parent

Gamates of male

Gamates of Female	t	t
T	Tt	Tt
t	tt	tt

Ratio = 2Tt : 2tt = 1 : 1

4. **(E)** Homozygous tall (TT) X Homozygous dwarf (tt) . . . Parent

Male gamates

Female gamates	T	T
t	Tt	Tt
t	Tt	Tt

Since T is dominant all plants will be tall, no dwarf plant, i.e. 4 tall (Tt) : O short (tt).

5. **(D)** Any character shown by an organism is due to the effects of hormones and genes.

6. **(A)** O blood group in man represents a universal donor because cellular antigens are absent.

7. **(E)** None of the children will be blind because blindness is not an inherited trait.

8. **(A)** O blood group can only receive blood during transfusion from O blood group alone because there are no cellular antigens on the cells.

9. **(D)** A pair of identical genes is referred to as homozygous and an organism having it is a homozygote.

10. **(E)** Character-producing factors in living organisms are genes.

11. **(D)** A child that can receive blood from all donors belongs to the blood group AB, i.e. universal recipient.

12. **(A)** Characters (externally shown) by which an organism is recognized are termed phenotype but the genetic basis is termed genotype.

13. **(E)** Hereditary material in a cell is known as DNA (Deoxyribonucleic acid).

14. **(C)** Carriers are Ss for sickle cell disease, so when carriers are crossed

Male gamates

Female Gamates	♀/♂	S	s
S		SS	Ss
s		Ss	ss

1 SS = homozygous dominant (i.e. not a carrier)

2 Ss = heterozygous dominant (i.e. carrier)

and 1 ss = homozygous recessive (i.e. suckler)

∴ Ratio is 2Ss : 1ss i.e. ratio of carriers to sicklers.

15. **(B)** There will be agglutination when the Group A serum mixes with Group B erthrocytes.

16. **(C)** Using the punnett square

Gamates	R	R
R	RR	RR
r	Rr	Rr

= 2RR and 2Rr

17. **(B)** The theory of natural selection was developed by Darwin and Wallace.

18. **(A)** Fossil records found in sedimentary rocks offer some explanation for the theory of evolution because the deposits have remains of organisms characteristic of when they were formed.

19. **(B)** Tongue rolling is an example of discontinuous variation observed in man because there are no intermediate stages, you are either able to do tongue rolling or not.

20. **(B)** The genotypes of the offsprings are:
1 homozygous smooth : 2 heterozygous smooth :
1 homozygous wrinkled.

21. **(A)** Elimination of wild animals so that other plants and animals can survive is not part of the objectives of conservation of natural resources.

22. **(E)** Gregor Mendel proved that the characteristics of an organism are controlled by genes occurring in pairs.

23. **(E)** Uracil is not a constituent of DNA but RNA.

24. **(E)** The haploid number of chromosomes in man is 23 and the diploid number is 46.

25. **(E)** Behaviour is not an evidence of evolution but fossil, embryology, anatomy and genetics are.

26. **(E)** Conservation is the term used by man in describing the effective management of natural resources.

27. **(B)** Jean Lamarch postulated the theory that new organs or characteristics develop in organisms when there is need for them.

28. **(C)** Whorl, compound, loop and arch are examples of finger prints and not simple.

29. **(A)** Genes are located on chromosomes.

30. **(D)** Nigerian Red Cross Society is not responsible for conservation anywhere in the world.

INSTRUCTIONS: Each of the following questions or incomplete statements below is followed by five suggested answers or completions. Choose the one that is best in each case and than shade the corresponding space on the answer sheet.

1. Which of the following groups of ecological terms is in correct order, from the simplest to the most complex?
   A. Organism, population, community, ecosystem, biosphere
   B. Organism, population, community, biosphere, ecosystem
   C. Population, organism, community, ecosystem, biosphere
   D. Biosphere, ecosystem, organism, community, population
   E. Ecosystem, biosphere, organism, community, population

2. A man who is normal for colour vision marries a normal heterozygous woman. What is the chance of their son being colour blind?
   A. 0% B. 25% C. 50%
   D. 75% E. 100%

3. If Ade's father has type A blood and his mother has type O blood, Ade's blood type will most likely be:
   A. A or B B. A or O C. A or AB
   D. B or O E. AB or O

4. In ecological succession, since lichens grow on bare rock, they are considered to be:
   A. primary consumers
   B. pioneer organisms
   C. climax organisms
   D. producers
   E. composers

5. At times, hyenas feed on remains of animals killed by other animals. At other times, hyenas themselves kill animals for food. Therefore, hyenas may best be described as:
   A. scavengers and herbivores
   B. scavengers and parasites
   C. scavengers and predators
   D. herbivores and predators
   E. herbivores and parasites

6. Which of the following statements is most true of producer organisms?
   A. They are eaten by carnivores
   B. They are eaten by scavengers
   C. They are parasitic
   D. They contain chlorophyll
   E. None of these

7. The innermost chamber of the respiratory system of land vertebrates into which air can be drawn is the:
   A. bronchiole B. branches
   C. alveolus D. bronchial tube
   E. sinus

8. Hormones are distributed through the body by
   A. blood plasma B. ducts
   C. lacteals D. endocrines
   E. enzymes

9. Why are insects and spiders classified as arthropods?
   A. They are carnivorous
   B. They have jointed legs
   C. They have a backbone in the embryonic stages
   D. They are terrestrial animals
   E. They have simple eyes

10. Viruses do not fit into our concept of the cell because they
    A. lack cytoplasm
    B. cannot multiply
    C. contain DNA
    D. are extremely small
    E. contain enzymes in their chlorophyll

11. The best description of an enzyme is that it
    A. becomes hydrolysed during chemical reactions
    B. becomes dehydrated during chemical synthesis
    C. speeds up the rate of chemical reactions
    D. serves as an inorganic catalyst
    E. is the source of energy for a chemical reaction

12. The region where two neurones are in close contact is known as
    A. mycelium B. a synapse

C. sori D. a sinus
E. a pellicle

13. In what way do valves in the veins aid in circulation?
A. They prevent the spurting action from becoming too uneven
B. They reduce the pressure with which the heart sends blood into the veins
C. They separate oxygenated from deoxygenated blood
D. They filter out blood cells
E. They keep the blood from flowing backward

14. A person with insufficient iron in his diet may become anaemic because
A. not enough oxyhaemoglobin is formed in the liver
B. the spleen removes too many leucocytes form the blood
C. the clotting reaction time is reduced
D. iron is needed to build up haemoglobin
E. iron is used in making fibrinogen

15. All of the following organelles are present in a **Paramecium** except
A. cilia B. a macronucleus
C. a food vacuole D. chloroplasts
E. a micronucleus

16. In which organelles of both plant and animal cells does aerobic respiration take place?
A. Mitochondria B. Lysosomes
C. Chloroplasts D. Golgi complex
E. Centrosomes

17. In animal cells, where does the energy to convert ADP to ATP come from?
A. Sunlight
B. Hormones
C. Organic molecules
D. Inorganic molecules
E. Carbon dioxide

18. Nutrients are digested to soluble form with the food vacuoles of the
A. hydra and earthworm
B. hydra and grasshopper
C. grasshopper and earthworm
D. Amoeba and Paramecium
E. Amoeba and human

19. Within which organelle are genes located?
A. Food vacuole
B. Contractiele vauole
C. Nucleus
D. Pinocytic vesicle
E. Cell wall

20. Special structures for absorption in the small intestine are called
A. alveoli B. tubules
C. mucous glands C. villi
E. peristalsis

21. A person who is exercising breathes more rapidly because
A. his lungs use up more energy
B. his calorie production goes down
C. he gives off more carbon dioxide, which stimulates the medulla
D. glycogen in his liver is converted to glucose
E. glycogen in his muscles is converted to glucose

22. Most marine fish cannot survive in fresh water because of
A. their inability to maintain homeostatic water balance
B. the lower oxygen concentration in fresh water
C. the higher carbon dioxide concentration in fresh water
D. the lack of food plants in fresh water
E. the great amount of pollution in fresh water

23. It has been determined that the volume of a cell varies as the cube of its diameter, and the surface area as the square of its diameter. As a cell grows in size, therefore, the ratio of surface area to volume
A. increases
B. decreases
C. remains the same
D. alternately increases and decreases
E. is determined by the cell's metabolic rate

24. In the ecosystem, which of the following is an abiotic factor?
A. Lichen B. Fungus
C. Alga D. Water
E. Blue-green alga

25. All of the following are examples of consumers except
A. carnivores B. herbivores
C. saprophytes D. green plants
E. parasites

26. Which of the following is the correct sequence in a food chain?
A. Mice, green plants, snake, hawk
B. Mice, snake, green plants, hawk
C. Green plants, mice, snake, hawk
D. Green plants, hawk, snake, mice
E. Hawk, green plants, snake, mice

27. In a pyramid of energy, which level represents the greatest amount of energy?
A. Producers
B. Primary consumers
C. Secondary consumers
D. Tertiary consumers
E. Decomposers

28. Oxyhaemoglobin is formed in the
A. lungs
B. spleen
C. right auricle
D. left auricle
E. bone marrow

Use the following instruction to answer questions **29–31.**

When Mendel self-pollinated many pea plants that were heterozygous for seed colour, he obtained about 8,000 seeds in the next generation (yellow seed colour is dominant; green is recessive).

29. Which of the following shows the number of yellow seeds and the number of green?
A. 4,000 – 4,000
B. 5,000 – 3,000
C. 3,000 – 5,000
D. 6,000 – 2,000
E. 2,000 – 6,000

30. Suppose that, instead of self-pollinating these plants, Mendel had cross-pollinated them. The ratio of yellow and green seeds he obtained in the next generation would most likely have been
A. 1:1
B. 1.5:1.75
C. 2:1
D. 3:1
E. 4:1

31. The results obtained in these experiments led Mendel to propose his Law of
A. Independent Assortment
B. Survival of the fittest
C. Segregation
D. Dominance
E. Blending

32. The production of glucose, a six-carbon sugar, is most closely associated with a chemical reaction that occurs during
A. aerobic respiration
B. anaerobic respiration
C. fermentation
D. the dark phase of photosynthesis
E. the light phase of photosynthesis

33. A characteristic of organisms that excrete uric acid as their main nitrogenous waste is that they
A. usually live in salt water
B. usually live in fresh water
C. are usually land dwellers
D. can carry on only anaerobic respiration
E. cannot metabolize proteins

34. Of the following chemical substances, the only one related to the nervous system is
A. gibberellic acid
B. acetylcholine
C. insulin
D. deoxyribonucleic
E. ptyalin

35. From what part of a plant does the seed develop?
A. Hilium
B. Pollen tube
C. Anther
D. Oviduct
E. Ovule

36. Red corpuscles are to haemoglobin as chloroplasts are to
A. guard cells
B. palisade cells
C. chlorophyll
D. photosynthesis
E. cytoplasm

37. Foods have to be digested before they can be used by the body because
A. the stomach is the centre of all digestion
B. the villi digest only nutrients they can absorb
C. only insoluble materials can pass through membranes
D. only soluble materials can pass through membranes
E. assimilation always takes place before digestion

38. The cowpea is classified as a dicot because
A. its leaves have parallel veins
B. its flower parts are in groups of three
C. when it germinates, the plumule remains underground
D. it has two cotyledons in its seed
E. its seed has the diploid number of chromosomes

39. In a large litter of rabbits, three quarters of the offspring were black (black is dominant). The genotypes of the parents were most likely

A. BB X bb B. Bb X bb
C. Bb X Bb D. Bb X BB
E. BB X BB

40. A cross between two black rabbits produces some black offspring and some white offspring. The genetic make-up of the two parents is most likely
A. homozygous black
B. heterozygous black
C. heterozygous white
D. homozygous white
E. recessive white

41. By which of the following can movement of materials across animal cell membranes be accomplished?
I. Active transport II. Diffusion
III. Pinocytosis
A. I only B. II only
C. III only D. I and II only
E. I, II and III

42. The bat is classified as a mammal for all the following reasons except
A. it has hair
B. it suckles its young offspring
C. its embryo develops internally
D. its fertilization occurs externally
E. its is warm-blooded

43. All of the following are classes of vertebrates except
A. Pisces B. Amphibia
C. Reptilia D. Mollusca
E. Mammalia

44. Animals that have jointed appendages include which of the following?
I. Spider II. Crab III. Snail
A. I only B. II only
C. III only D. I and II only
E. I, II and III

45. All of the following are functions of the skin except
A. protection B. sensation
C. excretion
D. manufacture of vitamin D
E. exhalation

46. An increase in the diameter of an iroko tree is caused chiefly by the activity of the
A. vascular bundles B. bark
C. lenticels D. cambium
E. annual rings

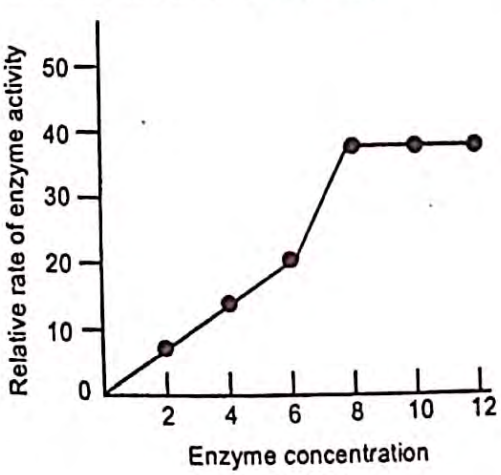
47. What effect does the hydrolytic action of enzymes have on organic molecules?
A. They are converted to more complex forms
B. Their hydrogen-ion concentration is increased
C. Their hydrogen-ion concentration is decreased
D. They become chemically inactive
E. They become smaller

48. Which of the following does not occur in mitosis?
A. Duplication of DNA
B. Duplication of the centriole
C. Crossing-over in homologous chromosomes
D. Production of diploid cells
E. Spindle formation

49. Which of the following explains why carnivorous animals depend on green plants for their food?
A. Carnivores need to achieve a balanced diet
B. Only herbivorous animals have grinding teeth
C. Only green plants can make food
D. Carnivores need protection against vitamin deficiency diseases
E. Only omnivores are both carnivores and herbivores

50. Although the island of Madagascar is separated from Africa only by a narrow strait, many plants and animals common on the mainland are unknown on the island. What principle does this fact illustrate?
A. Incomplete dominance
B. Independent assortment
C. Evolutionary equilibrium
D. Evolution in isolated populations
E. Ecological succession

51. Which of the following is indicated by the information on the accompanying graph when the substrate concentration is constant?



A. The rate of enzyme action is directly dependent upon the substrate concentration.
B. The rate of enzyme action increases constantly with an increase in enzyme concentration.
C. The rate of enzyme action becomes stabilized when a certain enzyme concentration is reached.
D. Enzyme concentration has no effect upon the rate of enzyme action.
E. When the substrate concentration is increased, the enzyme concentration is decreased.

Use the following instruction to answer questions **52–55.**

A branch is cut from a herbaceous plant and is inserted into a photometer, and sealed into a rubber tubing. All parts of the apparatus, including the funnel are completely filled with water. As water is lost from the leaves by transpiration, the level of the water in a calibrated capillary tube descends.

52. Which of the following is measured by the distance per minute that the water moves down the capillary tube?
A. Rate of photosynthesis
B. Rate of water loss from leaves
C. Amount of water absorbed by the leaves
D. Cohesiveness of the water molecule
E. The amount of water used in aerobic respiration

53. If the herbaceous plant is exposed to an actively operating electric fan, the rate of transpiration will
A. be proportionate to the rate of photosynthesis
B. be determined by the plants respiration rate
C. remain unchanged
D. increase
E. decrease

54. An experimenter coated the leaves of the herbaceous plant with vaseline. Which of the following effects would be observed?
A. The level of water in the calibrated capillary tube would rise
B. The level of water in the calibrated capillary tube would remain unchanged
C. The level of water in the calibrated capillary tube would descend
D. The level of water in the funnel would descend
E. Bubbles of air would appear in the water below the funnel

55. An experimenter enclosed the entire apparatus in a light proof box. What observation would be made about the level of the water in the calibrated capillary tube?
A. It would remain exactly the same
B. It would descend sharply
C. It would descend slightly
D. It would rise slightly
E. It would rise sharply

56. Development of an organism from an unfertilized egg is known as
A. fertilization
B. conjugation
C. maturation
D. parthenogenesis
E. Oogenesis

57. In what part of the cell are proteins synthesized?
A. Ribosomes
B. Nucleus
C. Centriole
D. Mitochondrion
E. Golgi complex

58. All of the following are found in a DNA molecule except
A. adenine
B. thymine
C. guanine
D. uracil
E. cytosine

59. In diabetes, excess sugar is found in the urine because
A. the basal metabolism is decreased
B. the kidneys are in a state of hypertension
C. the cells cannot utilize glucose
D. an excess insulin is produced
E. insufficient thyroxine is formed

60. Why can't a grasshopper be drowned by keeping its head under water?
A. It breathes through openings in its abdomen
B. The spiracles in its head filter out oxygen
C. It can hold its breath for hours
D. It is cold-blooded
E. The maxillae and mandibles serve as an airtight valve.

61. What is the reason that a blood clot in the coronary artery may be fatal?
A. It leads to atherosclerosis
B. Varicose veins invariably result as a by-product
C. The portal circulation is interfered with
D. The flow of blood to the heart muscle is blocked
E. The clot leads to coagulation of lymph

62. The fact that, under certain conditions, stomatal openings become smaller enables plants to avoid excessive loss of
A. carbon dioxide B. oxygen
C. water
D. essential plant hormones
E. chlorophyll

63. To the species involved, a major advantage of sexual reproduction over asexual reproduction is that sexual reproduction results in
A. greater number of offspring
B. greater variety of offspring
C. greater size of offspring
D. more rapid development of offspring
E. less vulnerability of offspring

64. Lactic acid, alcohol, and 2 ATPs are produced as a result of which of the following processes?
I. Aerobic respiration
II. Anaerobic respiration
III. Photosynthesis
A. I only B. II only
C. I and III only D. II and III only
E. I, II, and III

65. Digestive enzymes that hydrolyze molecules of fat into fatty acid and glycerol molecules are known as
A. lactases B. lipases
C. leucocytes D. lipids
E. lacteals

66. Which of the following conditions is not essential in a self-sustained ecosystem?
A. A means to permit the cycling of carbon between living organisms and their environment
B. A means to permit the cycling of water between living organisms and their environment
C. A constant supply of energy
D. A living system capable of incorporating energy into organic compounds
E. Equal numbers of plants and animals

67. The concentration of carbon dioxide in the atmosphere remains relatively constant at about 0.04 percent as a result of established equilibrium between the processes of
A. assimilation and excretion
B. oxidation and photosynthesis
C. photosynthesis and assimilation
D. photosynthesis and reproduction
E. respiration and reproduction

68. The process that most directly enables a root hair cell to absorb minerals by active transport, and enables a muscle cell to contract, is
A. circulation B. ingestion
C. digestion D. excretion
E. respiration

69. A person who has tested positive for the virus known as HIV is likely to develop
A. Lyme disease B. Elephantiasis
C. AIDS D. Malaria
E. Rickets

70. Which of the following represent(s) a carbohydrate?
I. $C_{12}H_{22}O_{11}$ II. $C_{17}H_{35}COOH$
III. $(C_6H_{10}O_5)_n$
A. I only B. II only
C. III only D. I and II only
E. I and III only

71. The cells of a *Drosophila* fly have been shown to contain 8 chromosomes. The number of chromosomes in the fertilized egg is
A. 2 B. 4 C. 8
D. 12 E. 16

72. A biochemist analysed an unknown organic compound and determined its chemical make-up. The accompanying chart summarizes the results of the biochemist's analysis:

Element	Number of Atoms per Molecule
C	16
H	32
O	2
N	0
S	0
P	0

On the basis of this chart, which class of organic compounds was the biochemist studying?
A. Carbohydrates B. Lipids
C. Proteins D. Nucleic acids
E. Amino acids

73. On the basis of their patterns of nutrition, most animals are classified as
A. decomposers B. autotrophic
C. heterotrophic D. photosynthesis
E. anaerobic

74. In the following food chain, what is the role of the rabbit

Cabbage plant ⟶ Rabbit ⟶ Fox

A. Saprophyte B. Consumer
C. Producer D. Decomposer
E. Carnivore

75. The basis of life in the ocean food chain is
A. heterotroph aggregates
B. lichen
C. very small fish
D. plankton
E. coral

76. The organs of excretion in human beings include which of the following?
I. Lungs B. Kidneys III. Skin

A. I only B. II only
C. I and II only D. II and III only
E. I, II and III

77. Which organ of the human digestive system produces hydrochloric acid?
A. Spleen B. Gall bladder
C. Stomach D. Small intestine
E. Large intestine

78. All the living things present in 30 cm^2 of soil make up a
A. community B. niche
C. species D. biome
E. biosphere

79. The body often has extra strength in an emergency because
A. the storage of glucose is stimulated
B. the activity of the digestive tract is stimulated
C. the production of pancreatic juice is stimulated
D. the rate of peristalsis is stimulated
E. the secretion of adrenaline is stimulated

80. The chamber of mammalian heart that receives blood from the lungs is
A. pulmonary artery
B. right auricle
C. pulmonary vein
D. left auricle
E. dorsal aorta

81. All of the following are physical factors of the environment except
A. light B. temperature
C. water D. predators
E. pH

82. The transfer of materials between organisms and the environment is known as a
A. niche B. cycle
C. synthesis D. free gas
E. solution

83. The use of predators or parasites to control pests in the farm is known as
A. predator control
B. chemical control
C. animal control
D. biological control
E. parasite control

84. All of the following are major causes of forest destruction except
A. excessive cutting B. forest fires
C. conservation D. insects
E. fungal diseases

85. Decayed organic matter in the soil is known as
A. aqueous humour
B. humus C. hilum
D. hydra E. hyphae

86. Terracing is practiced to
A. hold water back
B. enrich the soil
C. increase the nitrogen content of soil
D. add green manure to the soil
E. keep rodents away

87. All of the following are examples of good soil conservation measures except
A. contour ploughing
B. cultivation of soil
C. cover crops
D. strip cropping
E. crop rotation

88. The lichen is an example of
A. an alga and a fungus
B. a protozoan
C. bacteria
D. an alga and a slime mold
E. an alga and yeast

89. Yeast is useful in the baking industry because it carries on
A. aerobic respiration
B. fermentation

C. photosynthesis
D. the Krebs cycle
E. symbiosis

90. Two warm-blooded groups of vertebrates are
A. mammals and sharks
B. birds and amphibians
C. mammals and birds
D. amphibians and reptiles
E. mammals and amphibians

Answer Key to Model Questions Set 1

1. A
2. C
3. B
4. B
5. C
6. D
7. C
8. A
9. B
10. A
11. C
12. B
13. E
14. D
15. D
16. A
17. C
18. D
19. C
20. D
21. C
22. A
23. B
24. D
25. D
26. C
27. A
28. A
29. D
30. D
31. C
32. D
33. C
34. B
35. E
36. C
37. D
38. D
39. C
40. B
41. D
42. D
43. D
44. D
45. E
46. D
47. E
48. C
49. C
50. D
51. C
52. B
53. D
54. B
55. C
56. D
57. A
58. D
59. C
60. A
61. D
62. C
63. B
64. B
65. B
66. E
67. B
68. E
69. C
70. E
71. C
72. B
73. C
74. B
75. D
76. E
77. C
78. A
79. E
80. D
81. D
82. B
83. D
84. C
85. B
86. A
87. B
88. A
89. B
90. C

Explanations To Model Questions Set 1

1. **(A)** An organism is either a single-called or a many celled living thing. Large numbers of organisms of one species make up a population. A community consists of all the plant and animal populations interacting in a given environment. A ecosystem is a self-sustaining living community in relation to the physical environment. The biosphere is that portion of the earth in which ecosystems operate, and includes the biologically inhabited soil, water, and air.

2. **(C)** When XY and XX are crossed, the possible results are:

	X	Y
X	XX	XY
X	XX	XY

Male: 50% normal, 50% colour-blind
Female: 100% normal (50% heterozygous)

3. **(B)** Ade's father with a type A blood, can have either AA or AO genes, while his mother, with O blood, can have only OO genes, i.e.

	A	A
O	AO	AO
O	AO	AO

Result: 100% AO — Type A

	A	O
O	AO	OO
O	AO	OO

Result: 50% AO — Type A
50% OO — Type O

4. **(B)** Lichens are pioneer organisms because they first populate a given area, e.g. bare rock.

5. **(C)** As a scavenger, the hyena will loiter in the vicinity of an animal killed by a lion, and after the lion leaves, will hurry over to feast on the remains. At other times, as a predator, it will hunt and kill an antelope or other small animal.

6. **(D)** Producer organisms are green plants that contain chlorophyll and synthesize organic compounds by photosynthesis.

7. **(C)** After entering the nasal passages, air passes through the trachea into bronchi, the bronchial tubes, and finally into the alveoli, where oxygen enters the blood and where it picks up carbon dioxide given off by the blood.

8. **(A)** Hormones are produced by the ductless glands and are circulated in the blood plasma.

9. **(B)** Insects have six jointed legs; spiders have eight legs; they also have an exoskeleton.

10. **(A)** Viruses contain a central core of nucleic acid (DNA or RNA) surrounded by a protein covering, but no cytoplasm.

11. **(C)** An enzyme is an organic catalyst. If affects the rate of a chemical reaction without being changed.

12. **(B)** In a synapse, the nerve impulse passes from one neuron to another; there is no direct connection between the two neurons at this point.

13. **(E)** The blood is kept flowing in a forward direction by the pulsating pressure in the arteries. The flow in the veins is smooth, and may be against the force of gravity, as in the legs. The valves in the veins keep the blood moving toward the heart by preventing a backflow.

14. **(D)** An anaemic person has a lack of red blood cells. Haemoglobin is a major component of these cells, and contains iron.

15. **(D)** Chloroplasts are found in the cells of green plants and algae.

16. **(A)** Mitochondria are considered to be the powerhouses of the cell because they are the sites of aerobic respiration in which 36 ATP's are produced from each molecule of glucose.

17. **(C)** During respiration, organic molecules such as glucose are broken down, so that the energy released makes it possible for the addition of a phosphate group to ADP converting it to ATP.

18. **(D)** When *Amoeba* and *Paramecium* take in food it forms a food vacuole where enzymes digest it to a soluble form that then diffuses into the surrounding cytoplasm, where it is assimilated.

19. (C) Genes are located on DNA molecules which are contained within the chromosomes of the nucleus.

20. (D) Villi are tiny, fingerlike projections that line the insides of the small intestines, used for absorption.

21. (C) When the concentration of carbon dioxide increases in the blood passing through the medulla, the breathing rate is increased.

22. (A) Marine fish are not able to survive in fresh water because they cannot maintain homeostatic water balance. They are adapted to a water environment in which there is a relatively high concentration of salt.

23. (B) As the cell grows, the increase in surface lags behind the increase in volume.

24. (D) Abiotic factors are non-living factors in the environment like water, light, temperature, oxygen supply, minerals, pH and soil substrata, that affect living things.

25. (D) Consumers feed directly on green plants or on other consumers. Green plants are considered to be producers since they photosynthesize.

26. (C) In a food chain, green plants are the producers and the basis of the food. Mice as herbivores, live on green plants and are considered primary consumers. The snake preys on mice and is a secondary consumer. The hawk is a predator on the secondary consumer and frequently on the primary consumers.

27. (A) The producers are the basis of the pyramid of energy, since they carry on photosynthesis. They contain the greatest amount of energy. There is a loss of energy at each feeding level, from primary consumer through secondary to tertiary consumer.

28. (A) When blood passes through the lungs, oxygen of the air combines with the haemoglobin of the red blood cells to form oxyhaemoglobin.

29. (D) Hybrids produce dominant offspring in a ratio of 3:1 and since yellow is dominant over green there would be three times as many yellow (6,000) as green (2,000) seeds.

30. (D) The results would have been the same, i.e. 3 yellow : 1 green.

31. (C) The law of segregation states that, when large numbers of hybrids are crossed, the genes segregate and then recombine to produce dominant and recessive offsprings in a ratio of 3:1.

32. (D) Carbon fixation takes place during the dark phase of photosynthesis.

33. (C) This method of excretion serves as a water-conservation mechanism for certain land-dwelling organisms.

34. (B) Acetylcholine is produced along the length of a neuron and plays a part in the transmission of a nervous impulse. It is also formed in a synapse.

35. (E) The seed is a matured ovule.

36. (C) Chloroplasts are green because they contain chlorophyll; red blood cells are red because they have haemoglobin.

37. (D) Only simple and soluble form of food can diffuse into the blood stream after digestion.

38. (D) Cowpea has two cotyledons in its seed.

39. (C) When hybrids are crossed, the results are:

	B	b
B	BB	Bb
b	Bb	bb

Result: 25% homozygous black, 50% heteroxzygous black) 75% black
25% white

40. (B) In guinea pigs, black coat is dominant to white. Black guinea pigs that produce white offspring must carry a recessive gene and must be heterozygous black. This can be shown as follows:

	B	b
B	BB	Bb
b	Bb	bb

Result: 25% BB — Homozygous black
50% Bb — heterozygous black
25% bb — white

41. **(D)** During active transport, the cell uses energy to move soluble materials across cell membrane from a region of low concentration to a region of high concentration. By diffusion, molecules of soluble substances pass through cell membranes from a region of high concentration to a region of low concentration. This is passive transport since use of energy is not involved. In pinocytosis, relatively large particles, such as proteins, are engulfed by pockets of the cell membrane.

42. **(D)** As a mammal, the bat has internal fertilization, after which the embryo develops in the uterus and is nourished by the placent.

43. **(D)** Molusca are invertebrates, without a backbone.

44. **(D)** The spider and the crab are members of the Arthropoda phylum, whose characteristics include jointed appendages.

45. **(E)** The skin does not exhale air; this is done only through the lungs.

46. **(D)** Cambium consists of actively dividing cells that differentiate into xylem, phloem and additional cambium cells adding to the diameter of the trunk.

47. **(E)** A complex chain of organic molecules may be broken down during the process of hydrolysis into smaller molecules.

48. **(C)** Crossing-over takes place during meiosis, and not mitosis, when two homologous chromosomes pair up and exchange adjacent parts.

49. **(C)** Without green plants there would be no food. Carnivorous animals eat herbivorous animals, which depend on green plants for their food.

50. **(D)** Because of the water barrier, the plants and animals on the mainland were prevented from migrating to the islands and interbreeding. Therefore, changes that took place in one area were not transmitted to the other area. Over time, these changes became accentuated, making the mainland and island species quite different From each other. This illustrates the principle that evolution in isolated populations proceeds in different directions.

51. **(C)** The graph shows that the relative rate of enzyme activity rise as the enzyme concentration increases until the concentration is about 7. After that point, the relative rate of enzyme activity levels off at about 38, and remains the same.

52. **(B)** Water evaporates from the leaf by the process of transpiration.

53. **(D)** Exposure of the herbaceous plant to an actively operating electric fan will permit additional moisture to evaporate, leading to an increase in the rate of transpiration.

54. **(B)** When the leaves are coated with vaseline, the stomata are covered and are unable to give off any moisture. This brings transpiration to a halt, with no consequent water movement in the capillary tube.

55. **(C)** In the dark, the plant's activities are considerably reduced. Transpiration would proceed at a minimum level, resulting in a slight lowering of the level in the capillary tube.

56. **(D)** Parthenogenesis is the development of an organism from an infertilized egg.

57. **(A)** Ribosomes, which are tiny granules located on the endoplasmic reticulum of the cytoplasm, are the sites of protein synthesis in a cell.

58. **(D)** The RNA molecule contains uracil instead of thymine.

59. **(C)** In diabetes, where there is a shortage of insulin, the glucose remains unused and is excreted as waste.

60. **(A)** The grasshopper breathes by a system of tubes that connect with the openings or spiracles, in its abdomen.

61. **(D)** A clot in the coronary arteries which supply the cells of the heart with nourishment, will block the flow of blood through them causing heart attack. This condition is known as coronary thrombosis.

62. **(C)** During conditions of dryness, the guard cells reduce the size of the stomatal openings, thereby avoiding the excessive loss of water.

63. **(B)** When gamates are brought together during fertilization, there will be many different combinations of genes, resulting in a great deal of variety the offspring.

64. **(B)** Anaerobic respiration, also referred to as fermentation, takes place in the absence of oxygen. The only yield of energy is 2 ATP's plus lactic acid and alcohol.

65. **(B)** Lipases are digestive enzymes that break down the relatively large molecules of fat into smaller molecules of fatty acid and glycerol.

66. **(E)** An ecosystem is a community of plants and animals that are self-sustaining in relation to their physical environment. Equal numbers of plants and animals are not significant in these relationships.

67. **(B)** During oxidation (respiration) living things take in oxygen and give off carbon dioxide and water as wastes; during photosynthesis carbon dioxide and water are taken in by green plants and oxygen given off as a by-product. As a result, the concentrations of carbon dioxide and oxygen in the atmosphere remain relatively constant.

68. **(E)** Movement of minerals is against a concentration gradient and therefore requires the expenditure of energy which is made available during respiration. Also, in order to contract, a muscle cell requires energy, which is derived from respiration.

69. **(C)** The presence of HIV in a person's body leads to the development of AIDS.

70. **(E)** A carbohydrate contains the elements carbon, hydrogen, and oxygen, but with the ratio of hydrogen to oxygen being 2:1. Only I and III satisfy this condition.

71. **(C)** The fertilized egg multiplies by the process mitosis, resulting in all the somatic cells having the same number (in this case, 8) of chromosomes as the fertilized egg.

72. **(B)** Both lipids and carbohydrates contain carbon, hydrogen and oxygen. The ratio of hydrogen to oxygen in carbohydrates is 2:1 and in lipids not equal to 2:1. The ratio of hydrogen to oxygen in the analysed organic compound is not 2:1 therefore they are lipids.

73. **(C)** A heterotroph is an organism that cannot make its own food and depends directly or indirectly on green plants as the primary source of food. Animals are therefore heterotrophic as far as nutrition is concerned.

74. **(B)** The rabbit is a primary consumer that feeds directly on cabbage.

75. **(D)** The basic food in the marine biome is plankton, which is the basis of the food chain of the ocean.

76. **(E)** The lungs excrete carbon dioxide and water, kidneys excrete urine, containing complex molecules of wastes and the skin excretes water and some salts through its sweat glands.

77. **(C)** The stomach produces gastric juice, which consists of hydrochloric acid, water, and the enzymes pepsin and rennin.

78. **(A)** A community is defined as the populations of all the species living in a given area, here $30cm^2$ of soil.

79. **(E)** Adrenaline, produced by adrenal glands induce the muscles to have greater energy, so that the brain can think more clearly in an emergency.

80. **(D)** Oxygenated blood from the lungs flows through the pulmonary veins into the left auricle.

81. **(D)** Abiotic factors are non-living physical factors that affect living things. Predators are animals that kill other animals.

82. **(B)** Transfer of elements from inorganic sources to organic forms and back again to inorganic form forms a cycle.

83. **(D)** Biological control is the use of predators or parasites to control pests in the farm. The predator-prey relationship in the ecological community is exploited in checking the pests.

84. **(C)** Conservation is the planned use and protection of such natural resources as forests, soil and wildlife.

85. **(B)** Humus, the organic material of the soil, consists of decayed plant and animal matter.

86. **(A)** On steep slopes, terraces, or steps of soil and rock, are built across the hill and planted with soil-

binding plants. This method holds the water back and prevents the loss of soil.

87. **(B)** When soil is cultivated, it is ploughed and planted with crops. The binding action of roots on soil is weakened, leading to erosion. In addition, the mineral content is depleted by planting the same crop year after year.

88. **(A)** Lichen really consists of two organisms, a fungus and a one-celled alga, living together mutually. The alga makes food, while the fungus provides moisture and minerals.

89. **(B)** Yeast carries on anaerobic respiration, or fermentation, giving off the products carbon dioxide and alcohol. It is the carbondioxide that makes bread, cakes, etc., rise in bakeries.

90. **(C)** Fishes, amphibians and reptiles are cold-blooded animals whose body temperatures varies with the temperature of the environment.

MODEL QUESTIONS SET 2

INSTRUCTIONS: Each of the following questions or incomplete statements below is followed by five suggested answers or completions. Choose the one that is best in each case and then shade the corresponding space on the answer sheet.

1. In testing a substance for the presence of glucose, it should be heated with
   A. Benedict's solution
   B. Burrough's solution
   C. iodine solution
   D. iodine crystals
   E. glycogen

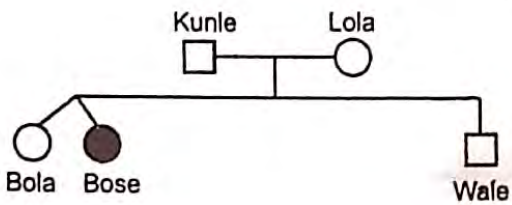
2. Which of the following would be most likely to occur in an ecosystem?
   A. As the number of prey decreases, the number of predators increases
   B. As the number of predators increases, the number of prey increases
   C. As the number of prey increases, the number of predators increases
   D. As the number of prey increases, the number of predators decreases
   E. As the number of predators decreases, the number of prey decreases

3. The transpiration rate of a tree would most probably be increased by
   A. increase in temperature only
   B. increase in both humidity and temperature
   C. increase in humidity and decrease in temperature
   D. increase in both humidity and air movement
   E. increase in both temperature and air movement

4. If a family has four children, three boys and a girl, what are the chances that the next child will be a girl?
   A. 0 B. 1 in 2 C. 1 in 3
   D. 1 in 4 E. 1 in 8

5. In which of the following organisms does aerobic respiration take place?
   I. Mammals
   II. Higher green plants
   III. Protozoa
   A. I only B. II only
   C. I and III only D. II and III only
   E. I, II and III

6. Which of the following statements about proteins is true?
   A. All proteins are enzymes
   B. All proteins have catalytic properties
   C. Proteins that are enzymes speed up chemical reactions dealing with the transformation of energy
   D. The 20-odd amino acids are synthesized from proteins
   E. DNA and RNA are highly structured proteins

7. 

In the pedigree of a family shown above, brown eyes are indicated as O and blue eyes as ● . Bola and Bose are twins. From this chart, it can be determined that:
   A. Kunle and Lola are homozygous for brown eyes
   B. Bola and Bose are identical twins
   C. Bose is heterozygous for blue eyes
   D. Bose is homozygous for blue eyes
   E. Bola and Wale are homozygous for brown eyes

8. A plant breeder obtained 252 red, 235 white and 503 pink flowers, when he crossed pink "four-o-clock". From this information, which of the following conclusions can be drawn?
   A. The phenotype of the hybrid is the same as its genotype
   B. The phenotype of the pure is different from its genotype
   C. The phenotype of the hybrid is different from its genotype
   D. The offspring show the phenotype of the dominant gene
   E. All of the above statements are false.

9. The following series of processes occurs in living things:

(i) $C_6H_{12}O_6 + 6O_2 \xrightarrow[\text{enzymes}]{\text{energy}} 6CO_2 + 6H_2O + \text{energy}$

(ii) $6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll + enzymes}]{\text{energy}} C_6H_{12}O_6 + 6H_2O + 6O_2 + \text{energy}$

Which of the following organisms is capable of performing the processes represented by both equations (i) and (ii)?

A. Amoeba B. An earthworm
C. Mucor D. Hibiscus plant
E. None of these organisms

10. Which statement concerning ribosomes is true?

A. They function in cell division
B. They contain DNA
C. They are the sites of carbohydrate synthesis
D. They are the sites of the protein synthesis
E. They permit cells to contract

11. In which of the following organisms does transpiration take place?

I. Cat II. Palm tree
III. *Paramecium*

A. I only B. II only
C. I and III only D. II and III only
E. I, II and III

12. During the process of respiration, energy is transferred from glucose molecules to molecules of

A. ACTH B. DNA C. RNA
D. ATP E. BCG

13. In humans, large amounts of carbohydrates are stored in the

A. liver B. gall bladder
C. spleen D. coccyx
E. pancreas

14. Which of the following is illustrated by an organism that exists in a mutually helpful relationship with another organism?

A. photosynthesis B. autotropism
C. saprophytism D. parasitism
E. mutualism

15. The tube that brings urine from the kidneys to the bladder is the

A. urethra B. ureter
C. uterus Eustachian tube
E. bile duct

16. A nerve cell that receives stimuli from the outside is known as a(n)

A. motor neuron B. sensory neuron
C. efferent neuron D. relay neuron
E. ganglion

17. Which of the following best describes the most important role of plankton on the surface of oceans?

A. It serves as the basis of the food chain for marine life
B. It is useful as human food
C. It is symbiotic to land forms of microscopic life
D. It serves directly as the basic diet of most fish
E. It is the basis of autotrophic nutrition

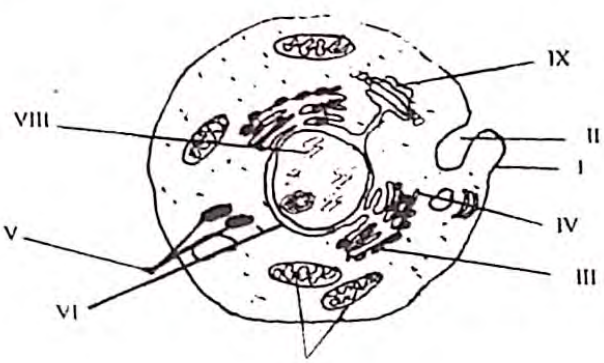
18. When solar energy reaches a green plant cell, which of the following occurs?

A. The solar energy is absorbed
B. All of the wavelengths of the solar energy are reflected by the green chlorophyll
C. An electron is created
D. An electron drops from its normal energy value to a lower level
E. An electron becomes "excited" and is destroyed

19. The thyroid gland

A. is known as the gland of emergency
B. stimulates the liver to store glycogen
C. controls the production of ACTH
D. regulates the rate of metabolism
E. contains the islets of Langerhans

Use the diagram below to answer questions **20–24**



20. Which of the following functions is the site of protein synthesis?

A. VIII B. V C. III
D. VII E. II

21. All of the following are organelles except
A. VII B. V C. IX
D. II E. III

22. Which of the following is believed to function in intracellular transport?
I. III B. VII C. VIII
D. IV E. V

23. Cellular respiration occurs at
A. VII B. IX C. VIII
D. V E. IV

24. Of the following, which contains molecules of DNA?
A. V B. III C. IX
D. IV E. VIII

25. The nucleus contains all of the following structures except
A. mitochondria B. chromatin
C. genes D. nucleolus
E. nuclear membrane

26. The most abundant substance in protoplasm is
A. protein B. fat
C. carbohydrate D. water
E. minerals

27. The conversion of non-living material into living protoplasm is known as
A. assimilation B. respiration
C. reproduction D. absorption
E. digestion

28. A cell obtains energy during the process of
A. ingestion B. respiration
C. irritability D. excretion
E. secretion

29. Animal cells do not possess
A. a cell membrane
B. a cell wall
C. cytoplasm
D. a nucleus
E. a nuclear membrane

30. Digestion in one-celled animals takes place in the
A. cyst B. contractile vacuole
C. food vacuole D. pellicle
E. gullet

31. Dissolved gases pass in and out of *Paramecium* through the
A. nuclear membrane
B. cell membrane
C. food vacuole
D. micronucleus
E. macronucleus

32. The *Euglena* is different from the *Amoeba* and *Paramecium* in possessing
A. cytoplasm B. a nucleus
C. a cell membrane D. chloroplast
E. spiral filament

33. The spiral structure in *Spirogyra* is the
A. Spireme B. Chloroplast
C. Chromatin D. Nucleus
E. Spiral filament

34. A combination of a phosphate, a ribose, and a nitrogen base is characteristic of a
A. carbohydrate B. lipid
C. protein D. nucleotide
E. fatty acid

35. The storehouse of energy in the cell is
A. ATP
B. the nucleus
C. the cell membrane
D. DNA
E. RNA

36. The release of the chemical-bond energy of food is called
A. digestion B. irritability
C. respiration D. ingestion
E. diffusion

37. A group of similar cells performing the same function is known as a(n)
A. organ B. system
C. tissue D. layer
E. membrane

38. An axon is part of a(n)
A. muscle cell B. nerve cell
C. epithelial cell D. bone cell
E. cartilage cell

39. Structures that conduct materials throughout a plant are called
A. lenticels B. meristem
C. vascular bundles D. cambium
E. epidermis

40. The growing region of a plant stem is called the
A. bark B. meristem
C. wood D. spongy layer
E. phloem

41. Cambium is located next to the
A. stomata B. guard cells
C. epidermis D. root hairs
E. xylem and phloem

42. A by-product of photosynthesis is
A. carbon dioxide B. water
C. carbon monoxide D. oxygen
E. nitrogen

43. The passage of gases into and out of a leaf is controlled by the
A. petiole B. guard cells
C. palisade layer D. spongy layer
E. veins

44. An autotroph makes food from
A. organic molecules
B. inorganic molecules
C. glucose
D. fats
E. proteins

45. The part of the spectrum in which photosynthesis is most active is
A. green-yellow B. yellow-orange
C. blue-violet D. blue-green
E. green-orange

46. Green plants carry on respiration
A. only during the day
B. only at night
C. during both day and night
D. only when photosynthesis is going on
E. only when photosynthesis is not going on

47. Mineral salts enter a plant largely through its
A. root hairs B. root cap
C. root tip D. stomata
E. guard cells

48. Green plants need nitrate for making
A. carbohydrates B. sucrose
C. starch D. fats
E. proteins

49. In the dark, a green plant excretes
A. hydrogen B. carbon dioxide
C. oxygen D. nitrogen
E. none of the above

50. All of the following are true of vitamin A except
A. it prevents night-blindness
B. it is formed from carotene
C. it is used in making visual purple
D. it is used in making haemoglobin
E. it is found in green, leafy vegetables

51. The proper clotting of blood depends on an adequate supply of vitamin
A. K B. B complex
C. D D. C
E. A

52. All of the following are parts of a tooth except
A. enamel B. dentine
C. root D. caries
E. pulp cavity

53. Pepsin is active only in the presence of
A. rennin B. erepsin
C. amylopsin D. hydrochloric acid
E. lipase

54. Bile is useful in the digestion of
A. starch B. sucrose
C. fat D. protein
E. amino acids

55. Food is moved along the alimentary canal by contractions known as
A. epiglottis B. circulation
C. secretion D. colic
E. peristalsis

56. The cells of the heart are nourished by the
A. carotid arteries B. coronary arteries
C. jugular vein D. portal vein
E. pulmonary arteries

57. All of the following statements about arteries are true except
A. they are thick-walled
B. they pulsate
C. they contain much elastic fibre tissue
D. the carry blood away from the heart
E. they contain valves

58. All of the following veins carry deoxygenated blood except the
A. superior vena cava
B. inferior vena cava
C. pulmonary vein
D. renal vein
E. hepatic vein

59. The liquid that bathes all the cells of the body is called
A. fibrinogen B. lymph
C. plasma D. blood
E. fibrin

60. Iron is needed for the
A. formation of white blood cells
B. formation of red blood cells
C. formation of platelets
D. prevention of goitre
E. prevention of diabetes

61. Red blood cells are formed in the
A. kidneys B. skeletal muscles
C. bone marrow D. cartilage
E. heart

62. An inherited disease in which blood does not clot is
A. thrombosis B. atherosclerosis
C. haemophilia D. heparin
E. anaemia

63. A person with Type A blood may safely receive a transfusion of
A. Type AB
B. Type A and Type AB
C. Type AB and Type O
D. Type A and Type O
E. none of these

64. Worn-out red blood cells are removed from the blood by the
A. lungs B. spleen
C. small intestine D. large intestine
E. pancreas

65. Serum is like plasma except that it has no
A. water B. fibrinogen
C. haemoglobin D. haemophlia
E. red blood cells

66. The correct order of structures (I-nasal cavity; II-bronchi; III-larynx; IV-alveoli; V-trachea) through which air passes is
A. I–V–III–II–IV
B. I–V–III–IV–II
C. I–III–IV–V–II
D. I–III–V–IV–II
E. I–III–V–II–IV

67. Oxyhaemoglobin is formed in the
A. lungs B. spleen
C. right auricle D. left auricle
E. bone marrow

68. During inhalation
A. the diaphragm flattens out, and he ribs are raised
B. the diaphragm is raised, and the ribs are lowered
C. the diaphragm flattens out, and the ribs are lowered
D. the diaphragm is raised, and the ribs are raised
E. the diaphragm is raised, and the ribs are stationary

69. Two wastes resulting from the oxidation of food are
A. oxygen and water
B. carbon dioxide and water
C. oxygen and carbon dioxide
D. oxygen and ATP
E. carbon dioxide and ATP

70. Urea is removed from the blood by the
A. lungs B. liver
C. kidneys D. spleen
E. urinary bladder

71. All of the following are functions of the human skin except
A. formation of vitamin D
B. sensation
C. protection against entrance of bacteria
D. excretion of carbon dioxide
E. regulation of body temperature

72. On a humid day, a person is uncomfortable because
A. he or she perspires more
B. there is less evaporation of sweat
C. sweat evaporates quickly
D. larger amounts of salt are excreted by the skin
E. the surface of the skin is too dry

73. Malpighian tubules are used for excretion in the
A. earthworm B. insect
C. hydra D. fish
E. frog

74. An element required in large amounts by the thyroid gland is
A. iron B. iodine
C. fluorine D. calcium
E. phosphorus

75. During a heart attack, a doctor may inject
A. thyroxin B. prolactin

C. insulin D. adrenalin
E. oestrogen

76. The action of the roots of a tree in growing into a drain pipe is an example of
A. positive hydrotropism
B. negative hydrotropism
C. positive geotropism
D. negative geotropism
E. negative chemotropism

77. The spinal cord serves as the centre of
A. subconscious thought
B. reflex actions
C. habits
D. tropisms
E. synpases

78. The sense of balance is maintained with the help of the
A. cochlea B. hammer
C. Anvil D. stirrup
E. semicircular canals

79. All of the following are examples of a reflex except
A. sneezing B. blinking
C. thinking D. coughing
E. contraction of the pupil of the eye

80. Vegetative propagation is a form of
A. sexual reproduction
B. asexual reproduction
C. conjugation
D. fertilization
E. vegetative pollination

81. All of the following are membranes in a bird's egg except the
A. chorion B. allantois
C. amnion D. yolk sac
E. albumen

82. The diploid number is restored as the result of
A. differentiation B. fertilization
C. cleavage D. meiosis
E. maturation

83. Spindle threads are attached to the chromosomes at the
A. asters B. mitochondria
C. prophase D. interphase
E. centromeres

84. When hybrids are crossed, the genotype ratio of the offspring is
A. 1:1 B. 3:1 C. 1:2:1
D. 4:1 E. 3:2

85. A change in genes is called a
A. genotype B. phenotype
C. cross-over D. back-cross
E. mutation

86. The presence of gill slits in a rabbit embryo indicates that
A. rabbits breathe by gills in the gastrula stage
B. rabbits descend from amphibians
C. vertebrates have a common ancestor
D. acquired characteristics can be inherited
E. the theory of regeneration may be true

87. The theory of use and disuse was based on
A. gene changes
B. chromosome changes
C. changes brought about by the environment
D. mitotic changes
E. mutations

88. According to Lamarck, evolution occurred as the result of
A. natural selection
B. the theory of recapitulation
C. overproduction
D. blending inheritance
E. inheritance of acquired characteristics

89. Green plants are the basis of life for
A. herbivorous animals
B. carnivorous animals
C. omnivorous animals
D. saprophytes
E. all of the above

90. All the members of a species inhabiting a given location comprise a(n)
A. community B. population
C. ecosystem D. biome
E. biosphere

Answer Key To Model Questions Set 2

1. A	19. D	37. C	55. E	73. B
2. C	20. C	38. B	56. B	74. B
3. E	21. D	39. C	57. E	75. D
4. B	22. D	40. B	58. C	76. A
5. E	23. A	41. E	59. B	77. B
6. C	24. E	42. D	60. B	78. E
7. D	25. A	43. B	61. C	79. C
8. A	26. D	44. B	62. C	80. B
9. D	27. A	45. C	63. D	81. E
10. D	28. B	46. C	64. B	82. B
11. B	29. B	47. A	65. B	83. E
12. D	30. C	48. E	66. E	84. C
13. A	31. B	49. B	67. A	85. E
14. E	32. D	50. D	68. A	86. C
15. B	33. B	51. A	69. B	87. C
16. B	34. D	52. D	70. C	88. E
17. A	35. A	53. D	71. D	89. E
18. A	36. C	54. C	72. B	90. B

Explanations To Model Questions Set 2

1. **(A)** When glucose is heated with Benedict's solution, a green to brick-red colour is obtained.

2. **(C)** The number of predators depends on their ability to catch prey. If more prey is available, the predators will be more successful and their numbers will increase.

3. **(E)** With an increase in temperature and air movement, there is an increase in the activities of the plant including transpiration.

4. **(B)** There is one chance out of two that a sperm with an X chromosome will unite with an egg, resulting in a girl.

5. **(E)** Aerobic respiration takes place in the presence of oxygen. Mammals, higher green plants and protozoa make use of oxygen.

6. **(C)** Enzymes are organic catalysts that can speed up the rate of a biochemical reaction without being changed.

7. **(D)** Blue eyes are a recessive trait; that is, two genes are needed for the trait to appear. Kunle and Lola are heterozygous for brown eyes, as seen from the appearance of blue-eyed Bose.

8. **(A)** Incomplete dominance, the hybrid shows a blending of the colours of both parents, and therefore is seen to have one gene from each. The phenotype is the pink colour of the hybrid The genotype is the genetic make up of the pink hybrid. This generation shows a 1:2:1 ratio.

9. **(D)** Equation (i) represents reactions in aerobic respiration. Equation (ii) represents the reactions involved in photosynthesis. ***Hibiscus*** plant carries on respiration all the time, and photo-synthesis in the light.

10. **(D)** Ribosomes are the sites of protein synthesis, including enzyme formation.

11. **(B)** Transpiration refers to evaporation of water vapour from the leaves of a plant, such as the palm tree.

12. **(D)** During cellular respiration, energy is released from glucose and then transferred to ADP with the addition of a high-energy phosphate molecule to convert the ADP to ATP.

13. **(A)** Glucose is stored in the liver as glycogen, or animal starch.

14. **(E)** Symbiotic organisms live together in an intimate relationship. In this relationship in which both organisms benefit is called mutualism.

15. **(B)** Each kidney is connected with the bladder by a ureter.

16. **(B)** A sensory or afferent neuron connects a sense organ with the brain or the spinal cord.

17. **(A)** Plankton is the basis of all life in the ocean, i.e. the basis of the food chain for marine life.

18. **(A)** In the green plant cell, solar energy of a particular range of wavelengths is absorbed by green pigment chlorophyll.

19. **(D)** The thyroid gland secretes thyroxin, which is carried to the cells of the body, where it controls the rate at which oxidation takes place and energy is released.

20. **(C)** Ribosome

21. **(D)** Pinocytic vesicle

22. **(D)** Endoplasmic reticulum

23. **(A)** Mitochondrion

24. **(E)** Chromosome

25. **(A)** Mitochondria are organelles located in the cytoplasm of the cell that are active in releasing energy from food.

26. **(D)** Water makes up from 65% to 90% of the protoplasm in various living things.

27. **(A)** Assimilation is the process by which non-living material in food is changed into the living material of the protoplasm.

28. **(B)** Energy is released in a cell during the process of tissue respiration.

29. **(B)** The cell wall is made largely of cellulose and is present only in plant cells.

30. (C) Digestion in one-celled animals takes place in the food vacuoles.

31. (B) The cell membrane is semi-permeable and controls the passage of dissolved materials into and out of the cell.

32. (D) The cell membrane is the outer layer of the cytoplasm in all types of living cells.

33. (B) The spiral chloroplast of spirogyra winds along the length of its cell.

34. (D) A nucleotide is made up of three types of molecular units viz: phosphate, ribose or deoxyribose and a nitrogen base.

35. (A) The storehouse of energy in a cell is ATP molecule.

36. (C) The release of the chemical-bond energy of food into energy that can be used for the life activities of the cell is known as respiration.

37. (C) A tissue is a group of similar cells performing the same function.

38. (B) An axon is a very large extension of a nerve cell or neuron.

39. (C) Vascular bundles conduct materials in plants and they are of two kinds of conducting tissues, i.e. xylem and phloem.

40. (B) The meristem contains actively growing and dividing cells. It is contained in the growing tip of a stem.

41. (E) Cambium contains actively growing cells and is found in the vascular bundles between xylem and phloem.

42. (D) Oxygen is the by-product of photosynthesis.

43. (B) The guard cells surround the stomata in the epidermis of a leaf. They control the passage of carbon dioxide and oxygen into and out of the leaf. They also control the passage of water vapour out of the leaf.

44. (B) An autotroph is an organism that can make its own food from inorganic molecules by converting it to organic molecules.

45. (C) Chlorophyll absorbs the blue-violet light of the spectrum most effectively, resulting in a high rate of photosynthesis.

46. (C) Green plants carry on respiration at all times as they release energy for their activities.

47. (A) Mineral salts are dissolved in soil water, which enters a plant through its root hairs.

48. (E) Nitrogen is one of the elements in the structure of proteins.

49. (B) In the dark, when photosynthesis is not going on, plants take in oxygen and give off carbon dioxide and water as a result of respiration.

50. (D) Iron is part of haemoglobin and not part of vitamin A.

51. (A) Vitamin K is necessary for the formation of prothrombin, one of the substances involved in the clotting of blood.

52. (D) Caries is a condition of tooth decay caused by bacteria in the mouth.

53. (D) Pepsin is an enzyme present in gastric juice in the stomach. Gastric juice contains hydrochloric acid.

54. (C) Bile does not act as an enzyme but it emulsifies fat breaking it into very small particles.

55. (E) Peristalsis is the alternate wavelike contractions and expansions of the involuntary muscles in the walls of the alimentary canal that move food along.

56. (B) The cells of the heart receive nourishment through the blood coming from the coronary arteries around it.

57. (E) Valves are found in veins, not in arteries. Valves prevent the blood from flowing backward.

58. (C) All veins carry deoxygenated blood except the pulmonary vein, which carries blood from the lungs to the left side of the heart. The blood becomes oxygenated in the lungs.

59. (B) All the cells of the body are surrounded by, and bathed by, a watery intercellular fluid called lymph.

60. **(B)** Iron is needed to form haemoglobin, a conjugated protein in red blood cells.

61. **(C)** Red blood cells are manufactured in the marrow of the bones.

62. **(C)** In people with haemophilia, the blood does not clot normally.

63. **(D)** A person with Type A blood can safely receive a transfusion from another person with Type A blood, which has the same agglutinogens or antibodies and from a person with Type O blood, who is called a universal donor because there are no agglutinogens in Type O blood.

64. **(B)** Worn-out red blood cells are removed from the body by the spleen and the liver.

65. **(B)** Serum is like plasma, but without fibrinogen, which was used up in the clotting process.

66. **(E)** As air passes through the respiratory system, it goes through the nose and then on to the nasal cavity, the pharynx, the larynx, the trachea, the brochi, the bronchial tubes and finally the alveoli in the lungs.

67. **(A)** As blood passes through the lungs, the haemoglobin in the red blood cells absorbs the oxygen in the alveoli to form oxyhaemoglobin.

68. **(A)** During inhalation, the diaphragm contracts and flattens out; at the same time the rib muscles contract, causing the ribs to be raised.

69. **(B)** Carbon dioxide and water are produced during respiration when food is oxidized by the cells. They are then sent out of the body in the exhaled air.

70. **(C)** Urea is removed from the blood by the kidneys by the process of excretion.

71. **(D)** Carbon dioxide is excreted into the alveoli of the lungs and then exhaled out of the body during breathing.

72. **(B)** On a humid day, there is more moisture in the air, allowing for less evaporation of sweat from the skin. This prevents the cooling of the skin and hence increase in body temperature. This results in uncomfortable feeling.

73. **(B)** The excretory system of insects consists of a number of small Malpighian tubes.

74. **(B)** The thyroid gland absorbs large amounts of iodine from the blood to make the hormone thyroxin.

75. **(D)** Adrenaline stimulates the heart to beat faster, and also acts to increase the breathing rate. For these reasons it may be given to a person suffering a heart attack.

76. **(A)** Roots show positive hydrotropism as they grow in the direction of water.

77. **(B)** The spinal cord serves as the centre of a number of reflex actions that take place without the action of the brain.

78. **(E)** The semicircular canals found in the inner ear, filled with liquid and with nerve endings connected to the cerebellum, are responsible for maintaining balance.

79. **(C)** Sneezing, blinking, coughing and pupil contraction are all reflexes. There is no thought involved in a reflex; it is inborn and automatic. Thinking is not a reflex.

80. **(B)** Vegetative propagation, a form of asexual reproduction, involves only one parent that can reproduce by a vegetative part of the plant — stem, leaf or root.

81. **(E)** Alubumen is the white of an egg; it is secreted by glands in the oviduct around the yolk as the egg passes through the tube.

82. **(B)** When the gamates, each of which contains the haploid number, unite in fertilization, the fertilized egg will contain the diploid number.

83. **(E)** During the late prophase stage, spindle threads become attached to the centromeres of the chromosomes.

84. **(C)** A cross of hybrids yields a genotype ratio of 1:2:1 in which the genes are distributed as 25% homozygous dominant, 50% heterozygous dominant, and 25% homozygous recessive.

85. **(E)** Genes sometimes change, altering the inheritance of the characteristics they control. Such a change is called a mutation.

86. (C) Although rabbit breathes by lungs, its early embryo develops gill slits that later disappear. This indicates descent from some ancestor with gills.

87. (C) According to the theory of use and disuse, an animal adapts itself to its environment and an organ that is used increases in size, one that is not used remains small, and such changes are inherited.

88. (E) According to Lamarck, evolution occurred as a result of the inheritance of acquired characteristics.

89. (E) Green plants make their own food. Herbivorous animals eat the plants. Carnivorous animals eat the herbivores. Omnivorous animals eat both plant and animal foods. Saprophytes live on both plant and animal foods.

90. (B) All the members of a species living in a given area comprise a population.

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